Carl Friedrich Gauss

Arithmetical Papers

English translation

Volume II: arithmetical papers, reviews and manuscript studies

This selection contains the historical title matter and first three complete papers of Volume II: A New Proof of an Arithmetical Theorem (Articles 1–7), presented on January 15, 1808; Summation of Certain Special Series (Articles 1–36), presented on August 24, 1808; and New Proofs and Extensions of the Fundamental Theorem in the Theory of Quadratic Residues, presented on February 10, 1817. The third paper contains the introduction, fifth proof (Articles 1–3), sixth proof (Articles 1–6) and new algorithm (Articles 1–6). The fourth paper, Theory of Biquadratic Residues, First Paper (Articles 1–23), presented on April 5, 1825, is included in full. The fifth paper, Theory of Biquadratic Residues, Second Paper, presented on April 15, 1831, is included in full, with its historical title and Articles 24–76. The self-reviews of May 12 and September 19, 1808, March 10, 1817, April 11, 1825, and April 23, 1831 are also included. Seven reviews of other writings follow: March 11, 1809; March 23, 1812; November 3, 1814; November 7, 1816; August 9, 1817; December 19, 1825; and July 9, 1831. They concern a work on irrational radicals, the factor tables of Chernac and Burckhardt, Erchinger's construction of the regular seventeen-sided polygon, and Seeber's positive ternary quadratic forms. Their German originals, like those of the self-reviews, are supplied separately; the German reader presents modern German renderings. The first manuscript work, Solution of a Pure Congruence, is included in full: Analysis of Residues, Chapter VI, first part, Articles 237–252. Its German division title is also supplied separately in the original. General Investigations of Congruences, Chapter VIII of Analysis of Residues, contains Articles 330–375 as printed, including the unfinished ending of Article 375. R. Dedekind's historical remarks follow; their German original is supplied separately, and the German reader gives a modern German rendering. The title and Articles 1–16 of Further Development of the Investigations of Pure Equations follow. Later texts of Volume II lie beyond this selection. Volume I accompanies it in the collected reader.

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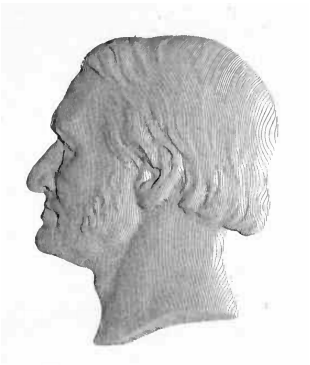
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A NEW PROOF

OF AN ARITHMETICAL THEOREM

BY

CARL FRIEDRICH GAUSS

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A NEW PROOF

OF AN ARITHMETICAL THEOREM.

1.

Questions in higher arithmetic very often present a singular phenomenon, which occurs much more rarely in analysis and contributes greatly to their charm. In analytic investigations we usually cannot reach new truths unless we have first thoroughly mastered the principles on which they rest and which must, as it were, open the way to them. In arithmetic, by contrast, most elegant new truths very frequently spring forth through induction and a kind of unexpected good fortune; their proofs lie so deeply hidden and are wrapped in such darkness that they elude every attempt and deny access to the keenest investigations. Moreover, so great and so wonderful is the connection among arithmetical truths that at first sight seem utterly different, that quite often, while we are seeking something quite different, we at last most happily reach the proof so greatly desired and previously sought in vain through long reflection, by a path far different from the one expected. Usually, however, truths of this kind are of such a nature that they can be approached by several very different paths, and the paths that first present themselves are not always the shortest. It must therefore certainly be highly valued if, after such a truth has long been investigated in vain, and has then indeed been proved, but by rather obscure detours, we finally succeed in discovering the simplest and genuine path.

2.

Among the questions discussed in the preceding article, a distinguished place is held by the theorem containing almost the whole theory of quadratic residues, which in the *Arithmetical Investigations* (Section IV) is distinguished by the name of the *fundamental theorem*. The illustrious LEGENDRE must undoubtedly be regarded as the *first* discoverer of this most elegant theorem, after the great geometers EULER and LAGRANGE had, long before, already discovered several of its special cases by induction. I do not pause here to enumerate these men's attempts at a proof; those who wish may consult the work just mentioned. Let me add only, in confirmation of what was said in the preceding article, what concerns my own attempts. I came upon the theorem itself independently in the year 1795, when I was entirely ignorant of everything that had already been worked out in higher arithmetic and wholly cut off from access to the literature. But for an entire year it tormented me and eluded my most strenuous efforts, until at last I obtained the proof given in the fourth section of that work. Afterwards three others, resting on wholly

different principles, presented themselves to me; I gave one in the fifth section, and the others, not inferior to it in elegance, I shall make public on another occasion. But all these proofs, although they seem to leave nothing to be desired in rigour, are derived from principles that are too different in kind, perhaps excepting the first, which nevertheless proceeds by more laborious reasoning and is burdened by rather lengthy operations. I therefore do not hesitate to declare that a *genuine* proof has so far been lacking. Let the experts now judge whether the one that I recently succeeded in discovering, and that the following pages present, deserves to be honoured with this name.

3.

THEOREM. *Let p be a positive prime number; k any integer not divisible by p; use the following definitions:*

A *collection of the numbers* $1,\ 2,\ 3\ \ldots.\ \frac{1}{2}(p-1)$

B *collection of these* $\frac{1}{2}(p+1),\ \frac{1}{2}(p+3),\ \frac{1}{2}(p+5)\ \ldots.\ p-1$

Take the products of k by the individual numbers in A, and their least positive residues modulo p, which are evidently all distinct and belong partly to A, partly to B. Now if altogether B contains μ of the residues, k will be either a quadratic residue or a quadratic nonresidue of p, according as μ is even or odd.

Proof. Let the residues belonging to A be $a,\ a'\ a''\ \ldots.$, and the remaining ones, belonging to B, be $b, b', b'' \ldots.$. It is clear that the complements of the latter, $p-b, p-b', p-b'' \ldots.$, are all distinct from the numbers $a,\ a',\ a''\ \ldots.$, but together with these fill out the collection A. Thus we have

$$1\,.\,2\,.\,3\ \ldots.\ \tfrac{1}{2}(p-1) = aa'a''\ \ldots.\ (p-b)(p-b')(p-b'')\ \ldots.$$

But the latter product evidently becomes

$$\equiv (-1)^{\mu} aa'a''\ \ldots.\ bb'b''\ \ldots.\ \equiv (-1)^{\mu} k\,.\,2k\,.\,3k\ \ldots.\ \tfrac{1}{2}(p-1)k$$
$$\equiv (-1)^{\mu} k^{\frac{1}{2}(p-1)}\,1\,.\,2\,.\,3\ \ldots.\ \tfrac{1}{2}(p-1) (\mathrm{mod}.p)$$

Hence we have

$$1 \equiv (-1)^{\mu} k^{\frac{1}{2}(p-1)}$$

or $k^{\frac{1}{2}(p-1)} \equiv \pm 1$, according as μ is even or odd; our theorem follows immediately.

4.

We can greatly shorten the following reasoning by introducing some suitable notation. Thus the symbol (k, p) will denote for us the number of products among

$$k,\ 2k,\ 3k\ \ldots.\ \tfrac{1}{2}(p-1)k,$$

whose least positive residues modulo p exceed half of it. Further, when x is any noninteger quantity, we shall use the symbol $[x]$ to denote the integer immediately smaller than x itself, so that $x - [x]$ is always a positive quantity lying between the limits 0 and 1. The following relations are now obtained with little trouble:

I. $[x] + [-x] = -1$.

II. $[x] + h = [x + h]$, whenever h is an integer.

III. $[x] + [h - x] = h - 1$.

IV. If $x - [x]$ is a fraction smaller than $\frac{1}{2}$, then $[2x] - 2[x] = 0$; but if $x - [x]$ is greater than $\frac{1}{2}$, then $[2x] - 2[x] = 1$.

V. Thus, when the least positive residue of the integer h modulo p lies below $\frac{1}{2}p$, we have $\left[\frac{2h}{p}\right] - 2\left[\frac{h}{p}\right] = 0$; but when that residue lies above $\frac{1}{2}p$, we have $\left[\frac{2h}{p}\right] - 2\left[\frac{h}{p}\right] = 1$.

VI. It follows at once that $(k, p) =$

$$\left[\frac{2k}{p}\right] + \left[\frac{4k}{p}\right] + \left[\frac{6k}{p}\right] \ldots. + \left[\frac{(p-1)k}{p}\right]$$
$$- 2\left[\frac{k}{p}\right] - 2\left[\frac{2k}{p}\right] - 2\left[\frac{3k}{p}\right] \ldots. - 2\left[\frac{\frac{1}{2}(p-1)k}{p}\right].$$

VII. From VI and I we obtain without difficulty

$$(k, p) + (-k, p) = \tfrac{1}{2}(p-1)$$

It follows that $-k$ has either the same or the opposite relation to p (insofar as it is a quadratic residue or nonresidue of it) as $+k$, according as p is of the form $4n + 1$ or of the form $4n + 3$. In the former case -1 is evidently a residue, in the latter a nonresidue, of p.

VIII. We shall transform the formula given in VI as follows. By III we have

$$\left[\frac{(p-1)k}{p}\right] = k - 1 - \left[\frac{k}{p}\right],$$
$$\left[\frac{(p-3)k}{p}\right] = k - 1 - \left[\frac{3k}{p}\right],$$
$$\left[\frac{(p-5)k}{p}\right] = k - 1 - \left[\frac{5k}{p}\right] \ldots.$$

Applying these substitutions to the last $\frac{p\mp 1}{4}$ terms of the upper series in that expression, we have

first, whenever p is of the form $4n+1$

$$(k,p) = \tfrac{1}{4}(k-1)(p-1)$$
$$- 2\left\{\left[\frac{k}{p}\right] + \left[\frac{3k}{p}\right] + \left[\frac{5k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-3)k}{p}\right]\right\}$$
$$- \left\{\left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$

second, whenever p is of the form $4n+3$

$$(k,p) = \tfrac{1}{4}(k-1)(p+1)$$
$$- 2\left\{\left[\frac{k}{p}\right] + \left[\frac{3k}{p}\right] + \left[\frac{5k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$
$$- \left\{\left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$

IX. For the special case $k = +2$, the formulas just given yield $(2,p) = \frac{1}{4}(p\mp 1)$, taking the upper or lower sign according as p is of the form $4n+1$ or $4n+3$. Thus $(2,p)$ is even, and hence $2Rp$, whenever p is of the form $8n+1$ or $8n+7$; on the other hand, $(2,p)$ is odd and $2Np$, whenever p is of the form $8n+3$ or $8n+5$.

5.

THEOREM. *Let x be a positive noninteger quantity, among whose multiples x, $2x$, $3x \ldots$. up to nx none is an integer; put $[nx] = h$, from which it is easily concluded that among the multiples of the reciprocal quantity $\frac{1}{x}$, $\frac{2}{x}$, $\frac{3}{x} \ldots\ldots$ up to $\frac{h}{x}$ an integer is likewise not found. Then I say that we have*

$$\left.\begin{array}{l} [x] + [2x] + [3x] \,.\, + [nx] \\ + \left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \left[\frac{3}{x}\right] \ldots. + \left[\frac{h}{x}\right] \end{array}\right\} = nh.$$

Proof. In the series $[x] + [2x] + [3x] \ldots. + [nx]$, which we shall put $= \Omega$, the first terms up to and including the term in position $\left[\frac{1}{x}\right]$ are evidently all $= 0$; the following ones up to the term in position $\left[\frac{2}{x}\right]$ are all $= 1$; the following ones up to the term in position $\left[\frac{3}{x}\right]$ are all $= 2$, and so on. Hence we have

$$\left.\begin{array}{l} \Omega = 0 \times \left[\frac{1}{x}\right] \\ \quad + 1 \times \left\{\left[\frac{2}{x}\right] - \left[\frac{1}{x}\right]\right\} \\ \quad + 2 \times \left\{\left[\frac{3}{x}\right] - \left[\frac{2}{x}\right]\right\} \\ \quad + 3 \times \left\{\left[\frac{4}{x}\right] - \left[\frac{3}{x}\right]\right\} \\ \qquad \text{etc.} \\ \quad + (h-1) \left\{\left[\frac{h}{x}\right] - \left[\frac{h-1}{x}\right]\right\} \\ \quad + h \left\{n - \left[\frac{h}{x}\right]\right\} \end{array}\right\} = hn - \left[\frac{1}{x}\right] - \left[\frac{2}{x}\right] - \left[\frac{3}{x}\right] \ldots. - \left[\frac{h}{x}\right]$$

Q. E. D.

6.

THEOREM. *If k, p denote any positive odd numbers relatively prime to each other, we have*

$$\left.\begin{aligned}&\left[\frac{k}{p}\right]+\left[\frac{2k}{p}\right]+\left[\frac{3k}{p}\right]\ldots.+\left[\frac{\frac{1}{2}(p-1)k}{p}\right]\\&+\left[\frac{p}{k}\right]+\left[\frac{2p}{k}\right]+\left[\frac{3p}{k}\right]\ldots.+\left[\frac{\frac{1}{2}(k-1)p}{k}\right]\end{aligned}\right\}=\tfrac{1}{4}(k-1)(p-1).$$

Proof. Supposing, as we may, that $k<p$, we have $\frac{\frac{1}{2}(p-1)k}{p}$ less than $\frac{1}{2}k$, but greater than $\frac{1}{2}(k-1)$, and therefore $\left[\frac{\frac{1}{2}(p-1)k}{p}\right]=\frac{1}{2}(k-1)$. Thus the present theorem evidently follows immediately from the preceding one by putting there $\frac{k}{p}=x$, $\frac{1}{2}(p-1)=n$, and hence $\frac{1}{2}(k-1)=h$.

Moreover, it can be proved in a similar manner that, if k is an *even* number relatively prime to p, we have

$$\left.\begin{aligned}&\left[\frac{k}{p}\right]+\left[\frac{2k}{p}\right]+\left[\frac{3k}{p}\right]\ldots.+\left[\frac{\frac{1}{2}(p-1)k}{p}\right]\\&+\left[\frac{p}{k}\right]+\left[\frac{2p}{k}\right]+\left[\frac{3p}{k}\right]\ldots.+\left[\frac{\frac{1}{2}kp}{k}\right]\end{aligned}\right\}=\tfrac{1}{4}k(p-1)$$

But we do not dwell on this proposition, which is not needed for our purpose.

7.

The fundamental theorem now follows immediately by combining the preceding theorem with proposition VIII of Article 4. Namely, if k, p denote any unequal positive prime numbers, and we put

$$(k,p) + \left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots . + \left[\frac{\frac{1}{2}(p-1)k}{p}\right] = L$$

$$(p,k) + \left[\frac{p}{k}\right] + \left[\frac{2p}{k}\right] + \left[\frac{3p}{k}\right] \ldots . + \left[\frac{\frac{1}{2}(k-1)p}{k}\right] = M$$

it is clear from VIII of Article 4 that L and M are always even numbers. But by the theorem of Article 6 we have

$$L + M = (k,p) + (p,k) + \tfrac{1}{4}(k-1)(p-1)$$

Thus whenever $\frac{1}{4}(k-1)(p-1)$ is even, which happens if either both k, p or at least one of them is of the form $4n+1$, (k,p) and (p,k) must necessarily be either both even or both odd. But whenever $\frac{1}{4}(k-1)(p-1)$ is odd, which happens if both k, p are of the form $4n+3$, one of the numbers (k,p), (p,k) must necessarily be even and the other odd. Thus in the former case the relation of k to p and the relation of p to k (insofar as one is a residue or nonresidue of the other) are identical, and in the latter case opposite.

Q. E. D.

SUMMATION

OF CERTAIN

SPECIAL SERIES

BY

CARL FRIEDRICH GAUSS

PRESENTED TO THE SOCIETY AUGUST XXIV, MDCCCVIII.

Commentationes societatis regiae scientiarum Gottingensis recentiores. Vol. I.
Gottingae MDCCCXI.

SUMMATION

OF CERTAIN SPECIAL SERIES.

1.

Among the more notable truths to which the theory of circle division has opened the way, the summation proposed in Disquiss. Arithmet., Article 356, holds a distinguished place. This is due both to its particular elegance and remarkable fruitfulness, which another investigation will later give us occasion to discuss more fully, and to the unusual difficulties that beset its rigorous and complete proof. These difficulties should have been all the less expected because they concern a certain restriction of the theorem more than the theorem itself: once this restriction is disregarded, a proof is immediately available and follows very easily from the theory developed in that work. The theorem was given there in the following form. Suppose that n is a prime number. Represent all the quadratic residues of n between 1 and $n-1$ inclusive by a, and all the nonresidues between the same limits by b; finally, let ω be the arc $\frac{360^0}{n}$, and k any fixed integer not divisible by n. Then we have

I. for a value of n of the form $4m+1$,

$$\begin{aligned}
\Sigma \cos ak\omega &= -\tfrac{1}{2} \pm \tfrac{1}{2}\sqrt{n} \\
\Sigma \cos bk\omega &= -\tfrac{1}{2} \mp \tfrac{1}{2}\sqrt{n}, \text{ and hence} \\
\Sigma \cos ak\omega - \Sigma \cos bk\omega &= \pm\sqrt{n} \\
\Sigma \sin ak\omega &= 0 \\
\Sigma \sin bk\omega &= 0
\end{aligned}$$

II. for a value of n of the form $4m+3$,

$$\begin{aligned}
\Sigma \cos ak\omega &= -\tfrac{1}{2} \\
\Sigma \cos bk\omega &= -\tfrac{1}{2} \\
\Sigma \sin ak\omega &= \pm\tfrac{1}{2}\sqrt{n} \\
\Sigma \sin bk\omega &= \mp\tfrac{1}{2}\sqrt{n} \\
\Sigma \sin ak\omega - \Sigma \sin bk\omega &= \pm\sqrt{n}
\end{aligned}$$

These summations were proved with full rigor in the place cited, and the only difficulty that remains here is to determine the *sign* to be placed before the radical. It can indeed be shown without difficulty that this sign depends on the number k to the following extent: the *same* sign must hold for all values of k that are quadratic residues of n, and the opposite sign for all values of k

that are quadratic nonresidues of n. Thus the whole question turns on the value $k = 1$: as soon as the sign for this value is known, the signs for all the other values of k are immediately available as well. Yet this very question, which at first seems to be among the easier ones, presents us with unforeseen difficulties, and the method that had guided us this far without impediment utterly refuses any further help.

2.

Before proceeding further, it will be useful to work out some examples of our summation numerically; we should, however, first make a few general observations.

I. In the case where n is a prime number of the form $4m + 1$, let all the quadratic residues of n between 1 and $\frac{1}{2}(n-1)$ inclusive be represented by a', and all the nonresidues between the same limits by b'. All the $n - a'$ are known to be among the a, and all the $n - b'$ among the b. Since all the a', b', $n-a'$, $n-b'$ together evidently make up the entire collection of numbers 1, 2, 3 .... $n-1$, all the a' together with all the $n - a'$ comprise all the a; likewise all the b' together with all the $n - b'$ comprise all the b. Hence

$$\begin{aligned}
\Sigma \cos ak\omega &= \Sigma \cos a'k\omega + \Sigma \cos(n-a')k\omega \\
\Sigma \cos bk\omega &= \Sigma \cos b'k\omega + \Sigma \cos(n-b')k\omega \\
\Sigma \sin ak\omega &= \Sigma \sin a'k\omega + \Sigma \sin(n-a')k\omega \\
\Sigma \sin bk\omega &= \Sigma \sin b'k\omega + \Sigma \sin(n-b')k\omega
\end{aligned}$$

Now, since $\cos(n-a')k\omega = \cos a'k\omega$, $\cos(n-b')k\omega = \cos b'k\omega$, $\sin(n-a')k\omega = -\sin a'k\omega$, $\sin(n-b')k\omega = -\sin b'k\omega$, it follows at once that

$$\begin{aligned}
\Sigma \sin ak\omega &= \Sigma \sin a'k\omega - \Sigma \sin a'k\omega = 0 \\
\Sigma \sin bk\omega &= \Sigma \sin b'k\omega - \Sigma \sin b'k\omega = 0
\end{aligned}$$

The summation of the cosines takes the form

$$\begin{aligned}
\Sigma \cos ak\omega &= 2\Sigma \cos a'k\omega \\
\Sigma \cos bk\omega &= 2\Sigma \cos b'k\omega
\end{aligned}$$

so that we must have

$$\begin{aligned}
1 + 4\Sigma \cos a'k\omega &= \pm\sqrt{n} \\
1 + 4\Sigma \cos b'k\omega &= \mp\sqrt{n} \\
2\Sigma \cos a'k\omega - 2\Sigma \cos b'k\omega &= \pm\sqrt{n}
\end{aligned}$$

II. In the case where n is of the form $4m + 3$, the complement of any residue a to n is a nonresidue, and the complement of any b is a residue. Thus all the $n - a$ coincide with all the b, and all the $n - b$ with all the a. This gives

$$\Sigma \cos ak\omega = \Sigma \cos(n - b)k\omega = \Sigma \cos bk\omega$$

Since all the a and b together make up all the numbers 1, 2, $3 \ldots . n - 1$, and hence $\Sigma \cos ak\omega + \Sigma \cos bk\omega = \cos k\omega + \cos 2k\omega + \cos 3k\omega + \text{etc.} + \cos(n - 1)k\omega = -1$, the summations

$$\Sigma \cos ak\omega = -\tfrac{1}{2}$$
$$\Sigma \cos bk\omega = -\tfrac{1}{2}$$

follow immediately. Likewise,

$$\Sigma \sin ak\omega = \Sigma \sin(n - b)k\omega = -\Sigma \sin bk\omega$$

which shows how the two summations

$$2\Sigma \sin ak\omega = \pm\sqrt{n}$$
$$2\Sigma \sin bk\omega = \mp\sqrt{n}$$

depend on one another.

3.

Here are the numerical calculations for several examples:

I. For $n = 5$ there is one value of a', namely $a' = 1$, and one value of b', namely $b' = 2$. We have

$$\cos \omega = +0{,}3090169944 \qquad \cos 2\omega = -0{,}8090169944$$

and hence $1 + 4 \cos \omega = +\sqrt{5}$, $1 + 4 \cos 2\omega = -\sqrt{5}$.

II. For $n = 13$ there are three values of a', namely 1, 3, 4, and as many values of b', namely 2, 5, 6. We calculate

$\cos \omega$	$= +0{,}8854560257$	$\cos 2\omega$	$= +0{,}5680647467$
$\cos 3\omega$	$= +0{,}1205366803$	$\cos 5\omega$	$= -0{,}7485107482$
$\cos 4\omega$	$= -0{,}3546048870$	$\cos 6\omega$	$= -0{,}9709418174$
Sum	$= +0{,}6513878190$	Sum	$= -1{,}1513878189$

Hence $1 + 4\Sigma \cos a'\omega = +\sqrt{13}$, $1 + 4\Sigma \cos b'\omega = -\sqrt{13}$.

III. For $n = 17$ we have four values of a', namely 1, 2, 4, 8, and as many values of b', namely 3, 5, 6, 7. The cosines are therefore calculated as follows:

$$\begin{array}{rcl} \cos\omega &=& +0{,}9324722294 \\ \cos 2\omega &=& +0{,}7390089172 \\ \cos 4\omega &=& +0{,}0922683595 \\ \cos 8\omega &=& -0{,}9829730997 \\ \hline \text{Sum} &=& +0{,}7807764064 \end{array} \qquad \begin{array}{rcl} \cos 3\omega &=& +0{,}4457383558 \\ \cos 5\omega &=& -0{,}2736629901 \\ \cos 6\omega &=& -0{,}6026346364 \\ \cos 7\omega &=& -0{,}8502171357 \\ \hline \text{Sum} &=& -1{,}2807764065 \end{array}$$

Hence $1 + 4\Sigma \cos a'\omega = +\sqrt{17}, \quad 1 + 4\Sigma \cos b'\omega = -\sqrt{17}$.

IV. For $n = 3$ there is just one value of a, namely $a = 1$, to which corresponds

$$\sin\omega = +0{,}8660254038$$

Hence $2 \sin\omega = +\sqrt{3}$.

V. For $n = 7$ there are three values of a, namely 1, 2, 4. These give the sines

$$\begin{array}{rcl} \sin\omega &=& +0{,}7818314825 \\ \sin 2\omega &=& +0{,}9749279122 \\ \sin 4\omega &=& -0{,}4338837391 \\ \hline \text{Sum} &=& +1{,}3228756556, \end{array}$$ and hence $2\Sigma \sin a\omega = +\sqrt{7}$.

VI. For $n = 11$ the values of a are 1, 3, 4, 5, 9, to which correspond the sines

$$\begin{array}{rcl} \sin\omega &=& +0{,}5406408175 \\ \sin 3\omega &=& +0{,}9898214419 \\ \sin 4\omega &=& +0{,}7557495744 \\ \sin 5\omega &=& +0{,}2817325568 \\ \sin 9\omega &=& -0{,}9096319954 \\ \hline \text{Sum} &=& +1{,}6583123952, \end{array}$$ and hence $2\Sigma \sin a\omega = +\sqrt{11}$.

VII. For $n = 19$ the values of a are 1, 4, 5, 6, 7, 9, 11, 16, 17, to which correspond the sines

$$\begin{aligned}
\sin\omega &= +0{,}3246994692\\
\sin 4\omega &= +0{,}9694002659\\
\sin 5\omega &= +0{,}9965844930\\
\sin 6\omega &= +0{,}9157733267\\
\sin 7\omega &= +0{,}7357239107\\
\sin 9\omega &= +0{,}1645945903\\
\sin 11\omega &= -0{,}4759473930\\
\sin 16\omega &= -0{,}8371664783\\
\sin 17\omega &= -0{,}6142127127\\
\text{Sum} &= +2{,}1794494718,
\end{aligned}$$

and hence $2\Sigma \sin a\omega = +\sqrt{19}$.

4.

In all these examples the radical takes the positive sign. This is also easily confirmed for the larger values $n = 23$, $n = 29$, etc., and already makes it highly probable that the same holds in general. But a proof of this phenomenon cannot be drawn from the principles set out in the place cited, and it must with full justification be regarded as a matter for deeper investigation. The purpose of this paper is therefore to present a rigorous proof of this most elegant theorem, attempted in vain in various ways over several years and finally achieved through unusual and quite subtle considerations. At the same time we shall bring the theorem to a far greater degree of generality, preserving, or rather increasing, its elegance. Finally, as a crowning addition, we shall show the remarkable and very close connection between this summation and another arithmetical theorem of the greatest importance. We hope that these investigations will themselves be welcome to geometers, and that the methods by which we have accomplished all this, and which may also be useful on other occasions, will be considered worthy of their attention.

5.

Our proof comes from considering a special kind of series whose terms depend on expressions such as

$$\frac{(1-x^m)\,(1-x^{m-1})\,(1-x^{m-2})\ldots.(1-x^{m-\mu+1})}{(1-x)\,(1-xx)\,(1-x^3)\ldots..(1-x^\mu)}$$

For brevity we shall denote such a fraction by (m, μ), and first make a few general observations about functions of this kind.

I. Whenever m is a positive integer less than μ, the function (m, μ) evidently vanishes, since the numerator contains the factor $1 - x^0$. For $m = \mu$, the factors in the numerator are identical, in reverse order, to those in the denominator, so that $(\mu, \mu) = 1$. Finally, for the case where m is a positive integer greater than μ, we have the formulas

$$(\mu+1,\mu) = \frac{1-x^{\mu+1}}{1-x} = (\mu+1,1)$$
$$(\mu+2,\mu) = \frac{(1-x^{\mu+2})\,(1-x^{\mu+1})}{(1-x)\qquad(1-xx)} = (\mu+2,2)$$
$$(\mu+3,\mu) = \frac{(1-x^{\mu+3})(1-x^{\mu+2})(1-x^{\mu+1})}{(1-x)\qquad(1-xx)\qquad(1-x^3)} = (\mu+3,3) \text{ etc.}$$

or, in general,

$$(m,\mu) = (m, m-\mu)$$

II. Further, it is easily confirmed that in general

$$(m,\mu+1) = (m-1,\mu+1) + x^{m-\mu-1}(m-1,\mu)$$

and since we likewise have

$$(m-1,\mu+1) = (m-2,\mu+1) + x^{m-\mu-2}(m-2,\mu)$$
$$(m-2,\mu+1) = (m-3,\mu+1) + x^{m-\mu-3}(m-3,\mu)$$
$$(m-3,\mu+1) = (m-4,\mu+1) + x^{m-\mu-4}(m-4,\mu), \text{ etc.}$$

and can continue this series as far as

$$\begin{aligned}(\mu+2,\mu+1) &= (\mu+1,\mu+1) + x(\mu+1,\mu)\\ &= (\mu,\mu) + x(\mu+1,\mu)\end{aligned}$$

provided that m is a positive integer greater than $\mu+1$, we obtain

$$\begin{aligned}(m,\mu+1) = (\mu,\mu) &+ x(\mu+1,\mu) + xx(\mu+2,\mu)\\ &+ x^3(\mu+3,\mu) + \text{etc.}\\ &+ x^{m-\mu-1}(m-1,\mu)\end{aligned}$$

This shows that if, for some fixed value of μ, every function (m, μ) is a polynomial whenever m is a positive integer, then every function $(m, \mu + 1)$ must also be a polynomial. Since that supposition holds for $\mu = 1$, it therefore also holds for $\mu = 2$, and hence for $\mu = 3$, etc. That is, in general, for any positive integral value of m, (m, μ) is a polynomial, or equivalently the product

$$(1 - x^m)(1 - x^{m-1})(1 - x^{m-2}) \dots (1 - x^{m-\mu+1})$$

is divisible by

$$(1 - x)(1 - x^2)(1 - x^3) \dots (1 - x^\mu)$$

6.

We shall now consider two series, both of which can lead to our goal. The first series is

$$1 - \frac{1-x^m}{1-x} + \frac{(1-x^m)\,(1-x^{m-1})}{(1-x)\,(1-xx)} - \frac{(1-x^m)(1-x^{m-1})(1-x^{m-2})}{(1-x)\quad(1-xx)\quad(1-x^3)} + \text{etc.}$$

or

$$1 - (m,1) + (m,2) - (m,3) + (m,4) - \text{etc.}$$

which for brevity we shall denote by $f(x,m)$. First, it is immediately evident that whenever m is a positive integer this series *terminates* after its term in position $m+1$ (with $= \pm 1$), and that in this case its sum must therefore be a finite polynomial in x. Further, Article 5, II, shows that in general, for any value of m,

$$\begin{aligned} 1 &= 1 \\ -(m,1) &= -(m-1,1) - x^{m-1} \\ +(m,2) &= +(m-1,2) + x^{m-2}(m-1,1) \\ -(m,3) &= -(m-1,3) - x^{m-3}(m-1,2), \text{ etc.} \end{aligned}$$

and hence

$$\begin{aligned} f(x,m) = 1 - x^{m-1} &- (1-x^{m-2})(m-1,1) \\ &+ (1-x^{m-3})(m-1,2) \\ &- (1-x^{m-4})(m-1,3) + \text{etc.} \end{aligned}$$

But evidently

$$\begin{aligned} (1-x^{m-2})(m-1,1) &= (1-x^{m-1})(m-2,1) \\ (1-x^{m-3})(m-1,2) &= (1-x^{m-1})(m-2,2) \\ (1-x^{m-4})(m-1,3) &= (1-x^{m-1})(m-2,3), \text{ etc.} \end{aligned}$$

from which we deduce the equation

$$f(x,m) = (1-x^{m-1})f(x,m-2) \ldots\ldots\ldots [1]$$

7.

Since for $m = 0$ we have $f(x, m) = 1$, the formula just found gives

$$\begin{aligned} f(x,2) &= 1 - x \\ f(x,4) &= (1-x)(1-x^3) \\ f(x,6) &= (1-x)(1-x^3)(1-x^5) \\ f(x,8) &= (1-x)(1-x^3)(1-x^5)(1-x^7), \text{ etc.} \end{aligned}$$

or, in general, for any even value of m,

$$f(x,m) = (1-x)(1-x^3)(1-x^5)\ldots.(1-x^{m-1}) \ldots [2].$$

Conversely, since for $m = 1$ we have $f(x, m) = 0$, we also have

$$\begin{aligned} f(x,3) &= 0 \\ f(x,5) &= 0 \\ f(x,7) &= 0 \text{ etc.} \end{aligned}$$

or, in general, for any odd value of m,

$$f(x,m) = 0$$

The latter summation could, moreover, already have been derived from the fact that in the series

$$1 - (m,1) + (m,2) - (m,3) + \text{etc.} + (m,m-1) - (m,m)$$

the last term cancels the first, the next to last cancels the second, etc.

8.

For our purpose the case where m is a positive odd integer is sufficient. But the unusual nature of the matter makes it worthwhile to add a few remarks about the cases where m is fractional or negative. Our series then evidently no longer terminates, but continues to infinity; it is also easily seen that it becomes divergent whenever x is given a value less than 1, so that its summation must be restricted to values of x greater than 1.

By formula [1] of Article 6 we have

$$f(x,-2) = \frac{1}{1-\frac{1}{x}}$$

$$f(x,-4) = \frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}}$$

$$f(x,-6) = \frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}} \cdot \frac{1}{1-\frac{1}{x^5}}, \text{ etc.}$$

so that the value of the function $f(x,m)$ can also be given in finite terms for a negative even integral value of m. For the remaining values of m, however, we shall convert the function $f(x,m)$ into an *infinite product* as follows.

As m tends to a negative *infinite* value, the function $f(x,m)$ becomes

$$1 + \frac{1}{x-1} + \frac{1}{x-1} \cdot \frac{1}{xx-1} + \frac{1}{x-1} \cdot \frac{1}{xx-1} \cdot \frac{1}{x^3-1} + \text{etc.}$$

This series is therefore equal to the infinite product

$$\frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}} \cdot \frac{1}{1-\frac{1}{x^5}} \cdot \frac{1}{1-\frac{1}{x^7}} \text{ etc. to infinity.}$$

Further, since in general

$$f(x,m) = f(x,m-2\lambda) \,.\, (1-x^{m-1})(1-x^{m-3}) \\ (1-x^{m-5}) \,..\, (1-x^{m-2\lambda+1})$$

we have

$$f(x,m) = f(x,-\infty) \,.\, (1-x^{m-1})(1-x^{m-3})(1-x^{m-5}) \\ \text{etc. to infinity.} \\ = \frac{1-x^{m-1}}{1-x^{-1}} \cdot \frac{1-x^{m-3}}{1-x^{-3}} \cdot \frac{1-x^{m-5}}{1-x^{-5}} \cdot \frac{1-x^{m-7}}{1-x^{-7}} \\ \text{etc. to infinity.}$$

whose factors evidently converge ever more closely to unity as they continue.

The case $m = -1$ deserves particular attention. Here

$$f(x,-1) = 1 + x^{-1} + x^{-3} + x^{-6} + x^{-10} + \text{etc.}$$

This series is therefore equal to the infinite product

$$\frac{1-x^{-2}}{1-x^{-1}} \cdot \frac{1-x^{-4}}{1-x^{-3}} \cdot \frac{1-x^{-6}}{1-x^{-5}} \text{ etc.}$$

or, writing x for x^{-1}, we obtain

$$1+x+x^3+x^6+\text{etc.} = \frac{1-xx}{1-x} \cdot \frac{1-x^4}{1-x^3} \cdot \frac{1-x^6}{1-x^5} \cdot \frac{1-x^8}{1-x^7} \text{ etc.}$$

This equality between two rather intricate expressions, to which we shall return on another occasion, is certainly very remarkable.

9.

Secondly, we shall consider the series

$$1 + x^{\frac{1}{2}}\frac{1-x^m}{1-x} + x\frac{(1-x^m)(1-x^{m-1})}{(1-x)(1-xx)} + x^{\frac{3}{2}}\frac{(1-x^m)(1-x^{m-1})(1-x^{m-2})}{(1-x)(1-xx)(1-x^3)} + \text{etc.}$$

or

$$1 + x^{\frac{1}{2}}(m,1) + x(m,2) + x^{\frac{3}{2}}(m,3) + xx(m,4) + \text{etc.}$$

which we shall denote by $F(x,m)$. We restrict this investigation to the case where m is a positive integer, so that this series too always ends at its term in position $m+1$ (with $= x^{\frac{1}{2}m}(m,m)$). Since

$$(m,m) = 1, \quad (m,m-1) = (m,1), \quad (m,m-2) = (m,2), \text{ etc.}$$

the series can also be written as follows:

$$F(x,m) = x^{\frac{1}{2}m} + x^{\frac{1}{2}(m-1)}(m,1) + x^{\frac{1}{2}(m-2)}(m,2) + x^{\frac{1}{2}(m-3)}(m,3) + \text{etc.}$$

This gives

$$\begin{aligned}(1+x^{\frac{1}{2}m+\frac{1}{2}})F(x,m) = 1 &+ x^{\frac{1}{2}}(m,1) + x(m,2) \\ &+ x^{\frac{3}{2}}(m,3) + \text{etc.} \\ &+ x^{\frac{1}{2}} \,.\, x^m + x \,.\, x^{m-1}(m,1) \\ &+ x^{\frac{3}{2}} \,.\, x^{m-2}(m,2) + \text{etc.}\end{aligned}$$

Since we have (Article 5, II)

$$\begin{aligned}(m,1)+x^m &= (m+1,1)\\ (m,2)+x^{m-1}(m,1) &= (m+1,2)\\ (m,3)+x^{m-2}(m,2) &= (m+1,3),\ \text{etc.}\end{aligned}$$

we obtain

$$(1+x^{\frac{1}{2}m+\frac{1}{2}})F(x,m) = F(x,m+1)\ \ldots\ldots\ [3]$$

But $F(x,0)=1$, so that

$$\begin{aligned}F(x,1) &= 1+x^{\frac{1}{2}}\\ F(x,2) &= (1+x^{\frac{1}{2}})(1+x)\\ F(x,3) &= (1+x^{\frac{1}{2}})(1+x)(1+x^{\frac{3}{2}}),\ \text{etc.}\end{aligned}$$

or, in general,

$$\begin{aligned}F(x,m) = (1+x^{\frac{1}{2}})&(1+x)(1+x^{\frac{3}{2}})\\ &\ldots.(1+x^{\frac{1}{2}m})\ \ldots.\quad [4]\end{aligned}$$

10.

With these preliminary investigations in place, let us now come closer to our aim. For a prime value of n, the squares 1, 4, $9\ldots.(\frac{1}{2}(n-1))^2$ are all incongruent to one another modulo n; their least residues for this modulus must therefore be identical to the numbers a, and hence

$$\begin{aligned}\Sigma\cos ak\omega &= \cos k\omega+\cos 4k\omega+\cos 9k\omega\\ &\quad+\text{etc.}+\cos(\tfrac{1}{2}(n-1))^2k\omega\\ \Sigma\sin ak\omega &= \sin k\omega+\sin 4k\omega+\sin 9k\omega\\ &\quad+\text{etc.}+\sin(\tfrac{1}{2}(n-1))^2k\omega\end{aligned}$$

Likewise, since the same squares 1, 4, $9\ldots.(\frac{1}{2}(n-1))^2$, in reverse order, are congruent to $(\frac{1}{2}(n+1))^2$, $(\frac{1}{2}(n+3))^2$, $(\frac{1}{2}(n+5))^2\ldots.(n-1)^2$, we also have

$$\begin{aligned}\Sigma\cos ak\omega &= \cos(\tfrac{1}{2}(n+1))^2k\omega\\ &\quad+\cos(\tfrac{1}{2}(n+3))^2k\omega+\text{etc.}+\cos(n-1)^2k\omega\\ \Sigma\sin ak\omega &= \sin(\tfrac{1}{2}(n+1))^2k\omega\\ &\quad+\sin(\tfrac{1}{2}(n+3))^2k\omega+\text{etc.}+\sin(n-1)^2k\omega\end{aligned}$$

Thus, putting

$$\begin{aligned} T &= 1 + \cos k\omega + \cos 4k\omega + \cos 9k\omega \\ &\quad + \text{etc.} + \cos(n-1)^2 k\omega \\ U &= \quad \sin k\omega + \sin 4k\omega + \sin 9k\omega \\ &\quad + \text{etc.} + \sin(n-1)^2 k\omega \end{aligned}$$

we obtain

$$\begin{aligned} 1 + 2\Sigma \cos ak\omega &= T \\ 2\Sigma \sin ak\omega &= U \end{aligned}$$

This shows that the summations proposed in Article 1 depend on summing the series T and U. We shall therefore leave the former aside and direct our investigation to these, carrying it out generally enough to include both prime values of n and any composite values. We shall, however, suppose that the number k is relatively prime to n: the case where k and n have a common divisor can be reduced to this one without difficulty.

11.

Let us denote the imaginary quantity $\sqrt{-1}$ by i and put

$$\cos k\omega + i \sin k\omega = r$$

Then $r^n = 1$, or r is a root of the equation $r^n - 1 = 0$. It is easy to see that all the numbers k, $2k$, $3k \ldots . (n-1)k$ are not divisible by n and are incongruent to one another modulo n. Thus the powers of r

$$1,\ r,\ rr,\ r^3 \ldots . r^{n-1}$$

are all unequal, while each also satisfies the equation $x^n - 1 = 0$. For this reason these powers represent *all* the roots of the equation $x^n - 1 = 0$.

These conclusions would not hold if k had a common divisor with n. For if ν were such a common divisor, $k \,.\, \dfrac{n}{\nu}$ would be divisible by n, and hence a power lower than r^n, namely $r^{\frac{n}{\nu}}$, would equal unity. In this case, therefore, the powers of r represent at most $\dfrac{n}{\nu}$ roots of the equation $x^n - 1 = 0$, and indeed give that many distinct roots if ν is the *greatest* common divisor of the numbers k, n. In our case, where k and n are supposed relatively prime, r can conveniently be called a *proper root* of the equation $x^n - 1 = 0$. In the other case, where k and n have a (greatest) common divisor ν, r would be called an *improper root* of that equation. Then, however, the same r would evidently be a proper root of the equation $x^{\frac{n}{\nu}} - 1 = 0$. The simplest improper root is unity, and when n is a prime number there are no other improper roots at all.

12.

Now, if we put

$$W = 1 + r + r^4 + r^9 + \text{etc.} + \text{etc.} + r^{(n-1)^2}$$

we evidently have $W = T + iU$, so that T is the real part of W, and U is obtained from the imaginary part of W by removing the factor i. The whole question is therefore reduced to finding the sum W. For this purpose we can use either the series considered in Article 6 or the one we showed how to sum in Article 9; the former, however, is less suitable when n is even. Nevertheless, we hope that readers will welcome our treating the case where n is odd by two methods.

First, then, suppose that n is an odd number, that r denotes any proper root of the equation $x^n - 1 = 0$, and that in the function $f(x, m)$ we put $x = r$ and $m = n - 1$. We evidently obtain

$$\begin{aligned}
\frac{1 - x^m}{1 - x} &= \frac{1 - r^{-1}}{1 - r} = -r^{-1} \\
\frac{1 - x^{m-1}}{1 - xx} &= \frac{1 - r^{-2}}{1 - rr} = -r^{-2} \\
\frac{1 - x^{m-2}}{1 - x^3} &= \frac{1 - r^{-3}}{1 - r^3} = -r^{-3}, \text{ etc.}
\end{aligned}$$

as far as

$$\frac{1 - x}{1 - x^m} = \frac{1 - r^{-m}}{1 - r^m} = -r^{-m}$$

(It is worth pointing out that these equations hold only insofar as r is supposed to be a proper root: if r were an improper root, the numerator and denominator of some of these fractions would vanish together, and the fractions would therefore become indeterminate).

From this we deduce the following equation

$$\begin{aligned}
f(r, n-1) &= 1 + r^{-1} + r^{-3} + r^{-6} + \text{etc.} + r^{-\frac{1}{2}(n-1)n} \\
&= (1 - r)(1 - r^3)(1 - r^5) \ldots . (1 - r^{n-2})
\end{aligned}$$

The same equation still holds if we replace r by r^λ, where λ denotes any integer relatively prime to n: for then r^λ too is a proper root of the equation $x^n - 1 = 0$. Let us therefore replace r by r^{n-2}, or equivalently r^{-2}. We obtain

$$\begin{aligned}
&1 + r^2 + r^6 + r^{12} + \text{etc.} + r^{(n-1)n} \\
&\quad = (1 - r^{-2})(1 - r^{-6})(1 - r^{-10}) \ldots (1 - r^{-2(n-2)})
\end{aligned}$$

Let us multiply both sides of this equation by

$$r \,.\, r^3 \,.\, r^5 \,.\,.\,.\,. \, r^{(n-2)} = r^{\frac{1}{4}(n-1)^2}$$

and, since

$$\begin{aligned}
r^{2+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-3)^2}, \\
r^{(n-1)n+\frac{1}{4}n-1)^2} &= r^{\frac{1}{4}(n+1)^2} \\
r^{6+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-5)^2}, \\
r^{(n-2)(n-1)+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n+3)^2} \\
r^{12+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-7)^2}, \\
r^{(n-3)(n-2)+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n+5)^2}, \text{ etc.}
\end{aligned}$$

the following equation results:

$$\begin{aligned}
&r^{\frac{1}{4}(n-1)^2} + r^{\frac{1}{4}(n-3)^2} + r^{\frac{1}{4}(n-5)^2} + \text{etc.} + r + 1 \\
&+ r^{\frac{1}{4}(n+1)^2} + r^{\frac{1}{4}(n+3)^2} + r^{\frac{1}{4}(n+5)^2} \\
&\qquad + \text{etc.} + r^{\frac{1}{4}(2n-2)^2} \\
&\qquad = (r - r^{-1})(r^3 - r^{-3})(r^5 - r^{-5}) \,.\,.\,.\,. \, (r^{n-2} - r^{-n+2})
\end{aligned}$$

or, arranging the terms on the left side differently,

$$\begin{aligned}
&1 + r + r^4 + \text{etc.} + r^{(n-1)^2} \\
&\qquad = (r - r^{-1})(r^3 - r^{-3}) \,.\,.\, (r^{n-2} - r^{-n+2}) \quad .. \quad [5]
\end{aligned}$$

13.

The factors on the right side of equation [5] can also be written as follows:

$$r - r^{-1} = -(r^{n-1} - r^{-n+1})$$
$$r^3 - r^{-3} = -(r^{n-3} - r^{-n+3})$$
$$r^5 - r^{-5} = -(r^{n-5} - r^{-n+5}), \text{ etc.}$$

up to

$$r^{n-2} - r^{-n+2} = -(r^2 - r^{-2})$$

The equation then takes this form:

$$W = (-1)^{\frac{1}{2}(n-1)}(r^2 - r^{-2})(r^4 - r^{-4})(r^6 - r^{-6}) \\ \ldots . (r^{n-1} - r^{-n+1})$$

Multiplying this equation by [5] in its original form gives

$$W^2 = (-1)^{\frac{1}{2}(n-1)}(r - r^{-1})(r^2 - r^{-2})(r^3 - r^{-3}) \\ \ldots . (r^{n-1} - r^{-n+1})$$

where $(-1)^{\frac{1}{2}(n-1)}$ is either $= +1$ or $= -1$, according as n is of the form $4\mu + 1$ or of the form $4\mu + 3$. Hence

$$W^2 = \pm r^{\frac{1}{2}n(n-1)}(1 - r^{-2})(1 - r^{-4})(1 - r^{-6}) \\ \ldots . (1 - r^{-2(n-1)})$$

But it is easy to see that r^{-2}, r^{-4}, $r^{-6} \ldots . r^{-2n+2}$ give all the roots of the equation $x^n - 1 = 0$ except the root $x = 1$. The following identity must therefore hold for an indeterminate:

$$(x - r^{-2})(x - r^{-4})(x - r^{-6}) \ldots . (x - r^{-2n+2}) \\ = x^{n-1} + x^{n-2} + x^{n-3} + \text{etc.} + x + 1$$

Thus, putting $x = 1$, we obtain

$$(1 - r^{-2})(1 - r^{-4})(1 - r^{-6}) \ldots . (1 - r^{-2n+2}) = n$$

and since evidently $r^{\frac{1}{2}n(n-1)} = 1$, our equation becomes

$$W^2 = \pm n \ldots . \ldots . \ldots . [6]$$

Thus, in the case where n is of the form $4\mu + 1$, we have

$$W = \pm\sqrt{n}, \quad \text{and hence} \quad T = \pm\sqrt{n}, \quad U = 0$$

In the other case, where n is of the form $4\mu + 3$, we instead have

$$W = \pm i\sqrt{n}, \quad \text{and so} \quad T = 0, \quad U = \pm\sqrt{n}$$

14.

The method of the preceding article gives only the absolute value of the sums T, U, and leaves it uncertain whether T in the first case and U in the second should be set $= +\sqrt{n}$ or $= -\sqrt{n}$. But, at least for the case where $k = 1$, this can be decided from equation [5] as follows. Since, for $k = 1$,

$$\begin{aligned} r - r^{-1} &= 2i \sin \omega \\ r^3 - r^{-3} &= 2i \sin 3\omega \\ r^5 - r^{-5} &= 2i \sin 5\omega \text{ etc.} \end{aligned}$$

that equation becomes

$$W = (2i)^{\frac{1}{2}(n-1)} \sin\omega \sin 3\omega \sin 5\omega \ldots \sin(n-2)\omega$$

Now, in the case where n is of the form $4\mu+1$, the sequence of odd numbers

$$1,\ 3,\ 5,\ 7 \ldots . \tfrac{1}{2}(n-3), \quad \tfrac{1}{2}(n+1) \ldots . (n-2)$$

contains $\frac{1}{4}(n-1)$ numbers smaller than $\frac{1}{2}n$, to which positive sines evidently correspond. The remaining $\frac{1}{4}(n-1)$, on the other hand, are greater than $\frac{1}{2}n$, and negative sines correspond to them. The product of all the sines must therefore equal a positive quantity multiplied by $(-1)^{\frac{1}{4}(n-1)}$. Thus W will equal a positive real quantity multiplied by i^{n-1}, or by 1, since $i^4 = 1$ and $n-1$ is divisible by 4: that is, W will be real and positive. Hence we must have

$$W = +\sqrt{n}, \quad T = +\sqrt{n}$$

In the other case, where n is of the form $4\mu + 3$, in the sequence of odd numbers

$$1,\ 3,\ 5,\ 7 \ldots . \tfrac{1}{2}(n-1), \quad \tfrac{1}{2}(n+3) \ldots . (n-2)$$

the first $\frac{1}{4}(n+1)$ will be smaller than $\frac{1}{2}n$, but the remaining $\frac{1}{4}(n-3)$ will be greater. Thus, among the sines of the arcs ω, 3ω, $5\omega \ldots . (n-2)\omega$, there will be $\frac{1}{4}(n-3)$ negative ones, and so W will be the product of $i^{\frac{1}{2}(n-1)}$, a positive real quantity and $(-1)^{\frac{1}{4}(n-3)}$. The third factor is $= i^{\frac{1}{2}(n-3)}$, which, multiplied by the first, gives $i^{n-2} = i$, since $i^{n-3} = 1$. We must therefore have

$$W = +i\sqrt{n}, \quad \text{and} \quad U = +\sqrt{n}$$

15.

We shall now show how the same conclusions can be derived from the series considered in Article 9. In equation [4], replace $x^{\frac{1}{2}}$ by $-y^{-1}$; we then obtain

$$1 - y^{-1}\frac{1-y^{-2m}}{1-y^{-2}} + y^{-2}\frac{(1-y^{-2m})(1-y^{-2m+2})}{(1-y^{-2})(1-y^{-4})} - y^{-3}\frac{(1-y^{-2m})(1-y^{-2m+2})(1-y^{-2m+4})}{(1-y^{-2})(1-y^{-4})(1-y^{-6})} + \text{etc.}$$

up to the term in position $m+1$

$$= (1-y^{-1})(1+y^{-2})(1-y^{-3})(1+y^{-4}) \ldots (1 \pm y^{-m}) \ldots . \text{ [7]}$$

If we take for y a proper root of the equation $y^n - 1 = 0$, say r, and at the same time put $m = n-1$, we obtain

$$\frac{1-y^{-2m}}{1-y^{-2}} = \frac{1-r^2}{1-r^{-2}} = -r^2$$
$$\frac{1-y^{-2m+2}}{1-y^{-4}} = \frac{1-r^4}{1-r^{-4}} = -r^4$$
$$\frac{1-y^{-2m+4}}{1-y^{-6}} = \frac{1-r^6}{1-r^{-6}} = -r^6$$

up to

$$\frac{1-y^{-2}}{1-y^{-2m}} = \frac{1-r^{2n-2}}{1-r^{-2n+2}} = -r^{2n-2}$$

Here we should note that none of the denominators $1-r^{-2}$, $1-r^{-4}$, etc. becomes $= 0$. Equation [7] therefore takes this form:

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} = (1-r^{-1})(1+r^{-2})(1-r^{-3}) \ldots . (1+r^{-n+1})$$

Multiplying, on the right side of this equation, the first term by the last, the second by the next to last, etc., we obtain

$$(1-r^{-1})(1+r^{-n+1}) = r - r^{-1}$$
$$(1+r^{-2})(1-r^{-n+2}) = r^{n-2} - r^{-n+2}$$
$$(1-r^{-3})(1+r^{-n+3}) = r^3 - r^{-3}$$
$$(1+r^{-4})(1-r^{-n+4}) = r^{n-4} - r^{-n+4} \text{ etc.}$$

It is easy to see that these partial products combine to form the product

$$(r - r^{-1})(r^3 - r^{-3})(r^5 - r^{-5}) \\ \ldots . (r^{n-4} - r^{-n+4})(r^{n-2} - r^{-n+2})$$

which will therefore be

$$= 1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} = W$$

This equation is identical with equation [5], derived in Article 12 from the first series; the remaining arguments are then established in the same way as in Articles 13 and 14.

16.

We turn to the other case, where n is an even number. First let n be of the form $4\mu + 2$, or twice an odd number. It is clear that the numbers $\frac{1}{4}nn$, $(\frac{1}{2}n + 1)^2 - 1$, $(\frac{1}{2}n + 2)^2 - 4$, etc., or in general $(\frac{1}{2}n + \lambda)^2 - \lambda\lambda$, give odd quotients when divided by $\frac{1}{2}n$, and are therefore congruent, modulo n, to $\frac{1}{2}n$. It follows that, if r is a proper root of the equation $x^n - 1 = 0$, and hence $r^{\frac{1}{2}n} = -1$, we have

$$\begin{aligned} r^{(\frac{1}{2}n)^2} &= -1 \\ r^{(\frac{1}{2}n+1)^2} &= -r \\ r^{(\frac{1}{2}n+2)^2} &= -r^4 \\ r^{(\frac{1}{2}n+2)^2} &= -r^9 \text{ etc.} \end{aligned}$$

Thus, in the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

the term $r^{(\frac{1}{2}n)^2}$ cancels the first, the next cancels the second, etc., so that

$$W = 0, \quad T = 0, \quad U = 0$$

17.

The case remains where n is of the form 4μ, or divisible by four. Here, in general, $(\frac{1}{2}n+\lambda)^2 - \lambda\lambda$ is divisible by n, and therefore

$$r^{(\frac{1}{2}n+\lambda)^2} = r^{\lambda\lambda}$$

Thus, in the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

the term $r^{(\frac{1}{2}n)^2}$ equals the first, the next equals the second, etc., so that

$$W = 2(1 + r + r^4 + r^9 + \text{etc.} + r^{(\frac{1}{2}n-1)^2})$$

Now suppose that in equation [7] of Article 15 we put $m = \frac{1}{2}n - 1$, and take for y a proper root of the equation $y^n - 1 = 0$, say r. Then, just as in Article 15, the equation takes the following form:

$$1 + r + r^4 + \text{etc.} + r^{(\frac{1}{2}n-1)^2} = (1 - r^{-1})(1 + r^{-2})(1 - r^{-3}) \ldots . (1 - r^{-\frac{1}{2}n+1})$$

or

$$W = 2(1 - r^{-1})(1 + r^{-2})(1 - r^{-3})(1 + r^{-4}) \ldots (1 - r^{-\frac{1}{2}n+1}) \quad .. \quad [8]$$

Further, since $r^{\frac{1}{2}n} = -1$, and therefore

$$1 + r^{-2} = -r^{\frac{1}{2}n-2}(1 - r^{-\frac{1}{2}n+2})$$
$$1 + r^{-4} = -r^{\frac{1}{2}n-4}(1 - r^{-\frac{1}{2}n+4})$$
$$1 + r^{-6} = -r^{\frac{1}{2}n-6}(1 - r^{-\frac{1}{2}n+6}) \text{ etc.}$$

and the product of the factors $-r^{\frac{1}{2}n-2}$, $-r^{\frac{1}{2}n-4}$, $-r^{\frac{1}{2}n-6}$, etc., up to $-r^2$ is $= (-1)^{\frac{1}{4}n-1} r^{\frac{1}{16}nn-\frac{1}{4}n}$, the preceding equation can also be written as

$$W = 2(-1)^{\frac{1}{4}n-1} r^{\frac{1}{16}nn-\frac{1}{4}n}(1 - r^{-1})(1 - r^{-2}) (1 - r^{-3})(1 - r^{-4}) \ldots . (1 - r^{-\frac{1}{2}n+1})$$

Since we have

$$1 - r^{-1} = -r^{-1}(1 - r^{-n+1})$$
$$1 - r^{-2} = -r^{-2}(1 - r^{-n+2})$$
$$1 - r^{-3} = -r^{-3}(1 - r^{-n+3}) \text{ etc.}$$

we obtain

$$(1 - r^{-1})(1 - r^{-2})(1 - r^{-3}) \dots (1 - r^{-\frac{1}{2}n+1})$$
$$= (-1)^{\frac{1}{2}n-1} r^{-\frac{1}{8}nn+\frac{1}{4}n}(1 - r^{-\frac{1}{2}n-1})$$
$$(1 - r^{-\frac{1}{2}n-2})(1 - r^{-\frac{1}{2}n-3}) \dots (1 - r^{-n+1})$$

and hence

$$W = 2(-1)^{\frac{3}{4}n-2} r^{-\frac{1}{16}nn}(1 - r^{-\frac{1}{2}n-1})$$
$$(1 - r^{-\frac{1}{2}n-2})(1 - r^{-\frac{1}{2}n-3}) \dots . (1 - r^{-n+1})$$

Multiplying this value of W by the one found earlier, and introducing the factor $1 - r^{-\frac{1}{2}n}$ on both sides, gives

$$(1 - r^{-\frac{1}{2}n})W^2 = 4(-1)^{n-3} r^{-\frac{1}{4}n}(1 - r^{-1}.)$$
$$(1 - r^{-2})(1 - r^{-3}) \dots . (1 - r^{-n+1})$$

But we have

$$1 - r^{-\frac{1}{2}n} = 2$$
$$(-1)^{n-3} = -1$$
$$r^{-\frac{1}{4}n} = -r^{\frac{1}{4}n}$$

$$(1 - r^{-1})(1 - r^{-2})(1 - r^{-3}) \dots . (1 - r^{-n+1}) = n$$

From this we finally conclude

$$W^2 = 2r^{\frac{1}{4}n} n \text{ . . . . . [9]}$$

Now it is easy to see that $r^{\frac{1}{4}n}$ is either $= +i$ or $= -i$, according as k is of the form $4\mu + 1$ or of the form $4\mu + 3$. And since

$$2i = (1 + i)^2, \quad -2i = (1 - i)^2$$

in the case where k is of the form $4\mu + 1$, we have

$$W = \pm(1 + i)\sqrt{n}, \quad \text{and so} \quad T = U = \pm\sqrt{n}$$

but in the other case, where k is of the form $4\mu + 3$,

$$W = \pm(1-i)\sqrt{n}, \quad \text{and so} \quad T = -U = \pm\sqrt{n}$$

18.

The method of the preceding article supplied the absolute values of the functions T, U, and specified the conditions under which they must be given equal or opposite signs; but it does not yet determine the signs themselves. For the case where $k = 1$, we shall supply what is missing as follows.

Let us put $\rho = \cos\frac{1}{2}\omega + i\sin\frac{1}{2}\omega$, so that $r = \rho\rho$. Since $\rho^n = -1$, it is clear that equation [8] can be written as

$$W = 2(1+\rho^{n-2})(1+\rho^{-4})(1+\rho^{n-6})(1+\rho^{-8})$$
$$\ldots(1+\rho^{-n+4})(1+\rho^2)$$

or, with the factors arranged in a different order,

$$W = 2(1+\rho^2)(1+\rho^{-4})(1+\rho^6)(1+\rho^{-8})$$
$$\ldots(1+\rho^{-n+4})(1+\rho^{n-2})$$

Now we have

$$1+\rho^2 = 2\rho\cos\frac{1}{2}\omega$$
$$1+\rho^{-4} = 2\rho^{-2}\cos\omega$$
$$1+\rho^{-6} = 2\rho^3\cos\frac{3}{2}\omega$$
$$1+\rho^{-8} = 2\rho^{-4}\cos 2\omega, \text{ etc.}$$

up to

$$1+\rho^{-n+4} = 2\rho^{-\frac{1}{2}n+2}, \cos(\frac{1}{4}n-1)\omega$$
$$1+\rho^{n-2} = 2\rho^{\frac{1}{2}n-1}\cos(\frac{1}{4}n-\frac{1}{2})\omega$$

Thus we have

$$W = 2^{\frac{1}{2}n}\rho^{\frac{1}{4}n}\cos\frac{1}{2}\omega\cos\omega\,\frac{3}{2}\omega$$
$$\ldots\cos(\frac{1}{4}n-\frac{1}{2})\omega$$

The cosines entering this product are evidently all positive, while the factor $\rho^{\frac{1}{4}n}$ is $= \cos 45^0 + i \sin 45^0 = (1+i)\sqrt{\frac{1}{2}}$. From this we conclude that W is the product of $1+i$ and a positive real quantity, so that necessarily

$$W = (1+i)\sqrt{n}, \quad T = +\sqrt{n}, \quad U = +\sqrt{n}$$

19.

It will be worthwhile to collect all the summations developed so far into a single overview here. In general, we have

$T =$	$U =$	for n of the form
$\pm\sqrt{n}$	$\pm\sqrt{n}$	4μ
$\pm\sqrt{n}$	0	$4\mu+1$
0	0	$4\mu+2$
0	$\pm\sqrt{n}$	$4\mu+3$

and, when we set $k = 1$, the radical must be given the positive sign. What we observed by induction in Article 3 for prime values of n has therefore now been proved with full rigor. Nothing remains except to show how to determine the signs in every case, for any values of k. But before we can tackle this task in full generality, we must first consider more closely the cases where n is a prime number or a power of a prime number.

20.

First let n be an odd prime number. From what we set out in Article 10, it is clear that $W = 1 + 2\Sigma r^a = 1 + 2\Sigma R^{ak}$ if we put $R = \cos\omega + i\sin\omega$, with a, as there, denoting indefinitely all the quadratic residues of n lying between 1 and $n-1$. If we likewise use b to denote indefinitely all the quadratic nonresidues between the same limits, we readily see that all the numbers ak are congruent modulo n either to all the a or to all the b (without regard to order), according as k is a residue or a nonresidue. Thus, in the first case, we have

$$W = 1 + 2\Sigma R^a = 1 + R + R^4 + R^9 + \text{etc.} + R^{(n-1)^2}$$

and hence $W = +\sqrt{n}$ if n is of the form $4\mu+1$, and $W = +i\sqrt{n}$ if n is of the form $4\mu+3$.

In the other case, where k is a nonresidue of n, we have

$$W = 1 + 2\Sigma R^b$$

Since all the a, b evidently make up the complete collection of numbers 1, 2, 3 …, and therefore

$$\Sigma R^a + \Sigma R^b = R + R^2 + R^3 + \text{etc.} + R^{n-1} = -1$$

we obtain

$$W = -1 - 2\Sigma R^a = -(1 + R + R^4 + R^9 + \text{etc.} + R^{(n-1)^2})$$

and hence $W = -\sqrt{n}$ if n is of the form $4\mu + 1$, and $W = -i\sqrt{n}$ if n is of the form $4\mu + 3$.

From this we conclude:

first, if n is of the form $4\mu + 1$ and k is a quadratic residue of n,

$$T = +\sqrt{n}, \quad U = 0$$

second, if n is of the form $4\mu + 1$ and k is a nonresidue of n,

$$T = -\sqrt{n}, \quad U = 0$$

third, if n is of the form $4\mu + 3$ and k is a residue of n,

$$T = 0, \quad U = +\sqrt{n}$$

fourth, if n is of the form $4\mu + 3$ and k is a nonresidue of n,

$$T = 0, \quad U = -\sqrt{n}$$

21.

Second, let n be the square or a higher power of an odd prime number p, and put $n = p^{2\varkappa}q$, with q either $= 1$ or $= p$. Here we should first note that, if λ is any integer not divisible by $p^{\varkappa}$, we have

$$\begin{aligned} & r^{\lambda\lambda} + r^{(\lambda+p^{\varkappa}q)^2} + r^{(\lambda+2p^{\varkappa}q)^2} + r^{(\lambda+3p^{\varkappa}q)^2} \\ & \quad + \text{etc.} + r^{(\lambda+n-p^{\varkappa}q)^3} \\ & = r^{\lambda\lambda}\{1 + r^{2\lambda p^{\varkappa}q} + r^{4\lambda p^{\varkappa}q} \\ & \quad + r^{6\lambda p^{\varkappa}q} + \text{etc.} + r^{2\lambda(n-p^{\varkappa}q)}\} \\ & = \frac{r^{\lambda\lambda}(1 - r^{2\lambda n})}{1 - r^{2\lambda p^{\varkappa}q}} = 0 \end{aligned}$$

It is then easy to see that

$$W = 1 + r^{p^{2\varkappa}} + r^{4p^{2\varkappa}} + r^{9p^{2\varkappa}} + \text{etc.} + r^{(n-p^{\varkappa})^2}$$

For the remaining terms of the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

can be divided into $(p^\varkappa - 1)q$ partial series, each with $p^\varkappa$ terms, whose sums vanish under the transformation just given.

It follows that in the case where $q = 1$, or where n is a power of a prime number with an even exponent, we have

$$W = p^\varkappa = +\sqrt{n}, \text{ and so } T = +\sqrt{n}, \quad U = 0$$

In the other case, where $q = p$, or where n is a power of a prime number with an odd exponent, we shall put $r^{p^{2\varkappa}} = \rho$. Then ρ is a proper root of the equation $x^p - 1 = 0$, namely $\rho = \cos \frac{k}{p} 360^0 + i \sin \frac{k}{p} 360^0$, and we obtain

$$\begin{aligned} W &= 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p^{\varkappa+1}-1)^2} \\ &= p^\varkappa (1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p-1)^2}) \end{aligned}$$

But the sum of the series $1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p-1)^2}$ is determined by the preceding article. It follows directly that

$$\begin{aligned} W &= \pm\sqrt{n} = T, \quad \text{if } p \text{ is of the form } 4\mu + 1 \\ W &= \pm i\sqrt{n} = iU, \quad \text{if } p \text{ is of the form } 4\mu + 3 \end{aligned}$$

with the positive or negative sign, according as k is a residue or a nonresidue of p.

22.

What we set out in Articles 20 and 21 also readily gives the following proposition, which will be of considerable use to us below. Put

$$W' = 1 + r^h + r^{4h} + r^{9h} + \text{etc.} + r^{h(n-1)^2}$$

with h denoting any integer not divisible by p. Then, in the case where $n = p$, or where n is a power of p with an odd exponent, we have

$$\begin{aligned} W' &= W, \text{ if } h \text{ is a quadratic residue of } p \\ W' &= -W, \text{ if } h \text{ is a quadratic nonresidue of } p \end{aligned}$$

For W' evidently arises from W if, for k, we substitute kh. In the first case, k and kh are alike, and in the second they are unlike, with respect to being residues or nonresidues of p.

But in the case where n is a power of p with an even exponent, we evidently have $W' = +\sqrt{n}$, and therefore always $W' = W$.

23.

In Articles 20, 21 and 22 we considered odd prime numbers and powers of such numbers. The case therefore remains where n is a power of two.

For $n = 2$, we evidently have $W = 1 + r = 0$.

For $n = 4$, we obtain $W = 1 + r + r^4 + r^9 = 2 + 2r$. Thus $W = 2 + 2i$ whenever k is of the form $4\mu + 1$, and $W = 2 - 2i$ whenever k is of the form $4\mu + 3$.

For $n = 8$, we have $W = 1+r+r^4+r^9+r^{16}+r^{25}+r^{36}+r^{49} = 2+4r+2r^4 = 4r$. Hence

$$\begin{aligned}
W &= (1+i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+1\\
W &= (-1+i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+3\\
W &= (-1-i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+5\\
W &= (1-i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+7
\end{aligned}$$

If n is a higher power of two, put $n = 2^{2\varkappa}q$, with q either $= 1$ or $= 2$, and $\varkappa$ greater than 1. Here we must first observe that, if λ is any integer not divisible by $2^{\varkappa-1}$, we have

$$\begin{aligned}
&r^{\lambda\lambda} + r^{(\lambda+2^{\varkappa}q)^2} + r^{(\lambda+2.2^{\varkappa}q)^2} + r^{(\lambda+3.2^{\varkappa}q)^2}\\
&\quad + \text{etc.} + r^{(\lambda+n-2^{\varkappa}q)^2}\\
&= r^{\lambda\lambda}\{1 + r^{2^{\varkappa+1}\lambda q} + r^{2.2^{\varkappa+1}\lambda q}\\
&\quad + r^{3.2^{\varkappa+1}\lambda q} + \text{etc.} + r^{(2n-2^{\varkappa+1}q)\lambda}\}\\
&= \frac{r^{\lambda\lambda}(1 - r^{2\lambda n})}{1 - r^{2^{\varkappa+1}\lambda q}} = 0
\end{aligned}$$

It is then easy to see that

$$W = 1 + r^{2^{2\varkappa-2}} + r^{4.2^{2\varkappa-2}} + r^{9.2^{2\varkappa-2}} + \text{etc.} + r^{(n-2^{\varkappa-1})^2}$$

Put $r^{2^{2\varkappa-2}} = \rho$. Then ρ is a root of the equation $x^{4q} - 1 = 0$, namely $\rho = \cos\frac{k}{4q}360^0 + i\sin\frac{k}{4q}360^0$; we then obtain

$$\begin{aligned}
W &= 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(2^{\varkappa+1}q-1)^2}\\
&= 2^{\varkappa-1}(1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(4q-1)^2})
\end{aligned}$$

But the sum of the series $1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(4q-1)^2}$ is determined by what we explained for the cases $n = 4$, $n = 8$. From this we conclude

that in the case where $q = 1$, or where n is a power of the number 4, we have

$$W = (1+i)2^{\varkappa} = (1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 4\mu + 1$$
$$W = (1-i)2^{\varkappa} = (1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 4\mu + 3$$

these are exactly the formulas given for $n = 4$;

but in the case where $q = 2$, or where n is a power of two with an odd exponent greater than 3, we have

$$W = (1+i)2^{\varkappa}\sqrt{2} = (1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 1$$
$$W = (-1+i)2^{\varkappa}\sqrt{2} = (-1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 3$$
$$W = (-1-i)2^{\varkappa}\sqrt{2} = (-1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 5$$
$$W = (1-i)2^{\varkappa}\sqrt{2} = (1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 7$$

these too agree exactly with those we gave for $n = 8$.

24.

Here too it will be worthwhile to determine the ratio of the sum of the series

$$W' = 1 + r^h + r^{4h} + r^{9h} + \text{etc.} + r^{h(n-1)^2}$$

to W, where h denotes any odd integer. Since W' arises from W by changing k into kh, the value of W' depends on the form of the number kh in the same way as W depends on the form of k. Put $\frac{W'}{W} = l$. It is clear that

I. in the case where $n = 4$, or it is a higher power of two with an even exponent, we have

$$l = 1, \quad \text{if } h \text{ is of the form } 4\mu + 1$$
$$l = -i, \quad \text{if } h \text{ is of the form } 4\mu + 3, \text{ and } k \text{ is of the form } 4\mu + 1$$
$$l = +i, \quad \text{if } h \text{ is of the form } 4\mu + 3, \text{ and } k \text{ is of the same form.}$$

II. in the case where $n = 8$, or it is a higher power of two with an odd exponent, we have

$l = 1$, if h is of the form $8\mu + 1$,
$l = -1$, if h is of the form $8\mu + 5$,
$l = +i$, if either h is of the form $8\mu + 3$, and k is of the form $4\mu + 1$,
or h is of the form $8\mu + 7$, and k is of the form $4\mu + 3$,
$l = -i$, if either h is of the form $8\mu + 3$, and k is of the form $4\mu + 3$,
or h is of the form $8\mu + 7$, and k is of the form $4\mu + 1$.

The preceding articles have completely determined the sum W for the cases where n is a prime number or a power of a prime number. It therefore remains to complete the cases where n is composed of several prime numbers; the following theorem will prepare the way for us:

25.

THEOREM. *Let n be the product of two coprime positive integers a, b, and put*

$$P = 1 + r^{aa} + r^{4aa} + r^{9aa} + \text{etc.} + r^{(b-1)^2 aa}$$
$$Q = 1 + r^{bb} + r^{4bb} + r^{9bb} + \text{etc.} + r^{(a-1)^2 bb}$$

Then I claim that $W = PQ$.

Proof. Let α denote indefinitely the numbers $0, 1, 2, 3 \ldots . a - 1$, β the numbers $0, 1, 2, 3 \ldots . b - 1$, and ν the numbers $0, 1, 2, 3 \ldots . n - 1$. Then evidently

$$P = \Sigma r^{aa\beta\beta}, \quad Q = \Sigma r^{bb\alpha\alpha}, \quad W = \Sigma r^{\nu\nu}$$

Thus $PQ = \Sigma r^{aa\beta\beta + bb\alpha\alpha}$, on substituting all the values of α and β, combined with one another in every way. Further, since $2ab\alpha\beta = 2\alpha\beta n$, we have $PQ = \Sigma r^{(a\beta + b\alpha)^2}$. But it is easy to see that the individual values of $a\beta + b\alpha$ are different from one another and equal to some value of ν. Hence $PQ = \Sigma r^{\nu\nu} = W$.

It should also be noted that r^{aa} is a proper root of the equation $x^b - 1 = 0$, and r^{bb} a proper root of the equation $x^a - 1 = 0$.

26.

Now let n be the product of three pairwise coprime numbers a, b, c. It is clear that, on putting $bc = b'$, a and b' will also be coprime; hence W is the product of the two factors

$$1 + r^{aa} + r^{4aa} + r^{9aa} + \text{etc.} + r^{(b'-1)^2 aa}$$
$$1 + r^{b'b'} + r^{4b'b'} + r^{9b'b'} + \text{etc.} + r^{(a-1)^2 b'b'}$$

But since r^{aa} is a proper root of the equation $x^{bc} - 1 = 0$, the first factor itself will be the product of

$$1 + \rho^{bb} + \rho^{4bb} + \rho^{9bb} + \text{etc.} + \rho^{(c-1)^2 bb}$$
$$1 + \rho^{cc} + \rho^{4cc} + \rho^{9cc} + \text{etc.} + \rho^{(b-1)^2 cc}$$

if we put $r^{aa} = \rho$. It is therefore clear that W is the product of the three factors

$$1 + r^{bbcc} + r^{4bbcc} + r^{9bbcc} + \text{etc.} + r^{(a-1)^2 bbcc}$$
$$1 + r^{aacc} + r^{4aacc} + r^{9aacc} + \text{etc.} + r^{(b-1)^2 aacc}$$
$$1 + r^{aabb} + r^{4aabb} + r^{9aabb} + \text{etc.} + r^{(c-1)^2 aabb}$$

where r^{bbcc}, r^{aacc}, r^{aabb} are respectively proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$, $x^c - 1 = 0$.

27.

It is easy to conclude in general that, if n is the product of any number of pairwise coprime factors a, b, c, etc., W is the product of the same number of factors, namely

$$1 + r^{\frac{nn}{aa}} + r^{\frac{4nn}{aa}} + r^{\frac{9nn}{aa}} + \text{etc.} + r^{\frac{(a-1)^2 nn}{aa}}$$
$$1 + r^{\frac{nn}{bb}} + r^{\frac{4nn}{bb}} + r^{\frac{9nn}{bb}} + \text{etc.} + r^{\frac{(b-1)^2 nn}{bb}}$$
$$1 + r^{\frac{nn}{cc}} + r^{\frac{4nn}{cc}} + r^{\frac{9nn}{cc}} + \text{etc.} + r^{\frac{(c-1)^2 nn}{cc}}, \text{ etc.}$$

where $r^{\frac{nn}{aa}}$, $r^{\frac{nn}{bb}}$, $r^{\frac{nn}{cc}}$, etc. are proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$. $x^c - 1 = 0$, etc.

28.

These principles now make the transition to a complete determination of W for any value of n immediately clear. Specifically, resolve n into factors a, b, c, etc., that are either distinct prime numbers or powers of distinct prime numbers, and put $r^{\frac{nn}{aa}} = A$, $r^{\frac{nn}{bb}} = B$, $r^{\frac{nn}{cc}} = C$, etc. Then A, B, C, etc. will be proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$, $x^c - 1 = 0$, etc., and W the product of the factors

$$1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}$$
$$1 + B + B^4 + B^9 + \text{etc.} + B^{(b-1)^2}$$
$$1 + C + C^4 + C^9 + \text{etc.} + C^{(c-1)^2}$$

Each of these factors can be determined from what we showed in Articles 20, 21, 23, so the value of the product will also become known. It will be useful to collect the rules for determining these factors in a single overview here. Since the root A is $= \frac{kn}{a} \cdot \frac{360^0}{a}$, the sum $1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}$, which we shall denote by L, will be determined by the number $\frac{kn}{a}$ in the same way as W is determined by k in our general investigation. Twelve cases must now be distinguished.

I. If a is a prime number of the form $4\mu + 1$, say $= p$, or a power of such a prime number with an odd exponent, and $\frac{kn}{a}$ is a quadratic residue of p, then $L = +\sqrt{a}$.

II. If the other conditions remain unchanged and $\frac{kn}{a}$ is a quadratic nonresidue of p, then $L = -\sqrt{a}$.

III. If a is a prime number of the form $4\mu + 3$, say $= p$, or a power of such a prime number with an odd exponent, and $\frac{kn}{a}$ is a quadratic residue of p, then $L = +i\sqrt{a}$.

IV. If, with the other conditions as in III, $\frac{kn}{a}$ is a quadratic nonresidue of p, then $L = -i\sqrt{a}$.

V. If a is the square, or a higher power with an even exponent, of an (odd) prime number, then $L = +\sqrt{a}$.

VI. If $a = 2$, then $L = 0$.

VII. If $a = 4$, or it is a higher power of two with an even exponent, and $\frac{kn}{a}$ is of the form $4\mu + 1$, then $L = (1 + i)\sqrt{a}$.

VIII. If, with the other conditions as in VII, $\frac{kn}{a}$ is of the form $4\mu + 3$, then $L = (1 - i)\sqrt{a}$.

IX. If $a = 8$, or it is a higher power of two with an odd exponent, and $\frac{kn}{a}$ is of the form $8\mu + 1$, then $L = (1 + i)\sqrt{a}$.

X. If, with the other conditions as in IX, $\frac{kn}{a}$ is of the form $8\mu + 3$, then $L = (-1 + i)\sqrt{a}$.

XI. If the other conditions remain unchanged and $\frac{kn}{a}$ is of the form $8\mu + 5$, then $L = (-1 - i)\sqrt{a}$.

XII. If the other conditions remain unchanged and $\frac{kn}{a}$ is of the form $8\mu+7$, then $L = (1-i)\sqrt{a}$.

29.

For example, let $n = 2520 = 8 \,.\, 9 \,.\, 5 \,.\, 7$ and $k = 13$. Here we obtain

for $a = 8$, by case XII, $L = (1-i)\sqrt{8}$
for the factor 9, by case V, the corresponding sum is $= \sqrt{9}$
for the factor 5, by case II, the corresponding sum is $= -\sqrt{5}$
for the factor 7, by case III, the corresponding sum is $= +i\sqrt{7}$.

Thus $W = (1-i) \,.\, (-i) \,.\, \sqrt{2520} = (-1-i)\sqrt{2520}$.
For the same value of n, let $k = 1$; then the corresponding sums are

for the factor 8 the sum $(-1+i)\sqrt{8}$
for the factor 9 the sum $\sqrt{9}$
for the factor 5 the sum $\sqrt{5}$
for the factor 7 the sum $-i\sqrt{7}$

This gives the product $W = (1+i)\sqrt{2520}$.

30.

Another method for determining the sum W in general follows from what we set out in Articles 22. 24. Put $\cos\omega + i\sin\omega = \rho$, and

$$\rho^{\frac{nn}{aa}} = \alpha, \quad \rho^{\frac{nn}{bb}} = \beta, \quad \rho^{\frac{nn}{cc}} = \gamma \text{ etc.}$$

so that $r = \rho^k$, $A = \alpha^k$, $B = \beta^k$, $C = \gamma^k$ etc. Then

$$1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(n-1)^2}$$

is the product of the factors

$$1 + \alpha + \alpha^4 + \alpha^9 + \text{etc.} + \alpha^{(a-1)^2}$$
$$1 + \beta + \beta^4 + \beta^9 + \text{etc.} + \beta^{(b-1)^2}$$
$$1 + \gamma + \gamma^4 + \gamma^9 + \text{etc.} + \gamma^{(c-1)^2}, \text{ etc.}$$

and therefore W is the product of the factors

$$w = 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(n-1)^2}$$

$$\mathfrak{A} = \frac{1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}}{1 + \alpha + \alpha^4 + \alpha^9 + \text{etc.} + \alpha^{(a-1)^2}}$$

$$\mathfrak{B} = \frac{1 + B + B^4 + B^9 + \text{etc.} + B^{(b-1)^2}}{1 + \beta + \beta^4 + \beta^9 + \text{etc.} + \beta^{(b-1)^2}}$$

$$\mathfrak{C} = \frac{1 + C + C^4 + C^9 + \text{etc.} + C^{(c-1)^2}}{1 + \gamma + \gamma^4 + \gamma^9 + \text{etc.} + \gamma^{(c-1)^2}}$$

The first factor w has already been determined by the investigations given above (Article 19); the remaining factors $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ etc. follow from the formulas of Articles 22. 24, which we collect here again to have everything together[*]. Twelve cases must be distinguished here, namely

I. If a is an (odd) prime number $= p$, or a power of such a number with an odd exponent, and k is a quadratic residue of p, the corresponding factor is $\mathfrak{A} = +1$.

II. If the other conditions remain unchanged and k is a quadratic non-residue of p, then $\mathfrak{A} = -1$.

III. If a is the square of an odd prime number, or a higher power of it with an even exponent, then $\mathfrak{A} = +1$.

IV. If a is $= 4$, or a higher power of two with an even exponent, and k is of the form $4\mu + 1$, then $\mathfrak{A} = +1$.

V. If, with the other conditions as in IV, k is of the form $4\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = -i$.

VI. If, with the other conditions as in IV, k is of the form $4\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = +i$.

VII. If a is $= 8$, or a higher power of two with an odd exponent, and k is of the form $8\mu + 1$, then $\mathfrak{A} = +1$.

VIII. If, with the other conditions as in VII, k is of the form $8\mu + 5$, then $\mathfrak{A} = -1$.

IX. If, with the other conditions as in VII, k is of the form $8\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = +i$.

X. If, with the other conditions as in VII, k is of the form $8\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = -i$.

XI. If, with the other conditions as in VII, k is of the form $8\mu + 7$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = -i$.

XII. If, with the other conditions as in VII, k is of the form $8\mu + 7$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = +i$.

[*] Clearly, what were k and h there are here $\frac{n}{a}$ and k for the second factor, $\frac{n}{b}$ and k for the third factor, etc.

We omit the case where $a = 2$; here $\mathfrak{A}$ would indeed be $= \frac{0}{0}$, or indeterminate, but then we always have $W = 0$.

The remaining factors $\mathfrak{B}, \mathfrak{C}$ etc. depend on b, c etc. in the same way as $\mathfrak{A}$ depends on a, insofar as these enter into their determination.

31.

By this second method, the first example of Article 29 runs as follows:

The factor w is $= (1+i)\sqrt{2520}$
For $a = 8$ the corresponding factor $\mathfrak{A}$ is, by case VIII, $= -1$
For the second factor of n, 9, the corresponding factor is $+1$ (by case III.)
For the factor 5 the corresponding factor is -1 (by case II.)
For the factor 7 the corresponding factor is -1 (by case II.)

This gives the product $W = (-1-i)\sqrt{2520}$, as in Article 29.

32.

Since the value of W can be determined by *two* methods, one of which rests on the relations of the numbers $\frac{nk}{a}$, $\frac{nk}{b}$, $\frac{nk}{c}$ etc. to the numbers a, b, c etc., while the other depends on the relations of k to the numbers a, b, c etc., there must be a conditional connection among all these relations, so that any one can be determined from the others. Suppose that all the numbers a, b, c etc. are odd prime numbers, and take $k = 1$. Divide the factors a, b, c etc. into two classes: one contains those of the form $4\mu + 1$, denoted by p, p', p'' etc.; the other consists of those of the form $4\mu+3$, denoted by q, q', q'' etc. We shall denote the number of the latter by m. With this done, we first observe that n is of the form $4\mu + 1$ if m is even (this also includes the case where there are no factors at all in the second class, or $m = 0$), whereas n is of the form $4\mu+3$ if m is odd. Now the determination of W by the first method is carried out as follows. Let the numbers $P, P'\ P''$ etc., Q, Q', Q'' etc. depend on the relations of the numbers $\frac{n}{p}$, $\frac{n}{p'}$, $\frac{n}{p''}$ etc., $\frac{n}{q}$, $\frac{n}{q'}$, $\frac{n}{q''}$ etc. to the numbers p, p', p'' etc., q, q', q'' etc., respectively, in such a way that we put

$$P = +1, \text{ if } \frac{n}{p} \text{ is a quadratic residue of } p$$
$$P = -1, \text{ if } \frac{n}{p} \text{ is a quadratic nonresidue of } p$$

and similarly for the others. Then W will be the product of the factors $P\sqrt{p}$, $P'\sqrt{p'}$, $P''\sqrt{p''}$ etc. $iQ\sqrt{q}$, $iQ'\sqrt{q'}$, $iQ''\sqrt{q''}$ etc., and therefore

$$W = PP'P'' \ldots QQ'Q'' \ldots i^m\sqrt{n}$$

By the second method, or rather directly from the rules of Article 19, we have

$$W = +\sqrt{n}, \text{ if } n \text{ is of the form } 4\mu + 1,$$
$$\text{or equivalently, if } m \text{ is even}$$
$$W = +i\sqrt{n}, \text{ if } n \text{ is of the form } 4\mu + 3,$$
$$\text{or if } m \text{ is odd.}$$

Both cases can be combined in the following formula:

$$W = i^{mm}\sqrt{n}$$

It therefore follows that

$$PP'P'' \ldots QQ'Q'' \ldots = i^{mm-m}$$

But i^{mm-m} is $= 1$ whenever m is of the form 4μ or $4\mu + 1$, and $= -1$ whenever m is of the form $4\mu+2$ or $4\mu+3$. From this we deduce the following most elegant

THEOREM. *Let a, b, c etc. denote distinct positive odd prime numbers, and put their product $= n$. Let m of them be of the form $4\mu + 3$, and the rest of the form $4\mu + 1$. The number of those among a, b, c etc. for which $\frac{n}{a}$, $\frac{n}{b}$, $\frac{n}{c}$ etc., respectively, are nonresidues will be even whenever m is of the form 4μ or $4\mu + 1$, but odd whenever m is of the form $4\mu + 2$ or $4\mu + 3$.*

Thus, for example, putting $a = 3$, $b = 5$, $c = 7$, $d = 11$, we have three numbers of the form $4\mu + 3$, namely 3, 7 and 11; and $5 \,.\, 7 \,.\, 11\, R3$; $3 \,.\, 7 \,.\, 11\, R5$; $3 \,.\, 5 \,.\, 11\, R7$; $3 \,.\, 5 \,.\, 7\, N11$, so that only $\frac{n}{d}$ is a nonresidue of d.

33.

The celebrated *fundamental theorem* about quadratic residues is nothing other than a special case of the theorem just developed. Specifically, by limiting the number of the numbers a, b, c etc. to *two*, it is clear that, if just one of them, or neither, is of the form $4\mu + 3$, we must have either both aRb, bRa, or both aNb, bNa; but if both are of the form $4\mu + 3$, one must be a nonresidue of the other, and the latter a residue of the former. Here, then, is a *fourth* proof of this most important theorem, whose first and second proofs we gave in the Disquisitiones Arithmeticae, and the third recently in a separate paper (*Commentt. T. XVI*); we shall later set out two others, again resting on entirely different principles. It is certainly most remarkable that this most beautiful theorem, which at first so stubbornly eluded every attempt, could later be reached by so many paths, worlds apart from one another.

34.

The remaining theorems, which form a kind of supplement to the fundamental theorem, namely those by which we distinguish the prime numbers for which $-1, +2$ and -2 are residues or nonresidues, can also be derived from the same principles. We shall begin with the residue $+2$.

Putting $n = 8a$, with a a prime number, and $k = 1$, the method of Article 28 makes W the product of two factors. One of them will be $+\sqrt{a}$, or $+i\sqrt{a}$, if 8, or equivalently 2, is a quadratic residue of a; but $-\sqrt{a}$ or $-i\sqrt{a}$ if 2 is a nonresidue of a. The second factor is

$$\begin{aligned}
&(1+i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+1\\
&(-1+i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+3\\
&(-1-i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+5\\
&(1-i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+7
\end{aligned}$$

But by Article 18 we always have $W = (1+i)\sqrt{n}$; dividing this value by the four values of the second factor shows that the first factor must be

$$\begin{aligned}
&+\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+1\\
&-i\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+3\\
&-\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+5\\
&+i\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+7
\end{aligned}$$

It follows immediately that 2 must be a residue of a in the first and fourth cases, but a nonresidue in the second and third.

35.

The prime numbers for which -1 is a residue or a nonresidue are easily distinguished with the aid of the following theorem, which is also quite remarkable in itself.

THEOREM. *The product of the two factors*

$$W' = 1 + r^{-1} + r^{-4} + \text{etc.} + r^{-(n-1)^2}$$
$$W = 1 + r + r^4 + \text{etc.} + r^{(n-1)^2}$$

is $= n$ *if* n *is odd; or* $= 0$ *if* n *is twice an odd number; or* $= 2n$ *if* n *is divisible by four.*

Proof. Since we evidently have

$$\begin{aligned} W &= r + r^4 + r^9 + \text{etc.} + r^{nn} \\ &= r^4 + r^9 + \text{etc.} + r^{(n+1)^2} \\ &= r^9 + \text{etc.} + r^{(n+2)^2} \quad \text{etc.} \end{aligned}$$

the product WW' can also be written as follows:

$$\begin{aligned} & 1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} \\ & + r^{-1}(r + r^4 + r^9 + r^{16} + \text{etc.} + r^{nn}) \\ & + r^{-4}(r^4 + r^9 + r^{16} + r^{25} + \text{etc.} + r^{(n+1)^2}) \\ & + r^{-9}(r^9 + r^{16} + r^{25} + r^{36} + \text{etc.} + r^{(n+2)^2}) \\ & \text{etc.} \\ & + r^{-(n-1)^2}(r^{(n-1)^2} + r^{nn} + r^{(n+1)^2} \\ & \qquad + r^{(n+2)^2} + \text{etc.} + r^{(2n-2)^2}) \end{aligned}$$

Summing this expression vertically gives

$$\begin{aligned} & n \\ & + r(1 + rr + r^4 + r^6 + \text{etc.} + r^{2n-2}) \\ & + r^4(1 + r^4 + r^8 + r^{12} + \text{etc.} + r^{4n-4}) \\ & + r^9(1 + r^6 + r^{12} + r^{18} + \text{etc.} + r^{6n-6}) \\ & + \text{etc.} \\ & + r^{(n-1)^2}(1 + r^{2n-2} + r^{4n-4} \\ & \qquad + r^{6n-6} + \text{etc.} + r^{2(n-1)^2}) \end{aligned}$$

Now, if n is odd, each part of this expression, except the first, n, will be $= 0$; for the second is evidently $\frac{r(1-r^{2n})}{1-rr}$, the third $\frac{r^4(1-r^{4n})}{1-r^4}$ etc. Whenever n is even, however, we must also except the part

$$r^{\frac{1}{4}nn}(1 + r^n + r^{2n} + r^{3n} + \text{etc.} + r^{nn-n})$$

which is $= nr^{\frac{1}{4}nn}$. In the first case, therefore, $WW' = n$, but in the second $= n + nr^{\frac{1}{4}nn}$; now $r^{\frac{1}{4}nn}$ is $= +1$ if n is divisible by four, and then we obtain $WW' = 2n$; on the other hand, $r^{\frac{1}{4}nn} = -1$ if n is twice an odd number, in which case $WW' = 0$. Q. E. D.

36.

Article 22 establishes that, if n is an odd prime number, $\frac{W'}{W}$ is $= +1$ or $= -1$ according as -1 is a residue or a nonresidue of n. Hence in the first case we must have $W^2 = +n$, and in the second $W^2 = -n$; by Article 13 we therefore conclude that the first case can occur only when n is of the form $4\mu + 1$, and the second when n is of the form $4\mu + 3$.

Finally, combining the conditions found for the residues $+2$ and -1 immediately shows that -2 is a residue of every prime number of the form $8\mu + 1$ or $8\mu + 3$, and a nonresidue of every prime number of the form $8\mu + 5$ or $8\mu + 7$.

NEW PROOFS AND EXTENSIONS OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES

BY

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NEW PROOFS AND EXTENSIONS OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

The fundamental theorem about quadratic residues, which ranks among the most beautiful truths of higher arithmetic, was readily discovered by induction, but proved with far greater difficulty. It often happens in this field that the proofs of the simplest truths, which seem to offer themselves to the investigator through induction of their own accord, lie deeply hidden and can finally be brought to light only after many unsuccessful attempts, perhaps by a path very different from the one along which they were sought. Then it often happens that, once one path has been found, *several* others gradually open up, leading to the same goal: some more briefly and directly, others as if by a roundabout way, starting from very different principles between which and the question at hand one would scarcely have suspected any connection. This remarkable connection among more hidden truths not only gives these investigations a particular charm; it also deserves careful study and unravelling, since it often yields new resources or advances for the science itself.

Although the arithmetical theorem considered here might therefore seem, through the earlier efforts that supplied four entirely different proofs[*)], to have been fully dealt with, I nevertheless return to the same subject and add two further proofs, which will certainly shed new light on it. The first is related to the third in a certain respect, since it starts from the same lemma; it then follows a different path, however, so that it can rightly be regarded as a new proof, which in elegance will seem, if not superior to that third proof, at least by no means inferior. The sixth proof rests on an entirely different and subtler principle, and gives a new example of the remarkable connection between arithmetical truths that at first sight seem very far apart. These two proofs are accompanied by a very simple new algorithm for deciding whether a given integer is a quadratic residue or a nonresidue of a given prime number.

[*)] Two are given in the fourth and fifth sections of the *Disquisitiones Arithmeticae*; the third in a separate paper (*Commentt. Soc. Gotting. Vol. XVI*), and the fourth in the paper *Summatio quarundam serierum singularium* (*Commentt. Recentiores, Vol. I*).

There was another reason that led me to publish now, in particular, the new proofs promised nine years ago. When I began in 1805 to investigate the theory of cubic and biquadratic residues, a far more difficult subject, I experienced much the same course of events as before in the theory of quadratic residues. The theorems that completely settle these questions, and display a remarkable analogy with the theorems about quadratic residues, were immediately discovered by induction as soon as they were sought along a suitable path; but all attempts to obtain proofs complete in every respect remained unsuccessful for a long time. This was precisely what prompted me to strive so eagerly to add more and more proofs to those already known for quadratic residues, sustained by the hope that, among many different methods, one or another might help to illuminate the related subject. This hope was by no means in vain, and tireless work was finally followed by good results. Soon it will be possible to bring the fruits of my nights of work before the public; but before I undertake this arduous task, I have resolved to return once more to the theory of quadratic residues, complete everything that still remains to be done in it, and thus, as it were, take leave of this part of higher arithmetic.

FIFTH PROOF OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

1.

We have already stated in the introduction that the fifth and third proofs start from the same lemma. For convenience, we have thought it appropriate to repeat it here in notation adapted to the present investigation.

LEMMA. *Let m be a prime number (positive and odd), and M an integer not divisible by m. Take the least positive residues of the numbers*

$$M,\ 2M,\ 3M,\ 4M\ \ldots\ldots\ \tfrac{1}{2}(m-1)M$$

modulo m. Some will be less than $\frac{1}{2}m$, and others greater; let the number of the latter be $= n$. Then M is a quadratic residue or a nonresidue of m, according as n is even or odd.

PROOF. Let those of these residues which are less than $\frac{1}{2}m$ be a, b, c, d etc., and the remaining ones, greater than $\frac{1}{2}m$, be a', b', c', d' etc. The complements of the latter to m, namely $m-a'$, $m-b'$, $m-c'$, $m-d'$ etc., are evidently all less than $\frac{1}{2}m$ and distinct both from one another and from the residues a, b, c, d etc. Thus, taken together with these, they are precisely all the numbers $1,\ 2,\ 3,\ 4\ldots.\frac{1}{2}(m-1)$, though in a different order. Putting the product

$$1\,.\,2\,.\,3\,.\,4\ldots..\tfrac{1}{2}(m-1) = P$$

we therefore have

$$P = abcd\ldots. \times (m-a')(m-b')(m-c')(m-d')\ldots.$$

and hence

$$(-1)^n P = abcd\ldots. \times (a'-m)(b'-m)(c'-m)(d'-m)\ldots.$$

Furthermore, modulo m, we have

$$\begin{aligned} PM^{\frac{1}{2}(m-1)} &\equiv abcd\,..\times a'b'c'd'\,..\\ &\equiv abcd\,..\times (a'-m)(b'-m)(c'-m)(d'-m)\,.. \end{aligned}$$

and hence

$$PM^{\frac{1}{2}(m-1)} \equiv P(-1)^n$$

It follows that $M^{\frac{1}{2}(m-1)} \equiv \pm 1$, taking the upper or lower sign according as n is even or odd. With the aid of the theorem proved in Article 106 of the *Disquisitiones Arithmeticae*, the lemma follows immediately.

2.

THEOREM. *Let m, M be positive odd integers coprime to one another, and let n be the number of those among the least positive residues of the numbers*

$$M,\ 2M,\ 3M \ldots\ldots \tfrac{1}{2}(m-1)M$$

modulo m which are greater than $\frac{1}{2}m$. Likewise, let N be the number of those among the least positive residues of the numbers

$$m,\ 2m,\ 3m \ldots\ldots \tfrac{1}{2}(M-1)m$$

modulo M which are greater than $\frac{1}{2}M$. Then the three numbers n, N, $\frac{1}{4}(m-1)(M-1)$ are either all even, or one is even and the other two odd.

PROOF. Let us denote

by f the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(m-1)$.
by f' the collection of numbers $m-1,\ m-2,\ m-3 \ldots\ldots \frac{1}{2}(m+1)$
by F the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(M-1)$
by F' the collection of numbers $M-1,\ M-2,\ M-3 \ldots\ldots \frac{1}{2}(M+1)$

Thus n indicates how many numbers Mf have their least positive residues modulo m in the collection f'. Likewise, N indicates how many numbers mF have their least positive residues modulo M in the collection F'. Finally, let

φ denote the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(mM-1)$
φ' denote the collection of numbers $mM-1,\ mM-2,\ mM-3 \ldots\ldots \frac{1}{2}(mM+1)$

Every integer not divisible by m must be congruent modulo m either to a residue from f or to one from f'. Likewise, every integer not divisible by M is congruent modulo M either to a residue from F or to one from F'. Thus all the numbers in φ, among which evidently none is divisible by both m and M, can be divided into eight classes as follows.

I. The first class contains the numbers congruent modulo m to a number from f, and modulo M to a number from F. We shall denote their number by α.

II. The numbers congruent modulo m, M, respectively, to numbers from f, F'; we put their number $= \beta$.

III. The numbers congruent modulo m, M, respectively, to numbers from f', F; we put their number $= \gamma$.

IV. The numbers congruent modulo m, M, respectively, to numbers from f', F'; let their number be $= \delta$.

V. The numbers divisible by m, but congruent modulo M to residues from F.

VI. The numbers divisible by m, but congruent modulo M to residues from F'.

VII. The numbers divisible by M, but congruent modulo m to residues from f.

VIII. The numbers divisible by M, but congruent modulo m to residues from f'.

Evidently classes V and VI together contain all the numbers mF. The number contained in VI is $= N$, so the number contained in V is $\frac{1}{2}(M-1)-N$. Likewise, classes VII and VIII together contain all the numbers Mf; class VIII contains n numbers, and class VII contains $\frac{1}{2}(m-1)-n$.

In just the same way, all the numbers in φ' are divided into eight classes IX . . XVI. If we keep the same order in doing this, it is easy to see that the numbers in the classes

$$\text{IX, X, XI, XII, XIII, XIV, XV, XVI}$$

are, respectively, the complements to mM of the numbers in the classes

$$\text{IV, III, II, I, VI, V, VIII, VII}$$

so that class IX contains δ numbers, class X contains γ, and so on. It is now clear that, if all the numbers of the first class are combined with all those of the ninth, we obtain all the numbers below mM which are congruent modulo m to a number from f, and modulo M to a number from F. It is easy to see that their number equals the number of all combinations of each individual f with each individual F. We therefore have

$$\alpha + \delta = \tfrac{1}{4}(m-1)(M-1)$$

and by a similar argument we also have

$$\beta + \gamma = \tfrac{1}{4}(m-1)(M-1)$$

Combining all the numbers of classes II, IV and VI evidently gives all the numbers below $\frac{1}{2}mM$ which are congruent to a residue from F' modulo M. These same numbers can also be written as follows:

$$F',\ M+F',\ 2M+F',\ 3M+F' \ldots .\ \tfrac{1}{2}(m-3)M+F'$$

Their total number is therefore $= \frac{1}{4}(m-1)(M-1)$, or we have

$$\beta + \delta + N = \tfrac{1}{4}(m-1)(M-1)$$

Likewise, combining all of classes III, IV and VIII gives

$$\gamma + \delta + n = \tfrac{1}{4}(m-1)(M-1)$$

These four equations give the following:

$$2\alpha = \tfrac{1}{4}(m-1)(M-1) + n + N$$
$$2\beta = \tfrac{1}{4}(m-1)(M-1) + n - N$$
$$2\gamma = \tfrac{1}{4}(m-1)(M-1) - n + N$$
$$2\delta = \tfrac{1}{4}(m-1)(M-1) - n - N$$

each of which establishes the theorem.

3.

If we now suppose that m and M are prime numbers, the fundamental theorem follows immediately by combining the preceding theorem with the lemma of Article 1. For it is clear that

I. whenever the numbers m, M both have the form $4k+1$, or just one of them does, the number $\frac{1}{4}(m-1)(M-1)$ is even. Thus n and N are either both even or both odd, and consequently either each of m and M is a quadratic residue of the other, or each is a quadratic nonresidue of the other.

II. Whenever both numbers m, M are of the form $4k+3$, however, $\frac{1}{4}(m-1)(M-1)$ is odd. Hence one of the numbers n, N is even and the other odd, and consequently one of the numbers m, M is a quadratic residue of the other, and the other a quadratic nonresidue of the first. Q. E. D.

SIXTH PROOF OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

1.

THEOREM. *Let p denote a prime number (positive and odd), n a positive integer not divisible by p, and x an indeterminate quantity. Then the function*

$$1 + x^n + x^{2n} + x^{3n} + \text{etc.} + x^{np-n}$$

is divisible by

$$1 + x + xx + x^3 + \text{etc.} + x^{p-1}$$

PROOF. Take a positive integer g such that $gn \equiv 1 \pmod{p}$, and put $gn = 1 + hp$. Then we have

$$\begin{aligned}\frac{1 + x^n + x^{2n} + x^{3n} + \text{etc.}\, x^{np-n}}{1 + x + xx + x^3 + \text{etc.}\, x^{p-1}} &= \frac{(1 - x^{np})(1 - x)}{(1 - x^n)(1 - x^p)} \\ &= \frac{(1 - x^{np})(1 - x^{gn} - x + x^{hp+1})}{(1 - x^n)(1 - x^p)} \\ &= \frac{1 - x^{np}}{1 - x^p} \cdot \frac{1 - x^{gn}}{1 - x^n} - \frac{x(1 - x^{np})}{1 - x^n} \cdot \frac{1 - x^{hp}}{1 - x^p}\end{aligned}$$

which is evidently a polynomial. Q. E. D.

Thus every polynomial in x divisible by $\frac{1-x^{np}}{1-x^n}$ is also divisible by $\frac{1-x^p}{1-x}$.

2.

Let α denote a positive primitive root modulo p; that is, let α be a positive integer such that the least positive residues of the powers $1, \alpha, \alpha\alpha, \alpha^3 \ldots\ldots \alpha^{p-2}$ modulo p are precisely the numbers $1, 2, 3, 4 \ldots\ldots p-1$, without regard to order. Further, denoting by fx the function

$$x + x^\alpha + x^{\alpha\alpha} + x^{\alpha^3} + \text{etc.} + x^{\alpha^{p-2}} + 1$$

it is clear that $fx - 1 - x - xx - x^3 - \text{etc.} - x^{p-1}$ is divisible by $1 - x^p$, and therefore all the more by $\frac{1-x^p}{1-x} = 1 + x + xx + x^3 + \text{etc.} + x^{p-1}$. This last function therefore also divides fx itself. Since x represents an indeterminate quantity, it follows that $f(x^n)$ is also divisible by $\frac{1-x^{np}}{1-x^n}$, and hence, by the preceding article, by $\frac{1-x^p}{1-x}$, whenever n is an integer not divisible by p. If, on the other hand, n is an integer divisible by p, the individual terms of the function $f(x^n)$, each decreased by one, are divisible by $1 - x^p$. In this case, therefore, $f(x^n) - p$ is also divisible by $1 - x^p$, and hence by $\frac{1-x^p}{1-x}$.

3.

THEOREM. *Putting*

$$x - x^{\alpha} + x^{\alpha\alpha} - x^{\alpha^3} + x^{\alpha^4} - \text{etc.} - x^{\alpha^{p-2}} = \xi$$

then $\xi\xi \mp p$ *is divisible by* $\frac{1-x^p}{1-x}$, *taking the upper sign whenever* p *is of the form* $4k+1$, *and the lower whenever* p *is of the form* $4k+3$.

PROOF. It is easy to see that, of the following $p-1$ functions

$$\begin{aligned}
&+ x\xi - xx + x^{\alpha+1} - x^{\alpha\alpha+1} \\
&\qquad + \text{etc.} + x^{\alpha^{p-2}+1} \\
&- x^{\alpha}\xi - x^{2\alpha} + x^{\alpha\alpha+\alpha} - x^{\alpha^3+\alpha} \\
&\qquad + \text{etc.} + x^{\alpha^{p-1}+\alpha} \\
&+ x^{\alpha\alpha}\xi - x^{2\alpha\alpha} + x^{\alpha^3+\alpha\alpha} - x^{\alpha^4+\alpha\alpha} \\
&\qquad + \text{etc.} + x^{\alpha^p+\alpha\alpha} \\
&- x^{\alpha^3}\xi - x^{2\alpha^3} + x^{\alpha^4+\alpha^3} - x^{\alpha^5+\alpha^3} \\
&\qquad + \text{etc.} + x^{\alpha^{p+1}+\alpha^3}
\end{aligned}$$

etc., as far as

$$\begin{aligned}
&- x^{\alpha^{p-2}}\xi - x^{2\alpha^{p-2}} + x^{\alpha^{p-1}+\alpha^{p-2}} - x^{\alpha^p+\alpha^{p-2}} \\
&\qquad\qquad + \text{etc.} + x^{\alpha^{2p-4}+\alpha^{p-2}}
\end{aligned}$$

the first is $= 0$, while each of the others is divisible by $1 - x^p$. Their sum is therefore also divisible by $1 - x^p$, and is found to be

$$\begin{aligned}
&= \xi\xi - (f(xx) - 1) + (f(x^{\alpha+1}) - 1) - (f(x^{\alpha\alpha+1}) - 1) \\
&\qquad + (f(x^{\alpha^3+1}) - 1) - \text{etc.} \\
&\qquad + (f(x^{\alpha^{p-2}+1}) - 1) \\
&= \xi\xi - f(xx) + f(x^{\alpha+1}) - f(x^{\alpha\alpha+1}) \\
&\qquad + f(x^{\alpha^3+1}) - \text{etc.} + f(x^{\alpha^{p-2}+1}) = \Omega
\end{aligned}$$

Thus this expression Ω is also divisible by $\frac{1-x^p}{1-x}$. Now, among the exponents $2,\ \alpha+1,\ \alpha\alpha+1,\ \alpha^3+1 \ldots\ldots \alpha^{p-2}+1$, only one is divisible by p, namely $\alpha^{\frac{1}{2}(p-1)}+1$. Hence, by the preceding article, the following individual parts of the expression Ω

$$f(xx), \quad f(x^{\alpha+1}), \quad f(x^{\alpha\alpha+1}), \quad (fx^{\alpha^3+1}) \quad \text{etc.}$$

with the sole exception of the term $f(x^{\alpha^{\frac{1}{2}(p-1)}+1})$, are divisible by $\frac{1-x^p}{1-x}$. Those parts may therefore be removed, leaving the following function still divisible by $\frac{1-x^p}{1-x}$:

$$\xi\xi \mp f(x^{\alpha^{\frac{1}{2}(p-1)}+1})$$

Here the upper or lower sign applies according as p is of the form $4k+1$ or of the form $4k+3$. Since, moreover, $f(x^{\alpha^{\frac{1}{2}(p-1)}+1})-p$ is divisible by $\frac{1-x^p}{1-x}$, $\xi\xi \mp p$ is also divisible by $\frac{1-x^p}{1-x}$. Q. E. D.

To prevent any ambiguity from the double sign, we shall use ε to denote the number $+1$ or -1, according as p is of the form $4k+1$ or $4k+3$. Thus $\frac{(1-x)(\xi\xi-\varepsilon p)}{1-x^p}$ is a polynomial in x, which we shall denote by Z.

4.

Let q be a positive odd number, so that $\frac{1}{2}(q-1)$ is an integer. Then $(\xi\xi)^{\frac{1}{2}(q-1)} - (\varepsilon p)^{\frac{1}{2}(q-1)}$ is divisible by $\xi\xi - \varepsilon p$, and hence also by $\frac{1-x^p}{1-x}$. Let us put $\varepsilon^{\frac{1}{2}(q-1)} = \delta$, and

$$\xi^{q-1} - \delta p^{\frac{1}{2}(q-1)} = \frac{1-x^p}{1-x} \cdot Y$$

Then Y is a polynomial in x, and $\delta = .+1$ whenever one of the numbers p, q, or indeed both, is of the form $4k+1$. On the other hand, $\delta = -1$ whenever both p, q are of the form $4k+3$.

5.

Let us now suppose that q too is a prime number, different from p. It is clear from the theorem proved in Article 51 of the *Disquisitiones Arithmeticae* that

$$. \quad \xi^q - (x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}})$$

is divisible by q, or is of the form qX, where X is a polynomial in x whose numerical coefficients are also integers (as is also to be understood of the other polynomials Z, Y, W occurring here). For the modulus p and primitive root α, let us denote the index of the number q by μ; that is, let $q \equiv \alpha^\mu \pmod{p}$. The numbers $q,\ q\alpha,\ q\alpha\alpha,\ q\alpha^3 \ldots\ldots q\alpha^{p-2}$ are therefore congruent modulo p, respectively, to the numbers $\alpha^\mu,\ \alpha^{\mu+1},\ \alpha^{\mu+2} \ldots\ldots \alpha^{p-2},\ 1,\ \alpha,\ \alpha\alpha \ldots\ldots \alpha^{\mu-1}$, so that

$$\begin{array}{l}
x^q - x^{\alpha^\mu} \\
x^{q\alpha} - x^{\alpha^{1+1}} \\
x^{q\alpha\alpha} - x^{\alpha^{\mu+2}} \\
x^{q\alpha^3} - x^{\alpha^{\mu+3}} \\
\quad . \\
\quad . \\
\quad . \\
\quad . \\
x^{q\alpha^{p-\mu-2}} - x^{\alpha^{p-2}} \\
x^{q\alpha^{p-\mu-1}} - x \\
x^{q\alpha^{p-\mu}} - x^\alpha \\
x^{q\alpha^{p-\mu+1}} - x^{\alpha\alpha} \\
\quad . \\
\quad . \\
\quad . \\
\quad . \\
x^{q\alpha^{p-2}} - x^{\alpha^{\mu-1}}
\end{array}$$

are divisible by $1 - x^p$. Taking these quantities alternately with positive and negative signs and adding them shows that $1 - x^p$ divides the function

$$x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}} \mp \xi$$

with the upper or lower sign applying according as μ is even or odd; that is, according as q is a quadratic residue or a nonresidue of p. We shall therefore put

$$x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}} - \gamma\xi = (1 - x^p)W$$

taking $\gamma = +1$ or $\gamma = -1$, according as q is a quadratic residue or a nonresidue of p. It is clear that W is a polynomial.

6.

With these preparations, we combine the preceding equations to obtain

$$\begin{aligned} q\xi X = \varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma) & \\ & + \frac{1 - x^p}{1 - x} \cdot (Z(\delta p^{\frac{1}{2}(q-1)} - \gamma) \\ & + Y\xi\xi - W\xi(1 - x)) \end{aligned}$$

Let us suppose that dividing the function ξX by

$$x^{p-1} + x^{p-2} + x^{p-3} + \text{etc.} + x + 1$$

yields the quotient U with remainder T, or that we have

$$\xi X = \frac{1 - x^p}{1 - x} \cdot U + T$$

where U, T are polynomials whose numerical coefficients are also integers, and T is certainly of lower degree than the divisor. We therefore have

$$\begin{aligned} qT - \varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma) & \\ & = \frac{1 - x^p}{1 - x} \cdot (Z(\delta p^{\frac{1}{2}(q-1)} - \gamma) \\ & \quad + Y\xi\xi - W\xi(1 - x) - qU) \end{aligned}$$

This equation evidently cannot hold unless the left side and the right side each vanish separately. Thus $\varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma)$ is divisible by q, as is $\delta p^{\frac{1}{2}(q-1)} - \gamma$; and therefore, since $\delta\delta = 1$, the number $p^{\frac{1}{2}(q-1)} - \gamma\delta$ is also divisible by q.

Now, if β denotes unity taken positively or negatively, according as p is a quadratic residue or a quadratic nonresidue of the number q, then $p^{\frac{1}{2}(q-1)} - \beta$ is divisible by q, and hence so is $\beta - \gamma\delta$. This cannot happen unless $\beta = \gamma\delta$. The fundamental theorem follows immediately. Namely,

I. Whenever the numbers p, q both have the form $4k + 1$, or just one of them does, so that $\delta = +1$, we have $\beta = \gamma$. Consequently either q is a quadratic residue of p and at the same time p a quadratic residue of q; or q is a nonresidue of p and at the same time p a nonresidue of q.

II. Whenever both p, q are of the form $4k + 3$, so that $\delta = -1$, we have $\beta = -\gamma$. Consequently either q is a quadratic residue of p and at the same time p a nonresidue of q; or q is a nonresidue of p and at the same time p a residue of q.

Q. E. D.

A new algorithm for deciding whether a given positive integer is a quadratic residue or a nonresidue of a given positive prime number.

1.

Before presenting a new solution to this problem, we shall briefly recall the solution given in the *Disquisitiones Arithmeticae*. It can be carried out quite readily with the help of the fundamental theorem and the following known theorems:

I. The relation of the number a to the number b (whether the former is a quadratic residue or a nonresidue of the latter) is the same as that of the number c to b, if $a \equiv c \pmod{b}$.

II. If a is the product of the factors $\alpha, \beta, \gamma, \delta$ etc., and b is prime, the relation of a to b depends on the relations of these factors to b as follows: a is a quadratic residue of b or a nonresidue according as an even or an odd number of those factors are nonresidues of b. Thus a factor that is a square can be disregarded entirely in this examination; if a factor is an integer raised to an odd power, that integer itself can take its place.

III. The number 2 is a quadratic residue of every prime of the form $8m+1$ or $8m+7$, and a nonresidue of every prime of the form $8m+3$ or $8m+5$.

Suppose, then, that a number a is given whose relation to a prime b is sought. If a is greater than b, first replace it by its least positive residue modulo b. Once this residue has been resolved into prime factors, theorem II reduces the question to finding the relation of each factor to b. The relation of the factor 2 (if it occurs once, three times, five times, etc.) is known from theorem III; by the fundamental theorem, the relation of each remaining factor depends on the relation of b itself to that factor. Thus, in place of the single relation of the given number to the prime b, we must now investigate several relations of the number b to other odd primes smaller than b. These problems are reduced in the same way to smaller moduli, and these successive reductions must evidently come to an end.

2.

To illustrate this solution, suppose that the relation of the number 103 to 379 is sought. Since 103 is already less than 379 and is itself prime, the fundamental theorem applies immediately: it tells us that the required relation is opposite to that of the number 379 to 103. This in turn is the same as the relation of the number 70 to 103, which itself depends on the relations of 2, 5, 7 to 103. The first of these is known from theorem III. By the fundamental theorem, the second depends on the relation of the number 103 to 5, which by theorem I is the same as that of 3 to 5; this in turn depends, by the fundamental theorem, on the relation of 5 to 3, which by theorem I is the same as that of 2 to 3, known from theorem III. Likewise, by the

fundamental theorem, the relation of 7 to 103 depends on that of 103 to 7, which by theorem I is the same as that of 5 to 7; this in turn depends, by the fundamental theorem, on the relation of 7 to 5, which by theorem I is the same as that of 2 to 5, known from theorem III. If we now wish to turn this analysis into a synthesis, the decision is reduced to fourteen steps. We give them all here, so that the greater elegance of the new solution may stand out more clearly.

1. The number 2 is a quadratic residue of the number 103 (theorem III).
2. The number 2 is a quadratic nonresidue of the number 3 (theorem III).
3. The number 5 is a quadratic nonresidue of the number 3 (from I and 2).
4. The number 3 is a quadratic nonresidue of the number 5 (the fundamental theorem and 3).
5. The number 103 is a quadratic nonresidue of the number 5 (I and 4).
6. The number 5 is a quadratic nonresidue of the number 103 (the fundamental theorem and 5).
7. The number 2 is a quadratic nonresidue of the number 5 (theorem III).
8. The number 7 is a quadratic nonresidue of the number 5 (I and 7).
9. The number 5 is a quadratic nonresidue of the number 7 (the fundamental theorem and 8).
10. The number 103 is a quadratic nonresidue of the number 7 (I and 9).
11. The number 7 is a quadratic residue of the number 103 (the fundamental theorem and 10).
12. The number 70 is a quadratic nonresidue of the number 103 (II, 1, 6, 11).
13. The number 379 is a quadratic nonresidue of the number 103 (I and 12).
14. The number 103 is a quadratic residue of the number 379 (the fundamental theorem and 13).

For brevity, in what follows we shall use the notation introduced in *Comment. Gotting. Vol. XVI.* By $[x]$ we shall mean the quantity x itself when x is an integer, and the greatest integer below x when x is not an integer. Thus $x - [x]$ is always nonnegative and less than unity.

3.

Problem. *Let a, b be positive integers coprime to one another, and put $[\frac{1}{2}a] = a'$. Find the sum*

$$\left[\frac{b}{a}\right] + \left[\frac{2b}{a}\right] + \left[\frac{3b}{a}\right] + \left[\frac{4b}{a}\right] + \text{etc.} + \left[\frac{a'b}{a}\right]$$

SOLUTION. For brevity, let us denote such a sum by $\varphi(a, b)$, so that we also have

$$\varphi(b, a) = \left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \left[\frac{3a}{b}\right] + \text{etc.} + \left[\frac{b'a}{b}\right]$$

if we put $[\frac{1}{2}b] = b'$. In the third proof of the fundamental theorem it was shown that, when a and b are odd,

$$\varphi(a, b) + \varphi(b, a) = a'b'$$

By following the same method, the truth of this proposition extends easily to the case in which one of the numbers $a,\ b$ is odd, as we already indicated there. Divide a by b, as in the method for finding the greatest common divisor of two integers, and let β be the quotient and c the remainder; then divide b by c, and so on, obtaining the equations

$$\begin{aligned} a &= \beta b + c \\ b &= \gamma c + d \\ c &= \delta d + e \\ d &= \varepsilon e + f \text{ etc.} \end{aligned}$$

In this way the continually decreasing sequence of numbers b, c, d, e, f etc. will eventually reach unity, since by hypothesis a and b are coprime. The last equation is therefore

$$k = \lambda l + 1$$

Since evidently

$$\begin{aligned} \left[\frac{a}{b}\right] &= \left[\beta + \frac{c}{b}\right] = \beta + \left[\frac{c}{b}\right] \\ \left[\frac{2a}{b}\right] &= \left[2\beta + \frac{2c}{b}\right] = 2\beta + \left[\frac{2c}{b}\right] \\ \left[\frac{3a}{b}\right] &= \left[3\beta + \frac{3c}{b}\right] = 3\beta + \left[\frac{3c}{b}\right] \end{aligned}$$

etc.. we have

$$\varphi(b, a) = \varphi(b, c) + \tfrac{1}{2}\beta(b'b' + b')$$

and consequently

$$\varphi(a, b) = a'b' - \tfrac{1}{2}\beta(b'b' + b') - \varphi(b, c)$$

By similar reasoning, on putting $[\frac{1}{2}c] = c'$, $[\frac{1}{2}d] = d'$ $[\frac{1}{2}e] = e'$ etc., we obtain

$$\begin{aligned}\varphi(b,c) &= b'c' - \tfrac{1}{2}\gamma(c'c' + c') - \varphi(c,d)\\ \varphi(c,d) &= c'd' - \tfrac{1}{2}\delta(d'd' + d') - \varphi(d,e).\\ \varphi(d,e) &= d'e' - \tfrac{1}{2}\varepsilon(e'e' + e') - \varphi(e,f)\end{aligned}$$

etc., as far as

$$\varphi(k,l) = k'l' - \tfrac{1}{2}\lambda(l'l' + l') - \varphi(l,1)$$

Since evidently $\varphi(l,1) = 0$, we obtain from this the formula

$$\begin{aligned}\varphi(a,b) = {} & a'b' - b'c' + c'd' - d'e' + \text{etc.} \pm k'l'\\ & - \tfrac{1}{2}\beta(b'b' + b') + \tfrac{1}{2}\gamma(c'c' + c') - \tfrac{1}{2}\delta(d'd' + d')\\ & + \tfrac{1}{2}\varepsilon(e'e' + e') - \text{etc.} \mp \tfrac{1}{2}\lambda(l'l' + l')\end{aligned}$$

4.

It now follows easily from the third proof that the relation of the number b to a, whenever a is prime, can be read directly from the value of the sum $\varphi(a, 2b)$. According as this sum is even or odd, b is a quadratic residue of a or a nonresidue. The sum $\varphi(a,b)$ itself can also be used for the same purpose, provided that we distinguish the case in which b is odd from that in which it is even. Namely,

I. When b is odd, b is a quadratic residue or a nonresidue of a according as $\varphi(a,b)$ is even or odd.

II. When b is even, the same rule holds if, in addition, a is of the form $8n+1$ or $8n+7$; but if, for an even value of b, the modulus a is of the form $8n+3$ or $8n+5$, the opposite rule must be applied: b is a quadratic residue of a if $\varphi(a,b)$ is odd, and a nonresidue if $\varphi(a,b)$ is even.

All this follows very easily from Article 4 of the third proof.

5.

Example. To find the relation of the number 103 to the prime 379, we have, for calculating the sum $\varphi(379, 103)$,

$$\begin{array}{r|r|l} a = 379 & a' = 189 & \\ b = 103 & b' = 51 & \beta = 3 \\ c = 70 & c' = 35 & \gamma = 1 \\ d = 33 & d' = 16 & \delta = 2 \\ e = 4 & e' = 2 & \varepsilon = 8 \end{array}$$

and hence

$$\begin{aligned} \varphi(379, 103) &= 9639 - 1785 + 560 - 32 - 3978 + 630 - 272 + 24 \\ &= 4786 \end{aligned}$$

so 103 is a quadratic residue of the number 379. If we prefer to use the sum $(379, 206)$ for the same purpose, we have the following scheme:

$$\begin{array}{r|r|l} 379 & 189 & \\ 206 & 103 & 1 \\ 173 & 86 & 1 \\ 33 & 16 & 5 \\ 8 & 4 & 4 \end{array}$$

from which we obtain

$$\begin{aligned} \varphi(379, 206) &= 19467 - 8858 + 1376 - 64 - 5356 + 3741 - 680 + 40 \\ &= 9666 \end{aligned}$$

Thus 103 is a quadratic residue of the number 379.

6.

To determine the relation of the number b to a, it is not necessary to calculate the individual terms of the sum $\varphi(a, b)$; it is enough to know how many of them are odd. Our rule can therefore also be stated as follows:

As above, take $a = \beta b + c,\ b = \gamma c + d,\ c = \delta d + e$ etc., until the sequence of numbers $a,\ b,\ c,\ d,\ e$ etc. reaches unity. Put $[\frac{1}{2}a] = a',\ [\frac{1}{2}b] = b',\ [\frac{1}{2}c] = c'$ etc., and let μ count the odd numbers in the sequence a', b', c' etc. that are immediately followed by an odd number. Further, let ν count the odd numbers in the sequence β, γ, δ etc. whose corresponding numbers in the sequence b', c', d' etc., respectively, have the form $4n + 1$ or $4n + 2$. Then b is a quadratic residue or a nonresidue of a according as $\mu + \nu$ is even or odd, with the sole exception that, when b is even and a is of the form $8n + 3$ or $8n + 5$, the opposite rule holds.

In our example the sequence a', b', c', d', e' has two instances in which an odd entry is immediately followed by an odd entry, so $\mu = 2$; the sequence $\beta', \gamma', \delta', \varepsilon'$ does contain two odd numbers, but their corresponding numbers in the sequence b', c', d', e' have the form $4n+3$, so $\nu = 0$. Thus $\mu+\nu$ is even, and consequently 103 is a quadratic residue of the number 379.

THEORY

OF BIQUADRATIC RESIDUES

FIRST PAPER

BY

CARL FRIEDRICH GAUSS

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THEORY OF BIQUADRATIC RESIDUES.

FIRST PAPER.

1.

The theory of quadratic residues reduces to a few fundamental theorems that rank among the most beautiful jewels of higher arithmetic. As is well known, these were first easily discovered through inductive observation and then proved in many different ways, leaving nothing further to be desired.

The theory of cubic and biquadratic residues, however, requires much deeper investigation. When we began to explore it in 1805, besides the results that lie, as it were, at the threshold, some special theorems presented themselves, remarkable both for their simplicity and for the difficulty of their proofs. But we soon found that the principles of arithmetic used up to then were wholly insufficient to establish the general theory. Rather, the theory necessarily requires the field of higher arithmetic to be enlarged, as it were, infinitely many times over. What this means will become entirely clear as these investigations continue. As soon as we entered this new field, inductive observation immediately opened a path to knowledge of the simplest theorems encompassing the entire theory. Their proofs, however, lay so deeply hidden that only after many unsuccessful attempts could they finally be brought to light.

Now that we are preparing to publish these studies, we shall begin with the theory of biquadratic residues. In this first paper we shall present the investigations that could already be completed on this side of the enlarged field of arithmetic. They pave the way to that field, as it were, and also contribute some new developments to the theory of the division of the circle.

2.

We introduced the notion of a biquadratic residue in Article 115 of the *Disquisitiones Arithmeticae*: an integer a, positive or negative, is called a biquadratic residue of the integer p if a can be congruent to a fourth power modulo p, and a biquadratic nonresidue if no such congruence exists. Throughout the following investigations, unless we expressly state otherwise, we shall assume that the modulus p is an odd positive prime and that a is not divisible by p, since all the remaining cases can very easily be reduced to this one.

3.

Clearly, every biquadratic residue of p is also a quadratic residue of the same number, and hence every quadratic nonresidue is also a biquadratic nonresidue. The converse also holds whenever p is a prime of the form $4n+3$.

For if, in this case, a is a quadratic residue of p, let us put $a \equiv bb \pmod{p}$. Here b is either a quadratic residue of p or a nonresidue. In the first case, let us put $b \equiv cc$, giving $a \equiv c^4$; that is, a is a biquadratic residue of p. In the second case, $-b$ is a quadratic residue of p (since -1 is a nonresidue of every prime of the form $4n + 3$). Putting $-b \equiv cc$ again gives $a \equiv c^4$, and a is a biquadratic residue of p. It is also easy to see that in this case the congruence $x^4 \equiv a \pmod{p}$ has no solutions besides $x \equiv c$ and $x \equiv -c$. Since these evident propositions exhaust the entire theory of biquadratic residues for prime moduli of the form $4n+3$, we shall exclude these moduli altogether from our investigation, restricting it to prime moduli of the form $4n + 1$.

4.

Thus, when p is a prime of the form $4n+1$, the converse of the proposition in the preceding article does not hold: there can be quadratic residues that are not also biquadratic residues. This happens whenever a quadratic residue is congruent to the square of a quadratic nonresidue. For put $a \equiv bb$, with b a quadratic nonresidue of p. If the congruence $x^4 \equiv a$ could be satisfied by a value $x \equiv c$, we would have $c^4 \equiv bb$, so that the product $(cc-b)(cc+b)$ would be divisible by p. Then p would have to divide either the factor $cc - b$ or the other factor $cc + b$; that is, either $+b$ or $-b$ would be a quadratic residue of p, and therefore both would be (since -1 is a quadratic residue), contrary to the hypothesis.

All integers not divisible by p could therefore be divided into three classes: the first would contain the biquadratic residues, the second the biquadratic nonresidues that are also quadratic residues, and the third the quadratic nonresidues. Clearly it suffices to classify only the numbers $1, 2, 3 \ldots . p - 1$. Half would belong to the third class, while the other half would be divided between the first and second classes.

5.

It will be better, however, to establish four classes as follows.

Let A be the collection of all biquadratic residues of p between 1 and $p-1$, inclusive, and let e be any chosen quadratic nonresidue of p. Further, let B be the collection of the least positive residues of the products eA modulo p, and let C, D respectively be the collections of the least positive residues of the products eeA, e^3A modulo p. It is then easy to see that the individual numbers in B are distinct, as are those in C and those in D; zero cannot occur among any of these numbers. Also, all numbers in A and C are quadratic residues of p, while all those in B and D are quadratic nonresidues. Thus neither of the collections A, C can have a number in common with B or D. But neither can A have a number in common with C, nor B with D. For suppose

I. that some number in A, say a, also occurs in C, where it arises from the congruent product eea', with a' a number in A. Put $a \equiv \alpha^4$, $a' \equiv \alpha'^4$, and choose an integer Θ such that $\Theta\alpha' \equiv 1$. We then have $ee\alpha'^4 \equiv \alpha^4$, and multiplying by Θ^4 gives

$$ee \equiv \alpha^4\Theta^4$$

That is, ee is a biquadratic residue, and hence e a quadratic residue, contrary to the hypothesis.

II. Similarly, suppose that a number belongs to both collections B, D, arising from the products ea, e^3a', with a, a' numbers in A. The congruence $ea \equiv e^3a'$ would imply $a \equiv eea'$. We would thus have a number arising from the product eea' that belonged to both C and A, which we have just proved impossible.

It is also easy to prove that *all* quadratic residues of p between 1 and $p-1$, inclusive, must occur either in A or in C, and all quadratic nonresidues of p between those limits must occur either in B or in D. For

I. Every such quadratic residue that is also a biquadratic residue belongs to A by hypothesis.

II. Write a quadratic residue h (less than p) that is also a biquadratic nonresidue as $\equiv gg$, where g is a quadratic nonresidue. Choose an integer γ such that $e\gamma \equiv g$. Then γ is a quadratic residue of p, which we write as $\equiv kk$. Hence

$$h \equiv gg \equiv ee\gamma\gamma \equiv eek^4$$

Since the least residue of k^4 belongs to A, the number h, which arises by multiplying it by ee, must belong to C.

III. Let h denote a quadratic nonresidue of p between 1 and $p-1$. Find an integer g between the same limits such that $eg \equiv h$. Then g is a quadratic

residue and so belongs to either A or C. In the first case, h evidently belongs to B, and in the second to D.

All this shows that the numbers $1, 2, 3 \ldots . p-1$ are distributed among the four lists A, B, C, D so that each occurs in one of them; each list must therefore contain $\frac{1}{4}(p-1)$ numbers. In this classification, the numbers belonging to A and C are intrinsic to those classes. The distinction between B and D, however, is arbitrary insofar as it depends on the choice of e, which itself always belongs to B. Thus, if a number from D is chosen instead, the classes B, D are interchanged.

6.

Since -1 is a quadratic residue of p, let us put $-1 \equiv ff \pmod{p}$. The four roots of the congruence $x^4 \equiv 1$ are then $1, f, -1, -f$. Thus if a is a biquadratic residue of p, say $\equiv \alpha^4$, the four roots of the congruence $x^4 \equiv a$ are $\alpha, f\alpha, -\alpha, -f\alpha$, which are easily seen to be pairwise incongruent. Hence, if we collect the least positive residues of the fourth powers $1, 16, 81, 256 \ldots . (p-1)^4$, they are always equal in groups of four, giving $\frac{1}{4}(p-1)$ distinct biquadratic residues that form the collection A. If we collect the least residues of the fourth powers only as far as $(\frac{1}{2}p - \frac{1}{2})^4$, each occurs twice.

7.

The product of two biquadratic residues is clearly a biquadratic residue: multiplying two numbers from class A always gives a product whose least positive residue belongs to the same class. Similarly, the product of a number from B and one from D, or of a number from C and one from C, has its least residue in A.

In B lie the residues of the products $A.B$ and $C.D$; in C those of $A.C$, $B.B$ and $D.D$; and finally in D those of $A.D$ and $B.C$.

The proofs are so evident that it suffices to indicate one. Let, for example, c and d be numbers from C and D, and put $c \equiv eea$, $d \equiv e^3a'$, with a, a' numbers from A. Then e^4aa' is a biquadratic residue, so its least residue belongs to A. For the product we have $cd \equiv e.e^4aa'$, so its least residue belongs to B.

We can now also easily decide which class contains a product of several factors. Assign to the classes A, B, C, D the characters $0, 1, 2, 3$, respectively. The character of a product is then either the sum of the characters of its individual factors or the least residue of that sum modulo 4.

8.

It seemed worthwhile to develop these elementary propositions without using the theory of power residues. With its help, everything can be proved much more easily still.

Let g be a primitive root for the modulus p: a number such that in the sequence of powers $g, gg, g^3 \ldots$., none before g^{p-1} is congruent to unity modulo p. Then the least positive residues of the numbers $1, g, gg, g^3 \ldots . g^{p-2}$ agree, apart from their order, with $1, 2, 3 \ldots . . p-1$, and are distributed among four classes as follows:

to	least residues of the numbers
A	$1,\ g^4, g^8, g^{12} \ldots . . g^{p-5}$
B	$g,\ g^5, g^9, g^{13} \ldots . . g^{p-4}$
C	$gg, g^6, g^{10}, g^{14} \ldots . . g^{p-3}$
D	$g^3, g^7, g^{11}, g^{15} \ldots . . g^{p-2}$

All the preceding propositions follow directly from this.

Just as the numbers $1, 2, 3 \ldots . p-1$ have here been divided among four classes, whose collections we denote by A, B, C, D, we may assign *any* integer not divisible by p to one of these classes according to its least residue modulo p.

9.

We denote by f the least residue of the power $g^{\frac{1}{4}(p-1)}$ modulo p. Since $ff \equiv g^{\frac{1}{2}(p-1)} \equiv -1$ (*Disquis. Arithm.*, Article 62), the symbol f clearly has the same meaning here as in Article 6. Thus the power $g^{\frac{1}{4}\lambda(p-1)}$, with λ a positive integer, is congruent modulo p to $1, f, -1, -f$, according as λ has the form $4m, 4m+1, 4m+2, 4m+3$, respectively, or according as the least residue of g^λ belongs to A, B, C, D, respectively. This gives us a very simple criterion for deciding which class contains a given number h not divisible by p: h belongs to A, B, C or D according as the power $h^{\frac{1}{4}(p-1)}$ is congruent modulo p to $1, f, -1$ or $-f$.

As a corollary, -1 always belongs to class A when p has the form $8n+1$, and to class C when p has the form $8n+5$. A proof of this theorem independent of the theory of power residues can easily be obtained from what we established in Article 115, III of the *Disquisitiones Arithmeticae*.

10.

Since *all* primitive roots for the modulus p arise as residues of the powers g^λ, on taking for λ all numbers coprime to $p-1$, it is easy to see that these roots are equally divided between B and D, with the base g always in B. If we replace g as base by another primitive root from B, the classification remains the same; if we use a primitive root from D as base, the classes B and D are interchanged.

If the classification is based on the criterion in the preceding article, the distinction between B and D depends on which root of the congruence $xx \equiv -1 \pmod{p}$ we choose as the characteristic number f.

11.

To make it easier to illustrate by examples the subtler investigations that we are about to undertake, we give here the construction of the classes for all moduli below 100. For each modulus we have chosen the least primitive root.

$$p = 5$$

$$g = 2,\ f = 2$$

A	1
B	2
C	4
D	3

$$p = 13$$

$$g = 2,\ f = 8$$

A	$1, 3, 9$
B	$2, 5, 6$
C	$4, 10, 12$
D	$7, 8, 11$

$$p = 17$$

$$g = 3,\ f = 13$$

A	$1, 4, 13, 16$
B	$3, 5, 12, 14$
C	$2, 8, 9, 15$
D	$6, 7, 10, 11$

$$p = 29$$

$$g = 2,\ f = 12$$

A	$1, 7, 16, 20, 23, 24, 25$
B	$2, 3, 11, 14, 17, 19, 21$
C	$4, 5, 6, 9, 13, 22, 28$
D	$8, 10, 12, 15, 18, 26, 27$

$$p = 37$$

$$g = 2,\ f = 31$$

A	$1, 7, 9, 10, 12, 16, 26, 33, 34$
B	$2, 14, 15, 18, 20, 24, 29, 31, 32$
C	$3, 4, 11, 21, 25, 27, 28, 30, 36$
D	$5, 6, 8, 13, 17, 19, 22, 23, 35$

$$p = 41$$

$$g = 6,\ \ f = 32$$

A	$1, 4, 10, 16, 18, 23, 25, 31, 37, 40$
B	$6, 14, 15, 17, 19, 22, 24, 26, 27, 35$
C	$2, 5, 8, 9, 20, 21, 32, 33, 36, 39$
D	$3, 7, 11, 12, 13, 28, 29, 30, 34, 38$

$$p = 53$$

$$g = 2,\ \ f = 30$$

A	$1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49$
B	$2, 3, 19, 20, 26, 30, 31, 32, 35, 39, 41, 45, 48$
C	$4, 6, 7, 9, 11, 17, 25, 29, 37, 38, 40, 43, 52$
D	$5, 8, 12, 14, 18, 21, 22, 23, 27, 33, 34, 50, 51$

$$p = 61$$

$$g = 2,\ \ f = 11$$

A	$1, 9, 12, 13, 15, 16, 20, 22, 25, 34, 42, 47, 56, 57, 58$
B	$2, 7, 18, 23, 24, 26, 30, 32, 33, 40, 44, 50, 51, 53, 55$
C	$3, 4, 5, 14, 19, 27, 36, 39, 41, 45, 46, 48, 49, 52, 60$
D	$6, 8, 10, 11, 17, 21, 28, 29, 31, 35, 37, 38, 43, 54, 59$

$$p = 73$$

$$g = 5,\ \ f = 27$$

A	$1, 2, 4, 8, 9, 16, 18, 32, 36, 37, 41, 55, 57, 64, 65, 69, 71, 72$
B	$5, 7, 10, 14, 17, 20, 28, 33, 34, 39, 40, 45, 53, 56, 59, 63, 66, 68$
C	$3, 6, 12, 19, 23, 24, 25, 27, 35, 38, 46, 48, 49, 50, 54, 61, 67, 70$
D	$11, 13, 15, 21, 22, 26, 29, 30, 31, 42, 43, 44, 47, 51, 52, 58, 60, 62$

$$p = 89$$

$$g = 3,\ \ f = 34$$

A	$1, 2, 4, 8, 11, 16, 22, 25, 32, 39, 44.\ 45,$
	$50, 57, 64, 67, 73, 78, 81, 85, 87, 88$
B	$3, 6, 7, 12, 14, 23, 24, 28, 33, 41, 43, 46,$
	$48, 56, 61, 65, 66, 75, 77, 82, 83, 86$
C	$5, 9, 10, 17, 18, 20, 21, 34, 36, 40, 42, 47,$
	$49, 53, 55, 68, 69, 71, 72, 79, 80, 84$
D	$13, 15.\ 19, 26, 27, 29.\ 30, 31, 35, 37, 38.\ 51,$
	$52, 54, 58, 59, 60, 62, 63, 70, 74, 76$

$$p = 97$$

$$g = 5,\ \ f = 22$$

A	$1, 4, 6, 9, 16, 22, 24, 33, 35, 36, 43, 47,$
	$50, 54, 61, 62, 64, 73, 75, 81, 88, 91, 93, 96$
B	$5, 13, 14, 17, 19, 20, 21, 23, 29, 30, 41, 45,$
	$52, 56, 67, 68, 74, 76, 77, 78, 80, 83, 84, 92$
C	$2, 3, 8, 11, 12, 18, 25, 27, 31, 32, 44, 48,$
	$49, 53, 65.\ 66, 70, 72, 79, 85, 86, 89, 94, 95$
D	$7, 10, 15, 26, 28, 34, 37, 38, 39.\ 40, 42, 46,$
	$51, 55, 57, 58, 59, 60, 63, 69, 71, 82, 87, 90$

12.

Since 2 is a quadratic residue of every prime of the form $8n + 1$, but a nonresidue of every prime of the form $8n + 5$, for prime moduli of the first form 2 belongs to class A or C, and for moduli of the second form to B or D. The distinction between B and D is not intrinsic, depending only on the choice of f, so we shall set moduli of the form $8n + 5$ aside for a while. An *inductive investigation* of moduli of the form $8n + 1$ gives 2 in A for $p = 73, 89, 113, 233, 257, 281, 337, 353$, etc.; in contrast, 2 belongs to C for $p = 17, 41, 97, 137, 193, 241, 313, 401, 409, 433, 449, 457$, etc.

Since, moreover, for a prime modulus of the form $8n+1$, -1 is a biquadratic residue, -2 clearly always belongs to the same class as $+2$.

13.

When we compare the examples in the preceding article, at least at first sight no simple criterion seems to offer itself for distinguishing the first moduli from the second. Nevertheless, there are *two* such criteria, remarkable for their elegance and simplicity. The following considerations will prepare the way to one of them.

The modulus p, being a prime of the form $8n+1$, can be expressed in the form $aa+2bb$ in exactly one way (*Disquiss. Arithm.*, Article 182, II). We assume that the roots a, b are taken positive. Clearly a is odd and b even; let us put $b = 2^\lambda c$, with c odd. We now observe:

I. Since $p \equiv aa \pmod{c}$, p is a quadratic residue of c, and hence also of each prime factor into which c is resolved. Conversely, by the fundamental theorem, each of these prime factors is a quadratic residue of p, and so their product c is also a quadratic residue of p. Since this also holds for 2, b is clearly a quadratic residue of p, and hence both bb and $-bb$ are biquadratic residues.

II. Thus $-2bb$ must belong to the same class as 2. Since $aa \equiv -2bb$, 2 clearly belongs to class A or C according as a is a quadratic residue of p or a quadratic nonresidue.

III. Now suppose that a has been resolved into prime factors. Denote those of the form $8m+1$ or $8m+7$ by $\alpha, \alpha', \alpha''$, etc., and those of the form $8m+3$ or $8m+5$ by β, β', β'', etc.; let the number of the latter be $= \mu$. Since $p \equiv 2bb \pmod{a}$, p is a quadratic residue of those prime factors of a of which 2 is a quadratic residue, that is, of $\alpha, \alpha', \alpha''$, etc.; it is a quadratic nonresidue of those factors of which 2 is a quadratic nonresidue, that is, of β, β', β'', etc. Conversely, by the fundamental theorem, each of $\alpha, \alpha', \alpha''$, etc., is a quadratic residue of p, while each of β, β', β'', etc., is a quadratic nonresidue. We therefore conclude that their product a is a quadratic residue of p or a nonresidue according as μ is even or odd.

IV. It is easy to verify that the product of all $\alpha, \alpha', \alpha''$, etc., has the form $8m+1$ or $8m+7$, and that the same holds for the product of all β, β', β'', etc., when their number is even. In this case the product a must therefore also have the form $8m+1$ or $8m+7$. In contrast, the product of all β, β', β'', etc., when their number is odd, has the form $8m+3$ or $8m+5$, and the same therefore holds for the product a in this case.

From all this we obtain the elegant theorem:

Whenever a has the form $8m+1$ or $8m+7$, 2 belongs to A; whenever a has the form $8m+3$ or $8m+5$, 2 belongs to C.

The examples listed in the preceding article confirm this. The first moduli decompose as follows: $73 = 1 + 2 \,.\, 36$, $89 = 81 + 2 \,.\, 4$, $113 = 81 + 2 \,.\, 16$, $233 = 225 + 2 \,.\, 4$, $257 = 225 + 2 \,.\, 16$, $281 = 81 + 2 \,.\, 100$, $337 = 49 + 2 \,.\, 144$, $353 = 225 + 2 \,.\, 64$; the second as follows: $17 = 9 + 2 \,.\, 4$, $41 = 9 + 2 \,.\, 16$, $97 = 25 + 2 \,.\, 36$, $137 = 9 + 2 \,.\, 64$, $193 = 121 + 2 \,.\, 36$, $241 = 169 + 2 \,.\, 36$, $313 = 25 + 2 \,.\, 144$, $401 = 9 + 2 \,.\, 196$, $409 = 121 + 2 \,.\, 144$, $433 = 361 + 2 \,.\, 36$, $449 = 441 + 2 \,.\, 4$, $457 = 169 + 2 \,.\, 144$.

14.

The decomposition of p into a square and twice a square has revealed such a remarkable connection with the classification of 2 that it seems worthwhile to ask whether a decomposition into two squares, which p is also known to admit, might give a similar result. Here are the decompositions of the numbers p for which 2 belongs to class

A	C
$9 + 64$	$1 + 16$
$25 + 64$	$25 + 16$
$49 + 64$	$81 + 16$
$169 + 64$	$121 + 16$
$1 + 256$	$49 + 144$
$25 + 256$	$225 + 16$
$81 + 256$	$169 + 144$
$289 + 64$	$1 + 400$
	$9 + 400$
	$289 + 144$
	$49 + 400$
	$441 + 16$

First we observe that, of the two squares into which p decomposes, one must be odd, which we put $= aa$, and the other even, which we put $= bb$. Since aa has the form $8n+1$, values of b that are twice an odd number clearly correspond to values of p of the form $8n + 5$, which are excluded from our inductive investigation here, since they would have 2 in class B or D. For values of p of the form $8n + 1$, however, b must be divisible by four. If we may trust the inductive evidence presented by this table, 2 belongs to class A for every modulus for which b has the form $8n$, and to class C for every modulus for which b has the form $8n + 4$. This theorem, however, requires much deeper investigation than the one obtained in the preceding article. Its proof must be preceded by several preliminary investigations concerning the order in which the numbers in A, B, C, D follow one another.

15.

Let us denote the number of numbers in A immediately followed by a number in A, B, C, D, respectively, by $(00), (01), (02), (03)$; likewise the number of numbers in B followed by a number in A, B, C, D, respectively, by $(10), (11), (12), (13)$. Similarly, let C contain respectively $(20), (21), (22), (23)$ numbers, and D contain $(30), (31), (32), (33)$ numbers, followed by a number in A, B, C, D. We propose to determine these sixteen counts a priori. To help readers compare the general reasoning with examples, it seemed useful to give here the numerical values of the entries in the scheme (S)

$$\begin{array}{llll} (00), & (01), & (02), & (03) \\ (10), & (11), & (12), & (13) \\ (20). & (21), & (22), & (23) \\ (30), & (31), & (32), & (33) \end{array}$$

for each modulus whose classification we gave in Article 11.

$p = 5$	$p = 13$	$p = 17$	$p = 29$
$0, 1, 0.\ 0$	$0, 1, 2, 0$	$0, 2, 1, 0$	$2.\ 3, 0.\ 2$
$0, 0, 0, 1$	$1, 1, 0, 1$	$2, 0, 1, 1$	$1, 1, 2, 3$
$0, 0, 0, 0$	$0, 1, 0, 1$	$1, 1, 1, 1$	$2, 1, 2.\ 1$
$0, 0, 1, 0$	$1, 0, 1, 1$	$0, 1, 1, 2$	$1, 2, 3, 1$

$p = 37$	$p = 41$	$p = 53$	$p = 61$
$2, 1, 2, 4$	$0, 4, 3, 2$	$2, 3, 6, 2$	$4, 3, 2, 6$
$2, 2, 4, 1$	$4, 2, 2, 2$	$4, 4, 2, 3$	$3, 3, 6, 3$
$2, 2, 2, 2$	$3, 2, 3, 2$	$2, 4, 2, 4$	$4, 3, 4, 3$
$2, 4, 1, 2$	$2, 2, 2, 4$	$4, 2, 3, 4$	$3, 6, 3, 3$

$p = 73$	$p = 89$	$p = 97$
$5, 6, 4, 2$	$3, 8, 6, 4$	$2, 6, 7, 8$
$6, 2, 5, 5$	$8, 4, 5, 5$	$6, 8, 5, 5$
$4, 5, 4, 5$	$6, 5, 6, 5$	$7, 5, 7, 5$
$2, 5, 5, 6$	$4, 5, 5, 8$	$8, 5, 5, 6$

Since moduli of the forms $8n + 1$ and $8n + 5$ behave differently, we must treat them separately. We begin with the first.

16.

The symbol (00) indicates how many different ways the equation $\alpha + 1 = \alpha'$ can be satisfied, with α, α' denoting arbitrary numbers in A. For a modulus of the form $8n + 1$, as understood here, α' and $p - \alpha'$ belong to the same

collection. We may therefore say more elegantly that (00) expresses the number of different ways of satisfying the equation $1 + \alpha + \alpha' = p$. Clearly the congruence $1 + \alpha + \alpha' \equiv 0 \pmod{p}$ can also take the place of this equation.

Likewise,

(01) indicates the number of solutions of the congruence $1+\alpha+\beta \equiv 0 \pmod{p}$
(02) the number of solutions of the congruence $1 + \alpha + \gamma \equiv 0$
(03) the number of solutions of the congruence $1 + \alpha + \delta \equiv 0$
(11) the number of solutions of the congruence $1 + \beta + \beta' \equiv 0$, etc.

where β and β' denote arbitrary numbers in B, γ numbers in C, and δ numbers in D. We immediately obtain the following six equations:

$$(01) = (10), \quad (02) = (20), \quad (03) = (30),$$
$$(12) = (21), \quad (13) = (31), \quad (23) = (32)$$

From any given solution of the congruence $1 + \alpha + \beta \equiv 0$ we obtain a solution of $1 + \delta + \delta' \equiv 0$ by taking for δ the number between $1 \ldots . p - 1$ that satisfies $\beta\delta \equiv 1$ (which clearly belongs to D), and for δ' the least positive residue of the product $\alpha\delta$ (which likewise belongs to D). The reverse passage from a given solution of $1 + \delta + \delta' \equiv 0$ to a solution of $1 + \alpha + \beta \equiv 0$ is equally clear: choose β so that $\beta\delta \equiv 1$, and at the same time put $\alpha \equiv \beta\delta'$. Thus the two congruences have the same number of solutions, or $(01) = (33)$.

Similarly, from the congruence $1 + \alpha + \gamma \equiv 0$ we derive $\gamma' + \gamma'' + 1 \equiv 0$ by choosing γ' in C so that $\gamma\gamma' \equiv 1$, and γ'' in the same collection congruent to the product $\alpha\gamma'$. It follows easily that these two congruences admit the same number of solutions, or $(02) = (22)$.

Likewise, from $1 + \alpha + \delta \equiv 0$ we derive $\beta + \beta' + 1 \equiv 0$ by choosing β, β' so that $\beta\delta \equiv 1$, $\beta\alpha \equiv \beta'$, giving $(03) = (11)$.

Finally, from $1 + \beta + \gamma \equiv 0$ we similarly derive both $\delta + 1 + \beta' \equiv 0$ and $\gamma' + \delta' + 1 \equiv 0$, and conclude that $(12) = (13) = (23)$.

We have thus obtained eleven equations among our sixteen unknowns, reducing them to five. The scheme S can therefore be written as

$$\begin{array}{cccc} h, & i, & k, & l \\ i, & l, & m, & m \\ k, & m, & k, & m \\ l, & m, & m, & i \end{array}$$

Three further equations of condition can easily be added. Every number in A, except the last, $p - 1$, must be followed by a number in one of the collections A, B, C or D. Thus

$$(00) + (01) + (02) + (03) = 2n - 1$$

and likewise

$$(10)+(11)+(12)+(13)=2n$$
$$(20)+(21)+(22)+(23)=2n$$
$$(30)+(31)+(32)+(33)=2n$$

In the notation just introduced, the first three equations give

$$h+i+k+l=2n-1$$
$$i+l+2m=2n$$
$$k+m=n$$

The fourth is identical to the second. These equations allow us to eliminate three unknowns, so that all sixteen have now been reduced to two.

17.

To determine them completely, it will be useful to investigate the number of solutions of the congruence

$$1+\alpha+\beta+\gamma\equiv 0 \pmod{p}$$

where α,β,γ denote arbitrary numbers in A,B,C. Clearly $\alpha=p-1$ is inadmissible, since $\beta+\gamma\equiv 0$ cannot hold. Substituting the remaining values for α in turn gives h,i,k,l values of $1+\alpha$ belonging to A,B,C,D, respectively. For each *given* value of $1+\alpha$ belonging to A, say $1+\alpha=\alpha^0$, the congruence $\alpha^0+\beta+\gamma\equiv 0$ admits as many solutions as $1+\beta'+\gamma'\equiv 0$ (put $\beta\equiv\alpha^0\beta'$, $\gamma\equiv\alpha^0\gamma'$), that is, $(12)=m$ solutions. Likewise, for each given value of $1+\alpha$ belonging to B, say $1+\alpha=\beta^0$, the congruence $\beta^0+\beta+\gamma\equiv 0$ has as many solutions as $1+\alpha'+\beta'\equiv 0$ (put $\beta\equiv\beta^0\alpha'$, $\gamma\equiv\beta^0\beta'$), that is, $(01)=i$ solutions. Similarly, for each given value of $1+\alpha$ belonging to C, say $1+\alpha=\gamma^0$, the congruence $\gamma^0+\beta+\gamma\equiv 0$ can be solved in as many different ways as $1+\delta+\alpha'\equiv 0$ (put $\beta\equiv\gamma^0\delta$, $\gamma\equiv\gamma^0\alpha'$), so the number of solutions is $(03)=l$. Finally, for each given value of $1+\alpha$ belonging to D, say $1+\alpha=\delta^0$, the congruence $\delta^0+\beta+\gamma\equiv 0$ has as many solutions as $1+\gamma'+\delta'\equiv 0$ (put $\beta\equiv\delta^0\gamma'$, $\gamma\equiv\delta^0\delta'$), that is, $(23)=m$ solutions. Adding all these counts, the congruence $1+\alpha+\beta+\gamma\equiv 0$ clearly admits

$$hm+ii+kl+lm$$

different solutions.

In exactly the same way we find that, on substituting for β the numbers in B in turn, the sum $1+\beta$ takes respectively $(10),(11),(12),(13)$, or i,l,m,m, values belonging to A,B,C,D. For each *given* value of $1+\beta$ belonging to these collections, the congruence $1+\beta+\alpha+\gamma\equiv 0$ admits respectively

$(02), (31), (20), (13)$, or k, m, k, m, different solutions. The total number of solutions is therefore

$$= ik + lm + km + mm$$

We obtain the same value by basing the calculation on the values of the sum $1 + \gamma$.

18.

These two expressions for the same count give the equation

$$0 = hm + ii + kl - ik - km - mm$$

and hence, eliminating h by means of $h = 2m - k - 1$,

$$0 = (k - m)^2 + ii + kl - ik - kk - m$$

The last two equations of Article 16 give $k = \frac{1}{2}(l + i)$. Substituting this value turns $ii + kl - ik - kk$ into $\frac{1}{4}(l - i)^2$, so the preceding equation, multiplied by 4, becomes

$$0 = 4(k - m)^2 + (l - i)^2 - 4m$$

Since $4m = 2(k + m) - 2(k - m) = 2n - 2(k - m)$, it follows that

$$2n = 4(k - m)^2 + 2(k - m) + (l - i)^2$$

or

$$8n + 1 = (4(k - m) + 1)^2 + 4(l - i)^2$$

Thus, putting

$$4(k - m) + 1 = a, \quad 2l - 2i = b$$

we have

$$p = aa + bb$$

It is known that p can be decomposed into two squares in exactly one way, with the odd square taken as aa and the even square as bb, so aa, bb are completely determined numbers. But a itself is also completely determined: the square root must be taken positive or negative according as the positive root has the form $4M+1$ or $4M+3$. We shall shortly discuss how to determine the sign of b.

Combining these new equations with the last three equations of Article 16 completely determines the five numbers h, i, k, l, m in terms of a, b and n, as follows:

$$\begin{aligned} 8h &= 4n - 3a - 5 \\ 8i &= 4n + a - 2b - 1 \\ 8k &= 4n + a - 1 \\ 8l &= 4n + a + 2b - 1 \\ 8m &= 4n - a + 1 \end{aligned}$$

If, in place of n, we prefer to use the modulus p, the scheme S, with every entry multiplied by 16 to avoid fractions, is

$p-6a-11$	$p+2a-4b-3$	$p+2a-3$	$p+2a+4b-3$
$p+2a-4b-3$	$p+2a+4b-3$	$p-2a+1$	$p-2a+1$
$p+2a-3$	$p-2a+1$	$p+2a-3$	$p-2a+1$
$p+2a+4b-3$	$p-2a+1$	$p-2a+1$	$p+2a-4b-3$

19.

It remains to show how to choose the sign of b. In Article 10 we pointed out that the distinction between B and D, which is not intrinsic, depends on the choice of f, for which either root of $xx \equiv -1$ must be taken, and that the two collections are interchanged when one root is replaced by the other. Inspection of the scheme just given shows that a similar interchange accompanies a change in the sign of b. We can therefore foresee that there must be a connection between the sign of b and f. To find it, we first observe that, if μ denotes a nonnegative integer and z takes all the values $1, 2, 3 \ldots . p-1$, then modulo p we have either $\Sigma z^\mu \equiv 0$ or $\Sigma z^\mu \equiv -1$, according as μ is not divisible by $p-1$ or is divisible by it. The latter part of the theorem follows from the fact that for a value of μ divisible by $p-1$, $z^\mu \equiv 1$. We prove the former part as follows. Let g denote a primitive root. All the values of z coincide with the least residues of all g^y when y takes all the values $0, 1, 2, 3 \ldots . p-2$, and hence $\Sigma z^\mu \equiv \Sigma g^{\mu . y}$. Now

$$\Sigma g^{\mu y} = \frac{g^{\mu(p-1)} - 1}{g^\mu - 1}, \quad \text{and hence}$$
$$(g^\mu - 1)\Sigma z^\mu \equiv g^{\mu(p-1)} - 1 \equiv 0$$

For a value of μ not divisible by $p-1$, g^μ cannot be congruent to 1, or equivalently $g^\mu - 1$ cannot be divisible by p. Hence $\Sigma z^\mu \equiv 0$. Q. E. D.

Now expand the power $(z^4+1)^{\frac{1}{4}(p-1)}$ by the binomial theorem. The preceding lemma gives

$$\Sigma(z^4+1)^{\frac{1}{4}(p-1)} \equiv -2 \text{ (mod. } p)$$

The least residues of all z^4 give all the numbers in A, each occurring four times. Among the least residues of z^4+1 we therefore have

$$\begin{array}{ll} 4(00) & \text{to } A \\ 4(01) & \text{to } B \\ 4(02) & \text{to } C \\ 4(03) & \text{to } D \end{array}$$

belonging to those classes, and four are $= 0$ (namely, when $z^4 \equiv p-1$). Using the criteria for the collections A, B, C, D, we obtain

$$\begin{aligned} \Sigma(z^4+1)^{\frac{1}{4}(p-1)} \equiv\ & 4(00) + 4f(01) \\ & - 4(02) - 4f(03) \end{aligned}$$

and hence

$$-2 \equiv 4(00) + 4f(01) - 4(02) - 4f(03)$$

or, substituting for $(00), (01)$, etc., the values found in the preceding article,

$$-2 \equiv -2a - 2 - 2bf$$

We therefore conclude that we must always have $a + bf \equiv 0$, or, multiplying by f,

$$b \equiv af$$

This congruence determines the sign of b when f has already been chosen, or determines f when the sign of b is prescribed in some other way.

20.

Having completely solved our problem for moduli of the form $8n+1$, we turn to the other case, where p has the form $8n+5$. We can treat it more briefly, since all the arguments differ little from the preceding ones.

Since, for such a modulus, -1 belongs to class C, the numbers in the collections A, B, C, D have their complements to the sum p in the classes C, D, A, B, respectively. It follows readily that

symbol	denotes the number of solutions of the congruence
(00)	$1+\alpha+\gamma \equiv 0$
(01)	$1+\alpha+\delta \equiv 0$
(02)	$1+\alpha+\alpha' \equiv 0$
(03)	$1+\alpha+\beta \equiv 0$
(10)	$1+\beta+\gamma \equiv 0$
(11)	$1+\beta+\delta \equiv 0$
(12)	$1+\beta+\alpha \equiv 0$
(13)	$1+\beta+\beta' \equiv 0$
(20)	$1+\gamma+\gamma' \equiv 0$
(21)	$1+\gamma+\delta \equiv 0$
(22)	$1+\gamma+\alpha \equiv 0$
(23)	$1+\gamma+\beta \equiv 0$
(30)	$1+\delta+\gamma \equiv 0$
(31)	$1+\delta+\delta' \equiv 0$
(32)	$1+\delta+\alpha \equiv 0$
(33)	$1+\delta+\beta \equiv 0$

from which we immediately obtain six equations:

$$(00)=(22), \quad (01)=(32), \quad (03)=(12),$$
$$(10)=(23), \quad (11)=(33), \quad (21)=(30)$$

Multiplying the congruence $1+\alpha+\gamma \equiv 0$ by a number γ' from the collection C, chosen so that $\gamma\gamma' \equiv 1$, and taking for γ'' the least residue of the product $\alpha\gamma'$, which must evidently also belong to the collection C, gives $\gamma'+\gamma''+1 \equiv 0$, from which we conclude that $(00)=(20)$.

In exactly the same way we obtain the equations $(01)=(13)$, $(03)=(31)$, $(10)=(11)=(21)$.

With these eleven equations we can reduce our sixteen unknowns to five and write the scheme S as follows:

$$\begin{array}{cccc} h, & i, & k, & l \\ m, & m, & l, & i \\ h, & m, & h, & m \\ m, & l, & i, & m \end{array}$$

We also have the equations

$$\begin{aligned}(00)+(01)+(02)+(03)&=2n+1\\(10)+(11)+(12)+(13)&=2n+1\\(20)+(21)+(22)+(23)&=2n\\(30)+(31)+(32)+(33)&=2n+1\end{aligned}$$

or, using the notation just introduced, these three (I):

$$\begin{aligned}h+i+k+l&=2n+1\\2m+i+l&=2n+1\\h+m&=n\end{aligned}$$

With their help we can therefore now reduce our unknowns to two.

We shall derive the remaining equations by considering the number of solutions of the congruence $1+\alpha+\beta+\gamma\equiv 0$ (here too, α,β,γ denote unspecified numbers from the collections A,B,C, respectively). Consider *first* that $1+\alpha$ gives h,i,k,l numbers belonging to A,B,C,D, respectively, and that for each given value of α in these four cases there are m,l,i,m solutions, respectively. The total number of solutions will be

$$=hm+il+ik+lm$$

Second, since $1+\beta$ gives m,m,l,i numbers belonging to A,B,C,D, and for each *given* value of β in these four cases there are h,m,h,m solutions, the total number of solutions will be

$$=hm+mm+hl+im$$

From this we derive the equation

$$0=mm+hl+im-il-ik-lm$$

which, using the equation $k=2m-h$ obtained from (I), becomes

$$0=mm+hl+hi-il-im-lm$$

Now from equations I we also have $l+i=1+2h$, and hence

$$\begin{aligned}2i&=1+2h+(i-l)\\2l&=1+2h-(i-l)\end{aligned}$$

Substituting these values into the preceding equation gives

$$0=4mm-4m-1-8hm+4hh+(i-l)^2$$

If, finally, we replace $4m$ here with $2(h+m)-2(h-m)$, or, by the last equation in I, with $2n-2(h-m)$, we obtain

$$0 = 4(h-m)^2 - 2n + 2(h-m) - 1 + (i-l)^2$$

and hence

$$8n + 5 = (4(h-m)+1)^2 + 4(i-l)^2$$

Thus, putting

$$4(h-m)+1 = a, \quad 2i-2l = b$$

gives

$$p = aa + bb$$

Since here too p can be split in only one way into two squares, one even and the other odd, aa and bb will be completely determined numbers: evidently aa must be set equal to the odd square and bb to the even one. Moreover, the *sign* of a must be chosen so that $a \equiv 1 \pmod 4$, and the sign of b so that $b \equiv af \pmod p$, as is readily proved by arguments entirely similar to those used in the preceding article.

With these preliminaries, the five numbers h, i, k, l, m are determined in terms of a, b and n as follows:

$$\begin{aligned} 8h &= 4n + a - 1 \\ 8i &= 4n + a + 2b + 3 \\ 8k &= 4n - 3a + 3 \\ 8l &= 4n + a - 2b + 3 \\ 8m &= 4n - a + 1 \end{aligned}$$

Or, if we prefer expressions in terms of p, the terms of the scheme S, multiplied by 16, will be as follows:

$p+2a-7$	$p+2a+4b+1$	$p-2a+1$	$p+2a-4b+1$
$p-2a-3$	$p-2a-3$	$p+2a-4b+1$	$p+2a+4b+1$
$p+2a-7$	$p-2a-3$	$p+2a-7$	$p-2a-3$
$p-2a-3$	$p+2a-4b+1$	$p+2a+4b+1$	$p-2a-3$

21.

Having solved our problem, we return to the main investigation, now undertaking the complete determination of the collection to which the number 2 belongs.

I. Whenever p has the form $8n+1$, we already know that the number 2 lies either in the collection A or in the collection C. In the former case we readily see that the numbers $\frac{1}{2}(p-1)$, $\frac{1}{2}(p+1)$ also belong to A, and in the latter to C. Now consider that, if α and $\alpha+1$ are consecutive numbers in the collection A, then $p-\alpha-1$, $p-\alpha$ are also such numbers; or, equivalently, numbers in the collection A that are followed by a number in the same collection are always paired (α and $p-1-\alpha$). Their number, (00), will therefore always be even, unless one is paired with itself, that is, unless $\frac{1}{2}(p-1)$ belongs to A, in which case their number will be odd. From this we conclude that (00) is odd whenever 2 belongs to the collection A, and even whenever 2 belongs to C. But we have

$$16(00) = aa + bb - 6a - 11$$

or, putting $a = 4q+1$, $b = 4r$ (see Article 14),

$$(00) = qq - q + rr - 1$$

Since $qq-q$ is evidently always even, (00) will be odd or even according as r is even or odd. Thus 2 will belong to A or to C according as b has the form $8m$ or the form $8m+4$. This is precisely the theorem found by inductive investigation in Article 14.

II. We can treat the other case, where p has the form $8n+5$, just as completely. Here the number 2 belongs either to B or to D. We readily see that in the former case $\frac{1}{2}(p-1)$ belongs to B and $\frac{1}{2}(p+1)$ to D, whereas in the latter $\frac{1}{2}(p-1)$ belongs to D and $\frac{1}{2}(p+1)$ to B. Now consider that, if β is a number from B followed by a number from D, then $p-\beta-1$ will also be a number from B and $p-\beta$ from D; that is, numbers with this property always occur in pairs. Their number, (13), will therefore be even, except when one of them is paired with itself, that is, when $\frac{1}{2}(p-1)$ belongs to B and $\frac{1}{2}(p+1)$ to D; in this case (13) will be odd. From this we conclude that (13) is even whenever 2 belongs to D, and odd whenever 2 belongs to B. But we have

$$16(13) = aa + bb + 2a + 4b + 1$$

or, putting $a = 4q+1$, $b = 4r+2$,

$$(13) = qq + q + rr + 2r + 1$$

Thus (13) will be odd whenever r is even; conversely, (13) will be even whenever r is odd. From this we conclude that 2 belongs to B whenever b has the form $8m+2$, and to D whenever b has the form $8m+6$.

The result of these investigations can be stated as follows:

The number 2 *belongs to the collection* A, B, C *or* D *according as the number* $\frac{1}{2}b$ *has the form* $4m$, $4m+1$, $4m+2$ *or* $4m+3$.

22.

In the *Disquisitiones Arithmeticae* we explained the general theory of circle division and of the solution of the equation $x^p - 1 = 0$. Among other things, we showed that, if μ is a divisor of the number $p-1$, the function $\frac{x^p-1}{x-1}$ can be split into μ factors of degree $\frac{p-1}{\mu}$ with the help of an auxiliary equation of degree μ. Besides the general theory of this factorization, we separately considered the special cases where $\mu = 2$ or $\mu = 3$ in Articles 356–358 of that work, and showed how to give the auxiliary equation a priori, that is, without working out the scheme of least residues of the powers of a primitive root for the modulus p. Even without our pointing it out, attentive readers will readily see the very close connection between the next case of that theory, namely $\mu = 4$, and the investigations developed here in Articles 15–20, with whose help that case too can be treated completely without difficulty. But we reserve this treatment for another occasion, and accordingly have preferred to carry out the investigation in this paper in a purely arithmetical form, without bringing in the theory of the equation $x^p - 1 = 0$ at all. Instead, as a finishing touch, we shall add some further new, purely arithmetical theorems very closely connected with the subject treated so far.

23.

If the power $(x^4+1)^{\frac{1}{2}(p-1)}$ is expanded by the binomial theorem, there will be three terms in which the exponent of x is divisible by $p-1$, namely

$$x^{2(p-1)}, \quad Px^{p-1} \quad \text{and } 1$$

where P denotes the middle coefficient

$$\frac{\frac{1}{2}(p-1) \,.\, \frac{1}{2}(p-3) \,.\, \frac{1}{2}(p-5) \,.\, \ldots \,.\, \frac{1}{4}(p+3)}{1 \,.\, 2 \,.\, 3 \,.\, \ldots \,.\, \frac{1}{4}(p-1)}$$

Thus, substituting for x the numbers $1, 2, 3 \ldots p-1$ in turn, we obtain by the lemma in Article 19

$$\Sigma(x^4+1)^{\frac{1}{2}(p-1)} \equiv -2 - P$$

But, considering what we explained in Article 19, and also that the numbers in the collections A, B, C, D, raised to the power with exponent $\frac{1}{2}(p-1)$, are congruent modulo p to the numbers $+1, -1, +1, -1$, respectively, we readily see that

$$\Sigma(x^4+1)^{\frac{1}{2}(p-1)} \equiv 4(00) - 4(01) + 4(02) - 4(03)$$

and hence, by the schemes given at the ends of Articles 18, 20,

$$\Sigma(x^4+1)^{\frac{1}{2}(p-1)} \equiv -2a - 2$$

Comparing these two values gives a particularly elegant theorem: we have

$$P \equiv 2a \pmod{p}$$

Denoting the four products

$$\begin{array}{l} 1 \,.\, 2 \,.\, 3 \,.\, \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \,.\, \frac{1}{4}(p-1) \\ \frac{1}{4}(p+3) \,.\, \frac{1}{4}(p+7) \,.\, \frac{1}{4}(p+11) \,.\, \ldots\ldots \,.\, \frac{1}{2}(p-1) \\ \frac{1}{2}(p+1) \,.\, \frac{1}{2}(p+3) \,.\, \frac{1}{2}(p+5) \,.\, \ldots\ldots \,.\, \frac{3}{4}(p-1) \\ \frac{1}{4}(3p+1) \,.\, \frac{1}{4}(3p+5) \,.\, \frac{1}{4}(3p+9) \,.\, \ldots\ldots \,.\, (p-1) \end{array}$$

by q, r, s, t, respectively, we can write the preceding theorem as

$$2a \equiv \frac{r}{q} \pmod{p}$$

Since each factor of q has its complement to p in t, we have $q \equiv t \pmod{p}$ whenever the number of factors is even, that is, whenever p has the form $8m+1$, and $q \equiv -t$ whenever the number of factors is odd, or p has the form $8m+5$. Likewise, in the former case $r \equiv s$, and in the latter $r \equiv -s$. In both cases $qr \equiv st$, and since we know that $qrst \equiv -1$, we have $qqrr \equiv -1$, and hence $qr \equiv \pm f \pmod{p}$. Combining this congruence with the theorem just found gives $rr \equiv \pm 2af$, and thus, by Articles 19, 20,

$$2b \equiv \pm rr \pmod{p}^{*)}$$

It is very remarkable that the splitting of the number p into two squares can be found by entirely direct operations: the root of the odd square will be the least absolute residue of $\frac{r}{2q}$, and the root of the even square the least absolute residue of $\frac{1}{2}rr$ modulo p. The expression $\frac{r}{2q}$, whose value for $p = 5$ is $= 1$, can also be written as follows for larger values of p:

$$\frac{6 \,.\, 10 \,.\, 14 \,.\, 18 \,.\, \ldots \,.\, (p-3)}{2 \,.\, 3 \,.\, 4 \,.\, 5 \,.\, \ldots \,.\, \frac{1}{4}(p-1)}$$

But since we also know the sign with which this formula gives the root of the odd square, namely the sign that always makes it of the form $4m+1$, it deserves particular attention that a similar general criterion for the sign of the root of the even square has not yet been found. If anyone finds such a criterion and communicates it to us, we shall be very grateful. Meanwhile, it seemed appropriate to include here the values of the numbers a, b, f obtained, for values of p below 200, from the least residues of the expressions $\frac{r}{2q}$, $\frac{1}{2}rr$, qr.

*) and $\{(a \mp b)q\}^2 \equiv a \equiv \left(\frac{r-qrr}{2}\right)^2$

p	a	b	f
5	+1	+2	2
13	−3	−2	5
17	+1	−4	13
29	+5	+2	12
37	+1	−6	31
41	+5	+4	9
53	−7	−2	23
61	+5	−6	11
73	−3	−8	27
89	+5	−8	34
97	+9	+4	22
101	+1	−10	91
109	−3	+10	33
113	−7	+8	15
137	−11	+4	37
149	−7	−10	44
157	−11	−6	129
173	+13	+2	80
181	+9	+10	162
193	−7	+12	81
197	+1	−14	183

THEORY

OF BIQUADRATIC RESIDUES

SECOND PAPER

BY

CARL FRIEDRICH GAUSS

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THEORY OF BIQUADRATIC RESIDUES.

SECOND PAPER.

24.

In the first paper we completely dealt with everything required for the biquadratic classification of the number $+2$. Namely, we think of all numbers not divisible by the modulus p (assumed to be a prime number of the form $4n+1$) as distributed among four collections A, B, C, D, according as each, raised to the power with exponent $\frac{1}{4}(p-1)$, becomes congruent modulo p to $+1$, $+f$, -1, $-f$, with f denoting either root of the congruence $ff \equiv -1 \pmod{p}$. We found that the decision as to which collection the number $+2$ belongs to depends on splitting the number p into two squares: if we put $p = aa + bb$, with aa denoting the odd square and bb the even one, and further assume that the *signs* of a, b are chosen so that $a \equiv 1 \pmod{4}$, $b \equiv af \pmod{p}$, then the number $+2$ must belong to the collection A, B, C, D according as $\frac{1}{2}b$ has the form $4n$, $4n+1$, $4n+2$, $4n+3$, respectively.

A rule for classifying the number -2 also follows immediately. Since -1 belongs to class A when $\frac{1}{2}b$ is even, and to class C when it is odd, the theorem in Article 7 shows that the number -2 belongs to class A, B, C, D according as $\frac{1}{2}b$ has the form $4n$, $4n+3$, $4n+2$, $4n+1$, respectively.

These theorems can also be expressed as follows:

Belongs	$+2$	-2
to the collection	if b, modulo 8, is congruent to	
A	0	0
B	$2a$	$6a$
C	$4a$	$4a$
D	$6a$	$2a$

We readily see that, stated in this way, the theorems no longer depend on the condition $a \equiv 1 \pmod{4}$, but still hold if $a \equiv 3 \pmod{4}$, provided that the other condition, $af \equiv b \pmod{p}$, is maintained.

It is equally easy to see that the content of these theorems can be elegantly condensed into a single formula, namely:

if a and b are taken positive, we always have

$$b^{\frac{1}{2}ab} \equiv a^{\frac{1}{2}ab}\, 2^{\frac{1}{4}(p-1)} \pmod{p}$$

25.

Let us now see how far inductive investigation points towards the classification of the number 3. The table in Article 11, continued further (always taking the least primitive root), shows that +3 belongs

to the collection

A for			B for			C for			D for		
p	a	b	p	a	b	p	a	b	p	a	b
13	−3	+2	17	+1	−4	37	+1	−6	5	+1	+2
109	−3	+10	29	+5	+2	61	+5	−6	41	+5	−4
181	+9	+10	53	−7	+2	73	−3	−8	149	−7	+10
193	−7	−12	89	+5	−8	97	+9	+4	173	+13	+2
229	−15	+2	101	+1	+10	157	−11	−6			
277	+9	+14	113	−7	−8	241	−15	−4			
			137	−11	−4						
			197	+1	−14						
			233	+13	+8						
			257	+1	−16						
			269	+13	+10						
			281	+5	+16						
			293	+17	+2						

At least at first sight, we see no simple connection between the values of the numbers a, b to which the same collection corresponds. But, considering that a similar decision in the theory of quadratic residues is made by a simpler rule for the number −3 than for the number +3, we may hope for equally favourable success in the theory of biquadratic residues. We find that −3 belongs to the collection

A for			B for			C for			D for		
p	a	b	p	a	b	p	a	b	p	a	b
37	+1	−6	5	+1	+2	13	−3	+2	29	+5	+2
61	+5	−6	17	+1	−4	73	−3	−8	41	+5	−4
157	−11	−6	89	+5	−8	97	+9	+4	53	−7	+2
193	−7	−12	113	−7	−8	109	−3	+10	101	+1	+10
			137	−11	−4	181	+9	+10	197	+1	−14
			149	−7	+10	229	−15	+2	269	+13	+10
			173	+13	+2	241	−15	−4	293	+17	+2
			233	+13	+8	277	+9	+14			
			257	+1	−16						
			281	+5	+16						

where the law suggested by inductive investigation presents itself at once. Namely, -3 belongs to the collection

A, whenever b is divisible by 3, equivalently $b \equiv 0 \pmod 3$
B, whenever $a + b$ is divisible by 3, equivalently $b \equiv 2a \pmod 3$
C, whenever a is divisible by 3, equivalently $a \equiv 0 \pmod 3$
D, whenever $a - b$ is divisible by 3, equivalently $b \equiv a \pmod 3$

26.

We find that the number $+5$ is to be assigned to the collection

A for $p =$	101, 109, 149, 181, 269
B for $p =$	13, 17, 73, 97, 157, 193, 197, 233, 277, 293
C for $p =$	29, 41, 61, 89, 229, 241, 281
D for $p =$	37, 53, 113, 137, 173, 257

Considering the values of the numbers a, b corresponding to each p, we find the law here as readily as for the classification of the number -3. Namely, we arrive at the collection

A, whenever $b \equiv 0 \pmod 5$
B, whenever $b \equiv a$
C, whenever $a \equiv 0$
D, whenever $b \equiv 4a$

These rules evidently cover all cases, since for $b \equiv 2a$ or $b \equiv 3a \pmod 5$ we would have $aa + bb \equiv 0$, Q. E. A., since by hypothesis p is a prime number different from 5.

27.

Likewise, inductive investigation applied to the numbers -7, -11, $+13$, $+17$, -19, -23 and carried far enough points to the following rules:

For the number -7

A	$a \equiv 0$, or $b \equiv 0 \pmod 7$
B	$b \equiv 4a$, or $b \equiv 5a$
C	$b \equiv a$, or $b \equiv 6a$
D	$b \equiv 2a$, or $b \equiv 3a$

For the number -11.

A	$b \equiv 0,\ 5a,$ or $6a \pmod{11}$
B	$b \equiv a,\ 3a$ or $4a$
C	$a \equiv 0,$ or $b \equiv 2a$ or $9a$
D	$b \equiv 7a,\ 8a$ or $10a$

For the number $+13$.

A	$b \equiv 0,\ 4a,\ 9a \pmod{13}$
B	$b \equiv 6a,\ 11a,\ 12a$
C	$a \equiv 0;\ b \equiv 3a,\ 10a$
D	$b \equiv a,\ 2a,\ 7a$

For the number $+17$.

A	$a \equiv 0;\ b \equiv 0,\ a,\ 16a \pmod{17}$
B	$b \equiv 2a,\ 6a,\ 8a,\ 14a$
C	$b \equiv 5a,\ 7a,\ 10a,\ 12a$
D	$b \equiv 3a,\ 9a,\ 11a,\ 15a$

For the number -19.

A	$b \equiv 0,\ 2a,\ 5a,\ 14a,\ 17a \pmod{19}$
B	$b \equiv 3a,\ 7a,\ 11a,\ 13a,\ 18a$
C	$a \equiv 0;\ b \equiv 4a,\ 9a,\ 10a,\ 15a$
D	$b \equiv a,\ 6a,\ 8a,\ 12a,\ 16a$

For the number -23.

A	$a \equiv 0;\ b \equiv 0,\ 7a,\ 10a,\ 13a,\ 16a \pmod{23}$
B	$b \equiv 2a,\ 3a,\ 4a,\ 11a,\ 15a,\ 17a$
C	$b \equiv a,\ 5a,\ 9a,\ 14a,\ 18a,\ 22a$
D	$b \equiv 6a,\ 8a,\ 12a,\ 19a,\ 20a,\ 21a$

28.

The special theorems obtained in this way by inductive investigation are found to be confirmed as far as that investigation is continued, and reveal a particularly beautiful form of the criteria. But when they are compared with one another to draw general conclusions, the following observations immediately present themselves.

The criteria for deciding which collection the prime number $\pm q$ belongs to (taking the upper or lower sign according as q has the form $4n+1$ or $4n+3$) depend on the forms of the numbers a, b compared with one another with respect to the modulus q. Namely,

I. Whenever $a \equiv 0 \pmod{q}$, $\pm q$ belongs to a definite collection, which is A for $q = 7,\ 17,\ 23$, and C for $q = 3,\ 11,\ 13,\ 19$. This suggests the conjecture that the former case holds generally whenever q has the form $8n \pm 1$, and the latter whenever q has the form $8n \pm 3$. Moreover, the collections B and D can already be excluded without inductive investigation when the value of a is divisible by q, in which case $p \equiv bb \pmod{q}$, that is, p is a quadratic residue of q; hence, by the fundamental theorem, $\pm q$ must be a quadratic residue of p.

II. When a is not divisible by q, the criterion depends on the value of the expression $\frac{b}{a} \pmod{q}$. This expression does admit q different values, namely $0,\ 1,\ 2,\ 3 \ldots . q - 1$; but whenever q has the form $4n + 1$, the two values of the expression $\sqrt{-1} \pmod{q}$ must be excluded. They evidently cannot be values of the expression $\frac{b}{a} \pmod{q}$, since $p = aa + bb$ is always assumed to be a prime number different from q. The number of admissible values of the expression $\frac{b}{a} \pmod{q}$ is therefore $= q - 2$ for $q \equiv 1 \pmod{4}$, while it remains $= q$ for $q \equiv 3 \pmod{4}$.

These values are now distributed among four classes: some, to be denoted without further specification by α, correspond to the collection A; others, denoted by β, to the collection B; others, γ, to the collection C; and finally the rest, δ, to the collection D. Thus $\pm q$ is to be assigned to the collection A, B, C, D according as $b \equiv \alpha a$, $b \equiv \beta a$, $b \equiv \gamma a$, $b \equiv \delta a \pmod{q}$.

But the *law* of this distribution seems more deeply hidden, although some general features are readily noticed. The number of values in three of the classes is found to be the same, namely $= \frac{1}{4}(q - 1)$ or $\frac{1}{4}(q + 1)$, while in one (specifically, the one corresponding to the collection with the criterion $a \equiv 0$) it is smaller by one. Thus the total number of different criteria for each collection is the same, namely $= \frac{1}{4}(q - 1)$ or $\frac{1}{4}(q + 1)$. We further observe that 0 is always found in the first class (among the α), and that the complements of the numbers $\alpha, \beta, \gamma, \delta$ to q, namely $q - \alpha$, $q - \beta$, $q - \gamma$, $q - \delta$, are found in the first, fourth, third and second classes, respectively. Finally, we see that the values of the expressions $\frac{1}{\alpha}$, $\frac{1}{\beta}$, $\frac{1}{\gamma}$, $\frac{1}{\delta} \pmod{q}$ belong to the first, fourth, third and second classes whenever the criterion $a \equiv 0$ corresponds to the collection A, but to the third, second, first and fourth classes, respectively, whenever the criterion $a \equiv 0$ refers to the collection C. But this is about the limit of what can be reached by inductive investigation, unless we rather boldly presume to anticipate what will later be drawn from the genuine sources.

29.

Before going further, we should note that the criteria for prime numbers (taken positively when they have the form $4n + 1$, negatively when they have the form $4n + 3$) suffice to decide the question for all other numbers, provided that the theorem of Article 7 and the criteria for -1 and ± 2 are also used.

Thus, for example, if criteria for the number $+3$ are wanted, the criteria given in Article 25, which concern -3, will also hold for $+3$ whenever $\frac{1}{2}b$ is even. The collections A, B, C, D must instead be exchanged with the collections C, D, A, B whenever $\frac{1}{2}b$ is odd, giving the following rules:

$+3$ belongs

to the collection	if
A	$b \equiv 0 \pmod{12}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 2 \pmod 4$
B	$b \equiv 8a$ or $10a \pmod{12}$
C	$b \equiv 6a \pmod{12}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 0 \pmod 4$
D	$b \equiv 2a$ or $4a \pmod{12}$

Likewise, the criteria for ± 6 are obtained by combining the criteria for ∓ 2 and -3; namely,

$+6$ belongs

to the collection	if
A	$b \equiv 0,\quad 2a, 22a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 4a \pmod 8$
B	$b \equiv 4a,\quad 6a,\quad 8a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 2a \pmod 8$
C	$b \equiv 10a, 12a, 14a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 0 \pmod 8$
D	$b \equiv 16a, 18a, 20a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 6a \pmod 8$

while -6 belongs

to the collection	if
A	$b \equiv 0,\quad 10a, 14a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 4a \pmod 8$
B	$b \equiv 4a,\quad 8a, 18a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 6a \pmod 8$
C	$b \equiv 2a, 12a, 22a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 0 \pmod 8$
D	$b \equiv 6a, 16a, 20a \pmod{24}$; or simultaneously $a \equiv 0 \pmod 3,\ b \equiv 2a \pmod 8$

In the same way, the criteria for the number $+21$ will be assembled from the criteria for -3 and -7; the criteria for -105 from those for -1, -3, $+5$, -7, etc.

30.

Inductive investigation thus opens up a very rich harvest of particular theorems related to the theorem for the number 2. But a common bond is lacking, rigorous proofs are lacking, since the method by which we completed the treatment of the number 2 in the first paper admits no further application. There are indeed various methods by which proofs could be obtained for particular cases, especially those concerning the distribution of quadratic residues between the collections A, C. We do not dwell on them, however, since a general theory embracing *all* cases must be our aim. Having begun to devote our thoughts to this matter in 1805, we soon became convinced that the true source of the general theory must be sought in an extension of the domain of arithmetic, as we already indicated in Article 1.

While higher arithmetic, in the questions treated so far, deals only with real integers, the theorems on biquadratic residues shine in their full simplicity and genuine beauty only when the domain of arithmetic is extended to *imaginary* quantities. Its objects are then, without restriction, numbers of the form $a + bi$, where i denotes, as usual, the imaginary quantity $\sqrt{-1}$, and a, b range without restriction over all real integers between $-\infty$ and $+\infty$. We shall call such numbers *complex integers*, treating real numbers as a species contained within the complex numbers rather than as opposed to them. The present paper will set out both the elementary theory of complex numbers and the first beginnings of the theory of biquadratic residues, which we shall undertake to complete in every respect in a subsequent continuation[*)].

[*)] We should at least mention here in passing that the domain thus defined is particularly suited to the theory of biquadratic residues. The theory of cubic residues must similarly be built on the consideration of numbers of the form $a + bh$, where h is an imaginary root of the equation $h^3 - 1 = 0$, say $h = -\frac{1}{2} + \sqrt{\frac{3}{4}} \,.\, i$; and likewise the theory of residues of higher powers will require the introduction of other imaginary quantities.

31.

First we introduce some names that will help us to be brief and clear. The domain of complex numbers $a + bi$ contains

I. real numbers, where $b = 0$, and among these, according to the nature of a,

1) zero,

2) positive numbers,

3) negative numbers;

II. imaginary numbers, where b is unequal to zero. Here we distinguish in turn

1) imaginary numbers without a real part, that is, where $a = 0$;

2) imaginary numbers with a real part, where b is nonzero and $a = 0$ also does not hold.

The former may, if desired, be called pure imaginary numbers, and the latter mixed imaginary numbers.

In this theory we use four units, $+1$, -1, $+i$, $-i$, which will be called simply positive, negative, positive imaginary and negative imaginary.

We shall call the three products of any complex number by -1, $+i$, $-i$ its *companions* or *numbers associated with it*. Thus, except for zero (which is associated with itself), four *distinct* numbers are always associated.

By contrast, we call the number obtained from a complex number by exchanging i with $-i$ its *conjugate*. Among imaginary numbers, therefore, two *distinct* numbers are always conjugate, while real numbers are their own conjugates, if we choose to extend the name to them.

We call the product of a complex number and its conjugate the *norm* of each. The square of a real number is therefore to be taken as its norm.

In general we have eight related numbers, namely,

$$\begin{array}{r|r} a+bi & a-bi \\ -b+ai & -b-ai \\ -a-bi & -a+bi \\ b-ai & b+ai \end{array}$$

in which we see two groups of four associated numbers and four pairs of conjugates; their common norm is $aa + bb$. But the eight numbers reduce to four distinct ones whenever either $a = \pm b$, or $a, b = 0$ for one of the two numbers.

The following results follow immediately from these definitions:

The conjugate of the product of two complex numbers is the product of their conjugates.

The same holds for a product of more factors, and also for quotients.

The norm of the product of two complex numbers is equal to the product of their norms.

This theorem also extends to products of any number of factors and to quotients.

The norm of any complex number (except zero, an exception that from now on we shall generally leave understood) is a *positive* number.

Moreover, nothing prevents us from extending our definitions to fractional or even irrational values of a, b; but $a + bi$ will be called a complex integer only when *each* of a, b is an integer, and rational only when *each* of a, b is rational.

32.

The rules for arithmetic operations on complex numbers are well known: introducing the norm reduces division to multiplication, since

$$\frac{a+bi}{c+di} = (a+bi)\,.\,\frac{c-di}{cc+dd} = \frac{ac+bd}{cc+dd} + \frac{bc-ad}{cc+dd}\,.\,i$$

The square root is extracted with the help of the formula

$$\surd(a+bi) = \pm\left(\surd\frac{\surd(aa+bb)+a}{2} + i\surd\frac{\surd(aa+bb)-a}{2}\right)$$

if b is positive, or of the formula

$$\surd(a+bi) = \pm\left(\surd\frac{\surd(aa+bb)+a}{2} - i\surd\frac{\surd(aa+bb)-a}{2}\right)$$

if b is negative. There is no need to dwell here on how transforming the complex quantity $a + bi$ into $r(\cos\varphi + i\sin\varphi)$ makes calculations easier.

33.

We call a complex integer that can be resolved into two factors different from the units[*)] a composite complex number; a number that admits no such factorization will be called a complex prime. It follows immediately that every real composite number is also a composite complex number. But a real prime may be a composite complex number, and this holds for the number 2 and all positive real primes of the form $4n+1$ (except the number 1), since they are known to be decomposable into two positive squares. For example, $2 = (1+i)(1-i)$, $5 = (1+2i)(1-2i)$, $13 = (3+2i)(3-2i)$, $17 = (1+4i)(1-4i)$, etc.

On the other hand, positive real primes of the form $4n+3$ are always complex primes. For if such a number q were $= (a+bi)(\alpha+\beta i)$, we should also have $q = (a-bi)(\alpha-\beta i)$, and therefore $qq = (aa+bb)(\alpha\alpha+\beta\beta)$. But qq can be resolved into positive factors greater than unity in only one way, namely into $q \times q$, so we should have to have $q = aa+bb = \alpha\alpha+\beta\beta$, Q. E. A.; for the sum of two squares cannot have the form $4n+3$.

Negative real numbers evidently retain the same names as positive ones, and the same holds for pure imaginary numbers.

It therefore remains to show how to distinguish composites from primes among mixed imaginary numbers. This is done by the following

THEOREM. *Every mixed imaginary integer $a+bi$ is a complex prime or a composite number according as its norm is a real prime or a composite number.*

Proof. I. Since the norm of a composite complex number is always composite, a complex number whose norm is a real prime must evidently be a complex prime. Q. E. P.

II. If, however, the norm $aa+bb$ is composite, let p be a positive real prime dividing it. We must now distinguish two cases.

1) If p has the form $4n+3$, we know that $aa+bb$ cannot be divisible by p unless p divides both a, b, so that $a+bi$ will be composite.

[*)] Or, equivalently, factors whose norms are greater than unity.

2) If p does not have the form $4n+3$, it can certainly be decomposed into two squares. We therefore put $p = \alpha\alpha + \beta\beta$. Since

$$(a\alpha + b\beta)(a\alpha - b\beta) = aa(\alpha\alpha + \beta\beta) - \beta\beta(aa + bb)$$

and this is therefore divisible by p, p must divide one of the factors $a\alpha + b\beta$, $a\alpha - b\beta$. Moreover, since

$$(a\alpha + b\beta)^2 + (b\alpha - a\beta)^2 = (a\alpha - b\beta)^2 + (b\alpha + a\beta)^2 = (aa + bb)(\alpha\alpha + \beta\beta)$$

and this is therefore divisible by pp, in the first case $b\alpha - a\beta$, and in the second $b\alpha + a\beta$, must also be divisible by p. Thus in the first case

$$\frac{a+bi}{\alpha+\beta i} = \frac{a\alpha + b\beta}{p} + \frac{b\alpha - a\beta}{p} \,.\, i$$

will be a complex integer, while in the second

$$\frac{a+bi}{\alpha-\beta i} = \frac{a\alpha - b\beta}{p} + \frac{b\alpha + a\beta}{p} \,.\, i$$

will be an integer. Since the given number is therefore divisible by $\alpha + \beta i$ or by $\alpha - \beta i$, and the norm of the quotient $= \frac{aa+bb}{p}$ differs from unity by hypothesis, $a + bi$ is evidently a composite complex number in either case. Q. E. S.

34.

The following four kinds therefore exhaust the entire range of complex primes:

1) the four units, $1, +i, -1, -i$, which, however, we shall usually leave understood to be excluded when discussing prime numbers;

2) the number $1 + i$ with its three associates $-1 + i$, $-1 - i$, $1 - i$;

3) the positive real primes of the form $4n + 3$ with their three associates each;

4) the complex numbers whose norms are real primes of the form $4n + 1$ greater than unity. To each given norm of this kind there always correspond eight complex primes and no more, since such a norm can be decomposed into two squares in only one way.

35.

Just as real integers are divided into even and odd, and the former in turn into those divisible by four and those that are twice an odd number, an equally essential distinction presents itself among complex numbers. They are either

not divisible by $1+i$, namely numbers $a+bi$ in which one of the numbers a, b is odd and the other even;

or divisible by $1+i$ but not by 2, whenever both a, b are odd;

or divisible by 2, whenever both a, b are even.

The numbers of the first class may suitably be called *odd complex numbers*, those of the second *half-even*, and those of the third *even*.

A product of several complex factors will always be odd when all the factors are odd; half-even when one factor is half-even and the others odd; and even when the factors include at least two half-even ones or at least one even one.

The norm of any odd complex number has the form $4n+1$; the norm of a half-even number has the form $8n+2$; finally, the norm of an even number is the product of a number of the form $4n+1$ and the number 4 or a higher power of two.

36.

Since the relation between four associated complex numbers is analogous to that between two opposite real numbers (that is, numbers equal in absolute value and bearing opposite signs), and the positive one of these is usually and rightly regarded as primary, the question arises whether a similar distinction can be established among four associated complex numbers, and whether it should be considered useful. To decide this, we must consider that the principle of distinction must be such that the product of two numbers regarded as primary among their respective associates always becomes primary among its own associates. But we soon become convinced that no such principle exists at all unless the distinction is restricted to integers; indeed, a *useful* distinction must be limited to odd numbers. For these, however, the proposed aim can be reached in two ways. Namely,

I. The product of two numbers $a+bi, a'+b'i$ such that a, a' have the form $4n+1$ and b, b' are even will have the same property: the real part will be $\equiv 1 \pmod 4$, and the imaginary part even. And it is easy to see that among four associated odd numbers only one has that form.

II. If a number $a+bi$ is such that $a-1$ and b are either both divisible by four or both twice an odd number, its product with a complex number of the same form will have the same form; and it is easy to see that only one of four associated odd numbers has this form.

Of these two principles, which are almost equally suitable, we shall adopt the latter: among four associated odd complex numbers we shall regard as primary the one congruent to the positive unit modulo $2+2i$. This will let us state several notable theorems more concisely. For example, the complex primes $-1+2i$, $-1-2i$, $+3+2i$, $+3-2i$, $+1+4i$, $+1-4i$, etc., are primary, as are the real numbers -3, -7, -11, -19, etc., which must evidently always be taken with a negative sign. The conjugate of an odd primary complex number will also be primary.

For half-even and even numbers in general, a similar distinction would be too arbitrary and of little use. Of the associated primes $1+i$, $1-i$, $-1+i$, $-1-i$, we can indeed choose one as primary in preference to the others, but we shall not extend such a distinction to composites.

37.

If some factors of a composite complex number are themselves composite, and these in turn are resolved into their factors, we evidently arrive eventually at prime factors; that is, every composite number can be resolved into prime factors. If any of these are not primary, replace each by the product of its primary associate and $i, -1$ or $-i$. In this way, every composite complex number M can evidently be reduced to the form

$$M = i^{\mu} A^{\alpha} B^{\beta} C^{\gamma} \dots .$$

where A, B, C, etc., are distinct primary complex primes, and $\mu = 0, 1, 2$ or 3. Concerning this factorization, we have the theorem that it can be carried out in only one way. On a passing consideration this theorem might seem evident in itself, but it certainly needs a proof. The following paves the way to that proof.

THEOREM. *The product $M = A^{\alpha} B^{\beta} C^{\gamma} \dots$, where A, B, C, etc., denote distinct primary complex primes, cannot be divisible by any primary complex prime that does not occur among A, B, C, etc.*

Proof. Let P be a primary complex prime not contained among A, B, C, etc., and let p, a, b, c, etc., be the norms of the numbers P, A, B, C, etc. It follows readily that the norm of M will be $= a^{\alpha} b^{\beta} c^{\gamma}$, etc.; hence, if M were divisible by P, this number would have to be divisible by p. Since the individual norms are either real primes (from the sequence $2, 5, 13, 17$, etc.) or squares of real primes (from the sequence $9, 49, 121$, etc.), this evidently cannot happen unless p is identical with one of the norms a, b, c, etc. We therefore suppose $p = a$. But since P, A are distinct primary complex primes by hypothesis, it is easy to see that these conditions cannot hold together unless P, A are conjugate imaginary complex numbers, and consequently $p = a$ is an odd real prime (not the square of a prime). We therefore suppose $A = k+li$, $P = k-li$.

Hence, extending the notion and notation of congruence to complex integers, we have $A \equiv 2k \pmod{P}$, from which it follows readily that

$$M \equiv 2^\alpha k^\alpha B^\beta C^\gamma \dots . \pmod{P}$$

Therefore, if M is assumed divisible by P, then

$$2^\alpha k^\alpha B^\beta C^\gamma \dots .$$

will also be divisible by P, and consequently the norm of this number, which is

$$= 2^{2\alpha} k^{2\alpha} b^\beta c^\gamma \dots .$$

will be divisible by p. But since 2 and k are certainly not divisible by p, it follows that p must be identical with one of the numbers b, c, etc.; let, for example, $p = b$. From this, however, we conclude that either $B = k+li$ or $B = k-li$, that is, either $B = A$ or $B = P$, both contrary to the hypothesis.

The other theorem, that factorization into primes can be carried out in only one way, follows very easily from this theorem, by reasoning entirely analogous to that which we used for real numbers in the *Disquisitiones Arithmeticae* (Article 16). It would therefore be superfluous to dwell on it here.

38.

We now turn to the congruence of numbers with respect to complex moduli. At the outset of this investigation, however, we should indicate how the domain of complex quantities can be made visible to intuition.

Every real quantity can be expressed by a segment of a line infinite in both directions, taken from an arbitrary starting point and measured against an arbitrary segment adopted as the unit. It can thus be represented by the other endpoint, with points on one side of the starting point representing positive quantities and those on the other negative quantities. In the same way, any complex quantity can be represented by a point in an infinite plane in which a fixed line corresponds to the real quantities: the complex quantity $x+iy$ by the point with abscissa $= x$ and ordinate, taken positively on one side of the line of abscissas and negatively on the other, $= y$. In this way we may say that any complex quantity measures the difference between the position of the point to which it corresponds and that of the starting point. The positive unit denotes an arbitrary fixed displacement in an arbitrary fixed direction; the negative unit an equally large displacement in the opposite direction; and the imaginary units equally large displacements in the two perpendicular lateral directions.

This sheds considerable light on the metaphysics of the quantities we call imaginary. If the starting point is denoted by (0), and two complex quantities m, m' correspond to points M, M', whose positions relative to (0) they

express, the difference $m-m'$ is nothing other than the position of M relative to M'. On the other hand, if the product mm' represents the position of N relative to (0), it is easy to see that this position is determined by the position of M relative to (0) in the same way as the position of M' is determined by the position of the point corresponding to the positive unit. Thus we may aptly say that the positions of the points corresponding to the complex quantities mm' $m, m', 1$ form a *proportion*. But we reserve a fuller treatment of this matter for another occasion. The difficulties in which the theory of imaginary quantities is thought to be entangled have largely arisen from unsuitable names (some have even used the discordant name of impossible quantities). Had we started from the concepts offered by two-dimensional manifolds, as seen in their greatest purity in our intuitions of space, and called positive quantities direct, negative quantities inverse and imaginary quantities lateral, simplicity would have taken the place of complications, and clarity the place of darkness.

39.

What was set out in the preceding article concerns continuous complex quantities. In arithmetic, which deals only with integers, the representation of complex numbers will be a system of equally spaced points arranged on equally spaced lines so as to divide the infinite plane into infinitely many equal squares. All numbers divisible by a given complex number $a + bi = m$ will likewise form infinitely many squares with sides $= \sqrt{(aa + bb)}$, or areas $= aa + bb$. The latter squares will be inclined to the former whenever neither of the numbers a, b is $= 0$. Each number not divisible by the modulus m will correspond to a point either inside such a square or on a side shared by two squares; the latter case, however, cannot occur unless a, b have a common divisor. It is also clear that numbers congruent modulo m occupy corresponding positions in their squares. Hence, if we collect all the numbers lying inside a specified square, together with all those that may lie on two of its non-opposite sides, and finally add a number divisible by m, we readily obtain a complete system of incongruent residues modulo m: every integer must be congruent to one of them, and to only one. Nor would it be difficult to show that the number of these residues equals the norm of the modulus, namely $= aa + bb$. But it seems advisable to prove this very important theorem in another, purely arithmetical way.

40.

THEOREM. *For a given complex modulus $m = a + bi$, whose norm is $aa + bb = p$ and for which a, b are relatively prime, every complex integer will be congruent to a residue in the sequence $0, 1, 2, 3 \ldots p - 1$, and to no more than one.*

Proof. I. Let α, β be integers such that $\alpha a + \beta b = 1$. Then

$$i = \alpha b - \beta a + m(\beta + \alpha i)$$

Thus, for a given complex integer $A + Bi$, we have

$$A + Bi = A + (\alpha b - \beta a)B + m(\beta B + \alpha B i)$$

Denote by h the least positive residue of $A + (\alpha b - \beta a)B$ modulo p, and put

$$A + (\alpha b - \beta a)B = h + kp = h + m(ak - bki)$$

Then

$$A + Bi = h + m(\beta B + ak + (\alpha B - bk)i)$$

or

$$A + Bi \equiv h (\text{mod. } m). \qquad \text{Q. E. P.}$$

II. Whenever two real numbers h, h' are congruent to the same complex number modulo m, they are also congruent to each other. We therefore put $h - h' = m(c + di)$, giving

$$(h - h')(a - bi) = p(c + di)$$

and hence

$$(h - h')a = pc, \quad (h - h')b = -pd$$

and also, since $a\alpha + b\beta = 1$,

$$h - h' = p(c\alpha - d\beta), \text{ i. e. } h \equiv h' (\text{mod. } p)$$

Thus h and h', if distinct, cannot both belong to the collection of numbers $0, 1, 2, 3 \ldots . p - 1$. Q. E. S.

41.

THEOREM. *For a complex modulus $m = .a+bi$, whose norm is $aa+bb = p$ and for which a, b are not relatively prime but have greatest common divisor λ (which we suppose taken positively), every complex number is congruent to a residue $x + yi$ such that x is one of the numbers $0, 1, 2, 3 \ldots . \frac{p}{\lambda} - 1$ and y one of the numbers $0, 1, 2, 3 \ldots . \lambda - 1$, and to exactly one of all the p residues of this form.*

Proof. I. Taking integers α, β such that $\alpha a + \beta b = \lambda$, we have $\lambda i = \alpha b - \beta a + m(\beta + \alpha i)$. Now let $A + Bi$ be a given complex number, y the least positive residue of B modulo λ, and x the least positive residue of $A + (\alpha b - \beta a) . \frac{B-y}{\lambda}$ modulo $\frac{p}{\lambda}$, and put

$$A + (\alpha b - \beta a) . \frac{B-y}{\lambda} = x + \frac{p}{\lambda} . k$$

Then

$$\begin{aligned} A + Bi - (x + yi) &= \frac{p}{\lambda} . k + (B - y)i - (\alpha b - \beta a)\frac{B-y}{\lambda} \\ &= \frac{p}{\lambda} . k + \frac{B-y}{\lambda} . m(\beta + \alpha i) \\ &= \left(\frac{a}{\lambda} - \frac{b}{\lambda} . i\right) km + \frac{B-y}{\lambda}(\beta + \alpha i)m \end{aligned}$$

which is divisible by m; equivalently, $A + Bi \equiv x + yi (\text{mod. } m)$ Q. E. P.

II. Suppose that, modulo m, two numbers $x + yi, x' + y'i$ are congruent to the same complex number, and therefore also to each other modulo m. All the more, they are congruent modulo λ, so $y \equiv y' (\text{mod.} \lambda)$. If both y, y' are therefore assumed to belong to the numbers $0, 1, 2, 3 \ldots . \lambda - 1$, we must have $y = y'$. We then also have $x \equiv x' (\text{mod. } m)$: $x - x'$ is divisible by m, and consequently the integer $\frac{x-x'}{\lambda}$ is divisible by $\frac{a}{\lambda} + \frac{b}{\lambda} . i$, or

$$\frac{x - x'}{\lambda} \equiv 0 \left(\text{mod.} \frac{a}{\lambda} + \frac{b}{\lambda} . i\right)$$

But since $\frac{a}{\lambda}, \frac{b}{\lambda}$ are relatively prime, the second part of the theorem in the preceding article shows that $\frac{x-x'}{\lambda}$ is also divisible by the norm of $\frac{a}{\lambda} + \frac{b}{\lambda} . i$, that is, by the number $\frac{p}{\lambda\lambda}$, and consequently $x - x'$ by $\frac{p}{\lambda}$. Thus, if both x, x' are also assumed to belong to the collection of numbers $0, 1, 2, 3 \ldots . \frac{p}{\lambda} - 1$, we must have $x = x'$, or the residues $x + yi, x' + y'i$ are identical. Q. E. S.

It is also evident that this includes the case where the modulus is real, namely $b = 0$ and consequently $\lambda = \pm a$, as well as the case where it is purely imaginary, namely $a = 0$ and consequently $\lambda = \pm b$. In either case we have $\frac{p}{\lambda} = \lambda$.

42.

Thus, assigning complex numbers congruent to one another modulo a given modulus to the same class, and incongruent numbers to different classes, we obtain altogether p classes exhausting the whole range of integers, where p denotes the norm of the modulus. A collection of that many numbers, one taken from each class, gives a complete system of incongruent residues, such as we specified in Articles 40, 41. In that system, the choice of residues representing their classes was based on the principle of adopting in each class a residue $x + yi$ for which y has its least value and, among all those with that same least value of y, the one for which x has its least value, excluding negative values for both x and y. But other purposes will call for other principles. In particular, we should note the method of choosing residues that give the simplest quotients when divided by the modulus. If $\alpha + \beta i, \alpha' + \beta' i, \alpha'' + \beta'' i$, etc., are quotients obtained by dividing congruent numbers by the modulus, the differences among $\alpha, \alpha', \alpha''$, etc., are evidently integers, as are the differences among β, β', β'', etc. It is clear that there is always one residue for which α and β lie between the bounds 0 and 1, the former included and the latter excluded. We simply call such a residue a least residue. If preferred, the bounds $-\frac{1}{2}$ and $+\frac{1}{2}$ may be adopted instead, one included and the other excluded; we shall call a residue satisfying this restriction *absolutely least*.

Concerning · these least residues, the following problems arise.

43.

The least residue of a given complex number $A+Bi$ modulo $a+bi$, of norm $= p$, is found as follows. If $x+yi$ is the required least residue, $(x+yi)(a-bi)$ will be the least residue of the product $(A+Bi)(a-bi)$ modulo $(a+bi)(a-bi)$, that is, modulo p. Put

$$aA + bB = Fp + f, \quad aB - bA = Gp + g$$

so that f, g are the least residues of $aA + bB, aB - bA$ modulo p. Then

$$x + yi = \frac{f + gi}{a - bi}$$

or

$$x = \frac{af - bg}{p} = A - aF + bG$$
$$y = \frac{ag + bf}{p} = B - aG - bF$$

Evidently, the least residues f, g must be taken either between the bounds 0 and $p - 1$ or between $-\frac{1}{2}p$ and $+\frac{1}{2}p$, according as the complex number's simply least or absolutely least residue is required.

44.

A complete system of least residues for a given modulus can be constructed in several ways. The first method begins by determining the bounds within which the real terms must lie, and then, for each value within those bounds, assigning bounds to the imaginary parts. The general criterion for a least residue $x+yi$ modulo $a+bi$ is that both $ax+by = \xi$ and $ay-bx = \eta$ lie between the bounds 0 and $aa+bb$ when simply least residues are considered, or between $-\frac{1}{2}(aa+bb)$ and $+\frac{1}{2}(aa+bb)$ when absolutely least residues are required, one bound being excluded. Special rules would require distinguishing the cases arising from the different signs of the numbers a, b. Since developing that distinction presents no difficulty, we refrain from dwelling on it here; it will suffice to show the nature of the method by a single example.

For the modulus $5+2i$, the simply least residues $x+yi$ must be such that both $5x+2y = \xi$ and $5y-2x = \eta$ equal one of the numbers $0, 1, 2, 3 \ldots . 28$. The equation $29x = 5\xi - 2\eta$ shows that positive values of x cannot exceed $\frac{5.28}{29}$, and negative values, disregarding their sign, cannot exceed $\frac{2.28}{29}$. All admissible values of x are therefore $-1, 0, 1, 2, 3, 4$. For $x = -1$, $2y$ must equal one of the numbers $5, 6, 7 \ldots . 33$, and $5y$ one of the numbers $-2, -1, 0, 1 \ldots . 26$; hence the least value of y is $+3$ and the greatest $+5$. Treating the remaining values of x in the same way gives the following table of all the least residues:

x	y
-1	$3, 4, 5$
0	$0, 1, 2, 3, 4, 5$
$+1$	$1, 2, 3 \,.\, 4, 5 \,.\, 6$
$+2$	$1, 2, 3, 4, 5, 6$
$+3$	$2, 3, 4 \,.\, 5, 6$
$+4$	$2, 3, 4$

Similarly, for absolutely least residues, ξ and η must equal one of the numbers $-14, -13, -12 \ldots . + 14$. Hence $29x$ cannot lie outside the bounds $-7 \,.\, 14$ and $+7 \,.\, 14$, so x must equal one of the numbers $-3, -2, -1, 0, 1, 2, 3$. For $x = -3$, $2y = \xi - 5x = \xi + 15$ will equal one of the numbers $1, 2, 3 \ldots . 29$, while $5y = \eta + 2x = \eta - 6$ will equal one of the numbers $-20, -19, -18 \ldots . + 8$; this gives for y the single value $+1$. Treating the remaining values of x in the same way, we have the table of all the absolutely least residues:

x	y
-3	$+1$
-2	$-2, -1, 0, +1, +2$
-1	$-3, -2, -1, 0, +1, +2$
0	$-2, -1, 0, +1, +2$
$+1$	$-2, -1, 0, +1, +2, +3$
$+2$	$-2, -1, 0, +1, +2$
$+3$	-1

45.

In applying the second method, it will be useful to distinguish two cases.

In the first case, where a and b have no common divisor, put $a\alpha + \beta b = 1$, and let k be the least positive residue of $\beta a - \alpha b$ modulo p. The identities

$$a(\beta a - \alpha b) = \beta p - b(\alpha a + \beta b),$$
$$b(\beta a - \alpha b) = -\alpha p + a(\alpha a + \beta b)$$

then show that $ak \equiv -b, bk \equiv a (\text{mod. } p)$. Thus, putting as above $ax + by = \xi$, $ay - bx = \eta$, we have $\eta \equiv k\xi$, $\xi \equiv -k\eta (\text{mod. } p)$. All the numbers $\xi + \eta i$ to which simply least residues $x + yi$ correspond are therefore obtained by taking either for ξ, successively, the values $0, 1, 2, 3 \ldots . p - 1$, and for η the least positive residues of the products $k\xi$ modulo p, or, in another order, for η those values and for ξ the least residues of the products $-k\eta$. From each $\xi + \eta i$ the corresponding $x + yi$ is then found by the formula

$$x + yi = \frac{\xi + \eta i}{a - bi} = \frac{a\xi - b\eta}{p} + \frac{a\eta + b\xi}{p} \,.\, i$$

It is also clear that η, as ξ increases by one, either increases by k or decreases by $p - k$, and consequently $x + yi$ undergoes

$$\text{either the change } \frac{a - kb}{p} + \frac{ak + b}{p} \,.\, i$$
$$\text{or this one } \frac{a - kb}{p} + b + \left(\frac{ak + b}{p} - a \right) i$$

This observation serves to make the construction easier.

Finally, if absolutely least residues $x + yi$ are required, these instructions evidently change only in that ξ must successively be given values between the bounds $-\frac{1}{2}p$ and $+\frac{1}{2}p$, while for η we must take the absolutely least residues of the products $k\xi$. Here is a table of least residues for the modulus $5 + 2i$, arranged in this way:

Simply least residues.

$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$
0	0	$10 + 25i$	$+5i$	$20 + 21i$	$+2 + 5i$
$1 + 17i$	$-1 + 3i$	$11 + 13i$	$+1 + 3i$	$21 + 9i$	$+3 + 3i$
$2 + 5i$	$+i$	$12 + i$	$+2 + i$	$22 + 26i$	$+2 + 6i$
$3 + 22i$	$+1 + 4i$	$13 + 18i$	$+1 + 4i$	$23 + 14i$	$+3 + 4i$
$4 + 10i$	$+2i$	$14 + 6i$	$+2 + 2i$	$24 + 2i$	$+4 + 2i$
$5 + 27i$	$-1 + 5i$	$15 + 23i$	$+1 + 5i$	$25 + 19i$	$+3 + 5i$
$6 + 15i$	$+3i$	$16 + 11i$	$+2 + 3i$	$26 + 7i$	$+4 + 3i$
$7 + 3i$	$+1 + i$	$17 + 28i$	$+1 + 6i$	$27 + 24i$	$+3 + 6i$
$8 + 20i$	$+4i$	$18 + 16i$	$+2 + 4i$	$28 + 12i$	$+4 + 4i$
$9 + 8i$	$+1 + 2i$	$19 + 4i$	$+3 + 2i$		

Absolutely least residues.

$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$
$-14 - 6i$	$-2 - 2i$	$-4 - 10i$	$-2i$	$+5 - 2i$	$+1$
$-13 + 11i$	$-3 + i$	$-3 + 7i$	$-1 + i$	$+6 - 14i$	$+2 - 2i$
$-12 - i$	$-2 - i$	$-2 - 5i$	$-i$	$+7 + 3i$	$+1 + i$
$-11 - 13i$	$-1 - 3i$	$-1 + 12i$	$-1 + 2i$	$+8 - 9i$	$+2 - i$
$-10 + 4i$	-2	0	0	$+9 + 8i$	$+1 + 2i$
$-9 - 8i$	$-1 - 2i$	$+1 - 12i$	$+1 - 2i$	$+10 - 4i$	$+2$
$-8 + 9i$	$-2 + i$	$+2 + 5i$	$+i$	$+11 + 13i$	$+1 + 3i$
$-7 - 3i$	$-1 - i$	$+3 - 7i$	$+1 - i$	$+12 + i$	$+2 + i$
$-6 + 14i$	$-2 + 2i$	$+4 + 10i$	$+2i$	$+13 - 11i$	$+3 - i$
$-5 + 2i$	-1			$+14 + 6i$	$+2 + 2i$

The second case, where a, b are not relatively prime, can easily be reduced to the preceding one. Let λ be the greatest common divisor of a, b, and let $a = \lambda a'$, $b = \lambda b'$. Let F denote any least residue modulo λ, with the modulus regarded as a complex number: that is, let it represent any number $x + yi$ such that x, y lie either between the bounds 0 and λ or between $-\frac{1}{2}\lambda$ and $+\frac{1}{2}\lambda$, according as simply least or absolutely least residues are considered. Further, let F' denote any least residue modulo $a' + b'i$. Then $(a' + b'i)F + F'$ will represent any least residue modulo $a + bi$, and a complete system of these residues results when all values of F are combined with all values of F'.

46.

Two complex numbers are called relatively prime if they have no common divisors other than the units. Whenever such common divisors do exist, those whose norm is greatest are called greatest common divisors.

If the factorization of two given numbers into prime factors is available, the greatest common divisor is determined in exactly the same way as for real numbers (*Disquiss. Ar.* Article 18). This also shows that every common divisor of two given numbers must divide their greatest common divisor found in this way. Since it is already evident that its three associates are also common divisors, there will always be four numbers, and no more, that are to be called greatest common divisors; their norm will be a multiple of the norm of any other common divisor.

If the factorization of two given numbers into simple factors is not available, the greatest common divisor is found by an algorithm similar to that for real numbers. Let m, m' be the two given numbers, and form by repeated division the sequence m'', m''', etc., so that m'' is the absolutely least residue of m modulo m', then m''' the absolutely least residue of m' modulo m'', and so on. Denoting the norms of m, m', m'', m''', etc., respectively by p, p', p'', p''', etc., $\frac{p''}{p'}$ will be the norm of the quotient $\frac{m''}{m'}$ and thus, by the definition of an absolutely least residue, certainly no greater than $\frac{1}{2}$; the same holds for $\frac{p'''}{p''}$, etc. The positive real integers p', p'', p''', etc., will therefore form a steadily decreasing sequence, so that a term 0 must eventually be reached. Equivalently, in the sequence m, m', m'', m''', etc., we shall eventually reach a term that divides the preceding term without a remainder. Let this be $m^{(n+1)}$, and put

$$\begin{aligned} m &= km' + m'' \\ m' &= k'm'' + m''' \\ m'' &= k''m''' + m'''' \end{aligned}$$

and so on, as far as

$$m^{(n)} = k^{(n)} m^{(n+1)}$$

Going through these equations in reverse order shows that $m^{(n+1)}$ divides each preceding term $m^{(n)} \ldots . m'', m', m$; going through the same equations in forward order makes it clear that every common divisor of m, m' also divides each subsequent term. The first conclusion shows that $m^{(n+1)}$ is a common divisor of m, m'; the second, that this divisor is greatest.

Whenever the last residue $m^{(n+1)}$ equals one of the four units $1, -1, i, -i$, this will show that m and m' are relatively prime.

47.

If the equations of the preceding article, omitting the last, are combined so as to eliminate $m'', m''', m'''' \dots . m^{(n)}$, an equation of the following form results:

$$m^{(n+1)} = hm + h'm'$$

where h, h' are integers; more specifically, if we use the notation introduced in *Disquiss. Ar.* Article 27,

$$\begin{aligned} h &= \pm[k', k'', k''' \dots . k^{(n-1)}] \\ &= \pm[k^{(n-1)}, k^{(n-2)} \dots . k'', k'] \\ h' &= \mp[k, k', k'', k''' \dots . k^{(n-1)}] \\ &= \mp[k^{(n-1)}, k^{(n-2)} \dots . k'', k', k] \end{aligned}$$

the upper or lower signs applying according as n is even or odd. We state this theorem as follows:

A greatest common divisor of two complex numbers m, m' can be put in the form $hm + h'm'$, with h, h' integers.

This evidently holds for the greatest common divisor to which the algorithm of the preceding article led, and also for its three associates. For these, the coefficients h, h' must be replaced either by $hi, h'i$, by $-h, -h'$ or by $-hi, -h'i$.

Thus, whenever the numbers m, m' are relatively prime, the following equation can be satisfied:

$$1 = hm + h'm'$$

For example, let the given numbers be $31 + 6i = m$, $11 - 20i = m'$. Here we find

$$\begin{aligned} k &= i, & m'' &= +11 - 5i \\ k' &= +1 - i, & m''' &= +5 - 4i \\ k'' &= +2, & m'''' &= +1 + 3i \\ k''' &= -1 - 2i, & m''''' &= +i \\ k'''' &= +3 - i \end{aligned}$$

and hence

$$\begin{aligned} [k', k'', k'''] &= -6 - 5i \\ [k, k', k'', k'''] &= +4 - 10i \end{aligned}$$

and therefore

$$m''''' = i = (6 + 5i)m + (4 - 10i)m'$$

as well as

$$1 = (5 - 6i)m + (-10 - 4i)m'$$

which is confirmed by carrying out the calculation.

48.

The foregoing has prepared everything required for the theory of congruences of the first degree in the arithmetic of complex numbers. But since this theory does not differ essentially from the one that applies to the arithmetic of real numbers and has been set out at length in the *Disquisitiones Arithmeticae*, it will suffice to set down the main points here.

I. The congruence $mt \equiv 1 \pmod{m'}$ is equivalent to the indeterminate equation $mt + m'u = 1$. If this equation is satisfied by the values $t = h, u = h'$, the general solution of the congruence is expressed by $t \equiv h \pmod{m'}$. The condition for solvability is that the modulus m' and the coefficient m have no common divisor.

II. The solution of the congruence $ax + b \equiv c \pmod{M}$, in the case where a, M are relatively prime, depends on the solution of

$$at \equiv 1 \pmod{M}$$

If $t = h$ satisfies this congruence, the general solution of the former is contained in the formula

$$x \equiv (c - b)h \pmod{M}$$

III, The congruence $ax + b \equiv c(\text{mod. } M)$, in the case where a, M have a common divisor λ, is equivalent to

$$\frac{a}{\lambda} \,.\, x \equiv \frac{c-b}{\lambda} \quad \left(\text{mod.}\frac{M}{\lambda}\right)$$

Thus, when λ is taken to be the greatest common divisor of a, M, the solution of the given congruence is reduced to the preceding case. It is clear that the necessary and sufficient condition for solvability is that λ also divide the difference $c - b$.

49.

So far we have touched only on elementary matters, which nevertheless could not be omitted for the sake of the connection. In deeper investigations, the arithmetic of complex numbers resembles the arithmetic of real numbers in that more elegant and simpler theorems result when we admit only moduli that are prime numbers. Indeed, extending them to composite moduli usually involves more length than difficulty, and more work than ingenuity. The following discussion will therefore deal chiefly with prime moduli.

50.

Let X denote a function of the indeterminate x of the following form:

$$Ax^n + Bx^{n-1} + Cx^{n-2} + \text{etc.} + Mx + N;$$

where n is a positive real integer, A, B, C, etc., real or imaginary integers, and m a complex integer. Here too we shall call any integer a *root* of the congruence $X \equiv 0 \pmod{m}$ if, when substituted for x, it gives X a value divisible by the modulus m. We shall not regard solutions given by roots congruent modulo the modulus as different.

Whenever the modulus is a prime number, here too such a congruence of degree n cannot have more than n different solutions. Let α denote any fixed integer (complex). X can be put by division by $x-\alpha$, with the indeterminate left free, in the form $X = (x - \alpha)X' + h$, where h is a fixed integer and X' a function of degree $n - 1$ with integer coefficients. Whenever α is a root of the congruence $X \equiv 0 \pmod{m}$, h is evidently divisible by m; equivalently, with the indeterminate left free, we have $X \equiv (x - \alpha)X' \pmod{m}$.

Similarly, if β denotes a fixed integer and X' is put in the form $(x - \beta)X'' + h'$, X'' will be a function of degree $n - 2$ with integer coefficients. If β is supposed to be a root of the congruence $X \equiv 0$, it must also satisfy $(\beta-\alpha)X' \equiv 0$, and also $X' \equiv 0$ if the roots α, β are incongruent. We conclude that h' must also be divisible by m; equivalently, with the indeterminate left free, $X \equiv (x - \alpha)(x - \beta)X'' \pmod{m}$.

In the same way, on adding a third root γ, incongruent to the preceding ones, we shall have, with the indeterminate left free, $X \equiv (x-\alpha)(x-\beta)(x-\gamma)X'''$, where X''' is a function of degree $n-3$ with integer coefficients. We may continue in the same way. It is also clear that the coefficient of the highest term in each function is $= A$, which we may suppose not divisible by m, since otherwise the congruence $X \equiv 0$ would essentially have to be assigned a lower degree. Thus, whenever there are n incongruent roots, say $\alpha, \beta, \gamma \ldots . \nu$, we have, with the indeterminate left free,

$$X \equiv A(x-\alpha)(x-\beta)(x-\gamma)\ldots.(x-\nu) \pmod{m}$$

Consequently, substituting a new value incongruent to each of $\alpha, \beta, \gamma \ldots . \nu$ would certainly give X a value not divisible by m, from which the theorem follows at once.

This proof essentially agrees with the one we gave in *Disq. Ar.* Article 43, every step of which applies to complex numbers just as it does to real numbers.

51.

Most of what was given in the third section of the *Disquisitiones Arithmeticae* concerning residues of powers also holds in the arithmetic of complex numbers, with slight changes; indeed, the proofs of the theorems could generally be retained. So that nothing is missing, however, we shall present the principal theorems supported by concise proofs. Throughout, the modulus is to be understood to be a prime number.

THEOREM. *If k denotes an integer not divisible by the modulus m, of norm $= p$, then $k^{p-1} \equiv 1 \pmod{m}$.*

Proof. Let a, b, c, etc., form a complete system of incongruent residues modulo m, but with the residue divisible by m omitted, so that the number of these numbers, whose collection we denote by C, is $= p-1$. Further, let C' be the collection of products ka, kb, kc, etc. By hypothesis, none of these products is divisible by m, so each has a congruent residue in the collection C: we may thus have $ak \equiv a', bk \equiv b', ck \equiv c'$, etc., (mod. m), where the numbers a', b', c', etc., themselves occur in the collection C. Denote the collection of a', b', c', etc., by C''. Let P, P', P'' be the products of the individual numbers in the collections C, C', C'', respectively; that is,

$$\begin{aligned} P &= abc\ldots. \\ P' &= k^{p-1}abc\ldots. = k^{p-1}P \\ P'' &= a'b'c'\ldots. \end{aligned}$$

Since the numbers in C'' are congruent, in succession, to the numbers in C', we have $P'' \equiv P'$, or $P'' \equiv k^{p-1}P$. But any two numbers in C'' are readily seen to be incongruent, and therefore all are distinct. The numbers in C'' must consequently coincide exactly with those in C, with only their order changed, giving $P'' = P$. Thus $(k^{p-1} - 1)P$ is divisible by m. Since m is a prime number dividing none of the individual factors of P, $k^{p-1} - 1$ must necessarily be divisible by m. Q. E. D.

52.

THEOREM. *If k, as in the preceding article, denotes an integer not divisible by the modulus m, and t the least exponent (apart from 0) for which $k^t \equiv 1 \pmod{m}$, then t divides every other exponent u for which $k^u \equiv 1 \pmod{m}$.*

Proof. If t did not divide u, let gt be the multiple of u next greater than u, so that $gt - u$ is a positive integer less than t. From $k^t \equiv 1, k^u \equiv 1$ it follows that $0 \equiv k^{gt} - k^u \equiv k^u(k^{gt-u} - 1)$, and hence $k^{gt-u} \equiv 1$: there is a power of k with an exponent less than t that is congruent to unity, contrary to the hypothesis.

It follows as a corollary that t certainly divides $p - 1$.

Here too we shall call numbers k for which $t = p - 1$ *primitive roots* for. the modulus m. We shall now show that such numbers really exist.

53.

Factor the number $p - 1$ into its prime factors, so that

$$p - 1 = a^\alpha b^\beta c^\gamma \ldots .$$

where a, b, c, etc., denote distinct positive real prime numbers. Let A, B, C, etc., be integers (complex) not divisible by m, and respectively *not* satisfying the congruences

$$x^{\frac{p-1}{a}} \equiv 1, \quad x^{\frac{p-1}{b}} \equiv 1, \quad x^{\frac{p-1}{c}} \equiv 1 \quad \text{etc.}$$

modulo m. The existence of such numbers is clear from the theorem of Article 50. Finally, let h be congruent modulo m to the product

$$A^{\frac{p-1}{a^\alpha}} B^{\frac{p-1}{b^\beta}} C^{\frac{p-1}{c^\gamma}} \ldots .$$

Then I say that h will be a primitive root.

Proof. Denoting by t the exponent of the lowest power h^t congruent to unity, if h were not a primitive root, t would be a proper divisor of $p - 1$; that is, $\frac{p-1}{t}$ would be an integer greater than unity. This integer evidently has its real prime factors among a, b, c, etc. Let us therefore suppose, as we may, that $\frac{p-1}{t}$ is divisible by a, and put $p - 1 = atu$. Since $h^t \equiv 1$, we then also have $h^{tu} \equiv 1$, or

$$A^{\frac{p-1}{a^\alpha} \cdot \frac{p-1}{a}} B^{\frac{p-1}{b^\beta} \cdot \frac{p-1}{a}} C^{\frac{p-1}{c^\gamma} \cdot \frac{p-1}{a}} \ldots . \equiv 1$$

But $\frac{p-1}{ab^\beta}$ is evidently an integer, so

$$B^{\frac{p-1}{b^\beta}\cdot\frac{p-1}{a}} = (B^{p-1})^{\frac{p-1}{ab^\beta}} \equiv 1$$

and similarly $C^{\frac{p-1}{c^\gamma}\cdot\frac{p-1}{a}} \equiv 1$, and so on; consequently we must have $A^{\frac{p-1}{a^\alpha}\cdot\frac{p-1}{a}} \equiv 1$ Now determine a positive integer λ such that

$$\lambda b^\beta c^\gamma \ldots . \equiv 1 \pmod{a}$$

This can be done, since the prime number a does not divide $b^\beta c^\gamma \ldots .$; put $\lambda b^\beta c^\gamma \ldots . = 1 + a\mu$. Evidently $A^{\lambda\cdot\frac{p-1}{a^\alpha}\cdot\frac{p-1}{a}} \equiv 1$; or, since $\lambda \,.\, \frac{p-1}{a^\alpha} \,.\, \frac{p-1}{a} = (1+a\mu)\frac{p-1}{a} = (p-1)\mu + \frac{p-1}{a}$, we have $A^{(p-1)\mu} \,.\, A^{\frac{p-1}{a}} \equiv 1$. Hence, since $A^{(p-1)\mu} \equiv 1$ holds of itself, we also have $A^{\frac{p-1}{a}} \equiv 1$, contrary to the hypothesis. The supposition that t is a proper divisor of $p-1$ therefore cannot stand, and h must necessarily be a primitive root.

54.

Let h be a primitive root for the modulus m, of norm $= p$. The terms of the progression

$$1, h, hh, h^3 \ldots . h^{p-2}$$

will be incongruent to one another. It follows easily that every integer not divisible by the modulus must be congruent to one of them: this sequence gives a complete system of incongruent residues with zero excluded. The exponent of the power to which a given number is congruent may be called its *index*, while h is regarded as the *base*. Here are some examples, in which we have placed the absolutely least residue beside each index.

First example.

$$m = 5 + 4i, \quad p = 41, \quad h = 1 + 2i$$

Ind.	Residue	Ind.	Residue	Ind.	Residue	Ind.	Residue	Ind.	Residue
0	$+1$	8	-4	16	$-2+2i$	24	$+2i$	32	$+1+i$
1	$+1+2i$	9	$-3+i$	17	$-1+2i$	25	$-3i$	33	$+1+3i$
2	$+1-i$	10	$-i$	18	$+4i$	26	$+2+2i$	34	$+2$
3	$+3+i$	11	$+2-i$	19	$+1+3i$	27	$+2+i$	35	-3
4	$-2i$	12	$-1-i$	20	-1	28	$+4$	36	$+2-2i$
5	$+3i$	13	$+1-3i$	21	$-1-2i$	29	$+3-i$	37	$+1-2i$
6	$-2-2i$	14	-2	22	$-1+i$	30	$+i$	38	$-4i$
7	$-2-i$	15	$+3$	23	$-3-i$	31	$-2+i$	39	$-1-3i$

Second example.

$$m = 7, \quad p = 49, \quad h = 1 + 2i$$

Ind.	Residue	Ind.	Residue	Ind.	Residue	Ind.	Residue	Ind.	Residue
0	$+1$	10	$-1-i$	20	$+2i$	30	$+2-2i$	40	$+3$
1	$+1+2i$	11	$+1-3i$	21	$+3+2i$	31	$-1+2i$	41	$+3-i$
2	$-3-3i$	12	$-i$	22	$-1+i$	32	$+2$	42	$-2-2i$
3	$+3-2i$	13	$+2-i$	23	$-3-i$	33	$+2-3i$	43	$+2+i$
4	$-3i$	14	$-3+3i$	24	-1	34	$+1+i$	44	$-2i$
5	$-1-3i$	15	$-2-3i$	25	$-1-2i$	35	$-1+3i$	45	$-3-2i$
6	$-2+2i$	16	-3	26	$+3+3i$	36	$+i$	46	$+1-i$
7	$+1-2i$	17	$-3+i$	27	$-3+2i$	37	$-2+i$	47	$+3+i$
8	-2	18	$+2+2i$	28	$+3i$	38	$+3-3i$		
9	$-2+3i$	19	$-2-i$	29	$+1+3i$	39	$+2+3i$		

55.

We add some observations about primitive roots and the calculation of indices, omitting their proofs because they are easy.

I. In a given system, indices congruent modulo $p-1$ correspond to residues congruent modulo m, and conversely.

II. Residues corresponding to indices relatively prime to $p-1$ are themselves primitive roots, and conversely.

III. Suppose that, with a primitive root h taken as the base, the index of another primitive root h' is t, and, conversely, t' is the index of h when h' is taken as the base. Then $tt' \equiv 1 \pmod{p-1}$. Under the same assumptions, if the indices of some other number in these two systems are respectively u, u', then $tu' \equiv u, t'u \equiv u' \pmod{p-1}$.

IV. Excluding the numbers $1, 1+i$ and their three associates each, as being too trivial, from the moduli we consider leaves the prime numbers that we placed third and fourth in Article 34. The norms of the latter are real prime numbers of the form $4n+1$; those of the former, however, are squares of odd real prime numbers. Thus, in either case, $p-1$ is divisible by 4.

V. Denote the index of -1 by u. Then $2u \equiv 0 \pmod{p-1}$, so either $u \equiv 0$ or $u \equiv \frac{1}{2}(p-1)$. But since the index 0 corresponds to the residue $+1$, the index of -1 must be $\frac{1}{2}(p-1)$.

VI. Similarly, denoting by u the index of i, we have $2u \equiv \frac{1}{2}(p-1) \pmod{p-1}$, and therefore either $u \equiv \frac{1}{4}(p-1)$ or $u \equiv \frac{3}{4}(p-1)$. Here, however, the ambiguity depends on the choice of primitive root. Specifically, if, with the primitive root h taken as the base, the index of i is $\frac{1}{4}(p-1)$, the index becomes $\frac{3}{4}(p-1)$ when h^μ is taken as the base, where μ denotes a positive integer of the form $4n+3$ relatively prime to $p-1$, for example the number $p-2$ itself; and conversely. Thus one half of the primitive roots gives i the index $\frac{1}{4}(p-1)$, and the other half the index $\frac{3}{4}(p-1)$. Evidently, for the former bases $-i$ has the index $\frac{3}{4}(p-1)$, and for the latter the index $\frac{1}{4}(p-1)$.

VII. Whenever the modulus is a positive real prime number of the form $4n+3$, say $= q$, so that $p = qq$, the indices of all real numbers are divisible by $q+1$. For, if t denotes the index of the real number k, then, because $k^{q-1} \equiv \pmod{q}$, we have $(q-1)t \equiv 0 \pmod{qq-1}$, and consequently $\frac{t}{q+1}$ is an integer. Similarly, the indices of purely imaginary numbers such as ki are divisible by $\frac{1}{2}(q+1)$. Thus primitive roots for such moduli must evidently be sought only among mixed numbers.

VIII. For a modulus m that is a mixed complex prime number, however, whose norm p is consequently a real prime number of the form $4n+1$, any primitive roots may also be chosen among real numbers, among which a complete system of incongruent residues can indeed be exhibited (Article 40). Every real number that is a primitive root for the complex modulus m is evidently also, in the arithmetic of real numbers, a primitive root for the modulus p, and conversely.

56.

Although the theory of quadratic residues and nonresidues in the arithmetic of complex numbers is contained within the theory of biquadratic residues itself, we shall present its principal theorems separately here before turning to the latter. For brevity, however, we shall discuss only the principal case, where the modulus is an odd complex prime number.

Let m be such a modulus, and p its norm. Evidently, any integer not divisible by m, a condition always to be understood here, can or cannot be congruent to a square modulo m, according as its index, with some primitive root taken as the base, is even or odd. In the former case that integer is called a quadratic residue of m, and in the latter a nonresidue. It follows that, among $p-1$ numbers forming a complete system of incongruent residues not divisible by m, one half belongs to the quadratic residues and the other half to the quadratic nonresidues. Any other number outside this system must be assigned the same character in this respect as the number in the system congruent to it.

It also follows that the product of two quadratic residues, as well as the product of two nonresidues, is a quadratic residue, whereas the product of a quadratic residue and a nonresidue is a nonresidue. In general, a product of any number of factors is a quadratic residue or a nonresidue according as the number of nonresidues among its factors is even or odd.

The following general criterion for distinguishing quadratic residues from nonresidues presents itself at once:

A number k not divisible by the modulus is a quadratic residue or a nonresidue of it according as $k^{\frac{1}{2}(p-1)} \equiv 1$ or $k^{\frac{1}{2}(p-1)} \equiv -1 \pmod{m}$.

This theorem follows at once from the fact that, with any primitive root taken as the base, the index of the power $k^{\frac{1}{2}(p-1)}$ becomes either $\equiv 0$ or $\equiv \frac{1}{2}(p-1)$, according as the index of k is even or odd.

57.

For a given modulus, it is easy to divide a complete system of incongruent residues into two classes, quadratic residues and nonresidues. This also assigns all the remaining numbers to their respective classes. But a much deeper investigation is needed to find criteria distinguishing the moduli for which a given number is a quadratic residue from those for which it is a nonresidue.

As for the real units $+1$ and -1, these are actual squares in the arithmetic of complex numbers, and therefore quadratic residues for *every* modulus. It follows just as easily from the criterion in the preceding article that i, and likewise $-i$, is a quadratic residue of every modulus whose norm p has the form $8n+1$, but a nonresidue of every modulus whose norm has the form $8n+5$. Since it evidently makes no difference whether m or one of its associates

$im, -m, -im$ is taken as the modulus, we may suppose that the modulus is the primary one among its associates (Article 36, II). Thus, putting the modulus $= a + bi$, we may suppose a odd and b even. We then always have $aa \equiv 1 \pmod 8$, while bb is either $\equiv 0$ or $\equiv 4 \pmod 8$, according as b is divisible by four or twice an odd number. Thus $+i$ and $-i$ are evidently quadratic residues of the modulus in the former case and nonresidues in the latter.

58.

Since the character of a composite number, whether it is a quadratic residue or a nonresidue, depends on the characters of its factors, it evidently suffices to restrict the development of criteria distinguishing the moduli for which a given number k is a quadratic residue from those for which it is a nonresidue to values of k that are prime numbers and, moreover, primary among their associates. In this investigation, *examining examples* immediately supplies very elegant theorems.

Let us begin with $1 + i$, which is found to be a quadratic residue of the moduli

$$-1 + 2i,\ +3 - 2i,\ -5 - 2i,\ -1 - 6i,\ +5 + 4i,\ +5 - 4i,$$
$$-7,\ +7 + 2i,\ -5 + 6i, \quad \text{etc.}$$

but a quadratic nonresidue of the following:

$$-1 - 2i,\ -3,\ +3 + 2i,\ +1 + 4i,\ +1 - 4i,\ -5 + 2i,$$
$$-1 + 6i,\ +7 - 2i,\ -5 - 6i,\ -3 + 8i,\ -3 - 8i,$$
$$+5 + 8i,\ +5 - 8i,\ +9 + 4i,\ +9 - 4i \quad \text{etc.}$$

Examining this list carefully, in which we have always entered the primary one of each four associated moduli, we readily notice that all the moduli $a+bi$ in the first class satisfy $a+b \equiv +1 \pmod 8$, while those in the second satisfy $a + b \equiv -3 \pmod 8$. If, in place of the primary modulus m, we take its associate $-m$, this criterion must evidently change so that $a + b \equiv -1$ for moduli in the first class, and $\equiv +3 \pmod 8$ for those in the second. Thus, provided our examination of examples has not misled us, in general, if $a + bi$ denotes a prime number in which a is odd. b even, $1+i$ is a quadratic residue or a quadratic nonresidue of it according as $a + b \equiv \pm 1$ or $\equiv \pm 3 \pmod 8$.

The same rule holds for $-1-i$ as for $1+i$. On the other hand, considering $1 - i$ as the product of $-i$ and $1 + i$, $1 - i$ evidently has the same character as must be assigned to $1 + i$ whenever b is divisible by four, but the opposite character whenever b is twice an odd number. It follows easily that $1 - i$ is a quadratic residue of the prime number $a + bi$ whenever $a - b \equiv \pm 1$, but a nonresidue whenever $a - b \equiv \pm 3 \pmod 8$, always supposing a odd and b even.

This second proposition can also be deduced from the first with the help of a more general theorem, which we state as follows:

In the theory of quadratic residues, the character of $\alpha+\beta i$ with respect to the modulus $a+bi$ is the same as that of $\alpha-\beta i$ with respect to the modulus $a-bi$.

The proof of this theorem follows from the fact that both moduli have the same norm p, and that whenever $(\alpha+\beta i)^{\frac{1}{2}(p-1)}-1$ is divisible by $a+bi$, $(\alpha-\beta i)^{\frac{1}{2}(p-1)}-1$ is also divisible by $a-bi$, while whenever $(\alpha+\beta i)^{\frac{1}{2}(p-1)}+1$ is divisible by $a+bi$, $(\alpha-\beta i)^{\frac{1}{2}(p-1)}+1$ must also be divisible by $a-bi$.

59.

Let us proceed to odd prime numbers.

We find that $-1+2i$ is a quadratic residue of the moduli $+3+2i$, $+1-4i$, $-5+2i$, $-5-2i$, $-1-6i$, $+7-2i$, $-3+8i$, $+5+8i$, $+5-8i$, $+9+4i$, etc.,

but a nonresidue of the moduli $-1-2i$, -3, $+3-2i$, $+1+4i$, $-1+6i$, $+5+4i$, $+5-4i$, -7, $+7+2i$, $-5+6i$, $-5-6i$, $-3-8i$, $+9-4i$, etc.

Reducing the moduli of the first class to their absolutely least residues modulo $-1+2i$, we find only $+1$ and -1, namely $+3+2i\equiv-1$, $+1-4i\equiv-1$, $-5+2i\equiv+1$, $-5-2i\equiv-1$, etc.

On the other hand, all moduli of the second class are found to be congruent modulo $-1+2i$ either to $+i$ or to $-i$.

But $+1,-1$ are themselves quadratic residues of the modulus $-1+2i$, while $+i$ and $-i$ are its nonresidues. Thus, as far as the examination of examples can be trusted, the following theorem emerges: $-1+2i$ is a quadratic residue or a quadratic nonresidue of the prime number $a+bi$ according as the latter is a quadratic residue or a quadratic nonresidue of $-1+2i$, provided that $a+bi$ is the primary one of its four associates, or rather, that a is odd and b even.

Analogous theorems concerning $+1-2i$, $-1-2i$, $+1+2i$ follow directly from this theorem.

60.

Examining examples similarly for -3 or $+3$, we find that each is a quadratic residue of the moduli $+3+2i$, $+3-2i$, $-1+6i$, $-1-6i$, -7, $-5+6i$, $-5-6i$, $-3+8i$, $-3-8i$, $+9+4i$, $+9-4i$, etc.,

but a nonresidue of $-1+2i$, $-1-2i$, $+1+4i$, $+1-4i$, $-5+2i$, $-5-2i$, $+5+4i$, $+5-4i$, $+7+2i$, $+7-2i$, $+5+8i$, $+5-8i$, etc.

The former are congruent modulo 3 to one of the four numbers $+1,-1,+i,-i$, the latter to one of $+1+i,+1-i,-1+i,-1-i$. The former are precisely the quadratic residues of 3, and the latter its nonresidues.

Thus this examination of examples shows that the prime number $a+bi$, supposing a odd and b even, has the same relation to -3, and also to $+3$,

as the latter has to it, in respect of whether one is a quadratic residue or a nonresidue of the other.

Extending a similar examination to other prime numbers, we find this most elegant law of reciprocity confirmed everywhere, and are led to the following fundamental theorem about quadratic residues in the arithmetic of complex numbers:

If $a + bi, A + Bi$ denote prime numbers such that a, A are odd and b, B even, either each is a quadratic residue of the other, or each is a nonresidue of the other.

Yet despite the theorem's great simplicity, its proof is beset by great difficulties. We shall not dwell on these here, however, since the theorem itself is only a special case of a more general theorem that, as it were, exhausts the substance of the theory of biquadratic residues. Let us now turn to that theory.

61.

We extend to the arithmetic of complex numbers what was said in Article 2 of the preceding paper about the notion of a biquadratic residue and nonresidue. Here, as there, we restrict the investigation to moduli that are prime numbers. It will also usually be tacitly understood that the modulus is chosen to be primary among its associates, that is, $\equiv 1$ modulo $2+2i$, and that the numbers whose character, as biquadratic residues or nonresidues, is under consideration are not divisible by the modulus.

For a given modulus, therefore, the numbers not divisible by it could be divided into three classes: the first containing biquadratic residues; the second, those biquadratic nonresidues that are quadratic residues; and the third, quadratic nonresidues.

But here too it is better to establish two classes in place of the third, so that there are four in all.

With any primitive root taken as the base, biquadratic residues will have indices divisible by 4, or of the form $4n$; those nonresidues that are quadratic residues will have indices of the form $4n+2$; finally, the indices of quadratic nonresidues will be partly of the form $4n+1$, partly of the form $4n+3$. Four classes do arise in this way, but the distinction between the latter two would not be absolute: it would depend on the choice of the primitive root taken as the base. For it is easy to see that one half of the primitive roots gives a given quadratic nonresidue an index of the form $4n+1$, and the other half an index of the form $4n+3$. To remove this ambiguity, we shall always suppose that a primitive root is chosen for which the index $\frac{1}{4}(p-1)$ belongs to the number $+i$ (compare Article 55, VI). This gives a classification that we can state more concisely, independently of primitive roots, as follows.

Let the *first* class contain those numbers k for which $k^{\frac{1}{4}(p-1)} \equiv 1$; these numbers are biquadratic residues of the modulus.

Let the *second* class contain those for which $k^{\frac{1}{4}(p-1)} \equiv i$.

The *third*, those for which $k^{\frac{1}{4}(p-1)} \equiv -1$.

Finally, the *fourth*, those for which $k^{\frac{1}{4}(p-1)} \equiv -i$.

The third class will contain those biquadratic nonresidues that are quadratic residues; quadratic nonresidues will be distributed between the second and fourth classes.

We shall assign to the numbers in these classes the *biquadratic characters* $0, 1, 2, 3$, respectively. If we define the character λ of a number k modulo m as the exponent of that power of i to which $k^{\frac{1}{4}(p-1)}$ is congruent, then evidently characters congruent modulo 4 must be regarded as equivalent. For the time being, this notion is restricted to moduli that are prime numbers: in a continuation of these investigations we shall show how it can also be applied to composite moduli.

62.

To make an extensive examination of examples concerning the characters of numbers easier, we append a compact table here. With its help, the character of any given number with respect to a modulus whose norm does not exceed 157 is obtained with little work, provided that the following observations are taken into account.

Since the character of a composite number is equal to, or congruent modulo 4 to, the sum of the characters of its individual factors, it suffices if we can assign the characters of prime numbers for a given modulus. Further, since the characters of the units $-1, i, -i$ are evidently congruent to the numbers $\frac{1}{2}(p-1), \frac{1}{4}(p-1), \frac{3}{4}(p-1)$ modulo 4, it will also suffice to give the characters of numbers that are primary among their associates. Finally, since moduli congruent modulo m have the same character, it suffices to enter in the table the characters of numbers contained in a system of absolutely least residues. In addition, reasoning similar to that in Article 58 shows that, if for the modulus $a+bi$ the character of $A+Bi$ is λ, while for the modulus $a-bi$ λ' is the character of $A-Bi$, then we always have $\lambda \equiv -\lambda' \pmod 4$, or, equivalently, $\lambda+\lambda'$ is divisible by 4. It therefore suffices to enter in the table moduli in which b is either 0 or positive.

Thus, for example, if the character of $11-6i$ with respect to the modulus $-5-6i$ is sought, we substitute the numbers $11+6i, -5+6i$ for them. We then determine (Article 43) the absolutely least residue of $11+6i$ modulo $-5+6i$, which is $-1-4i = -1\times(1+4i)$. Since for the modulus $-5+6i$ the character of -1 is 30, while the character of $1+4i$, from the table, is 2, we obtain 32, or 0, as the character of $11+6i$ for the modulus $-5+6i$; and, by the last observation, this is also the character of $11-6i$ for the modulus $-5-6i$. Likewise, if the character of $-5+6i$ with respect to the modulus $11+6i$ is sought, its absolutely least residue $1-5i$ is factored into $-i\ 1+i,\ 3-2i$, to which the characters $117, 0, 1$ correspond. Hence the character sought will be 118, or 2; the same character must also be assigned to $-5-6i$ with respect to the modulus $11-6i$.

Modulus.	Character.	Numbers.
-3	3	$1+i$
$+3+2i$	3	$1+i$
$+1+4i$	1	$-1+2i$
	3	$1+i$
$-5+2i$	0	$-1-2i$
	1	$1+i$
	2	$-1+2i$
$-1+6i$	0	-3
	1	$1+i,\ -1+2i$

Modulus.	Character.	Numbers.
$-1+6i$	2	$-1-2i$
$+5+4i$	0	$1+i$
	1	-3
	3	$-1+2i,\ -1-2i$
-7	0	-3
	1	$-1+2i,\ 3-2i$
	2	$1+i$
	3	$-1-2i,\ 3+2i$
$+7+2i$	0	$1+i,\ 3+2i,\ 3-2i,\ 1-4i$
	1	-3
	2	$-1-2i,\ 1+4i$
	3	$-1+2i$
$-5+6i$	0	$1+i,\ -3,\ 3+2i,\ 3-2i$
	1	$1-4i$
	2	$1+4i$
	3	$-1+2i,\ -1-2i$
$-3+8i$	0	$-1+2i,\ 3-2i,\ 1-4i$
	1	$1+i,\ 3+2i$
	2	-3
	3	$-1-2i,\ 1+4i,\ -5+2i$
$+5+8i$	0	$-1-2i$
	1	$-5-2i,\ -1+6i$
	2	$-1+2i,\ 3-2i$
	3	$1+i,\ -3,\ 3+2i,\ 1+4i,\ 1-4i$
$+9+4i$	0	$-1+2i,\ 3+2i$
	1	$1+i,\ -1-2i,\ 3-2i$
	2	$-3,\ 1+4i$
	3	$1-4i,\ -5+2i$
$-1+10i$	0	$1+i,\ -1+2i,\ -1-2i,\ 3+2i$
	1	-3
	2	$3-2i,\ -5+2i,\ 5-4i$
	3	$1+4i,\ 1-4i$

Modulus.	Character.	Numbers.
$+3+10i$	1	$1+i,\ -1-2i,\ 1-4i$
	2	$-3,\ 3+2i,\ 1+4i,\ -5-2i$
	3	$-1+2i,\ 3-2i$
$-7+8i$	0	$1+i,\ -7$
	1	$3+2i,\ 3-2i,\ 1-4i,\ -5-2i$
	2	$-1-2i,\ 1+4i,\ -5+2i,\ -1-6i$
	3	$-1+2i,\ -3,\ -1+6i$
-11	0	-3
	1	$1+i,\ 3-2i,\ 1+4i,\ -5+2i,\ 5+4i$
	2	$-1+2i,\ -1-2i$
	3	$3+2i,\ 1-4i,\ -5-2i,\ 5-4i$
$-11+4i$	0	$1+i,\ -1+2i,\ 3+2i,\ 5+4i$
	1	$-1-2i,\ -1+6i$
	2	$-5+2i$
	3	$-3,\ 3-2i,\ 1+4i,\ 1-4i,\ -5-2i$
$+7+10i$	0	$1+4i,\ 1-4i,\ -1+6i,\ -1-6i$
	1	$-1+2i,\ 3+2i,\ -5+2i$
	2	$1+i,\ 3-2i$
	3	$-1-2i,\ -3,\ -5-2i$
$+11+6i$	0	$1+i,\ -1+2i,\ -3,\ 1+4i,\ 1-4i,\ -7$
	1	$-1-2i,\ 3+2i,\ 3-2i$
	2	$-5-2i,\ -1+6i,\ 5-4i$
	3	$-5+2i,\ 5+4i,\ 7-2i.$

63.

We shall now seek, by examining examples, common criteria for the moduli for which a given prime number has the same character. We always suppose the moduli to be primary among their associates, that is, numbers $a+bi$ for which either $a \equiv 1, b \equiv 0$ or $a \equiv 3, b \equiv 2 \pmod 4$.

For the number $1+i$, with which we begin, the rule suggested by the examples is more easily grasped if we separate moduli of the first kind, for which $a \equiv 1, b \equiv 0$, from moduli of the second kind, for which $a \equiv 3, b \equiv 2$. With the help of the table in the preceding article, we find the following correspondences:

Character	Moduli of the first kind.
0	$5+4i,\ -7+8i,\ -7-8i,\ -11+4i$
1	$1-4i,\ -3+8i,\ -3-8i,\ 9+4i,\ -11$
2	$5-4i,\ -7,\ -11-4i$
3	$-3,\ 1+4i,\ 5+8i,\ 5-8i,\ 9-4i$

If we consider these seventeen examples carefully, in all of them we find the character $\equiv \frac{1}{4}(a-b-1) \pmod 4$.

Likewise, the correspondences are

Character	Moduli of the second kind.
0	$3-2i,\ -1-6i,\ 7+2i,\ -5+6i,\ -1+10i,\ 11+6i$
1	$-5+2i,\ -1+6i,\ 7-2i,\ -1-10i,\ 3+10i$
2	$-1+2i,\ -5-2i,\ 3-10i,\ 7+10i$
3	$-1-2i,\ 3+2i,\ -5-6i,\ 7-10i,\ 11-6i$

In all these twenty examples, with a little attention, the character is found to be $\equiv \frac{1}{4}(a-b-5) \pmod 4$.

These two rules can easily be combined into one valid for both kinds of moduli, if we consider that $\frac{1}{4}bb$ is $\equiv 0$ for moduli of the first kind and $\equiv 1 \pmod 4$ for those of the second. Thus the character of $1+i$ with respect to any prime modulus that is primary among its associates is $\equiv \frac{1}{4}(a-b-1-bb) \pmod 4$.

We should note in passing that, since $(b+1)^2$ always has the form $8n+1$, or, equivalently, $\frac{1}{4}(2b+bb)$ is even, this character is always even or odd according as $\frac{1}{4}(a+b-1)$ is even or odd. This agrees with the rule for the quadratic character stated in Article 58.

Since $\frac{1}{4}(a-b-1), \frac{1}{4}(a-b+3)$ are integers, one even and the other odd, their product will be even, or, equivalently, $\frac{1}{8}(a-b-1)(a-b+3) \equiv 0 \pmod 4$. Hence the following expression can also be used in place of the one given for the biquadratic character:

$$\frac{1}{4}(a-b-1-bb) - \frac{1}{8}(a-b-1)(a-b+3) = \frac{1}{8}(-aa+2ab-3bb+1)$$

This form also has the advantage that it is not restricted to primary moduli, but supposes only that a is odd and b even. For under this supposition either $a+bi$ or $-a-bi$ will evidently be primary among its associates, and the value of that formula is the same for either modulus.

64.

Starting from the last rule obtained in the preceding article, we find

Number	Character $\equiv$
$-1+i$	$\frac{1}{8}(aa+2ab-bb-1)$
$-1-i$	$\frac{1}{8}(-aa+2ab+bb+1)$
$+1-i$	$\frac{1}{8}(aa+2ab+3bb-1)$

This follows at once from the fact that the character of i is $\frac{1}{4}(aa+bb-1)$, while that of -1 is $\frac{1}{2}(aa+bb-1) \equiv \frac{1}{2}bb$, since $aa-1$ always has the form $8n$. These four rules, although so far obtained by examining examples, are evidently connected in such a way that, as soon as the proof of one has been completed, the other three will have been proved at the same time. It is scarcely necessary to point out that these rules too suppose only that a is odd and b even.

If we are willing to use formulas restricted to primary moduli, we can use the following form:

Number	Character $\equiv$
$-1+i$	$\frac{1}{4}(-a-b+1-bb)$
$-1-i$	$\frac{1}{4}(a-b-1+bb)$
$+1-i$	$\frac{1}{4}(-a-b+1+bb)$

The simplest formulas result if, as we did at the beginning of our examination of examples, we distinguish moduli of the first and second kinds. The characters are

Number	For moduli of the first kind	For moduli of the second kind
$-1+i$	$\frac{1}{4}(-a-b+1)$	$\frac{1}{4}(-a-b-3)$
$-1-i$	$\frac{1}{4}(a-b-1)$	$\frac{1}{4}(a-b+3)$
$+1-i$	$\frac{1}{4}(-a-b+1)$	$\frac{1}{4}(-a-b+5)$

65.

For the number $-1+2i$, to which we now proceed, we shall likewise use the same distinction between those moduli $a+bi$ for which $a \equiv 1, b \equiv 0$ and those for which $a \equiv 3, b \equiv 2$. The table in Article 62 shows that, with respect to that number, the following correspondences hold:

Character	Moduli of the first kind
0	$-3+8i,\ +5-8i,\ +9+4i,\ -11+4i$
1	$+1+4i,\ +5-4i,\ -7,\ -3-8i$
2	$+1-4i,\ +5+8i,\ -7-8i,\ -11$
3	$-3,\ +5+4i,\ +9-4i,\ -7+8i,\ -11-4i$

Reducing each of these moduli to its absolutely least residue modulo $-1+2i$, we observe that all those to which character 0 corresponds are $\equiv 1$; those to which character 1 corresponds are $\equiv i$; those whose character is 2 become $\equiv -1$; finally, all those whose character is 3 become $\equiv -i$. But the characters of $1, i, -1, -i$ for the modulus $-1+2i$ are themselves $0, 1, 2, 3$, respectively. Thus, in all these 17 examples, the character of $-1+2i$ with respect to a modulus $a+bi$ of the first kind is identical with the character of that number with respect to the modulus $-1+2i$.

Likewise, with the help of the table, we find the correspondences

Character	Moduli of the second kind
0	$+3+2i,\ -5-2i,\ -1+10i,\ -1-10i,\ +11+6i$
1	$+3-2i,\ -1+6i,\ -5-6i,\ +7+10i,\ +7-10i$
2	$-5+2i,\ -1-6i,\ +7-2i$
3	$-1-2i,\ +7+2i,\ -5+6i,\ +3+10i,\ +3-10i,\ +11-6i$

Reducing these moduli to their least residues modulo $-1+2i$, all those to which the characters $0, 1, 2, 3$ respectively correspond are found congruent to $-1, -i, +1, +i$. But when, conversely, $-1+2i$ is taken as the modulus, these very numbers have the characters $2, 3, 0, 1$, respectively. Consequently, in all these 19 examples, the character of $-1+2i$ with respect to a modulus of the second kind differs by two units from the character of that number with respect to $-1+2i$ taken as the modulus.

Moreover, it is readily seen that entirely similar results will hold for $-1-2i$.

66.

For the number -3, we omit the distinction between moduli of the first and second kinds, since the result shows that it is superfluous here. Thus the correspondences are

Character	Moduli
0	$-1+6i,\ -1-6i,\ -7,\ -5+6i,$ $-5-6i,\ -11,\ 11+6i,\ 11-6i$
1	$-1-2i,\ 1-4i,\ -5+2i,\ 5+4i,\ 7+2i,$ $5-8i,\ -1+10i,\ -7-8i,\ -11-4i,\ 7-10i$
2	$3+2i,\ 3-2i,\ -3+8i,\ -3-8i,$ $9+4i,\ 3+10i,\ 3-10i$
3	$-1+2i,\ 1+4i,\ -5-2i,\ 5-4i,\ 7-2i,$ $5+8i,\ -1-10i,\ -7+8i,\ -11+4i,\ 7+10i$

Reducing these moduli to their least residues modulo 3, we see that those to which character 0 corresponds are partly $\equiv 1$, partly $\equiv -1$; those whose character is 1 become either $\equiv 1 - i$ or $\equiv -1 + i$; those whose character is 2 become either $\equiv i$ or $\equiv -i$; finally, those to which character 3 belongs are either $\equiv 1 + i$ or $\equiv -1 - i$. From this examination of examples we therefore conclude that the character of -3 for a modulus that is a prime number primary among its associates is identical with the character of that very number when 3, or equivalently -3, is considered as the modulus.

67.

Examining examples similarly for other prime numbers, we find that $3 \pm 2i, -1 \pm 6i, 7 \pm 2i, -5 \pm 6i$, etc., give theorems similar to the one reached in Article 65 for $-1+2i$, whereas $1 \pm 4i, 5 \pm 4i, -3 \pm 8i, 5 \pm 8i, 9 \pm 4i$, etc., behave in the same way as -3. The examination of examples thus leads to a most elegant theorem,. which, by analogy with the theory of quadratic residues in the arithmetic of real numbers, may be called the FUNDAMENTAL THEOREM of the theory of biquadratic residues, namely:

Let $a + bi, a' + b'i$ denote distinct prime numbers, primary among their respective associates, that is, congruent to unity modulo $2+2i$. The biquadratic character of $a+bi$ with respect to the modulus $a'+b'i$ will be identical with the character of $a' + b'i$ with respect to the modulus $a + bi$, if either both numbers $a + bi, a' + b'i$, or at least one of them, belong to the first kind, that is, are congruent to unity modulo 4. On the other hand, those characters will differ from one another by two units if neither number $a + bi, a' + b'i$ belongs to the first kind, that is, if each is congruent modulo 4 to $3 + 2i$.

Yet despite the great simplicity of this theorem, its proof must be counted among the most deeply hidden mysteries of higher arithmetic. At least as matters now stand, it can be unravelled only through the most subtle investigations, which would go far beyond the limits of the present paper. We therefore reserve the publication of this proof, as well as the development of the connection between this theorem and those that we had begun to establish by examining examples at the beginning of this paper, for a third paper. Nevertheless, by way of conclusion, we shall give here what is required to prove the theorems proposed in Articles 63. 64.

68.

We begin with prime numbers $a + bi$ for which $b = 0$ (the third species in Article 34). Here, for the number to be primary among its associates, a must therefore be a negative real prime number of the form $-(4n + 3)$, for which we shall write $-q$, such as $-3, -7, -11, -19$, etc. Denoting by λ the character of $1 + i$, with that number taken as the modulus, we must have

$$i^{\lambda} \equiv (1+i)^{\frac{1}{4}(qq-1)} \equiv 2^{\frac{1}{8}(qq-1)} \,.\, i^{\frac{1}{8}(qq-1)} \pmod{q}$$

But it is known that 2 is a quadratic residue or a quadratic nonresidue of q according as q has the form $8n+7$ or the form $8n+3$. From this we conclude that, in general,

$$2^{\frac{1}{2}(q-1)} \equiv (-1)^{\frac{1}{4}(q+1)} \equiv i^{\frac{1}{2}(q+1)} \pmod{q}$$

and consequently, raising to the power with exponent $\frac{1}{4}(q+1)$,

$$2^{\frac{1}{8}(qq-1)} \equiv i^{\frac{1}{8}(q+1)^2} \pmod{q}$$

The preceding equation therefore takes the form

$$i^{\lambda} \equiv i^{\frac{1}{8}(q+1)^2 + \frac{1}{8}(qq-1)} \equiv i^{\frac{1}{4}(qq+q)} \pmod{q}$$

whence it follows that

$$\lambda \equiv \frac{1}{4}(qq+q) \equiv \frac{1}{4}(q+1)^2 - \frac{1}{4}(q+1) \pmod{4}$$

or, since $\frac{1}{4}(q+1)^2 \equiv 0 \pmod{4}$, $\lambda \equiv -\frac{1}{4}(q+1) \equiv \frac{1}{4}(a-1) \pmod{4}$. This is precisely the theorem of Article 63 for the case $b = 0$.

69.

Moduli $a+bi$ for which $b=0$ does not hold (numbers of the fourth species in Art. 34) are much harder to deal with, however, and several investigations must come first. We shall denote the norm $aa+bb$, which will be a real prime of the form $4n+1$, by p.

Let S denote the collection of all simply least residues for the modulus $a+bi=m$, with zero excluded, so that the count of the numbers contained in S is $=p-1$. Let $x+yi$ denote any number of this system, and put $ax+by=\xi, ay-bx=\eta$. Then ξ,η will be integers lying between the bounds 0 and p, *excluding the bounds*. For in the present case, where a,b are relatively prime, the formulas of Art. 45, namely $\eta\equiv k\xi, \xi\equiv -k\eta \pmod p$, show that neither of the numbers ξ,η can be $=0$ unless the other vanishes at the same time, so that $x=0, y=0$, a combination we have already excluded. Thus the criterion for a number $x+yi$ to be contained in S is that the four numbers $\xi,\eta,p-\xi,p-\eta$ be positive.

We further observe that $\xi=\eta$ cannot hold for any such number. For it would follow that $p(x+y)=a(\xi+\eta)+b(\xi-\eta)=2a\xi$, which is absurd, since none of the factors $2,a,\xi$ is divisible by p. Similarly, the equation $p(x-y+a+b)=2a\xi+(a+b)(p-\xi-\eta)$ shows that $\xi+\eta=p$ cannot hold. Since the numbers $\xi-\eta, p-\xi-\eta$ must therefore be either positive or negative, we obtain from this a subdivision of the system S into four collections C, C', C'', C''', assigning the numbers as follows:

Into collection	Numbers for which
C	$\xi-\eta$ positive, $p-\xi-\eta$ positive
C'	$\xi-\eta$ positive, $p-\xi-\eta$ negative
C''	$\xi-\eta$ negative, $p-\xi-\eta$ negative
C'''	$\xi-\eta$ negative, $p-\xi-\eta$ positive

The criterion for a number of the collection C is thus, strictly speaking, sixfold: the six numbers $\xi,\eta,p-\xi,p-\eta,\xi-\eta,p-\xi-\eta$ must be positive. But conditions 2, 5 and 6 evidently already imply the others. The same applies to the collections C', C'', C''', so that the complete criteria are threefold, namely:

For collection	The following numbers must be positive
C	$\eta,\quad \xi-\eta,\quad p-\xi-\eta$
C'	$p-\xi,\quad \xi-\eta,\quad \xi+\eta-p$
C''	$p-\eta,\quad \eta-\xi,\quad \xi+\eta-p$
C'''	$\xi,\quad \eta-\xi,\quad p-\xi-\eta$

Even without our pointing it out, anyone will readily understand that, in the geometric representation of complex numbers (see Art. 39), the numbers of the system S lie inside the square whose sides join the points representing the numbers $0, a+bi, (1+i)(a+bi), i(a+bi)$, and that the subdivision of the system S corresponds to the division of the square by its diagonals. Here, however, we have preferred to use purely arithmetical reasoning, leaving illustration by a geometric representation to the experienced reader for the sake of brevity.

70.

If four complex numbers $r = x + yi, r' = x' + y'i, r'' = x'' + y''i, r''' = x''' + y'''i$ are related in such a way that $r' = m+ir, r'' = m+ir' = (1+i)m-r, r''' = m + ir'' = im - ir$, and the first, r, is supposed to belong to the collection C, then the others r', r'', r''' will belong respectively to the collections C', C'', C'''. For, putting $\xi = ax + by, \eta = ay - bx, \xi' = ax' + by', \eta' = ay' - bx', \xi'' = ax'' + by'', \eta'' = ay'' - bx'', \xi''' = ax''' + by''', \eta''' = ay''' - bx'''$, we find

$$\begin{aligned} \eta = p - \xi' = p - \eta'' = \xi''' \\ \xi - \eta = \xi' + \eta' - p = \eta'' - \xi'' = p - \xi''' - \eta''' \\ p - \xi - \eta = \xi' - \eta' = \xi'' + \eta'' - p = \eta''' - \xi''' \end{aligned}$$

from which the truth of the theorem follows at once by the criteria. Since $r = m + ir'''$ holds again, it is easy to see that, if r is supposed to belong to C', the numbers r', r'', r''' belong respectively to C'', C''', C; if it belongs to C'', they belong to C''', C, C'; and finally, if it belongs to C''', they belong to C, C', C''.

At the same time, we conclude that the individual collections C, C', C'', C''' contain equally many numbers, namely $\frac{1}{4}(p-1)$.

37.

THEOREM. *Let k denote an integer not divisible by m. If each number of the collection C is multiplied by k, and the simply least residues of the products modulo m are distributed among the collections C, C', C'', C''', with the counts of the residues belonging to these individual collections denoted respectively by c, c', c'', c''', then the character of the number k with respect to the modulus m will be $\equiv c' + 2c'' + 3c'''$ (mod 4).*

Proof. Let those c least residues belonging to C be $\alpha, \beta, \gamma, \delta$, etc.; next, let the c' residues belonging to C' be $m+i\alpha', m+i\beta', m+i\gamma', m+i\delta'$, etc.; further, let the c'' residues belonging to C'' be $(1+i)m-\alpha'', (1+i)m-\beta'', (1+i)m-\gamma'', (1+i)m-\delta''$, etc.; and finally, let the c''' residues belonging to C''' be $im-i\alpha''', im-i\beta''', im-i\gamma''', im-i\delta'''$, etc. Now consider four products, namely:

1) the product of all $\frac{1}{4}(p-1)$ numbers constituting the collection C:

2) the product of the products obtained by multiplying each of these numbers by k;

3) the product of the least residues of these products, namely of the numbers $\alpha, \beta, \gamma, \delta$, etc., $m+i\alpha', m+i\beta'$, etc., etc.

4) the product of all $c+c'+c''+c'''$ numbers $\alpha, \beta, \gamma, \delta$, etc., $\alpha', \beta', \gamma', \delta'$, etc., $\alpha'', \beta'', \gamma'', \delta''$, etc.

Denoting these four products in order by P, P', P'', P''', we evidently have

$$P' = k^{\frac{1}{4}(p-1)}P, \quad P' \equiv P'', \quad P'' \equiv P'''i^{c'+2c''+3c'''} \pmod{m}$$

and therefore

$$Pk^{\frac{1}{4}(p-1)} \equiv P'''i^{c'+2c''+3c'''} \pmod{m}$$

It is easy to see, however, that the numbers $\alpha', \beta', \gamma', \delta'$, etc., $\alpha'', \beta'', \gamma'', \delta''$, etc., $\alpha''', \beta''', \gamma''', \delta'''$, etc., all belong to the collection C, and are distinct both from one another and from the numbers $\alpha, \beta, \gamma, \delta$, etc., just as the latter themselves are distinct from one another. All these numbers taken together, disregarding their order, must therefore be exactly the same as all the numbers constituting the collection C. Hence we conclude that $P = P'''$, and consequently

$$Pk^{\frac{1}{4}(p-1)} \equiv Pi^{c'+2c''+3c'''} \pmod{m}$$

Finally, since none of the individual factors of the product P is divisible by m, we conclude that

$$k^{\frac{1}{4}(p-1)} \equiv i^{c'+2c''+3c'''} \pmod{m}$$

so that $c'+2c''+3c'''$ will be the character of the number k with respect to the modulus m. Q. E. D.

72.

To apply the general theorem of the preceding article to the number $1+i$, we must subdivide the collection C again into two smaller collections G and G'. We shall assign to the collection G those numbers $x+yi$ for which $ax+by=\xi$ is less than $\frac{1}{2}p$, and to the other collection G' those for which ξ is greater than $\frac{1}{2}p$. We shall denote the counts of the numbers contained in the collections G, G' respectively by g, g', so that $g+g'=\frac{1}{4}(p-1)$.

The complete criterion for numbers belonging to G is therefore that the three numbers $\eta, \xi-\eta, p-2\xi$ be positive. For the third condition for the collection C, requiring $p-\xi-\eta$ to be positive, is already implicitly contained in these conditions, since $p-\xi-\eta=(\xi-\eta)+(p-2\xi)$. Similarly, the complete criterion for numbers belonging to G' consists in positive values of the three numbers $\eta, p-\xi-\eta, 2\xi-p$.

It follows easily that the product of any number of the collection G by the number $1+i$ belongs to the collection C'''. For, putting

$$(x+yi)(1+i)=x'+y'i, \quad \text{and} \quad ax'+by'=\xi', \ ay'-bx'=\eta', \ \text{we find}$$
$$\xi'=\xi-\eta, \quad \eta'-\xi'=2\eta, \quad p-\xi'-\eta'=p-2\xi$$

that is, the criterion for the number $x+yi$ to belong to the collection G is identical with the criterion for the number $x'+y'i$ to belong to the collection C'''.

In precisely the same way, it is shown that the product of any number of the collection G' by $1+i$ belongs to the collection C''.

Thus, if in the preceding article we give k the value $1+i$, we have $c=0, c'=0, c''=g', c'''=g$, and the character of the number $1+i$ becomes $3g+2g'=\frac{1}{2}(p-1)+g$. Since the characters of the numbers $i, -1$ are $\frac{1}{4}(p-1), \frac{1}{2}(p-1)$, the characters of the numbers $-1+i, -1-i, 1-i$ will respectively be $\frac{3}{4}(p-1)+g, g, \frac{1}{4}(p-1)+g$. The whole matter now turns, therefore, on determining the number g.

73.

What we have set out in Arts. 69—72 is, strictly speaking, independent of the assumption that m is a primary number. From now on, however, we shall at least suppose that a is odd and b even, and further that a, b and $a-b$ are positive numbers. First of all, we must establish the bounds on the values of x in the collection G.

Putting $ay - bx = \eta, (a+b)x - (a-b)y = \zeta, p - 2ax - 2by = \theta$, the criterion for numbers $x+yi$ to belong to the collection G consists in the three conditions that η, ζ, θ be positive numbers. Since $px = (a-b)\eta + a\zeta, p(a - 2x) = a\theta + 2b\eta$, it is evident that x and $2a - x$ must be positive numbers, or that x must equal one of the numbers $1, 2, 3 \ldots \frac{1}{2}(a-1)$. Further, since $(a-b)\theta = 2b\zeta + p(a-b-2x)$, it is clear that, as long as x is less than $\frac{1}{2}(a-b)$, the second condition (requiring ζ to be positive) already implies the third (that θ must be positive). Conversely, whenever x is greater than $\frac{1}{2}(a-b)$, the second condition is already contained in the third. Thus, for the values x takes in $1, 2, 3 \ldots \frac{1}{2}(a-b-1)$, we need only ensure that η and ζ are positive, or that y is greater than $\frac{bx}{a}$ and less than $\frac{(a+b)x}{a-b}$. For such a given value of x, the total count of numbers $x+yi$ is therefore

$$\left[\frac{(a+b)x}{a-b}\right] - \left[\frac{bx}{a}\right]$$

if we use the brackets in the same sense in which we have already used them repeatedly elsewhere (compare *Theorematis arithm. dem. nova*, Art. 4, and *Theorematis fund. in doctr. de residuis quadr.* etc., *Algorithm. nov.*, Art. 3). For the values x takes in $\frac{1}{2}(a-b+1), \frac{1}{2}(a-b+3) \ldots . \frac{1}{2}(a-1)$, on the other hand, it suffices to give η and θ positive values, or to make y greater than $\frac{bx}{a}$ and less than $\frac{p-2ax}{2b}$, or $\frac{1}{2}b + \frac{aa-2ax}{2b}$. Thus, for such a given value of x, the total count of numbers $x+yi$ is

$$\left[\frac{1}{2}b + \frac{aa-2ax}{2b}\right] - \left[\frac{bx}{a}\right]$$

Hence we conclude that the count of numbers in the collection G is

$$g = \Sigma\left[\frac{(a+b)x}{a-b}\right] + \Sigma\left[\frac{1}{2}b + \frac{aa-2ax}{2b}\right] - \Sigma\left[\frac{bx}{a}\right]$$

where the first sum extends over all integer values of x from 1 to $\frac{1}{2}(a-b-1)$, the second from $\frac{1}{2}(a-b+1)$ to $\frac{1}{2}(a-1)$, and the third from 1 to $\frac{1}{2}(a-1)$.

If we use the symbol φ in the same sense as in the place cited (*Theorematis fund.* etc., *Algor. nov.*, Art. 3), namely so that

$$\varphi(t,u) = \left[\frac{u}{t}\right] + \left[\frac{2u}{t}\right] + \left[\frac{3u}{t}\right] \ldots + \left[\frac{t'u}{t}\right]$$

where t, u denote any positive numbers and t' the number $[\frac{1}{2}t]$, then the first term becomes $= \varphi(a-b, a+b)$, and the third $= -\varphi(a,b)$; the second, however, becomes

$$= \frac{1}{4}bb + \Sigma\left[\frac{aa-2ax}{2b}\right]$$

But, writing the terms in reverse order, we have

$$\begin{aligned}\Sigma\left[\frac{aa-2ax}{2b}\right] &= \left[\frac{a}{2b}\right] + \left[\frac{3a}{2b}\right] + \left[\frac{5a}{5b}\right] + \ldots + \left[\frac{(b-1)a}{2b}\right] \\ &= \varphi(2b,a) - \varphi(b,a)\end{aligned}$$

Our formula therefore takes the following form:

$$g = \varphi(a-b, a+b) + \varphi(2b,a) - \varphi(a,b) - \varphi(b,a) + \frac{1}{4}bb$$

Let us first consider the term $\varphi(a-b, a+b)$, which immediately changes into $\varphi(a-b, 2b) + 1 + 2 + 3 +$ etc. $+ \frac{1}{2}(a-b-1)$, or into

$$\varphi(a-b,2b) + \frac{1}{8}((a-b)^2 - 1)$$

Next, since the general theorem gives $\varphi(t,u) + \varphi(u,t) = [\frac{1}{2}t] \,.\, [\frac{1}{2}u]$ when t, u are relatively prime positive integers, we have

$$\varphi(a-b,2b) = \frac{1}{2}b(a-b-1) - \varphi(2b, a-b)$$

and consequently

$$\varphi(a-b,a+b) = \frac{1}{8}(aa+2ab-3bb-4b-1) - \varphi(2b,a-b)$$

Let us arrange the parts of $\varphi(2b, a-b)$ as follows:

$$\begin{aligned}&\left[\frac{a-b}{2b}\right] + \left[\frac{3(a-b)}{2b}\right] + \left[\frac{5(a-b)}{2b}\right] + \text{ etc. } + \left[\frac{(b-1)(a-b)}{2b}\right] \\ &+ \left[\frac{a-b}{b}\right] + \left[\frac{2(a-b)}{b}\right] + \left[\frac{3(a-b)}{b}\right] + \text{ etc. } + \left[\frac{\frac{1}{2}b(a-b)}{b}\right]\end{aligned}$$

The second series evidently becomes

$$= \varphi(b, a-b) = \varphi(b,a) - 1 - 2 - 3 - \text{ etc. } - \frac{1}{2}b$$
$$= \varphi(b,a) - \frac{1}{8}(bb + 2b)$$

We write the first series, with the order of its terms reversed, as follows:

$$\left[\frac{1}{2}(a+1-b) - \frac{a}{2b}\right] + \left[\frac{1}{2}(a+3-b) - \frac{3a}{2b}\right]$$
$$+ \left[\frac{1}{2}(a+5-b) - \frac{5a}{2b}\right] + \text{etc.} + \left[\frac{1}{2}(a-1) - \frac{(b-1)a}{2b}\right]$$

This expression. Since, with t denoting an integer and u a fraction, we generally have $[t-u] = t - 1 - [u]$, it changes into the following:

$$\frac{1}{8}b(2a-4-b) - \left[\frac{a}{2b}\right] - \left[\frac{3a}{2b}\right] - \left[\frac{5a}{2b}\right] - \text{ etc. } - \left[\frac{(b-1)a}{2b}\right]$$
$$= \frac{1}{8}b(2a-4-b) - \varphi(2b,a) + \varphi(b,a)$$

Hence

$$\varphi(2b, a-b) = 2\varphi(b,a) - \varphi(2b,a) + \frac{1}{4}b(a-3-b)$$

and therefore

$$\varphi(a-b, a+b) = \varphi(2b,a) - 2\varphi(b,a) + \frac{1}{8}(aa - bb + 2b - 1)$$

Substituting this value in the formula for g given above, and also substituting $\varphi(a,b) + \varphi(b,a) = \frac{1}{4}b(a-1)$, we obtain

$$g = 2\varphi(2b,a) - 2\varphi(b,a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1)$$

74.

Precisely similar reasoning deals with the case where a, b remain positive but $a-b$ is negative, or $b-a$ positive. The equations $p(a-2x) = 2b\eta + a\theta$, $p(b-a+2x) = 2b\zeta + (b-a)\theta$ show that $\frac{1}{2}a - x$ and $x + \frac{1}{2}(b-a)$ must be positive, and hence that x must equal one of the numbers $-\frac{1}{2}(b-a-1), -\frac{1}{2}(b-a-3), -\frac{1}{2}(b-a-5) \ldots . + \frac{1}{2}(a-1)$. Further, the equation $px + (b-a)\eta = a\zeta$ shows that, for negative values of x, the condition requiring η to be positive is already contained in the condition requiring ζ to be positive, whereas the reverse holds whenever x is given a positive value. Thus the values of y, for a fixed negative value of x, must lie between $\frac{(a+b)x}{a-b}$ and $\frac{p-2ax}{2b}$, whereas for a positive value of x they must lie between $\frac{bx}{a}$ and $\frac{p-2ax}{2b}$. For $x = 0$, these bounds are evidently 0 and $\frac{p-2ax}{2b}$, with the value $y = 0$ itself excluded. Hence

$$g = -\Sigma\left[\frac{(a+b)x}{a-b}\right] + \Sigma\left[\frac{1}{2}b + \frac{aa-2ax}{2b}\right] - \Sigma\left[\frac{bx}{a}\right]$$

where the first sum extends over all negative values of. x from -1 to $-\frac{1}{2}(b-a-1)$; the second over all values of x from $-\frac{1}{2}(b-a-1)$ to $\frac{1}{2}(a-1)$; and the third over all positive values of x from $+1$ to $\frac{1}{2}(a-1)$. In this way the first sum gives $-\varphi(b-a, b+a)$, the second, as in the preceding article, $\frac{1}{4}bb + \varphi(2b,a) - \varphi(b,a)$, and finally the third $-\varphi(a,b)$; thus we have

$$g = -\varphi(b-a, b+a) + \varphi(2b,a) - \varphi(b,a) - \varphi(a,b) + \frac{1}{4}bb$$

Now, in the same way as in the preceding article, we obtain

$$\begin{aligned}\varphi(b-a, b+a) &= \varphi(b-a, 2b) - \frac{1}{8}((b-a)^2 - 1) \\ &= \frac{1}{8}(3bb - 2ab - aa - 4b + 1) - \varphi(2b, b-a)\end{aligned}$$

as well as

$$\varphi(2b, b-a) = \varphi(2b,a) - 2\varphi(b,a) + \frac{1}{4}b(b-1-a)$$

and consequently

$$\varphi(b-a, b+a) = 2\varphi(b,a) - \varphi(2b,a) + \frac{1}{8}(bb - aa - 2b + 1)$$

and finally

$$g = 2\varphi(2b,a) - 2\varphi(b,a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1)$$

It is therefore proved that the same formula for g holds whether $a-b$ is positive or negative, provided that a, b are positive.

75.

To obtain a further reduction, we put

$$L = \left[\frac{a}{2b}\right] + \left[\frac{2a}{2b}\right] + \left[\frac{3a}{2b}\right] + \text{etc.} + \left[\frac{\frac{1}{2}ba}{2b}\right]$$

$$M = \left[\frac{(\frac{1}{2}b+1)a}{2b}\right] + \left[\frac{(\frac{1}{2}b+2)a}{2b}\right] + \left[\frac{(\frac{1}{2}b+3)a}{2b}\right] + \text{etc.} + \left[\frac{ba}{2b}\right]$$

$$N = \left[\frac{a+b}{2b}\right] + \left[\frac{2a+b}{2b}\right] + \left[\frac{3a+b}{2b}\right] + \text{etc.} + \left[\frac{\frac{1}{2}ba+b}{2b}\right]$$

It is easy to see that $[u]+[u+\frac{1}{2}] = [2u]$ holds in general, whatever real quantity u denotes. Thus $L + N = \varphi(b, a)$, and since evidently $L + M = \varphi(2b, a)$, we have

$$\varphi(2b, a) - \varphi(b, a) = M - N$$

Further, it is evident that the sum of the first term of the series N and the next-to-last term of the series M, namely $\left[\frac{a+b}{2b}\right] + \left[\frac{(b-1)a}{2b}\right]$, is $= \frac{1}{2}(a-1)$, and that the same sum is formed by the second term of the series N and the third-to-last term of the series M, and so on. Since the last term of the series M is also $= \frac{1}{2}(a-1)$, while the last term of the series N is $= \left[\frac{a+2}{4}\right] = \frac{1}{4}(a \mp 1)$, with the upper or lower sign applying according as a is of the form $4n+1$ or $4n-1$, we have

$$M + N = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1)$$

and therefore

$$\varphi(2b, a) - \varphi(b, a) = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1) - 2N$$

Thus the formula for g found in Arts. 73 and 74 becomes

$$g = \frac{1}{8}((a+b)^2 - 1) + 2n - 4N$$

putting $a \mp 1 = 4n$, where n will be an integer. Since this gives $1 = 16nn - 8an + aa$, this formula can also be written as follows:

$$g = \frac{1}{8}(-aa + 2ab + bb + 1) + 4(\frac{1}{2}(a+1)n - nn - N)$$

Since g is the character of the number $-1-i$ for the modulus $a+bi$, this character is therefore $\equiv \frac{1}{8}(-aa + 2ab + bb + 1) \pmod 4$. This is precisely the theorem found above (Art. 64) by examining examples, and the theorems concerning the characters of the numbers $1+i$, $1-i$, $-1+i$ follow from it at once. Thus these four theorems have now been rigorously proved for the case where a and b are positive.

76.

If a remains positive and b is negative, put $b = -b'$, so that b' is positive. It is now proved that, for the modulus $a + b'i$, the character of the number $-1 - i$ is $\equiv \frac{1}{8}(-aa + 2ab' + b'b' + 1) \pmod 4$. Hence the character of the number $-1 + i$ for the modulus $a - b'i$, by the theorem stated in Art. 62, is $\equiv \frac{1}{8}(aa - 2ab' - b'b' - 1)$, that is, the character of the number $-1 + i$ for the modulus $a + bi$ becomes $\equiv \frac{1}{8}(aa + 2ab - bb - 1)$. This is precisely the theorem stated in Art. 64, from which the other three concerning the characters of the numbers $1 + i$, $1 - i$, $-1 - i$ follow at once. Thus these theorems are also proved for the case where b is negative, that is, for all cases where a is positive.

Finally, if a is negative, put $a = -a'$, $b = -b'$. By what has already been proved, the character of the number $1 + i$ with respect to the modulus $a' + b'i$ is $\equiv \frac{1}{8}(-a'a' + 2a'b' - 3b'b' + 1) \pmod 4$, and it makes no difference whether we take the number $a' + b'i$ or its opposite $-a' - b'i$ as the modulus. Thus the character of the number $1 + i$ with respect to the modulus $a + bi$ is evidently $\equiv \frac{1}{8}(-aa + 2ab - 3bb + 1)$, and the same applies to the characters of the numbers $1 - i$, $-1 + i$, $-1 - i$.

We therefore conclude that the proof of the theorems concerning the characters of the numbers $1 + i$, $1 - i$, $-1 + i$, $-1 - i$ (Arts. 63. 64) is no longer subject to any restriction.

REVIEWS

OF HIS OWN

WRITINGS.

Göttingische gelehrte Anzeigen. 1808 May 12.

A paper submitted by Professor Gauss to the Royal Society of Sciences on January 15 of this year,

Theorematis arithmetici demonstratio nova,

an account of whose contents we still owe here, concerns the famous fundamental theorem in the theory of quadratic residues, which plays so important a role both throughout *higher arithmetic* and in the neighbouring parts of analysis. As is well known, an integer a is called a *quadratic residue* of the integer b if there are numbers of the form $xx - a$ divisible by b, while in the opposite case a is called a *quadratic nonresidue* of b. The number a may be positive or negative, whereas b is always regarded as positive. Higher arithmetic teaches that all prime numbers b for which a given number a is a quadratic residue are included in certain linear forms, while other linear forms contain all the primes of which a is a nonresidue. Thus, for example, -1 is a quadratic residue of all primes of the form $4n + 1$ and a quadratic nonresidue of all primes of the form $4n + 3$; further, $+2$ is a quadratic residue of all primes of the forms $8n + 1$, $8n + 7$, but a quadratic nonresidue of all primes of the forms $8n + 3$, $8n + 5$. There are infinitely many similar special theorems, but all can be derived by combining the two just mentioned with the following general theorem: Two distinct positive (odd) primes, p, q, always bear the *same* relation to each other (that is, one is a quadratic residue or nonresidue of the other according as the other is a residue or nonresidue of the first) if either both have the form $4n + 1$, or at least one does. Their mutual relation is opposite, however (that is, one is a nonresidue of the other if the latter is a residue of the former, and conversely), whenever *both* have the form $4n + 3$. This is the fundamental theorem mentioned above, which can be expressed in more than one form. The form chosen here is the one in which it is newly proved in Professor Gauss's paper.

The most beautiful theorems of higher arithmetic, especially those discussed here, have the peculiarity that they are easily discovered by examining examples, whereas their proofs lie extremely well hidden and can be found only through very deep investigations. This is precisely what gives higher arithmetic the magical appeal that has made it the favourite field of the foremost mathematicians, quite apart from its inexhaustible wealth, in which it so far surpasses every other part of pure mathematics. The two special theorems mentioned above were already known to Fermat, who claimed also to possess their proofs. Whether he was mistaken in this we cannot decide, since he never published anything about them; but we can certainly regard

it as possible, since there are several examples of self-deception among other great mathematicians, notably EULER, LEGENDRE, and also FERMAT himself. EULER gave the first proof of the first of these theorems. But this great mathematician did not succeed in proving the other, despite his devoted efforts over many years; it was left to LAGRANGE to fill that gap. Both mathematicians also proved various other special theorems, but a greater number, which they found by examining examples, always eluded their attempts at proof. It is, however, a remarkable twist of chance that neither mathematician discovered in this way the general fundamental theorem, which admits of so simple a statement. It was first presented, though in a somewhat different form, by LEGENDRE in the *Histoire de l'Académie des Sciences de Paris* 1785. Both there and later in his *Essai d'une théorie des nombres*, this excellent analyst sought to base the proof on very ingenious investigations, which nevertheless failed to reach the desired goal; and, unless we are mistaken, that goal could not have been reached by this route either.

The author of the paper to which this review is devoted entered the field of higher arithmetic at a time when all earlier work by other mathematicians in this subject was entirely unknown to him. It is chiefly to this circumstance that we must attribute his having followed a wholly distinctive course throughout. He found that fundamental theorem very early by examining examples, but only a whole year later, after many difficulties and unsuccessful attempts, did he succeed in finding the first completely rigorous proof, developed in the fourth section of his *Disquisitiones Arithmeticae*. This proof, however, rests on very laborious and lengthy arguments. He subsequently found three other proofs, which are free of that inconvenience but instead require other very deep investigations, wholly different in content. One of these proofs is also given in the cited work, Art. 262; the other two will be made known in due course. The wish therefore remained that it might be possible to discover a shorter proof, independent of investigations of an unrelated kind. The author thus hopes that friends of higher arithmetic will take pleasure in a fifth proof, presented in the present paper in fewer than five pages, which seems to leave nothing to be desired in any respect. Given the concentrated brevity of this proof, we can admittedly give only an imperfect idea of its course here. Anything more would be all the more unnecessary, since Volume XVI of the *Commentationes*, in which it has already been printed, will appear shortly.

The proof rests on the following new theorem: Let p denote a (positive odd) prime and k any integer not divisible by p. Suppose further that, among the remainders obtained by dividing the $\frac{1}{2}(p-1)$ products k, $2k$, $3k \ldots . \frac{1}{2}(p-1)k$ by p, there are in all μ remainders greater than $\frac{1}{2}p$ (and hence $\frac{1}{2}(p-1)-\mu$ less than $\frac{1}{2}p$). Then k will be a quadratic residue of p if μ is even, but a quadratic nonresidue if μ is odd. The number μ, which depends only on k and p, may be represented by the symbol (k, p). A sequence of deductions that cannot be abridged now shows that, if k and p are two relatively prime odd numbers, $(k, p) + (p, k) + \frac{1}{4}(k-1)(p-1)$ is always an *even* number. It follows that, whenever k and p both have the form $4n+3$, one of the numbers (k, p), (p, k) must be even and the other odd; in all other cases, that is, whenever both numbers k and p, or at least one of them, have the form $4n+1$, either $.(k, p)$, (p, k) must both be even or both must be odd. Combined with the theorem above, this immediately gives the truth of the fundamental theorem.—The same route by which these results are obtained also gives, in the paper, a new proof of the two special theorems mentioned above. It is easy to show that $(-1, p) = \frac{1}{2}(p-1)$, hence even or odd according as p has the form $4n+1$ or $4n+3$; similarly, $(2, p) = \frac{1}{4}(p-1)$ if p has the form $4n+1$, and $(2, p) = \frac{1}{4}(p+1)$ if p has the form $4n+3$. Thus $(2, p)$ is even whenever p has the form $8n+1$ or $8n+7$, but odd whenever p has the form $8n+3$ or $8n+5$.

Göttingische Gelehrte Anzeigen. 1808 September 19.

A lecture submitted by Professor Gauss to the Royal Society of Sciences,

Summatio quarumdam serierum singularium,

aims to develop more fully and in greater generality a remarkable investigation concerning the division of the circle, whose foundation had already been laid in the *Disquisitionibus Arithmeticis*, to provide it with complete proofs, and to show its unexpected connection with other important truths. Let n denote a prime, k any integer not divisible by n, and ω the arc $\frac{1}{n}360^\circ$. Among the numbers $1, 2, 3, 4, \ldots . n-1$, let the distinct quadratic residues of n be represented by a, a', a'', and so on, while the numbers remaining after these have been excluded, or the quadratic nonresidues of n, are represented by b, b', b'', and so on. It is proved in the cited work, Art. 356, that, when n has the form $4n+1$,

$$\left.\begin{array}{l}\cos ak\omega + \cos a'k\omega + \cos a''k\omega + \text{etc.} \\ -\cos bk\omega - \cos b'k\omega - \cos b''k\omega - \text{etc.}\end{array}\right\} = \pm\sqrt{n}.$$

and

$$\left.\begin{array}{l}\sin ak\omega + \sin a'k\omega + \sin a''k\omega + \text{etc.} \\ -\sin bk\omega - \sin b'k\omega - \sin b''k\omega - \text{etc.}\end{array}\right\} = 0$$

whereas, when n has the form $4m+3$, the sum of the first series becomes $= 0$ and that of the second $= \pm\sqrt{n}$. The sign to be placed before the radical depends on the value of the number k, or rather on its relation to n, and is easily determined for *all* values of k at a given value of n once it is determined for *one*. For it can be shown that *the same* sign applies to all values of k that are quadratic residues of n, and the opposite sign to all those that are quadratic nonresidues of n. Since the investigation had already been carried this far in the cited work, and only the determination of the sign for any one value of k remained, one would have thought that, with the main matter settled, this further determination could easily be supplied. This seemed all the more likely because examining examples immediately gives an extremely simple result: for $k = 1$, or for all values that are quadratic residues of n, the radical in the formulas above must always be taken *positively*. Yet in seeking a proof of this observation we encounter wholly unexpected difficulties, and the procedure that led so satisfactorily to the absolute value of those series proves entirely inadequate when the complete determination of the signs is required. The *metaphysical* reason for this phenomenon (to use the expression customary among French mathematicians) must be sought in the

fact that, in dividing the circle, analysis makes no distinction between the arcs $\omega, 2\omega, 3\omega \ldots (n-1)\omega$, but treats them all alike. As this gives the investigation a new interest, Professor GAUSS found in it, as it were, an invitation to leave nothing untried in overcoming the difficulty. Only after many and varied unsuccessful attempts did he succeed, by a route remarkable in itself. He starts from the summation of certain series whose terms have the following form:

$$\frac{(1-x^m)(1-x^{m-1})(1-x^{m-2})\ldots(1-x^{m-\mu+1})}{(1-x)\ (1-xx)\ (1-x^3)\ldots.(1-x^\mu)}$$

If, for brevity, we denote such a function by (m, μ), which, as the paper shows, is always a *polynomial* in x, then the series

$$1-(m,1)+(m,2)-(m,3)+\text{etc.}$$

$$1+x^{\frac{1}{2}}(m,1)+x(m,2)+x^{\frac{3}{2}}(m,3)+\text{etc.}$$

terminate after the term in position $m+1$, provided that m denotes a positive integer. For even values of m the sum of the first series becomes

$$=(1-x)(1-x^3)(1-x^5)\ldots(1-x^{m-1})$$

and $=0$ for odd values of m, whereas the sum of the second series always becomes

$$=(1+x^{\frac{1}{2}})(1+x)(1+x^{\frac{3}{2}})\ldots(1+x^{\frac{1}{2}m})$$

For fractional and negative values of m, too, the summation of these series leads to interesting results, though they are not needed for the present purpose. We content ourselves with giving just one of them here. The infinite series

$$1+x+x^3+x^6+x^{10}+\text{etc.}$$

whose exponents are the triangular numbers is the product of the factors

$$\frac{1-xx}{1-x}\times\frac{1-x^4}{1-x^3}\times\frac{1-x^6}{1-x^5}\times\frac{1-x^8}{1-x^7}\ \text{etc.}$$

or, if preferred, of

$$(1+x)^2(1+xx)^2(1+x^3)^2(1+x^4)^2\ \text{etc.}$$

multiplied by

$$(1-x)(1-xx)1-x^3)(1-x^4)\ \text{etc.}$$

An account of how these summations are applied to the main subject would take us too far here. We may refer readers to the paper itself all the more

readily since it will soon appear in print. The summations quoted above are only a special application of the summation of the following series:

$$1 + \cos k\omega + \cos 4k\omega + \cos 9k\omega + \text{etc.} + \cos(n-1)^2 k\omega = T$$

$$\sin k\omega + \sin 4k\omega + \sin 9k\omega + \text{etc.} + \sin(n-1)^2 k\omega = U$$

which the paper gives for all values of k, and without the restriction that n be prime. It is shown that

$$T = \pm\sqrt{n}, \quad T = \pm\sqrt{n}, \quad T = 0, \quad T = 0$$

and

$$U = \pm\sqrt{n}, \quad U = 0, \quad U = 0, \quad U = \pm\sqrt{n}$$

according as n has the forms $4m, 4m+1, 4m+2, 4m+3$, respectively. Here again the sign of the radical depends on k. Its determination requires many individual cases to be distinguished, and is developed and proved by two different routes in such a way that nothing will remain to be desired. Comparing these two routes also leads to the following very remarkable theorem: Let n be the product of any number of distinct odd primes a, b, c, d, and so on, of which altogether μ have the form $4m+3$. Suppose further that among those factors there are altogether ν for each of which the product of the others (respectively $\frac{n}{a}, \frac{n}{b}, \frac{n}{c}, \frac{n}{d}$, and so on) is a quadratic nonresidue. Then ν will be even whenever μ has the form $4m$ or $4m+1$, but odd whenever μ has the form $4m+2$ or $4m+3$. The known fundamental theorem concerning quadratic residues is only a special case of this theorem; conversely, the theorem just stated is easily derived from that fundamental theorem. Thus these investigations also provide a fourth proof of this important theorem, which the author first proved by two entirely different routes in the *Disquisitionibus Arithmeticis*, and recently by a third, equally different route in a separate paper.

Göttingische gelehrte Anzeigen. 1817 March 10.

On 10 February Court Councillor Gauss submitted a lecture to the Royal Society, entitled:

Theorematis fundamentalis in doctrina de residuis quadraticis demonstrationes et ampliationes novae.

A distinctive feature of higher arithmetic is that so many of its most beautiful theorems can be discovered with the greatest ease by examining examples, while their proofs are far from readily apparent. Often they are found only after many unsuccessful attempts, through deeply probing investigations and fortunate combinations. This remarkable phenomenon arises from the often wonderful interconnection between the different branches of this part of mathematics. For the same reason, theorems for which a proof was initially sought in vain for years can often later be proved in several entirely different ways. Once a new theorem has been discovered through examples, finding *some* proof must of course be the first requirement; but after this has succeeded, in higher arithmetic we must not always regard the investigation as finished and the search for other proofs as a needless luxury. For one thing, we do not usually find the most beautiful and simplest proofs first; for another, insight into the wonderful interconnection of the truth of higher arithmetic is precisely one of the chief attractions of this study, and not infrequently leads in turn to the discovery of new truths. For these reasons, finding new proofs of truths already known must here often be considered at least as important as discovering the truth itself. These reflections are familiar to those who know higher arithmetic: much of Euler's contribution to it is known to consist in finding proofs of theorems which Fermat had already discovered, apparently through examples.

The theory of quadratic residues provides a clear illustration of these remarks. It rests chiefly on the so-called fundamental theorem, which states that the mutual relations of two (odd positive) primes to one another (insofar as one is a quadratic residue or nonresidue of the other) are the same whenever one or both primes have the form $4k + 1$, but opposite whenever both primes have the form $4k + 3$. For readers less familiar with higher arithmetic, we recall that an integer is called a quadratic residue of another if adding a multiple of the latter to the former can give a square; it is called a nonresidue if this is impossible. We do not intend to repeat the whole history of this beautiful theorem, so easily discovered through examples. We merely observe that the author of the present paper, after investigations initially pursued in vain for quite some time, has already given, one after another, four

entirely different proofs. Two are contained in the *Disquisitionibus Arithmeticis*; the third is the subject of a separate paper in the sixteenth volume of the Commentationes; and the fourth is interwoven with the paper *summatio quarumdam serierum singularium* in the first volume of the *Commentationes recentiores*. For these two papers readers may consult our reviews of 1808, May 12 and September 19, which also give fuller historical references. That the author did not stop at these four proofs, although each leaves nothing to be desired as regards rigour, needs no justification among the friends of higher arithmetic. Still, he would probably not have worked so eagerly to add further proofs if a particular circumstance had not prompted him to do so, and this must be mentioned here. In the year 1805 he began to study the theories of cubic and biquadratic residues, which are much richer and more interesting still than the theory of quadratic residues. These investigations revealed the same phenomena as the latter theory, only, as it were, on a larger scale. As soon as the right path was taken, examining examples immediately yielded a number of extremely simple theorems which exhaust those theories, bear a surprising resemblance to the theorems for quadratic residues, and in particular also provide a counterpart to the fundamental theorem. But the difficulty of finding entirely satisfactory proofs of these theorems proved greater still, and only after many attempts, continued over a considerable number of years, did the author finally reach his goal. The close analogy between the theorems for quadratic and higher residues suggested that there must also be analogous proofs for the two. Yet the methods of proof first found for quadratic residues could not be applied to the higher residues at all; this was precisely what led him to keep seeking further new proofs for the former. The author therefore wishes the present paper, which opens up some further resources for the theory of quadratic residues, to be regarded as a precursor to the theory of cubic and biquadratic residues. He intends to publish that theory in the future; its subjects are among the most difficult in higher arithmetic.

The present paper consists of three independent parts. It contains the fifth and sixth proofs of the fundamental theorem, and a new method, connected with the third proof, · for deciding whether a given integer is a quadratic residue or nonresidue of a given prime. Of the first four proofs, the third was unquestionably the one that combined the greatest simplicity with independence from investigations of a different kind; LEGENDRE therefore also included it in the new edition of his *Essai d'une théorie des nombres*. The *fifth* proof seems at least to equal the third in both respects. The two proofs are related insofar as they start from the same theorem, but in their subsequent development they are completely different from one another.

The theorem is as follows. Let m be a (positive odd) prime and M an integer not divisible by m. If, among the remainders obtained by dividing the products

$$M,\ 2M,\ 3M,\ 4M \ldots\ldots \tfrac{1}{2}(m-1)M$$

by m, the number of those greater than $\frac{1}{2}m$ is denoted by n, then M is a quadratic residue or nonresidue of m according as n is even or odd. To obtain a proof of the fundamental theorem, it is now assumed that M, too, is an odd positive prime, and that N has the same meaning with respect to M and m as n has with respect to m and M. Thus whether N is even or odd determines whether m is a quadratic residue or nonresidue of M. By a very short chain of arguments the author shows that the number of all positive integers which are less than $\frac{1}{2}mM$ and at the same time on division by m leave a remainder less than $\frac{1}{2}m$, and on division by M a remainder less than $\frac{1}{2}M$,

$$= \tfrac{1}{8}(m-1)(M-1) + \tfrac{1}{2}n + \tfrac{1}{2}N$$

and consequently that

$$\tfrac{1}{4}(m-1)(M-1) + n + N$$

is always an even number. Thus, whenever at least one of the numbers m, M has the form $4k+1$, so that $\frac{1}{4}(m-1)(M-1)$ is even, $n+N$ will also be even; hence n and N are either both even or both odd. If, on the other hand, both m and M have the form $4k+3$, then $n+N$ must be odd, so one of the numbers n, N is even and the other odd. Together with the theorem above, this gives the fundamental theorem at once.

The *sixth* proof is as short and elegant as the fifth, but rests on somewhat more intricate combinations. The limited space of these pages allows us only to touch on its main point, omitting the details. Let the following symbols denote:

p, q two (distinct positive odd) primes,

α a so-called *primitive root* for the modulus p, that is, an integer not divisible by p (here taken positive), such that no power lower than α^{p-1} is congruent to one modulo p

x an indeterminate quantity

ξ the function

$$x - x^{\alpha} + x^{\zeta} - x^{\eta} + x^{\theta} - \text{etc.} - x^{\lambda}$$

where, for convenience in printing, $\zeta, \eta, \theta \ldots \lambda$ stand for the numbers $\alpha\alpha, \alpha^3, \alpha^4 \ldots \alpha^{p-2}$;

ε one, taken positively if p has the form $4k+1$, negatively if p has the form $4k+3$;

δ one, taken positively if at least one of the numbers p, q has the form $4k+1$, negatively if both have the form $4k+3$;

γ one, taken positively if q is a quadratic residue of p, negatively if q is a quadratic nonresidue of p;

β one, taken positively if p is a quadratic residue of q, negatively if p is a quadratic nonresidue of q.

With these preparations it follows readily from Article 51 of the *Disquisitiones Arithmeticae* that the function

$$\xi^q - x^q + x^{q\alpha} - x^{q\zeta} + x^{q\eta} - x^{q\theta} + \text{etc.} + x^{q\lambda}$$

has, when expanded, coefficients all divisible by q. Hence, if this function is set $= qX$, then X is a *polynomial* whose coefficients, too, are integers. By arguments which it would take too much space to discuss here, the paper proves that dividing the function $qX\xi$ by $x^{p-1}+x^{p-2}+x^{p-3}+x^{p-4}+\text{etc.}+x+1$ leaves the remainder

$$\varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma)$$

and therefore that dividing the function $X\xi$ by the same divisor leaves the remainder

$$\frac{\varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma)}{q}$$

This quantity must therefore be an integer. Since $\delta\delta = 1$, it follows readily that

$$p^{\frac{1}{2}(q-1)} - \gamma\delta$$

must be divisible by q. A known theorem also gives that $p^{\frac{1}{2}(q-1)} - \beta$ is divisible by q, so necessarily $\beta = \gamma\delta$, from which the fundamental theorem again follows at once.

The fundamental theorem, together with some known theorems, can indeed provide a fairly short way of deciding whether a given positive integer is a quadratic residue or nonresidue of a given prime, as the paper shows in detail. But further reflection on the third proof of the fundamental theorem led the author to a much more flexible solution. This forms the paper's third part; here we give only its final rule, omitting the reasoning behind it for brevity. To decide whether the positive integer b, not divisible by the prime a, is a quadratic residue or nonresidue of that prime, form, in just the same way as when finding the greatest common divisor of a and b, the equations

$$\begin{aligned} a &= \beta b + c \\ b &= \gamma c + d \\ c &= \delta d + e \\ d &= \varepsilon e + f \text{ etc.} \end{aligned}$$

until the sequence of numbers a, b, c, d, e, f and so on reaches .the unit. Let. the numbers $\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c, \frac{1}{2}d$ and so on, with their fractional part $\frac{1}{2}$ discarded whenever any of the numbers a, b, c, d and so on is odd, be denoted by a', b', c', d' and so on. Let μ be the number of occurrences of two odd numbers immediately following one another in the sequence a', b', c', d' and so on. Finally, let ν be the number of odd numbers in the sequence $\beta, \gamma, \delta, \varepsilon$ and so on to which, in the sequence b', c', d', e' and so on, a number of the form $4k + 1$ or $4k + 2$ respectively corresponds. Then b is a quadratic residue or nonresidue of a according as $\mu + \nu$ is even or odd, except in the single case where b is even and a has the form $8k + 3$ or $8k + 5$. In that case the rule is reversed: an even $\mu + \nu$ indicates that b is a quadratic nonresidue of a, while an odd $\mu + \nu$ indicates that b is a quadratic residue of a.

.Göttingische gelehrta Anzeigen. 1825 April 11.

On 5 April Court Councillor Gauss presented a lecture to the Royal Society, entitled:

Theoria Residuorum Biquadraticorum, Commentatio prima.

The theory of quadratic residues is known to form one of the most interesting parts of higher arithmetic. After repeated investigations it can now be regarded as complete and closed. Accounts of its history can be found in these pages for 1808 May 12 and September 19, and 1817 March 10. The last of these also gave some preliminary information about the investigations which the author of the present paper had been pursuing since the year 1805 into the related theory of cubic and biquadratic residues, equally fruitful and interesting but far more difficult. Although he already possessed the essential elements of this theory at that time, other work has so far prevented him from publishing anything about it. Only now has he been able to begin writing up part of these investigations. He has begun with the theory of biquadratic residues, which is more closely related to the theory of quadratic residues than is that of cubic residues. The present paper, however, is by no means intended to exhaust this exceptionally rich subject. The development of the *general* theory, which requires a quite distinctive extension of the domain of higher arithmetic, is reserved for a future continuation. This first paper contains those investigations which could be set out completely without such an extension. Only some of the results can be selected for this review.

An integer a is called a biquadratic residue of the integer p if there are numbers of the form $x^4 - a$ divisible by p; a biquadratic nonresidue if no numbers of that form can be divisible by p. Clearly all biquadratic residues of p are also quadratic residues of the same number, and hence all quadratic nonresidues are also biquadratic nonresidues. But not all quadratic residues are also biquadratic residues. It suffices to restrict the investigations to the case where p is a prime of the form $4n + 1$ and a is not divisible by p, since all other cases can easily be reduced to this one.

The investigations of this subject divide into two parts, according as p or a is regarded as given. The first is much less difficult than the second and, compared with it, is to be regarded as entirely elementary. The paper contains a complete account of everything essential to be said about it.

From the second part, by contrast, only a few special cases are treated here. These could be dealt with without too much preparation and can serve as preliminaries to the general theory to be given in the future. They are the cases where $a = -1$ and $a = \pm 2$ are set.

The first presents no difficulty at all: the *Disquisitiones Arithmeticae* had already shown that -1 is a biquadratic residue of p whenever p has the form $8n+1$, but only a quadratic residue and a biquadratic nonresidue of p when p has the form $8n+5$. The case $a = \pm 2$ is quite different. It has long been known that $+2$ and -2 are quadratic, and therefore also biquadratic, nonresidues of p when p has the form $8n+5$, and at least quadratic residues when p has the form $8n+1$. It is also known that for this form of p, $+2$ and -2 are either both biquadratic residues or both biquadratic nonresidues: but deciding which of these two cases occurs is an investigation of a much higher order. The paper develops two different criteria for this purpose.

The first criterion is connected with the decomposition of the number p into one square plus twice another, which, as is known (since, as already remarked, p is assumed to be prime), is always possible and possible in only one way. If we set $p = gg + 2hh$, then ± 2 is a biquadratic residue of p when g has the form $8n+1$ or $8n+7$, but a biquadratic nonresidue when g has the form $8n+3$ or $8n+5$.

The second criterion is connected with the decomposition of the number p into two squares, which is also known to be always possible and possible in only one way. If we set $p = ee + ff$ and suppose that ee denotes the odd square and ff the even square, then the assumed form $p = 8n+1$ already implies that $\frac{1}{2}f$, too, is even. Thus f has either the form $8m$ or the form $8m+4$: in the first case ± 2 is a biquadratic residue, in the other a biquadratic nonresidue of p.

We only touch here on an observation for which higher arithmetic so often provides an occasion: it is not so much the beauty and simplicity of the theorems themselves as the difficulty of proving them that makes them especially remarkable. Once there is reason to suspect a connection between the behaviour of the number ± 2 and the two decompositions of the number p just described, it is extremely easy to discover that connection by examining examples. But even for the first criterion its proof is not quite easy; for the second it lies much deeper, closely bound up with other subtle auxiliary investigations, which in turn lead to a remarkable extension of the theory of circle division. It is chiefly this wonderful interconnection of truths. that, as has often been observed, gives higher arithmetic such a distinctive appeal. Naturally these proofs themselves cannot be excerpted here; readers must consult the paper itself. But a couple of other new arithmetical theorems, also closely connected with the proof of the second criterion, deserve to be singled out here because of their simplicity.

Let p be a prime of the form $4k+1$, and put this number $= ee + ff$ so that ee denotes the odd square and ff the even square. If we further set

$$1 \,.\, 2 \,.\, 3 \ldots\ldots \; k = q$$

$$(k+1)(k+2)(k+3) \ldots\ldots 2k = r$$

then $\pm e$ is always the least remainder obtained by dividing $\frac{r}{2q}$ by p, and $\pm f$ the least remainder obtained by dividing $\frac{1}{2}rr$ by p (where the least remainder is always understood to be taken between the bounds $-\frac{1}{2}p$ and $+\frac{1}{2}p$). The number $\frac{r}{2q}$, whose value for $p = 5$ is 1, can for larger values of p also be put in the following form:

$$\frac{6 \,.\, 10 \,.\, 14 \,.\, 18 \ldots\ldots (p-3)}{2 \,.\, 3 \,.\, 4 \,.\, 5 \ldots\ldots k}$$

It is very remarkable that the decomposition of the number p into two squares can thus be obtained quite directly. But an accompanying circumstance is almost more remarkable still. This procedure always gives the root of the odd square, e, with a positive sign if e, taken positively, has the form $4m+1$, and with a negative sign if e, taken positively, has the form $4m+3$. By contrast, no general rule whatever has yet been found for the sign with which the root of the even square, f, emerges from this operation, either a priori or by examining examples. At the end of the paper the author therefore recommends this subject to the friends of higher arithmetic for further investigation, convinced that success would also open up a fruitful source of new extensions of this beautiful part of mathematics.

Göttingische gelehrte Anzeigen. 1831, April 23.

A paper presented to the Royal Society on April 15 by Court Councillor Gauss:

Theoria residuorum biquadraticorum, commentatio secunda,

continues the paper already printed in the sixth volume of the *Commentationes novae*, which was also reviewed in these pages at the time, in 1825, on April 11. Even this continuation, although more than twice as long as the first paper, does not yet exhaust this exceptionally rich subject, and completion of the whole will remain reserved for a future third paper.

Although the basic concepts of these theories and the contents of the first paper may be assumed known to everyone who has made higher arithmetic a subject of study, we nevertheless wish to recall those concepts briefly here for the convenience of readers interested in this part of mathematics who do not have the first paper immediately to hand. With respect to any integer p, another integer k is called a biquadratic residue if there are numbers of the form $x^4 - k$ that are divisible by p; in the opposite case it is called a biquadratic nonresidue of p. It is sufficient here to restrict attention to the case where p is a prime of the form $4n + 1$, and k is not divisible by that prime, since all other cases are either clear in themselves or can be reduced to this one.

For such a *given* value of p, all numbers not divisible by p fall into *four* classes. One contains the biquadratic residues, a second contains those biquadratic nonresidues that are quadratic residues of p, and the biquadratic nonresidues that are also quadratic nonresidues are divided between the remaining two. The principle of this division is that either $k^n - 1$, or $k^n + 1$, or $k^n - f$, or $k^n + f$ will always be divisible by p, where f denotes an integer that makes $ff + 1$ divisible by p. Anyone familiar with the elementary terminology will see how these verbal definitions can be expressed in it.

The theory of this classification is completed in the first paper, both for the immediately evident case $k = -1$ and for the cases $k = \pm 2$, which require subtle auxiliary investigations. The beginning of the present paper now proceeds to larger values of k. At first, however, only values that are themselves primes need be considered, and the outcome shows that the results take their simplest form when the values are taken positively or negatively according as their absolute values have the form $4m + 1$ or $4m + 3$. Examining examples immediately yields a rich harvest of new theorems with great ease; we give only a few of them here. The numbering of the classes as $1, 2, 3, 4$ refers to the cases where k^n is congruent to $1, f, -1, -f$, respectively. At the same

time, the value always chosen for f is the one that makes $a + bf$ divisible by p, where $aa + bb$ represents the decomposition of p into an odd and an even square. Thus examining examples shows that -3 always belongs to class $1, 2, 3, 4$ according as $b, a + b, a, a - b$, respectively, is divisible by 3; that $+5$ belongs to these classes in order according as $b, a - b, a, a + b$, respectively, is divisible by 5; and that -7 falls into class 1 when a or b; into class 2 when $a - 2b$ or $a - 3b$; into class 3 when $a - b$ or $a + b$; and into class 4 when $a + 2b$ or $a + 3b$ is divisible by 7, divisibility being the condition in each of these cases. Similar theorems result for $-11, +13, +17, -19, -23$, and so on. Yet, easy as it is to discover all such particular theorems by examining examples, it seems difficult to find a general law for these forms by that route, even though some common features soon become apparent; and finding proofs of these theorems is much more difficult still. The methods used for $+2$ and -2 in the first paper no longer apply here. Although other methods might also settle what concerns the first and third classes, they prove inadequate for establishing *complete* proofs.

One therefore soon recognizes that this rich region of higher arithmetic can be entered only by entirely new routes. In the first paper, the author had already indicated that a particular extension of the whole field of higher arithmetic was essential for this purpose, without then explaining more closely what it consisted in. The present paper is intended to bring this subject to light.

This consists in nothing other than the following: to give the theory of biquadratic residues its true foundation, the field of higher arithmetic, which had otherwise been extended only to the real integers, must also be extended to the imaginary ones, and these must be granted entirely equal rights of citizenship with the former. Once this is understood, that theory appears in an entirely new light, and its results acquire a most surprising simplicity.

Before the theory of biquadratic residues itself can be developed in this extended domain of numbers, however, the preceding parts of higher arithmetic, which have so far been treated only in relation to real numbers, must share in this extension. We can mention only a few of these preliminary investigations here. The author calls a quantity $a + bi$, where a and b denote real quantities and i is written for $\sqrt{-1}$ for brevity, a complex integer when a and b are both integers. Complex quantities thus do not stand opposed to real ones: they include them as the special case where $b = 0$. For convenient use, several concepts relating to complex quantities needed particular names, but we shall try to avoid those names in this review.

Just as the arithmetic of real numbers speaks of only two units, the positive and the negative, so in the arithmetic of complex numbers we have four units, $+1$, -1, $+i$, $-i$. A complex integer is called *composite* if it is the product of two integer factors that are not units; a complex number that does not

admit *such* a factorization is called a complex prime. For example, the real number 3 is prime even when considered as a complex number, whereas 5 is composite as a complex number $= (1 + 2i)(1 - 2i)$. Just as in the higher arithmetic of real numbers, primes play a principal role in this extended field of the science.

If a complex integer $a + bi$ is taken as modulus, one can give $aa + bb$ pairwise incongruent complex numbers, and no more, to one of which any given complex integer must be congruent; these can be called a complete system of incongruent residues. The so-called least and absolutely least residues in the arithmetic of real numbers have their exact counterparts here as well. For example, for the modulus $1+2i$, the complete system of absolutely least residues consists of 0, 1, i, -1 and $-i$. Almost all the investigations in the first four sections of the *Disquisitiones Arithmeticae* also have their place, with some modifications, in the extended arithmetic. The celebrated theorem of FERMAT, for example, takes the following form here: if $a + bi$ is a complex prime, and k a complex number not divisible by it, then always $k^{aa+bb-1} \equiv 1$ for the modulus $a + bi$. Especially remarkable, however, is that the Fundamental Theorem for quadratic residues has its exact counterpart in the arithmetic of complex numbers, in an even simpler form: if $a + bi$, $A + Bi$ are complex primes, with a and A odd, b and B even, then the first is a quadratic residue of the second if the second is a quadratic residue of the first, whereas the first is a quadratic nonresidue of the second if the second is a quadratic nonresidue of the first.

After these preliminary investigations, the paper turns to the theory of biquadratic residues itself. In place of the mere distinction between biquadratic residues and nonresidues, it first establishes a division of the numbers not divisible by the modulus into four classes. If the modulus is a complex prime $a + bi$, with a always assumed odd and b even, and p written for $aa + bb$ for brevity, and if k is a complex number not divisible by $a + bi$, then $k^{\frac{1}{4}(p-1)}$ will always be congruent to one of $+1, +i, -1, -i$. This establishes a division of all numbers not divisible by $a + bi$ into four classes, to which the biquadratic characters $0, 1, 2, 3$ are assigned in order. Clearly, character 0 refers to the biquadratic residues, the others to the biquadratic nonresidues, with character 2 also corresponding to quadratic $\cdot$ residues, and characters 1 and 3 to quadratic nonresidues.

It is easy to see that the main task is to determine this character only for values of k that are themselves complex primes, and here examining examples immediately leads to exceedingly simple results.

If first $k = 1 + i$, the character of this number is always found to be $\equiv \frac{1}{8}(-aa + 2ab - 3bb + 1) (\text{mod. } 4)$, and similar expressions are found for the cases $k = 1 - i$, $k = -1 + i$, $k = -1 - i$.

If, on the other hand, $k = \alpha + \beta i$ is a prime with α odd and β even, examining examples very easily yields a law of reciprocity entirely analogous to the Fundamental Theorem for quadratic residues. It can be expressed most simply as follows:

If both $\alpha + \beta - 1$ and $a + b - 1$ are divisible by 4 (all other cases can easily be reduced to this one), and the character of $\alpha + \beta i$ with respect to the modulus $a + bi$ is denoted by λ, whereas the character of $a + bi$ with respect to the modulus $\alpha + \beta i$ is denoted by l, then $\lambda = l$ if one of β, b (or both) is divisible by 4. whereas $\lambda = l \pm 2$ if neither of β, b is divisible by 4.

These theorems contain essentially everything of substance in the theory of biquadratic residues. Yet, easy as it was to discover them by examining examples, it is difficult to give rigorous proofs of them, especially of the second, the Fundamental Theorem of biquadratic residues. Because of the great length the present paper has already reached, the author was compelled to leave the presentation of the proof of the latter theorem, which he has possessed for 20 years, for a future third paper. The present paper does, however, give the complete proof of the former theorem concerning $1 + i$ (on which the other theorems for $1 - i$, $-1 + i$, $-1 - i$ depend), and this can already give some idea of the subject's complexity.

We still have a few general remarks to add. Moving the theory of biquadratic residues into the domain of complex numbers might seem objectionable and unnatural to someone less familiar with the nature of imaginary quantities and caught up in false ideas about them. It might suggest that the investigation is thereby left, as it were, hanging in the air, becomes unstable, and moves entirely beyond intuitive understanding. Nothing could be less well founded than such an opinion. On the contrary, the arithmetic of complex numbers admits the clearest visual representation. Although the author has followed a purely arithmetical treatment in this presentation, he has also given the necessary indications for this visual representation, which makes understanding more vivid and is therefore highly to be recommended. these will suffice for readers who think for themselves. Just as absolute integers are represented by a row of equally spaced points on a straight line, in which the starting point represents 0, the next point 1, and so on; and just as representing negative numbers then requires only an unbounded extension of this row on the opposite side of the starting point: so representing complex integers requires only the addition that this row be regarded as lying in a fixed unbounded plane, and that an unlimited number of similar rows be taken parallel to it on both sides, at equal distances from one another. In place of a row of points, we then have a system of points that can be arranged as rows of rows in two ways, and that serves to divide the whole plane into equal squares. The point nearest 0 in the first adjacent row on one side of the row representing the real numbers then corresponds to i, just as the point

nearest 0 in the first adjacent row on the other side corresponds to $-i$, and so on. With this representation, the execution of arithmetical operations on complex quantities, congruence, the formation of a complete system of incongruent numbers for a given modulus, and so on, admit a visual representation that leaves nothing to be desired.

On the other hand, this places the true metaphysics of imaginary quantities in a new, bright light.

Our general arithmetic, whose scope far surpasses the geometry of the ancients, is entirely a creation of modern times. Starting originally from the concept of absolute integers, it has extended its domain step by step: fractions were added to integers, irrational numbers to rational numbers, negative numbers to positive numbers, imaginary numbers to real numbers. Yet each advance was initially made with timid, hesitant steps. The first algebraists still called the negative roots of equations false roots, and they are indeed such when the problem to which they relate is presented in a form where the nature of the quantity sought admits no opposite. But just as there is no hesitation about including fractions in *general* arithmetic, although there are so many countable things for which a fraction has no meaning, so negative numbers could not be denied equal rights with positive ones in that arithmetic because countless things admit no opposite: the reality of negative numbers is sufficiently justified, since they find an adequate substrate in countless other cases. All this has, of course, long been understood. But imaginary quantities, set against real ones—formerly. and here and there even now, though inappropriately, called *impossible*—are still more tolerated than naturalized as citizens. They thus appear more like a play of signs empty of content in itself, to which any conceivable substrate is unconditionally denied, without anyone wishing to spurn the rich tribute this play of signs ultimately contributes to the treasury of relations among real quantities.

For many years the author has viewed this highly important part of mathematics from a different standpoint, from which imaginary quantities can be given an object just as well as negative ones. Until now, however, there has been no occasion to state it explicitly in public, although attentive readers will readily find traces of it in the paper on equations published in 1799 and in the prize essay on the transformation of surfaces. The present paper briefly indicates its main features, which are as follows.

Positive and negative numbers can have an application only where what is counted has an opposite that, conceived as combined with it, is equivalent to cancellation. Strictly considered, this condition holds only where what is counted consists of relations between pairs of objects, rather than substances (objects conceivable in themselves). It is postulated that these objects are arranged in a row in a definite way, for example $A, B, C, D \ldots$., and that the relation of A to B can be regarded as equal to the relation of B to C, and so

on. Here the concept of opposition requires nothing more than *interchanging* the members of the relation, so that if the relation (or transition) from A to B counts as $+1$, the relation from B to A must be represented by -1. Thus, insofar as such a row is unbounded on both sides, every real integer represents the relation of a member chosen arbitrarily as starting point to a definite member of the row.

But if the objects are of a kind that cannot be arranged in a single row, even an unbounded one, and can only be arranged in rows of rows, or, equivalently, form a manifold of two dimensions; and if the relations of one row to another, or the transitions from one to another, behave similarly to the earlier transitions from one member of a row to another member of the same row, then measuring the transition from one member of the system to another clearly requires, besides the former units $+1$ and -1, two further units $+i$ and $-i$, likewise opposite to one another. But it must also be postulated that the unit i always denotes the transition from a given member of a row to a *definite* member of the immediately adjacent row. In this way the system can therefore be arranged as rows of rows in two ways.

The mathematician abstracts entirely from the nature of the objects and the content of their relations; the sole concern is to count the relations and compare them with one another. In this respect, just as the relations denoted by $+1$ and -1, considered in themselves, are taken to be of the same kind, the mathematician is entitled to extend this to all four elements $+1$, -1, $+i$ and $-i$.

These relations can be brought within intuition only by a representation in space. The simplest case is where there is no reason to arrange the symbols of the objects in any way other than a square pattern: an unbounded plane is divided into squares by two systems of parallel lines crossing at right angles, and the intersections are chosen as the symbols. Each such point A has four neighbours. If the relation of A to one neighbouring point is denoted by $+1$, the relation to be denoted by -1 is thereby determined, whereas either of the other two may be chosen for $+i$, or the point corresponding to $+i$ may be taken at will to the *right* or to the *left*. Once forward and backward *in* the plane, and above and below with respect to its two sides, have been fixed (at will), this distinction between right and left is entirely determined *in itself*, although we can communicate our intuition of this distinction to others *only* by pointing to actually existing material things[*)]. Even after we have decided the latter point, however, we see that it depended on our arbitrary choice which of the two rows crossing at a point we wished to regard as the

[*)] KANT has already made both observations, but one cannot understand how this perceptive philosopher could believe he found in the first a proof of his opinion that space is *only* the form of our outer intuition, since the second so clearly proves the opposite, and that space must have a real meaning independent of our mode of intuition.

principal row, and which direction in it as corresponding to positive numbers. We also see that if we wish to take the relation previously treated as $+i$ for $+1$, we must necessarily take the relation previously denoted by -1 for $+i$. In mathematical language, this says that $+i$ is a mean proportional between $+1$ and -1, or corresponds to the sign $\sqrt{-1}$: we deliberately do not say *the* mean proportional, for $-i$ clearly has an equal claim. The possibility of giving $\sqrt{-1}$ an intuitive meaning is thus fully justified here, and nothing more is required to admit this quantity into the domain of the objects of arithmetic.

We believed that this brief presentation of the main points of a new theory of so-called imaginary quantities would serve readers interested in mathematics. If this subject has hitherto been viewed from a mistaken standpoint and mysterious obscurity has been found in it, this is largely attributable to the unsuitable names. Had $+1$, -1, $\sqrt{-1}$ been called direct, inverse and lateral units, for example, instead of positive, negative and imaginary (or even impossible) units, such obscurity could scarcely have been spoken of. The author has reserved a fuller treatment of the subject, which is really touched on only incidentally in the present paper, for the future. The question why relations among things that form a manifold of more than two dimensions cannot yield further kinds of quantities admissible in general arithmetic will then also receive its answer.

REVIEWS

OF OTHER AUTHORS'

WRITINGS.

Göttingische gelehrte Anzeigen. 1809, March 11.

Recherches sur l'irréductibilité Arithmétique et Géométrique des nombres et de leurs puissances. 1808. (No place of publication. 25 pages in large quarto.)

A work whose aim is to represent irrational radical quantities in the form of rational quantities. We must content ourselves with having drawn the attention of readers interested in mathematics to this little work, since the limits of these pages do not allow us to give a fuller account and examination here of the author's distinctive standpoint and treatment of radical quantities, which departs entirely from the usual one.

Göttingische Gelehrte Anzeigen. 1812, March 23.

Cribrum Arithmeticum, sive tabula continens numeros primos a compositis segregatos, occurrentes in serie numerorum ab unitate progredientium usque ad decies centena milia et ultra haec ad viginti millia (1020000). *Numeris compositis, per* 2, 3, 5 *non dividuis, adscripti sunt divisores simplices, non minimi tantum, sed omnino omnes. Confecit* Ladislaus Chernac, *Pannonius, A. L. M. Philos. et Medic. Doctor, in almo lyceo Daventriensi philosophiae professor. Daventriae* 1811. (At the author's expense, printed by J. H. Lange. XXII and 1022 pages, large quarto.)

The full title of this important and highly meritorious work already sufficiently indicates its contents: it is a table of all the prime factors of all numbers not divisible by 2, 3 and 5, from 1 to 1020000, calculated through several years of work as careful as it was laborious, and printed neatly and, as far as we have found by occasional checks, very correctly. Anyone who frequently carries out calculations with larger numbers can readily judge the value of such a gift to arithmetic. The author deserves double thanks: both for the exceedingly laborious work itself, by which he has placed his name beside the unforgettable names of Rhaeticus, Pitiscus, Brigg, Vlacq, Wolfram, Taylor and others, and for the certainly very considerable expense incurred in printing it, for which a publisher would otherwise scarcely have been found. Such tables have often been calculated before, though mostly on a smaller scale, but have either remained entirely in manuscript or have not been completely printed. As is well known, Lambert formerly did all he could to encourage the continuation of Pell's often reprinted table, which extends to 100000. According to a report by Bernoulli in Lambert's correspondence,

Oberreit had extended it to 500000, and a copy had come into Schulze's hands. Anton Felkel, as reported in the Monatl. Correspondenz, volume 2, page 223, had completed it in manuscript up to two million and intended later to provide it up to 2460000; but what had already been printed in Vienna at public expense was used for cartridges in the Turkish war because no buyers could be found! Thus many years of meritorious work were lost to the public: all the more, therefore, we considered it our duty to announce the publication of the present work here. The first million is now available for everyone's use; anyone with the opportunity and enthusiasm for this subject may therefore direct their efforts to what follows.

Göttingische gelehrte Anzeigen. 1814, November 3.

Tables des diviseurs pour tous les nombres du deuxième million, ou plus exactement depuis 1020000 *à* 2028000, *avec les nombres premiers qui s'y trouvent. Par* J. Ch. Burckhardt, *membre de l'institut impérial, du bureau des longitudes de France, et de plusieurs autres sociétés savantes. Paris*, 1814. *Mme Ve Courcier.* (VIII. and 112 pages in folio.)

Sooner than we would have dared to hope when reviewing Chernac's factor table covering the first million, we can already report the completion and publication of a similar table for the second million. Through this laborious work the distinguished author, whose name itself guarantees the greatest care and accuracy, has placed all those interested in arithmetic deeply in his debt. Chernac's table for the first million gives all the prime factors; Burckhardt's for the second, however, gives only the least divisor in each case. The complete factorization of a number in the second million therefore requires division by the least divisor and looking up the quotient in Chernac's table. But this small effort is of no significance compared with the great advantage of having the table occupy so much less space, while the prospect remains of seeing it extended in time to ten million. The author made it possible to fit the table into this small volume partly by restricting it to the least divisor and partly by printing it as economically as possible. If a stands for any of the eighty numbers below 300 that are not divisible by 2, 3 and 5, then every number not divisible by 2, 3 and 5 is included in the form $300n + a$. All eighty numbers for which n has the same value are found in a single vertical column, and each page contains thirty such columns. Thus each page includes, out of nine thousand numbers proceeding in their natural order, all those not divisible by 2, 3 or 5.

The method by which Mr. BURCKHARDT constructed his table deserves special mention here. He had a grid engraved on copper, in which 81 horizontal and 78 vertical lines formed a rectangle divided into 80×77, that is, 6160 small squares, and had the necessary number of impressions made from it. At the side, the eighty values of a could also be engraved immediately; the values of $300n$, in continuous order, were written with a pen above the 77 vertical columns. Thus each sheet represents all the numbers not divisible by 2, 3 and 5 found among each set of 23100 numbers proceeding in their natural order, and 44 sheets suffice to cover a whole million. It is easy to see that the numbers whose least divisor is 7 or 11 recur in the same order on every following sheet. These divisors could therefore be engraved directly on the copperplate and consequently appeared automatically in their proper places on every sheet. To enter the following divisors, for example 13, Mr. B. took only 13 columns across the width of a spare sheet and, treating it as the beginning of his table, cut out all the squares that had to contain the divisor 13. He therefore needed only to lay this stencil over the first thirteen columns of the first sheet, then over the next thirteen, and so on, to see at once all the places that had to be filled, insofar as they did not already contain 7 or 11, with 13. The same procedure was then used for the divisor 17, and so on. The spare sheets sufficed in this way up to the divisor 73; for the larger divisors 79, 83, and so on, Mr. B. appears to have assembled the frame from two or more pieces. For divisors exceeding 500, however, Mr. B. preferred to find the multiples by addition, needing to take only primes for the other factor. We find this whole procedure highly suitable, and would recommend it for imitation to anyone inclined to extend the table further. For the third and fourth millions, the author himself has meanwhile already carried out a large part of the calculations, so we have good reason to hope to see these too made available in print soon.

Göttingische gelehrte Anzeigen. 1816, November 7.

Tables des diviseurs pour tous les nombres du troisième million, ou plus exactement, depuis 2028000 *à* 3036000, *avec les nombres premiers qui s'y trouvent; par* J. Chr. Burckhardt, *membre de l'académie royale des sciences, du bureau des longitudes de France et de plusieurs autres sociétés savantes. Paris* 1816. *Mme Ve Courcier.* (112 pages in folio.)

Since in our review of the table of factors for the second million we have already described in detail both the method of calculation used by the distinguished author and the arrangement of the table itself, we can here content ourselves with announcing the publication of the table for the third million. Shortly we may also expect from the author the table for the *first* million, presented in the same way, so that the whole table extending beyond three million will then occupy only a moderately sized volume. For this the author deserves the thanks of all those interested in arithmetic, who see this laborious work meeting a need on a scale far exceeding anything one could have dared to hope for only a few years ago.

Göttingische gelehrte Anzeigen. 1817, August 9.

Tables des diviseurs, pour tous les nombres du premier million, ou plus exactement depuis 1 *à* 1020000, *avec les nombres premiers qui s'y trouvent; par* J. Chr. Burckhardt, *membre de l'académie des sciences dans l'institut royal, du bureau des longitudes de France, et de plusieurs autres sociétés savantes. Paris* 1817. *Mme Ve Courcier.* (114 pages in folio.)

Referring here to our reviews of the tables for the second and third millions, we now merely announce the actual · publication of these factor tables for the first million. We thus now possess a connected whole for the first three millions. For the present first million, the author used partly Chernac's *Cribrum Arithmeticum* and partly a manuscript table by Schenmark held by the library of the Royal Institute. The latter had not, however, been constructed with quite all the care one could wish for, and deciding which was right in cases where the two disagreed was often rather laborious. Only a very small number of errors were found in Chernac's table, which Mr. Burckhardt has reported here.

The author also already has most of the materials ready for the fourth, fifth and sixth millions, and offers to supply this continuation if the publisher is encouraged by sufficient sales of the first three millions. It would indeed be very regrettable if the fruits of such laborious and useful work were withheld from the world.

Göttingische gelehrte Anzeigen. December 19, 1825.

A short paper has been presented to the Royal Society on behalf of Mr. Erchinger of Thuningen in the Kingdom of Würtemberg, on the

Geometric Construction of the Regular Seventeen-sided Polygon.

As is well known, the general theory of regular polygons has acquired a new form and a wider scope through its close connection with higher arithmetic. The theory of those polygons that can be constructed geometrically is one part of it, though a comparatively small one. Since the time of the Greeks it had been known that the triangle, pentagon, fifteen-sided polygon and all polygons obtained from these by doubling or repeatedly doubling the number of sides have this property, and it was believed, and sometimes expressly asserted, that these were the only ones. Higher arithmetic has shown this to be an error: by laying open the true sources of the entirely general theory, it revealed of itself that, besides the polygons mentioned, there are countless others that can be constructed geometrically, of which the seventeen-sided polygon is the simplest. This shows in the clearest light the superiority of analysis, which encompasses the most general and the particular with equal ease, over geometry, which must always remain with the particular, is held back by ever greater complications as it advances from simpler cases to more complex ones, and would hardly ever have reached, without help from elsewhere, that next case after the known ones. Nevertheless, it remains important, interesting and desirable that purely geometric treatments should also continue to be cultivated, and that geometry should take possession of at least part of the new fields conquered by analysis. The reviewer knows of no one who has so far publicly treated the construction of the seventeen-sided polygon except Mr. Pauker, in the writings of the Courland Society and in his geometry. Mr. Erchinger's construction differs from this and is carried out more in the purely geometric spirit. It is as follows. (Anyone can draw the accompanying figure: a straight line on which the points $DBGAIFCE$ lie in that order.) Take any straight line AB and extend it backwards and forwards to C and D, so that $AC \times BC = AD \times BD = 4AB \times AB$; next determine the points E, G on the two sides of the extended line CA, so that $AE \times EC = AG \times CG = AB \times AB$, and the point F on the A side of the extended line BA, so that $AF \times DF = AB \times AB$; finally divide AE at I, so that $AI \times EI = AB \times AF$, where AI is the smaller and EI the larger part of AE. Then construct a triangle in which two sides are each $= AB$, and the third $= AI$. If a circle is drawn around this triangle, AI will be the side of the regular seventeen-sided polygon inscribed in that circle.

If one checks the correctness of this construction by comparing it with the theory of the seventeen-sided polygon given as an example in Article 354 of the *Disquisitiones Arithmeticae*, one readily sees that it is simply the geometric translation of the equations to which the application of the general theory leads. Indeed, the distances of the points C, D, E, F, G, I from A are precisely the quantities denoted there by (8 . 1), (8 . 3), (4 . 1), (4 . 3), (4 . 9), (2 . 1), if positive and negative signs are expressed by position, and the distance of the point B from A, taken in the same sense, is set $= -1$. But the real merit of Mr. ERCHINGER's paper lies less in setting out the construction itself, since analysis had already marked out the simplest path, than in giving a purely geometric proof of its correctness. This is carried out with such exemplary, painstaking care to avoid everything that is not purely elementary that it does the author credit, and prompts the wish that his indeed uncommon mathematical talent may receive every encouragement.

Göttingische gelehrte Anzeigen. July 9, 1831.

Investigations into the Properties of Positive Ternary Quadratic Forms, by Ludwig August Seeber, *Doctor of Philosophy, Full Professor of Physics at the University in Freiburg. Freiburg im Breisgau* 1831. (248 pages in 4.)

Functions of two indeterminate quantities x and y of the form $axx+2bxy+cyy$, where a, b, c denote specified integers, are known as *quadratic forms*, or, when a further distinction is needed, *binary quadratic forms*. They form one of the most interesting and richest subjects of higher arithmetic. The problems that arise first are these: to decide whether a given form of this kind contains another, $a'x'x' + 2b'x'y' + c'y'y'$, that is, whether it can be transformed into that form by a substitution $x = \alpha x' + \beta y'$, $y = \gamma x' + \delta y'$, in which $\alpha, \beta, \gamma, \delta$ are integers; whether such a relation between two forms is mutual, in which case the forms are called equivalent; further, in both cases, to give all possible transformations of one into the other; and finally, to find all possible representations of a given integer by a given form with integer values of the indeterminate quantities. These problems are completely solved in the *Disquisitiones Arithmeticae*, but constitute by far the smaller part of that work's section on quadratic forms. The finer investigations that follow require in part a preliminary treatment of a field one level higher and presenting much greater difficulties: the theory of similar functions of three indeterminate quantities x, y, z, which therefore have the form $axx + byy + czz+2a'yz+2b'xz+2c'xy$ and are called ternary quadratic forms. The solution of the principal problems concerning these ternary forms is developed in the work mentioned, though only as far as was necessary for the stated purpose. After an interval of thirty years, the author of the present work has now been the first to take up these investigations again and, for one principal kind of ternary form, namely the positive forms, has brought to completion what was left unfinished in the *Disquisitiones Arithmeticae*. For those who have studied higher arithmetic more deeply, we could describe in a few words what is newly achieved in the present work. But to be understood by others too, we must allow ourselves somewhat more detail, and we do this all the more willingly because these investigations have an interest of their own outside higher arithmetic as well.

The properties of a binary form $axx + 2bxy + cyy$ depend chiefly on the number $bb - ac$, which is therefore called the determinant of that form. Two equivalent forms always have equal determinants. But not all forms having a given determinant are therefore equivalent. Rather, such forms fall into a smaller or larger, but always finite, number of classes, with the forms belonging to one class equivalent to one another, and those belonging to different classes not equivalent. Forms whose determinant is positive can represent

positive and negative numbers alike; forms with a negative determinant, however, can represent only numbers having the same sign as a and c, so that positive and negative forms are distinguished here. The simplest forms in each class have definite criteria, are called reduced forms, and can be regarded as representatives of the whole class.

Similar relations concerning ternary forms are established in the *Disquisitiones Arithmeticae*. The determinant of the ternary form

$$axx + byy + czz + 2a'yz + 2b'xz + 2c'xy$$

is the number

$$aa'a' + bb'b' + cc'c' - abc - 2a'b'c'$$

Here too, equality of determinants is necessary for two forms to be equivalent, but is not sufficient. Rather, all forms with a specified determinant fall into a finite number of classes, in each of which the simplest forms may be called reduced and may, as it were, represent all the others. The distinction between positive and negative forms, however, behaves differently here than for binary forms. For every given determinant, whether positive or negative, there are some forms that can represent positive and negative numbers alike (indifferent forms), and others that can represent either only positive or only negative numbers (positive or negative forms). But positive forms exist only for negative determinants, and negative forms only for positive ones. It is evident, moreover, that a form's character as indifferent, positive or negative belongs at the same time to the whole class to which it belongs. The present work is confined to positive forms, whose determinants must therefore be negative. Everything that holds for these evidently transfers of itself to negative forms, whereas the indifferent forms, entirely excluded from the work, require a quite different treatment.

As already mentioned, the theory of ternary forms in the *Disquisitiones Arithmeticae* was developed only as far as was needed for its purpose there. Consequently, the problem of deciding whether two given ternary forms are equivalent had not yet been solved in full generality. It was indeed shown there how to find, for any given form, an equivalent form of the simplest kind, and that there can be only finitely many such reduced forms for any given determinant. But since each class contains several such reduced forms, which do not in all cases turn out *immediately* to be equivalent, a criterion was still lacking by which their equivalence or non-equivalence could be determined with certainty. The author of the present work has now met this need for positive forms completely and with exemplary thoroughness. His procedure is, however, presented somewhat differently from the account we have just given, and from how the matter would stand if the concept of reduced positive forms included only the essential conditions of greatest simplicity. For positive

forms, these conditions are that the numbers a, b, c, which are positive by their nature, must be no smaller than, respectively, b' or c', a' or c', and a' or b', without regard to signs. Mr. SEEBER has added further qualifications to the concept of reduced forms such that each class can contain only one form of this kind, and must contain one. In view of a beautiful theorem found by Mr. SEEBER from observed cases, to be mentioned further below, we give here the principal conditions that Mr. S. has incorporated into the concept of reduced forms: 1) among the numbers a', b', c', no two may have opposite signs; 2) without regard to sign, $2b'$ and $2c'$ must be no greater than a, further a and $2a'$ no greater than b, and b no greater than c; 3) when a', b', c' are all negative, twice the sum of these numbers must be no greater than $a + b$. We may omit here the remaining qualifications that must be added for certain special cases.

The principal content of the work consists first in solving the problem of finding, for any given positive form, an equivalent form having the character of a reduced form according to the established definition; and then in the rigorous proof of the theorem that two non-identical reduced forms cannot be equivalent, or, what is the same thing, that each class contains only one reduced form. We must do full justice to the spirit of thoroughness with which these matters are carried through. And although we must regret the very great length of treatment involved, perhaps discouraging to some, since the solution of the problem takes 41 pages and the proof of the theorem 91 pages, we would by no means have this taken as a criticism. When a difficult problem or theorem is to be solved or proved, the first step, always to be acknowledged with due thanks, is that a solution or proof should be found at all. The question whether this could have been done more easily and simply remains idle so long as the possibility is not at the same time established by actually doing it. We therefore consider it premature to dwell on this question here. — The remainder of the work chiefly contains equally thorough solutions of the following problems: to decide whether a given form contains another given form not equivalent to it; to find all possible transformations of a given form into a given equivalent form or one merely contained in it; and finally, for a given determinant, to give all possible classes of positive ternary forms.

We must also note that Mr. SEEBER has expressed ternary forms somewhat differently from the *Disquisitiones Arithmeticae*, where the coefficients of the products yz, xz, xy were deliberately assumed to be even integers, whereas Mr. S. also allows odd ones, and therefore denotes by a', b', c' what was denoted above by $2a'$, $2b'$, $2c'$. The greater generality obtained in this way is evidently only apparent, or at least unnecessary, since everything that can be said about such forms with odd coefficients also follows of itself if their doubles are considered instead. We therefore cannot approve this change,

which also entails some loss of simplicity. One consequence has been that what Mr. SEEBER calls the determinant is always four times the number bearing that name in the *Disquisitiones Arithmeticae*. In the present review we have retained the terminology of the *Disquisitiones Arithmeticae*.

For the last-mentioned problem (to give all possible reduced forms for each given determinant), Mr. SEEBER used a theorem to obtain bounds for the first three coefficients. According to this theorem, their product abc cannot exceed three times the determinant. Mr. SEEBER rigorously proves this theorem. In the preface, however, he remarks that among more than 600 cases he examined, he found not a single one in which that product exceeded twice the determinant. He therefore considers it highly probable that this tighter bound holds universally, but says that he did not succeed in finding a rigorous proof. Since this theorem, found by Mr. SEEBER from observed cases, is both remarkable in itself and important for shortening the solution of the stated problem, we wish to add a very simple proof here, so that we too may contribute in this review to the further development of the theory. Two cases must be distinguished.

I. If none of the numbers a', b', c' is negative, put

$$\begin{aligned} b - 2a' &= d, \quad c - 2b' = e, \quad a - 2c' = f \\ c - 2a' &= g, \quad a - 2b' = h, \quad b - 2c' = i \end{aligned}$$

The definition of reduced positive forms immediately implies that if

$$axx + byy + czz + 2a'yz + 2b'xz + 2c'xy$$

is such a form, none of those six numbers is negative; it is also evident that a, b, c are positive. If we now denote the form's (negative) determinant by $-D$, we have the identity, readily checked by expansion,

$$2D - abc = aa'd + bb'e + cc'f + a'hi + b'gi + c'gh + ghi$$

in which none of the seven terms on the right can be negative, and consequently abc is no greater than $2D$. The same conclusion follows in the same way from the identity

$$2D - abc = aa'g + bb'h + cc'i + a'ef + b'df + c'de + def$$

II. If none of the numbers a', b', c' is positive, put

$$\begin{aligned} b + 2a' &= d, \quad c + 2b' = e, \quad a + 2c' = f \\ c + 2a' &= g, \quad a + 2b' = h, \quad b + 2c' = i \end{aligned}$$

$$\begin{aligned} b + c + 2a' + 2b' + 2c' &= k \\ a + c + 2a' + 2b' + 2c' &= l \\ a + b + 2a' + 2b' + 2c' &= m \end{aligned}$$

and take the determinant of the form, as before, $= -D$. By the definition of reduced positive forms, none of the nine numbers $d, e, f, g, h, i, k, l, m$ can be negative. Thus the identity

$$\begin{aligned} 6D - 3abc = & - aa'(d + 2k) - bb'(e + 2l) - cc'(f + 2m) \\ & - a'hi - b'gi - c'gh + def + 2ghi \end{aligned}$$

in which all terms on the right are positive or zero, since a', b', c' are not positive but negative or zero, shows that $3abc$ cannot exceed $6D$, or abc exceed $2D$. The same conclusion follows likewise from the identity

$$\begin{aligned} 6D - 3abc = & - aa'(g + 2k) - bb'(h + 2l) - cc'(i + 2m) \\ & - a'ef - b'df - c'de + 2def + ghi \end{aligned}$$

Both equations are symmetric. If full symmetry is given up, the proof can be carried out with an even smaller number of terms, for example by the identity

$$\begin{aligned} 8D - 4abc = & - 2aa'(g + k) - 2bb'(e + l) - 4cc'm \\ & + (c + e)df + (c + g)hi \end{aligned}$$

We now wish to add a few remarks on the meaning of positive binary and ternary quadratic forms outside higher arithmetic. It is unnecessary to treat negative forms separately, while indifferent forms lie wholly outside this treatment.

The positive binary form $axx+2bxy+cyy$ represents, in general, the square of the distance between two unspecified points in a plane whose coordinates, relative to two axes inclined to one another at an angle with cosine $= \frac{b}{\sqrt{ac}}$, differ by $x\sqrt{a}$, $y\sqrt{c}$. If x and y are to mean only *integers*, the form therefore refers to a system of points arranged in parallelograms, lying at the intersections of two systems of parallel lines. The lines in each system are equally spaced: the spacings in one, measured parallel to the lines in the second, are $= \sqrt{a}$, and the spacings in the other, measured parallel to the lines in the first, $= \sqrt{c}$; the inclination of the two systems is the one given above. The plane thus appears divided entirely into equal parallelograms whose vertices

make up the system of points, with none of these points lying inside a parallelogram. The determinant taken with positive sign, thus $ac - bb$, is the square of the area of an elementary parallelogram. One and the same system of such points can be divided into parallelograms in infinitely many different ways and thus referred to equally many different forms. All these different forms are, however, what is technically called equivalent, and the area of an elementary parallelogram always remains the same. Two forms that are not equivalent, but of which one contains the other, refer to the same system of points: the first form to the whole system, the second to a part. Two forms that are technically called improperly equivalent (improprie aequivalentes) refer to two equal but oppositely placed systems of points, the plane being thought of as turned over, and so on.

In the same way, the positive ternary form

$$axx + byy + czz + 2a'yz + 2b'xz + 2c'xy$$

represents, in general, the square of the distance between two unspecified points in space whose coordinates relative to three axes (1), (2), (3) give the differences $x\sqrt{a}$, $y\sqrt{b}$, $z\sqrt{c}$: the cosines of the angles between axes (2) and (3), (1) and (3), and (1) and (2) are respectively $\frac{a'}{\sqrt{bc}}$, $\frac{b'}{\sqrt{ac}}$, $\frac{c'}{\sqrt{ab}}$. If x, y, z are to mean only integers, the form refers to a system of points arranged in parallelepipeds, that is, points arising as the intersections of three systems of parallel, equally spaced planes. The whole space thus appears divided into equal parallelepipeds whose vertices make up that system of points, and the square of the volume of an elementary parallelepiped equals the determinant of the ternary form taken with positive sign·. Equivalent forms represent one and the same system of points, referred only to other axes or fundamental planes. In the same way, all the other principal aspects of the theory of ternary forms find their geometric meaning here: one form being contained in another; the representation of a specified number or an indeterminate binary form by a ternary form; the theory of the adjoint ternary forms (formae adiunctae); the disappearance of the distinction between proper and improper equivalence; the nature of reduced forms; and so on. We must, however, confine ourselves to the indications above, especially since the present work, which considers ternary forms solely from a purely arithmetic standpoint, has given only indirect occasion for them. One will at least see from this what a rich field is opened here to investigations that have great theoretical interest in themselves and can also be used for a treatment of all relations among crystal forms that is both convenient and general. This is not the place to enter into the details of this application. We must nevertheless observe that, although a, b, c, a', b', c' were originally assumed to denote integers, the greater part of the theory of ternary forms, and in particular what is required for that application, remains valid independently of this assumption. Indeed, HAÜY's

specifications lead, for most kinds of crystal, to very simple integer values of the coefficients in the ternary forms referring to the corresponding arrangement of the system of points. But the more accurate later measurements of Wollaston, Malus, Biot, Kupffer and others contradict this and make it doubtful whether rational ratios of those coefficients agree with nature in every case. In any case, however, if one does not wish to drop from the theory the restriction to integer values of the coefficients, integers can always be found that come as close as one wishes to the results of measurement, since what matters here is their ratios to one another, rather than their absolute values.

Finally, we wish to give the geometric meaning of Seeber's theorem mentioned above. If a parallelepiped is such that none of its twelve edges (of which four at a time are equal) is greater than any of the twelve face diagonals (which are equal in pairs) or any of the four diagonals of the parallelepiped, then its volume multiplied by $\sqrt{2}$ is no smaller than the volume of a rectangular parallelepiped formed from the same edges.

MANUSCRIPT

PAPERS.

SOLUTION OF THE CONGRUENCE $X^m - 1 \equiv 0$.
ANALYSIS OF RESIDUES. CHAPTER SIX. FIRST PART.

237.

In Chapter III we showed that the congruence $x^n \equiv 1$, when a prime number p is taken as modulus, has μ roots, where μ is the greatest common divisor of the numbers n and $p-1$, and that these roots agree exactly with the roots of the congruence $x^\mu \equiv 1$. It therefore suffices to consider the case in which n is a divisor of $p-1$. That the solution not only of the congruence $x^n - 1 \equiv 0$ but of any other congruence for arbitrary moduli can be derived from its solution for prime moduli has already been shown at various places and will be developed more fully below (Chapter VIII).

238.

But there is no need to stop even here. In the same Chapter III we explained that the solution of the congruence $x^n \equiv 1$ depends on the solution of similar congruences $x^a \equiv 1$, $x^b \equiv 1$, and so on, where a, b, and so on, are primes or powers of primes, and n is the product of these numbers. Namely, if A, B, and so on, are respectively any roots of the congruences $x^a \equiv 1$, $x^b \equiv 1$, and so on, their product $AB\ldots$ will be one of the roots of the congruence $x^n \equiv 1$. Our investigations will therefore be confined to solving the congruence $x^n \equiv 1 \pmod{p}$, where p is a prime, and n a prime or a power of a prime and also a divisor of $p-1$.

239.

Furthermore, Chapter III shows that among the roots of the congruence $x^n \equiv 1$ there are always some whose powers can represent all the others. Thus, if r denotes a root of this kind (we called it *primitive* above when $n = p-1$, and shall retain this expression here, though in a broader sense), all the roots of the proposed congruence will be

$$1,\ r,\ rr,\ r^3 \quad \ldots\ldots \quad r^{n-1}$$

We must therefore direct every effort towards finding roots of this kind, since once they have been found the others will become evident of themselves. For brevity we denote any power of r by its exponent enclosed in parentheses, so that (0) denotes unity, (1) any primitive root of the congruence $x^n \equiv 1$, (2) the square of (1), and so on. Thus the sequence (0), (1), (2), (3), $\ldots$ $(n-1)$ includes all the roots. It is also known that (k) will always be such a primitive root whenever k is coprime to n; that is, in our case, where n is a power of the prime t, $= t^\nu$, whenever t does not divide k. The symbols (1), (2), and so

on, are evidently undetermined in themselves; but as soon as a definite value is assigned to (1), all the others become determined.

240.

Since our aim is to investigate primitive roots before the others, we must first separate them from the rest. This is done by removing from the sequence $(0), (1), (2) \ldots (n-1)$ all terms (k) for which k is divisible by t. If n is prime, or $\nu = 1$, only (0) is to be removed. Before proceeding to investigate the remaining roots, however, we earnestly advise the reader to work out several examples, so as to see concretely everything that might otherwise seem too general. We give one example; but this will not make it unnecessary to work out others independently.

Let $p = 29$; $n = 7$. The seven roots of the congruence $x^7 \equiv 1 \pmod{29}$ will be $1, 7, 16, 20, 23, 24, 25$. Since n is prime, all these roots except 1 will be primitive. Thus, if $7 = (1)$, the symbols will have the following meanings:

(0)	(1)	(2)	(3)	(4)	(5)	(6)
1	7	20	24	23	16	25

Everyone will also remember that the symbols (n) and (0), $(n+1)$ and (1), and so on, and in general (a) and (b), are equivalent whenever $a \equiv b \pmod{n}$.

241.

For our purpose, however, we must proceed in yet another way. We retain only those terms (k) for which k is not divisible by t, whose number is $\frac{t-1}{t}.n = \lambda$. All these numbers, or numbers congruent to them modulo n, can be represented by successive powers of some number. Let this be $= \rho$; then all the primitive roots of the congruence $x^n \equiv 1$ will be denoted as follows:

$$(1) \quad (\rho) \quad (\rho^2) \quad (\rho^3) \quad \ldots. \quad (\rho^{\lambda-1})$$

By this device we completely exclude all nonprimitive roots; the reasons and advantages will become clearer below. In our example we may therefore put $\rho = 3$, and the primitive roots of the congruence $x^7 \equiv 1$ are arranged as follows:

	(1)	(3)	(3^2)	(3^3)	(3^4)	(3^5)
or	(1)	(3)	(2)	(6)	(4)	(5)
which are	7	24	20	25	23	16

242.

To let the reader know the aim of the following investigations, it will be useful to state the theorem that we propose to prove and clarify.

If the number λ, which is $= t^{\nu-1}.t-1$, has prime factors a, b, c, d, and so on, and $\lambda = a^\alpha b^\beta c^\gamma \ldots$, the solution of the congruence $x^n - 1 \equiv 0$ depends on solving $\alpha + \beta + ..$ congruences of lower degree, of which α have degree a, β degree b, γ degree c, and so on.

Thus, in our example, solving the congruence $x^7 \equiv 1$ depends on a congruence of degree two and another of degree three; and it is apparent in general that the degrees of these congruences never depend on the modulus p. To reach the proof of this theorem, however, we must first give some propositions concerning the connection between congruences and their roots, although these investigations properly belong to Chapter eight, where they are to be pursued further.

243.

THEOREM. *Suppose the congruence*

$$x^m + Ax^{m-1} + Bx^{m-2} + \ldots. + N \equiv 0 \text{ (prime modulus)}$$

is such that, when the product of the m factors $x-r$, $x-r'$, $x-r''$, $x-r'''\ldots$ is expanded to give $x^m + ax^{m-1} + bx^{m-2} \ldots + n$, we have $A \equiv a$, $B \equiv b$, $C \equiv c$, and so on, modulo p. Then the quantities $r, r', r'' \ldots$ will be roots of the proposed congruence, and it will have no others.

Proof. I. We always have

$$\begin{aligned} &x^m + Ax^{m-1} + Bx^{m-2} + \ldots \\ &\quad \equiv x^m + ax^{m-1} + bx^{m-2} + \ldots \pmod{p} \end{aligned}$$

But the latter side of the congruence becomes $= 0$ on setting $x = r$, $x = r'$, $x = r''$, and so on; therefore, for these values of x, the former side becomes $\equiv 0 \pmod{p}$. This proves the first part.

II. If another value ρ, congruent to none of r, r', and so on, also satisfied the proposed congruence, we would have

$$\begin{aligned} 0 &\equiv \rho^m + A\rho^{m-1} + B\rho^{m-2} + \ldots \\ &\equiv \rho^m + a\rho^{m-1} + b\rho^{m-2} + \ldots \\ &\equiv (\rho - r)(\rho - r')(\rho - r'')(\rho - r''')\ldots \end{aligned}$$

But since none of the factors $\rho - r$, $\rho - r'$, $\rho - r''$, and so on, is $\equiv 0$, their product being $\equiv 0$ is impossible, because p is prime. There are therefore no roots besides r, r', and so on. This proves the second part.

244.

PROBLEM. *Let $r, r', r'' \ldots$ be unknown quantities, whose number is $= m$, whose sum is $= \alpha$, sum of squares $= \beta$, sum of cubes $= \gamma \ldots$, and sum of powers with exponent m is $= \mu$. Suppose that these numbers, also $= m$ in number, are not themselves given, but instead other numbers α', β', γ', and so on, congruent to them individually modulo p, where this is a prime and $> m$. Find a congruence of degree m whose roots are r, r', r'', and so on.*

Solution. Regard r, r', r'', and so on, as the roots of an equation

$$x^m + Ax^{m-1} + Bx^{m-2} + Cx^{m-3} + \ldots = 0$$

and determine its coefficients A, B, C, and so on, by the known method, using congruence in place of equality; namely, put

$$\begin{aligned}
-A &\equiv \alpha' \\
-2B &\equiv \beta' + A\alpha' \\
-3C &\equiv \gamma' + A\beta' + B\alpha' \\
-4D &\equiv \delta' + A\gamma' + B\beta' + C\alpha' \\
&\text{etc.} \\
-mN &\equiv \mu' + A\lambda' + \text{etc.}
\end{aligned}$$

These coefficients cannot be undetermined, since all the numbers $1, 2, 3 \ldots m < p$. I say that the congruence

$$x^m + Ax^{m-1} + Bx^{m-2} + \ldots + N \equiv 0$$

is the one required.

Proof. Let the equation whose roots are r, r', r'', r''', and so on, be

$$x^m + ax^{m-1} + bx^{m-2} + \ldots = 0$$

Then

$$\begin{aligned}
-a &= \alpha \\
-2b &= \beta + a\alpha \\
-3c &= \gamma + a\beta + b\alpha \\
-4d &= \delta + a\gamma + b\beta + c\alpha \\
&\text{etc.}
\end{aligned}$$

It will be evident to everyone from this that

$$a \equiv A, \quad b \equiv B, \quad c \equiv C \text{ etc. } (\text{mod. } p)$$

so, by the preceding section, the numbers r, r', r'', and so on, which are roots of the *equation*

$$x^m + ax^{m-1} + bx^{m-2} + \ldots = 0$$

will also be roots of the *congruence*

$$x^m + Ax^{m-1} + Bx^{m-2} + \ldots \equiv 0 \qquad \text{Q. E. D.}$$

We leave the construction of examples to the readers.

245.

We return to our purpose. Retaining the symbols used in sections 242 and earlier, we undertake to show that, if λ is the product of any factors efg, and so on, the primitive roots of the congruence $x^n \equiv 1$, whose number is λ, can be divided into e classes in such a way that the sums of the roots assigned to each class are given by a congruence of degree e. Once these are taken as known, each class can be subdivided into f groups so that the sum of each group is given by a congruence of degree f. These groups can be subdivided again, and so on, until the individual roots are reached.

246.

Definition. We shall call the collection of *all* terms contained in a form $(\rho^{ke+\alpha})$ of this kind (section 241) a *complete period*, or simply a *period*. Here e denotes any divisor of λ; α any given number; and k all integers from 0 to $\frac{\lambda}{e} - 1$. For brevity we denote such a period by $(e * \alpha)$. Thus, in our example, the terms

	$(1), (2), (4)$	form the periods	$(2 * 0)$,
	$(3), (6), (5)$		$(2 * 1)$
and	$(1), (6)$	form these	$(3 * 0)$
	$(3), (4)$		$(3 * 1)$
	$(2), (5)$		$(3 * 2)$

Now, if all terms are distributed into periods in any way, and each period is again divided into smaller periods, and so on, we say that what we promised in the preceding section is achieved.

Before entering into this exposition itself, however, we shall show that, although the formation of such a period depends on the two somewhat arbitrary quantities r, ρ, it contains nothing indeterminate: however these quantities are chosen, the same terms always come together in the same period, provided that the number of terms the period is to contain has been prescribed.

The criterion that two terms A, B belong to the same period is that each is contained in a form $(\rho^{ke+\alpha})$ of this kind, or that $A \equiv r^{\rho^{ke+\alpha}}$, $B \equiv r^{\rho^{k'e+\alpha}} \pmod{p}$. Here r is a primitive root of the congruence $x^n \equiv 1 \pmod{p}$, and ρ a primitive root of the congruence $x^\lambda \equiv 1 \pmod{n}$; see above.

We must show that, if, in place of the numbers r, ρ, other numbers, say s, σ, are chosen, then A and B are contained in similar forms $s^{\sigma^{le+\beta}}$, $s^{\sigma^{l'e+\beta}}$.

Let $s^m \equiv r \pmod{p}$; $\sigma^\mu \equiv \rho \pmod{n}$ and $m \equiv \sigma^\zeta \pmod{n}$, which is possible because r, ρ are primitive roots. Moreover, m will be coprime to n, and μ to λ (Chapter III). By the appropriate substitutions we obtain

$$A \equiv s^{\sigma^{\mu ke+\mu\alpha+\zeta}}, \qquad B = s^{\sigma^{\mu k'e+\mu\alpha+\zeta}} \qquad \text{Q. E. D.}$$

247.

THEOREM. *The product of two similar periods can be formed, independently of the number p, by adding similar periods and given numbers.*

(We call periods similar when they contain equally many terms, or when the number e is the same.)

Example. Let $n = 7$. The product of the periods $(1)+(6)$ and $(2)+(5)$ is, because $(a)\times(b) = (a+b)$, $(3)+(6)+(8)+(11)$, or it consists of the periods $(3)+(4)$ and $(1)+(6)$.

Proof. Let $\frac{\lambda}{e} = f$, and let the given periods be $(e*\alpha)$ and $(e*\beta)$, or the sums

$$(\rho^\alpha)+(\rho^{\alpha+e})+(\rho^{\alpha+2e})+\ldots+(\rho^{\alpha+(f-1)e}) \;\ldots\ldots\; P$$
$$(\rho^\beta)+(\rho^{\beta+e})+(\rho^{\beta+2e})+\ldots+(\rho^{\beta+(f-1)e}) \;\ldots\ldots\; Q$$

The product PQ consists of f^2 terms. Arrange them as follows. Form f rows, each consisting of f terms. Let the first contain the product of P with (ρ^β), the second the product $P\,.\,(\rho^{\beta+e})$, and so on, and so on. In the first row let the product arising from the term (ρ^α) occupy the first place, that from $(\rho^{\alpha+e})$ the second, and the others follow in succession. In the second row assign the first place to the product arising from $(\rho^{\alpha+e})$, the second to that from $(\rho^{\alpha+2e})$, and so on, and finally the last place to that from (ρ^α). Let the third begin with the product arising from $(\rho^{\alpha+2e})$, and continue in this way, following the product from the last term with the products from the first and second terms, and so on, and so on. Thus, denoting the successive terms of the period P by $1, 2, 3 \ldots z$ and those of the period Q by $\mathrm{I}, \mathrm{II}, \mathrm{III}, \ldots . Z$, arrange the terms of PQ as

$$1\,.\,\mathrm{I}+2\,.\,\mathrm{I}+3\,.\,\mathrm{I}+4\,.\,\mathrm{I}+\ldots+z\,.\,\mathrm{I}$$
$$2\,.\,\mathrm{II}+3\,.\,\mathrm{II}+4\,.\,\mathrm{II}+\ldots.+1\,.\,\mathrm{II}$$
$$3\,.\,\mathrm{III}+4\,.\,\mathrm{III}+\ldots+1\,.\,\mathrm{III}+2\,.\,\mathrm{III}$$
$$\text{etc.} \quad \text{etc.}$$

Now collect all terms occupying the same place in the individual rows into f groups; and I say
1° if any term is $\equiv 1$, all the other terms in the same group will also be $\equiv 1$
2° every group in which no term is $\equiv 1$ forms a period. — Once these statements have been proved, we will evidently have achieved our purpose.

The general form of such a group is

$$(\rho^{\alpha+ke}+\rho^{\beta}),\quad (\rho^{\alpha+(k+1)e}+\rho^{\beta+e}),\quad (\rho^{\alpha+(k+2)e}+\rho^{\beta+2e}),$$
$$\dots\ (\rho^{\alpha+(k+f-1)e}+\rho^{\beta+(f-1)e})$$

Indeed, in place of $\rho^{\alpha+(k-1)e}$ one can also write $\rho^{\alpha+(k+f-1)e}$, since $ef = \lambda$ and $\rho^{\lambda} \equiv 1 \pmod{n}$, and similarly for the preceding terms. Put $\rho^{\alpha+ke}+\rho^{\beta} \equiv \rho^{\chi} \pmod{n}$, which is permissible unless $\rho^{\alpha+ke}+\rho^{\beta}$ happens to be divisible by n[*]. The group can then be represented as (ρ^{χ}), $(\rho^{\chi+e})$, $(\rho^{\chi+2e})\dots(\rho^{\chi+(f-1)e})$, which is evidently the period $(e*\chi)$. But if $\rho^{\alpha+ke}+\rho^{\beta}$ is divisible by n, all terms of the group will be $\equiv (0)$, that is, $\equiv 1$. Q. E. D.

Note. This proof also gives a very easy method of expanding the product. We shall give another below which lacks this advantage, but seems worthy of consideration for its simplicity.

248.

We call all the smaller periods that make up a larger period a system of periods. Thus this name will be given to the periods

$$(ef*\alpha),\quad (ef*f+\alpha),\quad (ef*2f+\alpha)\ \dots.\ (ef*(e-1)f+\alpha)$$

that make up the period $(f*\alpha)$. It will be *properly ordered* if the numbers placed after the sign $*$, here α, $f+\alpha$, $2f+\alpha$, follow an arithmetic progression with difference f. Finally, systems are *similar* if both their smaller and their larger periods are similar.

THEOREM. *If the periods of two similar, properly ordered systems are multiplied together, namely the first by the first, the second by the second, the third by the third, and so on, the sum of all the products can be formed from given numbers and periods similar to the larger period.*

Proof. Let the systems be

$$(ef*\alpha),\quad (ef*\alpha+f),\quad (ef*\alpha+2f)\dots$$
$$(ef*\beta),\quad (ef*\beta+f),\quad (ef*\beta+2f)\dots$$

The products of the individual periods of the first system with the corresponding periods of the second consist, by the preceding section, of integers

[*] The proposition will have to be expressed somewhat differently if n denotes a power of a prime in general; when it is a prime number, however, nothing needs changing.

and similar periods. A little attention to how these periods arise shows that, if $(ef*\alpha)\times(ef*\beta)$ consists of an integer N and periods $(ef*A)$, $(ef*B)$, $(ef*C)$, and so on, the products are then formed as follows:

$$\begin{aligned}&(ef*\alpha+f)\times(ef*\beta+f) \text{ from } N \text{ and periods}\\ &\qquad (ef*A+f),\quad (ef*B+f),\quad (ef*C+f) \text{ etc.}\\ &(ef*\alpha+2f)\times(ef*\beta+2f) \text{ from } N \text{ and periods}\\ &\qquad (ef*A+2f),\quad (ef*B+2f),\quad (ef*C+2f) \text{ etc.}\end{aligned}$$

and in general

$$\begin{aligned}&(ef*\alpha+\mu f)\times(ef*\beta+\mu f) \text{ from } N \text{ and periods}\\ &\qquad (ef*A+\mu f),\quad (ef*B+\mu f),\quad (ef*C+\mu f) \text{ etc.}\end{aligned}$$

It follows directly that the sum of all the periods will be

$$eN+(f*A)+(f*B)+(f*C) \text{ etc.} \qquad \text{Q. E. D.}$$

This proof also supplies a method for finding that sum.

249.

It is easy to make this theorem still more general. Namely, if there are any number of similar, properly ordered systems, and the products of all the first periods, all the second periods, and so on, are formed, the sum of all these products consists of numbers and larger periods. If all these systems are taken to be equal, the sum of powers of all the periods, whatever the exponent, will consist of numbers and periods similar to the larger one. The aim of these results is now apparent. Let $\lambda = efgh\ldots$. Divide all the primitive roots into e periods A, A', A'', and so on, each of these again into f: B, B', B'', and so on, and each of these into g: C, C', C'', and so on. The sum of all the periods is given: it is $\equiv -1$. But according to what we have just said, we shall also have

$$\begin{gathered}(A)^2+(A')^2+(A'')^2+(A''')^2+\text{etc.}\\ (A)^3+(A')^3+(A'')^3+(A''')^3+\text{etc.}\\ \text{etc. etc.}\end{gathered}$$

Hence, by section 244, a congruence of degree e can be found whose roots are A, A', A'', and so on. Taking these as known, let each period be divided into smaller ones:

$$\begin{array}{lllll} A & \text{into} & B, & B', & B''\ldots\\ A' & \text{into} & B^{(n)}, & B^{(n+1)}, & B^{(n+2)}\ldots\\ & \text{etc.} & & & \end{array}$$

Thus $B + B' + B'' + \ldots \equiv A$ is given. But

$$(B)^2 + (B')^2 + (B'')^2 + \ldots$$
$$(B)^3 + (B')^3 + (B'')^3 + \ldots$$
$$\text{etc.}$$

consists of units and the periods A, A', A'', and so on. Therefore B, B', B'', and so on, will be given by a congruence of degree f, from which they can be found; and similarly the periods making up A', A'', and so on, can be determined. Anyone will now see that every period can be subdivided into smaller ones by precisely the same method, until the roots themselves are reached.

250.

But a difficulty occurs in applying these rules, and we must remove it. Since each congruence has several roots, we must decide which label to assign to each, so that they can be properly distinguished from one another. The labels of the periods depend on the numbers r, ρ, which can be chosen at will, so their labelling must also involve something arbitrary. The number ρ must, however, be fixed at the outset. The main feature of our method is that we derive smaller periods from larger ones. But this cannot be done without the proper order of the periods, which we have obtained through their *labels*. We must therefore ensure that all periods are distinguished by their labels as soon as they have been found.

Let the period A be denoted by $(e * \alpha)$ and divided into f periods B, C, D, and so on, which must be labelled. Each is clearly contained in a form $(ef * ke + \alpha)$ of this kind. But I say that, for one of them, B, the number k can be chosen at will, and the placement of the others derived from it.

Let R be any primitive root of the congruence $x^n \equiv 1$, and let B consist of the terms $R^\mu + R^\nu + \text{etc.}$. Put $\frac{1}{\mu}\rho^{ke+\alpha} \equiv \frac{1}{\upsilon} \pmod{n}$, and, since the value of r is arbitrary, provided only that A receives the label $(e * \alpha)$, which evidently happens of itself, put $r \equiv R^\upsilon \pmod{p}$. Then the first term of B will be $r^{\rho^{ke+\alpha}}$, and B may be denoted by $(ef * ke + \alpha)$. If, in place of R^μ, we had considered the term R^ν, we would have obtained a different value of r. But it is readily seen that, for any root ρ, the root r can have $\frac{\lambda}{ef}$ different values.

251.

Let us now see how, once one period is labelled, the others are distinguished by their labels. For this purpose, however, we must seek another method of finding the remaining periods: insofar as they, like A itself, are roots of a congruence, no order is discernible among them. Suppose that A is denoted by $(ef*0)$. It follows from the preceding discussion that

$$\begin{aligned} A^2 &\text{ has the form } M + N(ef*0) + O(ef*1) + P(ef*2) + \dots \\ A^3 &\text{ has the form } M' + N'(ef*0) + O'(ef*1) + \dots \\ &\text{etc.} \\ A^{ef-1} &\text{ has the form } M^* + N^*(ef*0) + O^*(ef*1) + \dots \end{aligned}$$

To these is added the congruence

$$(ef*0) + (ef*1) + \dots + (ef*ef-1) \equiv -1$$

We thus have $cf - 1$ linear congruences and the same number of unknown quantities, which can therefore be determined by elimination.

Note. A case may occur in which the unknown quantities are given by expressions of the form $\frac{V}{Wp}$. We shall show below how this difficulty can be remedied. Since this case can occur only very rarely, we do not wish to dwell on it here.

252.

Let this suffice in general for solving pure congruences. Much more will be said about them at various places below; in particular, many things can be taken over from the solution of pure equations, which we shall not neglect to note in their proper places. We add another example: comparing it with the rules will make everything clearer to those with less experience.

Let $n = 31$, $p = 311$; that is, we seek the roots of the congruence $x^{31} - 1 \equiv 0 \pmod{311}$. We must first find a primitive root of the congruence $y^{30} - 1 \equiv \pmod{31}$, such as $y \equiv 3$. Thus put $\rho \equiv 3$, and first divide all the primitive roots of the proposed congruence into 5 periods, namely

$$\begin{array}{ll} (5*0) & \dots\dots \quad (1) + (26) + (25) + (30) + (5) + (6) \\ (5*1) & \dots\dots \quad (3) + (16) + (13) + (28) + (15) + (18) \\ (5*2) & \dots\dots \quad (9) + (17) + (8) + (22) + (14) + (23) \\ (5*3) & \dots\dots \quad (27) + (20) + (24) + (4) + (11) + (7) \\ (5*4) & \dots\dots \quad (19) + (29) + (10) + (12) + (2) + (21) \end{array}$$

The required calculations give the sum of the periods $\equiv -1$, of their squares $\equiv 25$, cubes $\equiv 26$, fourth powers $\equiv 249$, and fifth powers $\equiv 564$.

Thus the periods will be the roots of the congruence

$$x^5 + x^4 - 12x^3 - 21x^2 + x + 5 \equiv 0$$

Further, one finds

$$(5*0)^2 \equiv 6 + 2(5*0) + 2(5*3) + (5*4)$$
$$(5*0)^3 \equiv 12 + 15(5*0) + 4(5*1) + 3(5*2) + 6(5*3) + 6(5*4)$$
$$(5*0)^4 \equiv 90 + 60(5*0) + 28(5*1) + 26(5*2) + 49(5*3) + 38(5*4)$$

and hence, by elimination,

$$5(5*1) \equiv 3(5*0)^4 - (5*0)^3 - 33(5*0)^2 - 24(5*0) + 15$$
$$5(5*2) \equiv -2(5*0)^4 - (5*0)^3 + 22(5*0)^2 + 31(5*0)$$
$$5(5*3) \equiv (5*0)^4 - 2(5*0)^3$$
$$5(5*4) \equiv -2(5*0)^4 + 4(5*0)^3$$

One root of the congruence found above is $\equiv 17$. Thus, if $(5*0) \equiv 17$, then $(5*1) \equiv 183$, $(5*2) \equiv 263$, $(5*3) \equiv 91$, $(5*4) \equiv 67$.

Now divide each of the periods found into three again, namely

$(5*0)$	into $(15*0),\ (15*5),\ (15*10)$
	or into $(1)+(30),\ (26)+(5),\ (25)+(6)$
$(5*1)$	into $(15*1),\ (15*6),\ (15*11)$
	or into $(3)+(28),\ (16)+(15),\ (13)+(18)$
	etc. etc.

Suppose that the periods into which the following are divided are roots of the respective congruences:

$(5*0)$	are roots of	$x^3 + Ax^2 + Bx + C \equiv 0$
$(5*1)$		$x^3 + A'x^2 + B'x + C' \equiv 0$
$(5*2)$		$x^3 + A''x^2 + B''x + C'' \equiv 0$
	etc.	

Then

$A \equiv -(5*0),$	$B \equiv (5*0) + (5*3),$	$C \equiv -2 - (5*4)$
$A' \equiv -(5*1),$	$B' \equiv (5*1) + (5*4),$	$C' \equiv -2 - (5*0)$
etc.	etc.	etc.

Therefore

$(15*0),\ (15*5),\ (15*10)$	will be roots of congruences	$x^3 - 17x^2 + 108x - 60 \equiv 0$
$(15*1),\ (15*6),\ (15*11)$		
$(15*2),\ (15*7),\ (15*12)$		
$(15*3),\ (15*8),\ (15*13)$		
$(15*4),\ (15*9),\ (15*14)$		

Here, however, we have

$$(15*0)^3 - 3(15*0) \equiv (15*1)$$
$$(15*1)^3 - 3(15*1) \equiv (15*2)$$
$$\text{etc.}$$

Hence, if one of the roots of the first congruence, 10, is taken as $(15*0)$, we have

$(15*0) \equiv$	10	$(15*5) \equiv$	$(15*10) \equiv$
$(15*1) \equiv$	37	$(15*6) \equiv$	$(15*11) \equiv$
$(15*2) \equiv$	-151	$(15*7) \equiv$	$(15*12) \equiv$
$(15*3) \equiv$	-39	$(15*8) \equiv$	$(15*13) \equiv$
$(15*4) \equiv$	-112	$(15*9) \equiv$	$(15*14) \equiv$

Finally, take the terms making up each of these periods. Then

(1), (30)	roots of congruences	$x^2 - (15*0)x + 1 \equiv 0$
(3), (28)		$x^2 - (15*1)x + 1 \equiv 0$
	etc.	

The roots of the first congruence are 126 and 195, which will therefore be primitive roots of the congruence $x^{31} \equiv 1$, and from these the remaining roots can readily be derived.

GENERAL INVESTIGATIONS OF CONGRUENCES.

ANALYSIS OF RESIDUES, CHAPTER EIGHT.

330.

The results on congruences presented in the preceding sections concern only the simplest cases and were mostly obtained by particular methods. In this section we shall try to bring the theory of congruences under higher principles, as far as is yet possible, much as the theory of *equations* is usually treated. As we have observed several times, there is a remarkable analogy between the two. Since all algebraic congruences involving a single unknown can be reduced to the form

$$X \equiv 0$$

where X is an algebraic function of the unknown x involving no fractions, functions of this kind must be considered first.

331.

If P, Q are functions of the indeterminate x of the forms

$$A + Bx + Cxx + Dx^3 + \ldots$$
$$H + Ix + Kxx + Lx^3 + \ldots$$

(which we shall henceforth always mean when we simply say *functions*), and the coefficients of like powers of x in the two functions are congruent for any given modulus, the *functions will be called congruent for that modulus.* Congruent functions clearly take congruent values when equal or congruent values are substituted for the indeterminate. Most of what we proved about *numbers* in Chapters I and II also holds for functions. Thus, if $P \equiv P'$, $Q \equiv Q'$, $R \equiv R'$, and so on, we clearly have $P + Q + R$ etc. $\equiv P' + Q' + R' +$ etc.; $P - Q \equiv P' - Q'$; $PQ \equiv P'Q'$; PQR etc. $\equiv P'Q'R'$ etc. The proofs are very easy and can be given in the same way as in Chapter I.

If $PQ \equiv R$, we shall denote the function Q by $\frac{R}{P}$, with the modulus indicated, and say that Q is the quotient when R is divided by P for this modulus. All functions congruent to Q can clearly be taken in its place; we shall regard them all as a *single* value. Below, however, we shall show in which cases such a quotient can have several values, that is, incongruent ones.

332.

If the modulus is a prime number and the divisor Q contains just one term Hx^h whose coefficient H is not divisible by the modulus, that is, provided only that H is not $\equiv 0$, the quotient cannot have several values. For if $QA \equiv P$ and $QB \equiv P$, we would have $Q(A-B) \equiv 0$. Now let

$$Q \equiv .. + Hx^h + Ix^{h+1} + \text{etc.}$$

where H is not divisible by p, and

$$A - B \equiv Lx^l + Mx^{l+1} + \text{etc.}$$

where L is not divisible by p (the function $A-B$ will have this form because we assume that A is not $\equiv B$). We would then have $Q(A-B) \equiv HLx^{h+l} + \text{etc.} \equiv 0$, which is absurd, since HL is not $\equiv 0$.

Rules for dividing the function P by Q, when this is possible, are now easily given. Let

$$P \equiv ax^\alpha + bx^{\alpha+1} + cx^{\alpha+2} + \text{etc.} + kx^\kappa$$
$$Q \equiv mx^\mu + nx^{\mu+1} + qx^{\mu+2} + \text{etc.} + tx^\tau$$

where a, k, m, t are not divisible by the modulus. We must have α not $< \mu$, and κ not $< \tau$. Division can be carried out as in ordinary algebraic calculation, provided that an integer is always taken for the quotient; namely, the quotient will always have the form $\frac{r}{m}$, which must be determined for the modulus. Now, if a remainder is left after $\kappa + \mu - \alpha - \tau + 1$ terms have been found, it will have the form

$$Ax^{\kappa+\mu-\tau+1} + Bx^{\kappa+\mu-\tau+2} + .. + Cx^\kappa$$

and if not all the coefficients $A, B, C..$ are $\equiv 0$, then P cannot be divided by Q.

It is also clear that division could have begun with the terms of highest degree, kx^κ, tx^τ. The calculation will be easier if Q is put in the form

$$mx^\mu(1 + qx + rxx + \text{etc.})$$

which, on putting $mv \equiv 1$, gives

$$\frac{P}{Q} \equiv \frac{vP : x^\mu}{1 + qx + \text{etc.}}$$

Division can then be completed by the usual methods.

333.

THEOREM. *If $x \equiv a$ is a root of the congruence $\xi \equiv 0$, then ξ can be divided by $x - a$ for the modulus of the congruence.*

Proof. For if it could not be divided, we would have $\xi \equiv (x-a)\xi' + b$, where b could not be divided by the modulus. Now, if x is set $\equiv a$, then ξ becomes $\equiv 0$ by hypothesis, so $(x-a)\xi' + b \equiv 0$. But then also $(x-a)\xi' \equiv 0$, and therefore b must be $\equiv 0$.

334.

PROBLEM. *Given two functions, find their common divisor (of greatest degree) for a given modulus.*

Solution. Let the functions be A, B, and let the degree of A be at least that of B. Divide A by B. If this is possible without a remainder, B will be the common divisor sought. If a remainder C is left, it will have lower degree than B. Thus let

$$A \equiv aB + C, \quad B \equiv bC + D, \quad C \equiv cD + E, \quad \text{etc.}$$

where A, B, C, D, a, b, c, and so on, are functions, and the degrees of the functions A, B, C, D, and so on, form a decreasing sequence. If some division eventually succeeds, for example $D \equiv dE$, the last divisor will be the common divisor sought. If none succeeds, a remainder will eventually be reached which has no degree, that is, a *number*; in this case, however, the functions A, B have no common divisor.

Proof. If the divisor E divides the preceding function without a remainder, it is easily seen to divide all the preceding ones. Hence E will be a common divisor of the functions A, B, proving the first assertion. But if there were a divisor of higher degree, say E', then, because $C \equiv A - aB$, it would also divide C, and by the same argument D, and so on, and therefore E: a function of higher degree would divide one of lower degree. This is absurd, proving the second assertion. It is also clear from this that, if a common divisor of any degree exists, a remainder with no degree cannot be reached. Otherwise a function with no degree would be divided by a function of some degree, which is absurd.

335.

THEOREM. *If A, B are coprime functions for the modulus p, and A has degree α, while B has degree β, then functions P, Q can be found whose degrees are respectively $< \beta$, $< \alpha$, such that*

$$PA + QB \equiv 1 \ (mod.\, p)$$

Proof. For in this case we have

$$A \equiv aB + C, \quad B \equiv bC + D, \quad \text{etc.} \quad K \equiv kL + M$$

where the degrees of the functions $A, B, C, D, \ldots K, L, M$ continually decrease, and M has no degree. Now form the sequences

$$\begin{array}{cccccc} a, & a', & a'', & a''', & \ldots & a^{(\kappa)} \\ 1, & b, & b', & b'', & \ldots & b^{(\kappa-1)} \end{array}$$

so that

$$\begin{array}{llll} a' \equiv ba + 1 & a'' \equiv ca' + a & a''' \equiv da'' + a' & \text{etc.} \\ b' \equiv cb + 1 & b'' \equiv db' + b & b''' \equiv eb'' + b' & \text{etc.} \end{array}$$

We then have

$$A - aB \equiv +C, \quad bA - a'B \equiv -D, \quad b'A - a''B \equiv +E, \quad \text{etc.}$$

as is readily seen, and hence, finally,

$$b^{(\kappa-1)}A - a^{(\kappa)}B \equiv \pm M$$

Now let $\frac{1}{\pm M} \equiv \mu$. Then, on putting $P \equiv \mu b^{(\kappa-1)}$, $Q \equiv -\mu a^{(\kappa)}$, we obtain

$$PA + QB \equiv 1$$

Further, it is clear that

$$\begin{array}{c} \text{degree of } B + \text{degree of } a = \text{degree of } A. \\ \text{degree of } C + \text{degree of } b = .\text{degree of } B \\ \text{etc.} \\ \text{degree of } L + \text{degree of } k = \text{degree of } K \end{array}$$

Therefore

$$\text{degree of } L + \text{sum of the degrees of } a, b, \ldots k = \text{degree of } A$$

The degree of $a^{(\kappa)}$ is evident, and hence also

$$\text{degree of } Q = \text{sum of the degrees of } a, b, c, \ldots \text{ that is, } = \alpha - \text{degree of } L$$

Thus

$$\text{degree of } P = \beta - \text{degree of } L \qquad \text{Q. E. D.}$$

336.

It follows that, if M is the common divisor of greatest degree of the functions A, B, we can always put

$$AP + BQ \equiv M$$

For brevity I omit examples of the preceding theorem, but readers will not neglect to gain skill in handling problems of this kind by working through examples. It is also worth pointing out that the preceding theorem holds for functions considered without a modulus, provided their coefficients are rational numbers. This will be evident from the method of proof itself. We cannot, however, dwell on this matter. The reader will observe similar things in what follows even without being prompted.

If A has no common divisor of any degree with either B or C, it will also have no common divisor with the product BC. For let

$$PA + QB \equiv 1, \quad \text{then} \quad PAC + QBC \equiv C$$

Now, if A had with BC a common divisor M, this would also divide C, contrary to the hypothesis. Thus, in general, if a function A is coprime to B, C, D, and so on, it will also be coprime to the product of them all.

If A, B, C, D, and so on, have no divisor common to them all, we can obtain

$$PA + QB + RC + SD + \text{etc.} \equiv 1$$

Let the common divisor of greatest degree of A and B be M; that of M and C be M'; that of M' and D be M'', and so on. The last term of this sequence clearly has no degree, by hypothesis. We can therefore put

$$aA + bB \equiv M, \quad mM + cC \equiv M', \quad m'M' + dD \equiv M'', \quad \text{etc.}$$

and after substitution the truth of the theorem is apparent.

337.

THEOREM. *If A, B, C, and so on, are functions coprime to one another (any two of them having no common divisor) for the modulus p, and the function M is divisible by each of them for the same modulus, it will also be divisible by the product of them all.*

Proof. For we can put $PA + QB \equiv 1$, so that

$$\frac{M}{A}Q + \frac{M}{B}P \equiv \frac{M}{AB}$$

Now, since C is coprime to AB, the function M will also be divisible by ABC, and by the same reasoning by $ABCD$, and so on.

338.

If the congruence $\xi \equiv 0$ has roots $x \equiv a$, $x \equiv b$, $x \equiv c$, and so on, then ξ can be divided by the product of $(x-a)$, $(x-b)$, $(x-c)$, and so on. For, since $a, b, c.$, and so on, are assumed incongruent, the functions $x-a, x-b, x-c$, and so on, will be coprime to one another. Since ξ is divisible by each of them, it will also be divisible by their product. Thus the number of roots cannot exceed the degree of the congruence: this is the proof of the theorem that we promised.

At the same time, however, this shows how the solution of congruences forms only one part of a much deeper investigation, namely the factorization of functions. The congruence $\xi \equiv 0$ clearly has no real roots if ξ has no factors of degree one. Yet nothing prevents ξ from being resolved into factors of degree two, three or more, from which roots that are, as it were, *imaginary* could be attributed to it. Indeed, had we wished to take a freedom similar to that taken by recent mathematicians and to introduce such imaginary quantities, we could have shortened all our following investigations incomparably. Nevertheless, we preferred to derive everything from principles[*)].

339.

Functions are called *prime* for a given modulus when they cannot be divided by any functions of lower degree for that modulus.

Thus all functions of degree one will be prime, while functions of degree two will either be prime or be composed of two of degree one. Hence ξ will be a prime function of degree two if the congruence $\xi \equiv 0$ admits no real roots. For example, $xx + x + 1$ is prime for the modulus 5, because

$$xx + x + 1 \equiv (x-2)^2 - 3 \pmod 5$$

and 3 is a quadratic nonresidue of 5.

These prime functions demand our attention above all. Although those other than functions of degree one cannot serve to find real roots, a fuller study of them recommends itself both for their remarkable properties and for other excellent truths to be deduced from them.

[*)] Perhaps on another occasion we shall explain our opinion on this matter more fully.

340.

THEOREM. *Every function is either prime or composed of prime functions; in the latter case it can be composed of prime functions in only one way.*

Proof. For unless the proposed function A is prime, it will be divisible by another function B of lower degree. If B is not a prime function, it will be divisible by another function C of lower degree. Continuing in this way, a prime function must eventually be reached. Otherwise this sequence would be infinite, which is absurd, since the degrees continually decrease. Now, if the last prime function is L, it will divide all the preceding ones. Hence $A \equiv LA'$, and A' has lower degree than A. Repeating this, $A' \equiv L'A''$, and so on, a prime function must clearly eventually be reached. Thus A will be $\equiv$ a product of prime functions L, L', L'', and so on. . This proves the first assertion.

Now, if we also had $A \equiv MM'M''$, and so on, and the functions L, L', L'', and so on, were not all the same as M, M', M'', and so on, remove those common to both sequences. Let $\lambda, \lambda', \lambda''$ . .; $\mu, \mu', \mu'', \ldots$ remain. Then μ will be coprime to $\lambda, \lambda', \lambda''$, and so on, and hence also to the product $\lambda\lambda'\lambda''$, and so on. Yet we must have

$$\lambda\lambda'\lambda'' \ldots \equiv \mu\mu'\mu'' \ldots \text{ that is, } \frac{\lambda\lambda'\lambda'' \ldots}{\mu} \equiv \mu'\mu'' \ldots \quad \text{a contradiction}$$

341.

The first main task of these investigations will be to determine the number of prime functions of any given degree. For a given modulus, the number of all different (incongruent) functions of any given degree is finite. Some of these are composed of prime functions of lower degrees, while others are prime; hence the number of the latter is also finite. A rigorous development of this matter is rather delicate; we shall begin with simpler cases.

With the modulus set $= p$, the number of all different functions of degree n of the form

$$x^n + Ax^{n-1} + Bx^{n-2} + Cx^{n-3} + \text{etc.}$$

will be p^n. For the number of coefficients $.A, B\,.\,C$, and so on, is n, and each can independently of the others be $\equiv 0, 1, 2, 3 \ldots (p-1) (\text{mod.}\, p)$. The theory of combinations therefore gives p^n different combinations, which determine the collection of all different functions of this degree.

Thus there will be p functions of degree one, namely $x, x+1, x+2$ up to $x+p-1$; there will be pp functions of degree two, and so on.

342.

We have already pointed out that all functions of degree one must be regarded as prime. If, therefore, as is sufficient for our purpose, we restrict ourselves to functions whose highest term has coefficient 1, there will be p functions of the first degree, or of one dimension.

All functions of degree two are either composed of two of degree one or are prime. The theory of combinations shows that p different things, with repetitions allowed, can be combined in $\frac{p.p+1}{1.2}$ different ways. Thus there will be that many functions composed of two prime functions of degree one, and hence $pp - \frac{p.p+1}{1.2} = \frac{1}{2}(pp - p)$ prime functions of degree two.

Similarly, from all functions of degree three, whose number is p^3, we must exclude those composed of three prime functions of degree one, whose number is $\frac{p.p+1.p+2}{1.2.3}$, and also those composed of a prime function of degree one and another of degree two, whose number is $p \,.\, \frac{1}{2}(pp - p)$. Removing these leaves $\frac{1}{3}(p^3 - p)$; this, then, is the number of prime functions of degree three. Clearly we can always continue in this way.

343.

To carry out these operations more easily and at the same time prepare the way for developing the general law, we shall consider the matter generally. For brevity we denote by (1) the number of prime functions of degree one, by (2) the number of prime functions of degree two, and further by (1^2) the number of functions composed of two prime functions of degree one, and so on, and so on. In general, $(1^\alpha 2^\beta 3^\gamma \ldots)$ denotes the number of all functions composed of prime functions, namely α of degree one, β of degree two, γ of degree three, and so on. Their degree will therefore be $\alpha + 2\beta + 3\gamma + \text{etc.}$ It follows from the preceding results and the theory of combinations that

$$(1^\alpha 2^\beta 3^\gamma 4^\delta \ldots) = (1^\alpha)(2^\beta)(3^\gamma)(4^\delta)\,..$$

$$(1^\alpha) = \frac{(1)\,.\,(1) + 1\,.\,(1) + 2\,.\,(1) + 3 \ldots (1) + \alpha - 1}{1\,.\,2\,.\,3\,.\,4 \ldots \alpha}$$

or, in general,

$$(a^\alpha) = \frac{(a)\,.\,(a) + 1\,.\,(a) + 2\,.\,(a) + 3 \ldots (a) + \alpha - 1}{1\,.\,2\,.\,3\,.\,4 \ldots \alpha}$$

Finally, if all the different ways of composing the number n by addition from the numbers $1, 2, 3, \ldots$ are collected, and these ways are denoted by $\alpha \,.\, 1 + \beta \,.\, 2 + \gamma \,.\, 3 + \text{etc.}$, then the sum of all these expressions $(1^\alpha 2^\beta 3^\gamma \,..)$ is clearly equal to the number of all functions of degree n, that is, $= p^n$. Thus

$$\begin{aligned} p &= (1) \\ pp &= (1^2) + (2) \\ p^3 &= (1^3) + (1 \,.\, 2) + (3) \\ p^4 &= (1^4) + (1^2 \,.\, 2) + (1 \,.\, 3) + (2^2) + (4) \\ &\text{etc.} \end{aligned}$$

The expression p^n clearly contains, besides the quantities $(1), (2), (3)$, and so on, also (n), which shows how all these quantities are to be determined from the preceding ones. Thus we find

$$\begin{array}{lll} (1) = p & (4) = \frac{1}{4}(p^4 - pp) & (7) = \frac{1}{7}(p^7 - p) \\ (2) = \frac{1}{2}(pp - p) & (5) = \frac{1}{5}(p^5 - p) & (8) = \frac{1}{8}(p^8 - p^4) \\ (3) = \frac{1}{3}(p^3 - p) & (6) = \frac{1}{6}(p^6 - p^3 - pp + p) & \text{etc.} \end{array}$$

344–346.

From the beginning of this sequence we observe that the highest term of the expression (n) is $\frac{1}{n}p^n$. If n is prime, $-\frac{1}{n}p$ is added to it; if n is composite, however, the law is less evident. But if we consider the matter more closely, we see that

$$\begin{array}{ll} p = (1) & p^5 = 5(5) + (1) \\ pp = 2(2) + (1) & p^6 = 6(6) + 3(3) + 2(2) + (1) \\ p^3 = 3(3) + (1) & p^7 = 7(7) + (1) \\ p^4 = 4(4) + 2(2) + (1) & p^8 = 8(8) + 4(4) + 2(2) + (1) \text{ etc.} \end{array}$$

Here the law of progression is apparent: if all the divisors of the number n are $\alpha, \beta, \gamma, \delta$, and so on, then

$$p^n = \alpha(\alpha) + \beta(\beta) + \gamma(\gamma) + \delta(\delta) + \text{etc.}$$

We now set out to prove that this observation holds in general.

We showed that the sum of all expressions of the form $(1^\alpha)(2^\beta)(3^\gamma)\ldots$, where always $\alpha + 2\beta + 3\gamma + \ldots = n$, exhausts all functions of degree n and is therefore $= p^n$. It follows that, — — —. If

$$\left(\frac{1}{1-x}\right)^{(1)}\left(\frac{1}{1-xx}\right)^{(2)}\left(\frac{1}{1-x^3}\right)^{(3)}\ldots$$

is expanded into the series $1 + Ax + Bx^2 \ldots = P$, then

$$A = p, \quad B = p^2, \quad C = p^3 \text{ etc.}$$

$$\frac{x\,\mathrm{d}P}{P\,\mathrm{d}x} = \frac{(1)x}{1-x} + \frac{2(2)x^3}{1-x^2} + \frac{3(3)x^3}{1-x^3}\ldots$$

— — — —

[From this, on substituting $\frac{px}{1-px}$ for $\frac{x\,\mathrm{d}P}{P\,\mathrm{d}x}$ and expanding the individual fractions into infinite series, the truth of the theorem follows of itself.]

347.

This theorem can also be expressed in another way. If all the divisors of the number n are $n, 1, \delta, \delta', \delta'', \delta'''$, and so on, the theorem states that

$$p^n = n(n) + (1) + \delta(\delta) + \delta'(\delta') + \text{etc.}$$

Now the product of the (n) prime functions of degree n clearly has degree $n(n)$, and likewise for the others. Hence

The product of all prime functions of degree one and of degrees n, δ, δ', and so on, will have degree p^n.

It is now easy to deduce the value of the expression (n) itself from this theorem, but for brevity we omit the analysis, which is not difficult. Let $n = a^\alpha b^\beta c^\gamma$, and so on, where a, b, c, and so on, are distinct prime numbers. Then

$$n(n) = p^n - \Sigma p^{\frac{n}{a}} + \Sigma p^{\frac{n}{ab}} - \Sigma p^{\frac{n}{abc}} \text{ etc.}$$

Here $\Sigma p^{\frac{n}{abc..}}$ denotes the collection of all expressions similar to $p^{\frac{n}{abc..}}$ when the quantities $a, b, c..$ are permuted among themselves in every way. Thus for $n = 36$ we have $36(36) = p^{36} - p^{18} - p^{12} + p^6$.

Let us add one more observation. If n has the form a^α, and a is prime, then $n(n) = p^n - p^{\frac{n}{a}}$. Since (n) must be an integer, we therefore have, whatever p may be,

$$p^n \equiv p^{\frac{n}{a}} \pmod{n}$$

Thus, if p is coprime to a,

$$p^{n-\frac{n}{a}} \equiv 1 \pmod{n}$$

and for $\alpha = 1$,

$$p^{a-1} \equiv 1 \pmod{a}$$

It is remarkable that these theorems can be obtained in such different ways.

348.

PROBLEM. *Given the equation*

$$x^m + Ax^{m-1} + Bx^{m-2} + Cx^{m-3} + \text{etc.} + M = 0$$

whose roots are $x = a$, $x = b$, $x = c$, and so on, find an equation whose roots are $x = a^\tau$, $x = b^\tau$, $x = c^\tau$, and so on.

First solution. Find, by the known theorem, the sums of the roots of the given equation, of their squares, cubes, and so on, through the power with exponent $m\tau$. This also gives the sums of the roots of the required equation, and of their squares, and so on, namely Σa^τ, $\Sigma a^{2\tau}$, and so on. The coefficients can then be determined by the same theorem.

For practical calculation this solution is easier; but the following is better suited to our purpose, and also to showing that the coefficients of the required equation will be integers if the coefficients of the given equation are integers.

Second solution. Let θ be a primitive root of the equation $x^\tau = 1$, and form the product of

$$\begin{aligned}
&x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.}\\
&x^m + A\theta x^{m-1} + B\theta\theta x^{m-2} + \text{etc.}\\
&x^m + A\theta\theta x^{m-1} + B\theta^4 x^{m-2} + \text{etc.}\\
&\text{etc.}\\
&x^m + A\theta^{\tau-1} x^{m-1} + B\theta^{2\tau-2} x^{m-2} + \text{etc.}
\end{aligned}$$

The roots of this product will therefore be

$$\begin{array}{llll}
a, & \theta a, & \theta\theta a & \text{etc.}\\
b, & \theta b, & \theta\theta b & \text{etc.}\\
c, & \theta c, & \theta\theta c & \text{etc.}
\end{array}$$

that is, the product will equal

$$(x^\tau - a^\tau)(x^\tau - b^\tau)(x^\tau - c^\tau)\ldots$$

and hence have the form

$$x^{\tau m} + A' x^{\tau(m-1)} + B' x^{\tau(m-2)} + \text{etc.}$$

Now, replacing x^τ by x, we have

$$x^m + A'x^{m-1} + B'x^{m-2} + \text{etc.} = (x - a^\tau)(x - b^\tau)(x - c^\tau)\dots$$

and hence

$$x^m + A'x^{m-1} + B'x^{m-2} + \text{etc.} = 0$$

is the required equation. That A', B', and so on, are not only rational numbers but also integers follows easily from the theory of the equation $x^\tau = 1$.

Since we shall often use this operation below, by (P, ρ^τ) we shall denote the function which, on being set equal to zero, gives an equation whose roots are the powers with exponent τ of the roots of the equation $P = 0$.

If $P \equiv Q$ for any modulus, then also $(P, \rho^\tau) \equiv (Q, \rho^\tau)$ for the same modulus.

349.

THEOREM. *The coefficient of the term x^n in (P, ρ^τ) is congruent, modulo τ, to the coefficient of the term $x^{\tau n}$ in P^τ, provided that τ is a prime number* (which, in this case, gives a third solution of the preceding problem).

Proof. It follows from Chapter Six that every coefficient of the product

$$(x^m + Ax^{m-1} + \text{etc.})(x^m + A\theta x^{m-1} + \text{etc.})\dots$$

has the following form after θ^τ is replaced by unity:

$$E + (1 + \theta + \theta\theta + \text{etc.} + \theta^{\tau-1})F$$

Now, if θ is regarded as a primitive root of the equation $x^\tau = 1$, the whole product becomes E; but if $\theta = 1$, the whole product becomes $P^\tau = E + \tau F$. Thus the coefficient of the term $x^{n\tau}$ in P^τ is congruent, modulo τ, to the coefficient of the term $x^{n\tau}$ in E, that is, to the coefficient of the term x^n in (P, ρ^τ).

350.

THEOREM. *If τ is a prime number, then*

$$(P, \rho^\tau) \equiv P \ (mod.\, \tau)$$

Proof. Let the coefficient of the term x^n in (P, ρ^τ) be $= N'$, and let the coefficient of the same term in P be $= N$. Then, on putting

$$P = x^m + Ax^{m-1} + \text{etc.} + Nx^n + \text{etc.}$$

we have

$$P^\tau \equiv x^{m\tau} + A^\tau x^{(m-1)\tau} + \text{etc.} + N^\tau x^{n\tau} + \text{etc.} \ (\text{mod.}\, \tau)$$

and hence, by the preceding article, $N' \equiv N^\tau \ (\text{mod.}\, \tau)$. Thus, since $N^\tau \equiv N$, we obtain $N' \equiv N$. Q. E. D.

It is also clear from this that $(P, \rho^{\alpha}) \equiv (P, \rho^{\alpha\tau})$ and $(P, \rho^{\tau}) \equiv (P, \rho^{\tau\tau})$, whence, in general,

$$(P, \rho^{\alpha}) \equiv (P, \rho^{\alpha\tau^k}) \,(\mathrm{mod.}\, \tau)$$

351.

THEOREM. *There is a value of the number ν less than p^m such that the function $x^{\nu} - 1$ can be divided by the given function P of degree m, whose lowest term does not involve the indeterminate x, modulo p.*

Proof. Divide by P the sequence of functions $1, x, xx \ldots$ through x^{p^m-1} as soon as their degree exceeds m. Since none can be divided by P without a remainder, all the remainders can be put in the form

$$Ax^{m-1} + Bx^{m-2} + \ldots + N$$

with all coefficients positive and $< p$. But since they can never all be $= 0$, there are clearly only $p^m - 1$ functions, one of which each remainder must equal. Since, through the power of x whose exponent is $p^m - 1$, there are p^m remainders, at least two must necessarily be the same. Let the same remainder result when x^a and $x^{a+\nu}$ are divided by P, with $a + \nu < p^m$. Thus $x^{a+\nu} - x^a$ can be divided by P. Since, by hypothesis, x, and hence also x^a, is a function coprime to P, it follows that $x^{\nu} - 1$ can also be divided by P. Q. E. D.

Corollary. If $x^{\nu} - 1$ can be divided by P, then $x^{k\nu} - 1$ can also be divided by P, where k denotes any integer.

352.

THEOREM. *Keeping the notation of the preceding article, if P is a prime function and x^{ν} is the lowest power which, after unity is subtracted, can be divided by P, then ν is either $= p^m - 1$ or a divisor of this number, with the sole exception of the case $P = x$.*

Proof. Since P is a prime function of degree m, there are $p^m - 1$ different functions of degree less than m (excluding the function 0 from the total), all of which are coprime to P. Now, since x^ν is assumed to be the lowest power which leaves unity as remainder on division by P, it is clear that dividing all lower powers, from $1, x, ..$ through $x^{\nu-1}$, by P gives ν different remainders, which we denote generally by A. If these exhaust all possible remainders, the theorem is proved; but if some are still absent from their number, let B be any one of them. It is clear that the function Bx^ν, divided by P, gives remainder B, and that, in general, $Bx^{\nu+k} \equiv Bx^k (\mathrm{mod.}\, P)$. But all the functions from B through $Bx^{\nu-1}$ give remainders different from one another and from the remainders A. For if $Bx^\lambda \equiv Bx^{\lambda+\delta} (\mathrm{mod.}\, P)$, we would also have $1 \equiv x^\delta (\mathrm{mod.}\, P)$, with $\delta < \nu$, contrary to the hypothesis; while if $Bx^\lambda \equiv x^\mu (\mathrm{mod.}\, P)$, we would have $B \equiv x^{\mu+\nu-\lambda} (\mathrm{mod.}\, P)$, and hence B would be one of the remainders A, contrary to the hypothesis. We thus clearly have ν further new remainders. We may proceed in the same way (exactly as above, §. .), and it will become apparent that the number $p^m - 1$ of all possible remainders is either $= \nu$, or $= 2\nu$, or $= 3\nu$, or, in general, a multiple of the number ν. Q. E. D.

353.

It follows from the preceding theorem and the corollary to §. 351 that every prime function of degree n divides the function $x^{p^n-1} - 1$ modulo p. Thus all functions of degree one, except the single one which is $\equiv x$, divide $x^{p-1} - 1$, which is FERMAT'S THEOREM; all prime functions of degree two, that is, of the form $xx + Ax + B$, divide the function $x^{pp-1} - 1$, and so on. Now let all divisors of the number n be $n, \delta, \delta', \delta''$, and so on, . 1. It is clear that $p^n - 1$ is also divisible by $p^\delta - 1$, $p^{\delta'} - 1$, $p^{\delta''} - 1$, and so on, $p - 1$. Hence the function $x^{p^n-1} - 1$ can be divided by *all prime functions of degrees $n, \delta, \delta', \delta''$, and so on, down to the prime functions of degree one (excluding the function x)*, and therefore also, since all these functions are prime in themselves and hence coprime to one another, by *the product of them all.* But this same product has degree $p^n - 1$ (§. 347), because the one function x is absent. Thus it is clear that *this product itself must be congruent to* $x^{p^n-1} - 1 (\mathrm{mod.}\, p)$.

354.

THEOREM. *If the function $x^\nu - 1$ is divisible by the function P, then*

$$(P, \rho^{k\nu+t}) \equiv (P, \rho^t)$$

where k, t denote any integers.

Proof. Let

$$P = x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.}$$

It is known that, if

$$\frac{mx^{m-1} + (m-1)Ax^{m-2} + \text{etc.}}{x^m + Ax^{m-1} + \text{etc.}}$$

is expanded in an infinite series of the form

$$m\frac{1}{x} + \alpha\frac{1}{xx} + \beta\frac{1}{x^3} + \gamma\frac{1}{x^4} + \text{etc.}$$

then α is the sum of the roots of the equation $P = 0$, β the sum of their squares, and so on. From this it follows without difficulty that the sums of the powers with exponents $\nu + 1$, $\nu + 2$, and so on, are congruent to the sum of the roots, of their squares, and so on. Hence, unless the modulus is equal to or less than the degree of the function P, we have

$$(P, \rho^{\nu+1}) \equiv P, \quad (P, \rho^{\nu+2}) \equiv (P, \rho^2), \quad (P, \rho^{\nu+3}) \equiv (P, \rho^3) \text{ etc.}$$

We shall consider that case below.

355.

THEOREM. *If, in the sequence*

$$(P,\rho^0),\quad (P,\rho)\quad (P,\rho^2),\quad (P,\rho^3)\text{ etc.}$$

the terms following the term in position ν are successively congruent to the first terms, then $x^\nu - 1$ can be divided by P, provided that P contains no factor more than once.

Proof. On putting $\frac{\mathrm{d}P}{\mathrm{d}x} = Q$, Q will be a function coprime to P. Let

$$\frac{Q}{P} \equiv \frac{A}{x} + \frac{B}{xx} + \frac{C}{x^3} + \text{etc.}$$

Then, after the term $\frac{N}{x^\nu}$, there follows, by hypothesis,

$$\frac{A}{x^{\nu+1}} + \frac{B}{x^{\nu+2}} + \frac{C}{x^{\nu+3}} + \text{etc.}$$

Thus

$$\frac{Q}{P} \equiv \frac{Ax^{\nu-1} + Bx^{\nu-2} + \text{etc.}}{x^\nu - 1}$$

which shows that the function $x^\nu - 1$ can be divided by P. Q. E. D.

356.

THEOREM. *If P is a prime function of x of degree m, and X is a function of $x, x^p, x^{pp}, x^{p^3} \,..\, x^{p^{m-1}}$ in which all these quantities occur in the same way, that is, which remains unchanged however they are permuted among themselves, then dividing the function X by P gives a remainder which is a number.*

Proof. Let the remainder be

$$Ax^{m-1} + Bx^{m-2} + \,..\, + N \equiv \xi$$

All coefficients $A, B, C \ldots$ up to, but excluding, N will be $\equiv 0$. This is proved as follows. Since $X - \xi$ is divisible by P, $X^p - \xi^p$ can also be divided by P. But it is easy to see that X^p is what X becomes when x is replaced by x^p, x^p by x^{pp}, and so on, . and $x^{p^{m-1}}$ by x^{p^m}, or, equivalently, x. Thus it is clear that $X^p \equiv X \pmod{P}$; hence, since $X^p \equiv \xi^p$ and $X \equiv \xi (\mathrm{mod.}\, P)$, we also have $\xi^p \equiv \xi (\mathrm{mod.}\, P$, or

$$\xi^p - \xi \equiv 0 (\mathrm{mod.}\, P)$$

But $\xi^p - \xi$ is congruent, modulo p, to the product of $\xi, \xi+1, \xi+2, ..$ through $\xi + p - 1$. These factors are all coprime to P unless ξ is simply a number. Hence $\xi^p - \xi$ cannot be divisible by P in any other way. Q. E. D.

Functions of this kind are the sum of all the quantities, the sum of their squares, cubes, and so on, and the sums of their products taken two at a time, three at a time, and so on. We shall determine what that number is in the following article:

357.

THEOREM. *Let the prime function of the preceding article be*

$$P \equiv x^m - Ax^{m-1} + Bx^{m-2} - Cx^{m-3} + \text{etc.}$$

Then the remainder when the sum of the quantities x, x^p, *and so on,* $x^{p^{m-1}}$ *is divided by* P *is* $\equiv A$*; for the sum of their products taken two at a time it is* $\equiv B$*; for the sum of their products taken three at a time it is* $\equiv C$*, and so on.*

Proof. Let those functions be X, Y, Z, and so on, and let their remainders be the numbers A', B', C', and so on, in their respective order. It is easy to see that x, x^p, x^{pp}, and so on, are roots of the equation

$$z^m - Xz^{m-1} + Yz^{m-2} - Zz^{m-3} + \text{etc.} = 0$$

Thus, on putting $z = x$, we have

$$x^m - Xx^{m-1} + Yx^{m-2} - Zx^{m-3} + \text{etc.} = 0$$

But the functions $X - A'$, $Y - B'$, $Z - C'$, and so on, are divisible by P, and hence so is the function

$$x^m - A'x^{m-1} + B'x^{m-2} - C'x^{m-3} + \text{etc.}$$

This cannot happen unless $A' \equiv A$, $B' \equiv B$, $C' \equiv C$, and so on. Q. E. D.

It is also known that any other function X of x, x^p, and so on, [in which all these quantities occur in the same way,] can always be derived from these. Thus

$$x^2 + x^{2p} + x^{3pp} + \text{etc.} \equiv AA - 2B(\text{mod.}\, P) \text{ etc. etc.}$$

Example. Let $p = 5$ and $P \equiv x^2 + 2x + 3$. Then the function $x + x^5$, divided by P, gives $\equiv -2$, and $x^6 \equiv 3$, and so on, and so on.

358. 359.

THEOREM. *Let P be a prime function and x^ν the lowest power of x which, divided by P, gives remainder 1. Further, let $P \equiv (P, \rho^n)$. Then n is congruent to some power of the number p modulo ν.*

Proof. We showed above that, if P

$$= x^m + Ax^{m-1} + Bx^{m-2} + \text{etc.}$$

then

$$z^m + Az^{m-1} + Bz^{m-2} + \text{etc.} - (z-x)(z-x^p)\ldots(z-x^{p^{m-1}})$$

is divisible by P. In the same way it would follow that

$$z^m + Az^{m-1} + Bz^{m-3} + \text{etc.} - (z-x^n)(z-x^{np})\ldots(z-x^{np^{m-1}})$$

is divisible by P. Since these factors are coprime to one another, the individual factors must necessarily be congruent to the corresponding individual factors modulo P, p. Thus $z - x^n$ must be $\equiv z - x^{p^\kappa}$, that is, $p^\kappa \equiv n (\text{mod.}\, \nu)$. Q. E. D.*)

Finding the prime divisors of the function $x^\nu - 1$ for a prime modulus.

360.

If ν is divisible by the modulus p or by one of its powers, put $\nu = p^k\lambda$. Then

$$x^\nu - 1 \equiv (x^\lambda - 1)^{p^k} (\text{mod.}\, p)$$

This makes it clear that only the case where ν is not divisible by p needs to be considered.

*) If $(P, \rho^a) \equiv (P, \rho^b) (\text{mod.}\, p)$, then $a \equiv p^\kappa b (\text{mod.}\, \nu)$.

Proof. Let $z^m + Az^{m-1} + Bz^{m-2} + \ldots = \Pi$. Then $(\Pi, \rho^a) \equiv (\Pi, \rho^b) (\text{mod.}\, P)$; but

$$(\Pi, \rho^a) \equiv (z-x^a)(z-x^{ap})(z-x^{app})\ldots(z-x^{ap^{m-1}}),$$
$$(\Pi, \rho^b) \equiv (z-x^b)(z-x^{bp})(z-x^{bpp})\ldots(z-x^{bpm-1})$$

which establishes the proposition.

The product of $\Pi, (\Pi, \rho^2), (\Pi, \rho^3)$, and so on, (Π, ρ^ν) is $\equiv (z^\nu - 1)^m (\text{mod.}\, P)$, for

$$(z-x)(z-x^2)(z-x^3)\ldots(z-x^\nu)$$
$$\equiv (z-x^p)(z-x^{2p})(z-x^{3p})\ldots(z-x^{\nu p}) \equiv \text{etc.} \equiv z^\nu - 1$$

In the sequence $P, (P, \rho^2), (P, \rho^3)$, and so on, $\ldots (P, \rho^\nu)$, all prime divisors of the function $x^\nu - 1$ occur, each m times. It follows that the product of them all is $\equiv (x^\nu - 1)^m$.

If $p^m \equiv 1 (\mathrm{mod.}\, \nu)$, with m as small as possible, then $x^{p^m-1} - 1$ is clearly divisible by $x^\nu - 1$. Thus $x^\nu - 1$ cannot have any divisors other than those of $x^{p^m-1} - 1$. But this expression has prime divisors of degree m and others whose degrees divide the number m. Thus $x^\nu - 1$ also has divisors of these kinds. We shall show how many it has of each kind by an example, from which the general law can easily be deduced.

Let $\nu = 63$ and $p = 13$. Then $m = 6$. Thus $x^{63} - 1$ has prime factors of degrees six, three, two and one modulo 13. It is clear that the product of the factors of degree one is the common divisor of greatest degree of the functions $x^{63} - 1$ and $x^{12} - 1$, that is, $x^3 - 1$; hence there are three prime factors of degree one. The product of all prime factors of degrees two and one is the common divisor of the functions $x^{63} - 1$ and $x^{168} - 1$, that is, $x^{21} - 1$; hence there are $\frac{21-3}{2}$, or 9, factors of degree two. The product of the prime factors of degrees three and one is the common divisor of the functions $x^{63} - 1$ and $x^{2196} - 1$, that is, $x^9 - 1$; hence there are $\frac{9-3}{3}$, that is, 2, divisors of degree three. Finally, the remaining ones have degree six, and their number is therefore $= \frac{63-6-18-3}{6}$, that is, 6.

Careful consideration of this matter easily gives the following general rule:

Let δ be a divisor of m, and let all divisors of the number δ less than δ be $\delta', \delta'', \delta'''$, and so on. Let the greatest common divisors of ν with $p^\delta - 1$, $p^{\delta'} - 1$, $p^{\delta''} - 1$, and so on, be μ, μ', μ'', and so on, respectively. Put $\frac{\mu}{\mu'}; \frac{\mu}{\mu''}, \frac{\mu}{\mu'''}$, and so on, $= \lambda', \lambda'', \lambda'''$, and so on. Then $x^\nu - 1$ has $\frac{1}{\delta}$ times as many prime divisors of degree δ as there are numbers below μ divisible by none of the numbers $\lambda', \lambda'', \lambda'''$, and so on.

361.

THEOREM. *If a function X of the indeterminate x is divisible by another function ξ, and X, when x is replaced by x^k, becomes X', then X' is divisible by $(\xi, \rho^{\frac{1}{k}})$.*

Proof. Let $X \equiv \xi\upsilon$, and let ξ, υ become ξ', υ' when x is replaced by x^k. It is clear that $X' \equiv \xi'\upsilon'$. But ξ' is divisible by $(\xi, \rho^{\frac{1}{k}})$. Hence so is X'. Q. E. D.

362.

With these principles established, we can easily determine the prime divisors of the function $x^\nu - 1$. We assume that all those divisors which also divide some function $x^{\nu'} - 1$, where $\nu' < \nu$, have already been found, and propose to investigate the remaining ones. They can all be included in the expression (P, ρ^k), if P is one of them and k takes all values less than ν and coprime to it.

In Chapter VI we showed how the primitive roots of the equation $x^\nu = 1$ can be divided into classes so that, when all are expressed as powers of one of them, the same division into classes results whichever primitive root is taken as the base; we called such collections of roots *periods*. It is clear that the functions $x, x^\alpha, x^\beta, x^\gamma$, and so on, where α, β, γ, and so on, denote all numbers coprime to ν, can be divided into periods in the same way, and each larger period again into smaller ones, until periods of the form $x^k, x^{kp}, x^{kpp} \ldots x^{kp^{m-1}}$ are finally reached. Once this is done, the following is clear:

1. Since every period is composed of smallest periods of the form $x^k + x^{kp} +$ etc., division by any prime function of degree m gives a remainder which is a number.

2. Since all terms of a period can always be put in the form $x^{\kappa . a^\alpha b^\beta c^\gamma \ldots}$, where $\kappa.\ a, b, c \ldots$ are fixed numbers, while all values may be substituted for $\alpha, \beta, \gamma \ldots$, it is clear that the period becomes itself when x is replaced by x^k and k has the form $a^\alpha b^\beta c^\gamma \ldots (\mathrm{mod.}\, \nu)$. From this it is easy to see that all the functions $P, (P\ \rho^k)$, and so on, with k denoting a number of this kind, give the same remainder when the period is divided by them.

3. Thus, after such a remainder has been subtracted, the period can be divided by the product of all the functions (P, ρ^k).

363.

The main task is to determine these remainders. First, find the remainder that the largest period gives on division by the product of all the appropriate prime functions. If this product is

$$\equiv x^\lambda - Ax^{\lambda-1} + \text{etc.}$$

then this remainder is $\equiv A$. The form of this product is readily found, and Chapter VI shows that $A = 0$ if ν is divisible by a square; otherwise A is either $= +1$ or $= -1$, according as the number of prime factors of ν is even or odd.

Now split this largest period into smaller periods, and represent the terms of any one period by $x^{kp^\pi u}$, so that k is a number fixed within each period but varying between periods, while π and u vary within each period and can take in the other periods the same values that they take in any one period. For the moment, take some prime function P as the base, and let the

periods $\Sigma x^{p^\pi u}$, $\Sigma x^{k' p^\pi u}$, etc., on division by it, give the remainders A, A', etc., respectively. Then $\Sigma x^{p^\pi u} - A$ is divisible by the product of all the functions (P, ρ^u); $\Sigma x^{k' p^\pi u} - A'$ by the product of all the functions $(P, \rho^{k' u})$; etc., etc. It is readily seen that the quantities A, A', etc., are roots of a known congruence. Namely, let the periods of the roots of the equation $x^\nu = 1$ corresponding to the preceding periods be roots of the equation $Q = 0$; then A, A', etc., will be roots of the congruence $Q \equiv 0$. For we have

$$A + A' + \text{etc.} \equiv \text{the sum of the periods},$$
$$AA + A'A' + \text{etc.} \equiv \text{the sum of the squares of the periods}$$

etc., etc. The calculation is entirely similar to the one we set out in Chapter VI, if in place of ρ we substitute x, since here too we can put unity for x^ν, as we did there for ρ^ν.

Once the roots A, A', etc., have been found, choose one as the remainder of the period $\Sigma x^{p^\pi u}$, and use it to order the remainders of the others in the same way as in Chapter VI. This choice is left free here too, since the function P is still entirely undetermined. The subsequent calculation is wholly analogous to the one we treated in Chapter VI; setting out every detail would take us too far. Finally, once we have reached Σx^{p^π}, the main task is complete. For, on putting

$$P \equiv x^m + ax^{m-1} + bx^{m-2} + \text{etc.}$$

we have $-a \equiv \Sigma x^{p^\pi}$. The second coefficient of the other functions (P, ρ^k) is obtained in the same way, and from this the remaining coefficients of P can be determined. It may often happen that we reach identical congruences from which nothing seems to follow. How this difficulty can be met will be shown below.

364.

An example will make all this much clearer. The proposed task is to resolve the function $x^{15} - 1$ into factors modulo 17. We have $m = 4$, and since the product of all the elementary functions is

$$\equiv \frac{x^{15} - 1 \,.\, x - 1}{x^3 - 1 \,.\, x^5 - 1} = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$$

there will therefore be only two prime factors of degree four, P and P'. Now distribute $x, xx, x^4, x^7, x^8, x^{11}, x^{13}, x^{14}$ into these two periods:

$$\Sigma x^{17^\alpha} \equiv x + xx + x^4 + x^8,$$
$$\Sigma x^{7.17^\alpha} \equiv x^7 + x^{11} + x^{13} + x^{14}$$

Modulo one of the functions P, P', let

$$\Sigma x^{17^\alpha} \equiv A, \quad \Sigma x^{7.17^\alpha} \equiv A'$$

Then

$$\begin{aligned} A + A' &\equiv 1 \\ AA &\equiv \Sigma x^{2.17^\alpha} + \Sigma x^{3.17^\alpha} + \Sigma x^{5.17^\alpha} + \Sigma x^{9.17^\alpha} \\ A'A' &\equiv \Sigma x^{14.17^\alpha} + \Sigma x^{6.17^\alpha} + \Sigma x^{5.17^\alpha} + \Sigma x^{3.17^\alpha} \end{aligned}$$

and hence

$$\begin{aligned} AA + A'A' &\equiv \Sigma x^{17^\alpha} + \Sigma x^{7.17^\alpha} + 4\Sigma x^{3.17^\alpha} + 2\Sigma x^{5.17^\alpha} \\ &\equiv 1 - 4 - 4 \equiv -7 \end{aligned}$$

Thus A and A' will be the roots of the congruence

$$xx - x + 4 \equiv 0 \pmod{17}$$

which are $6, 12$. It follows that P divides

$$x^8 + x^4 + xx + x - 6$$

and is

$$\equiv x^4 - 6x^3 - 2xx - 12x + 1$$

Further, P' will be $\equiv (P, \rho^7)$, and will be

$$\equiv x^4 - 12x^3 - 2xx - 6x + 1$$

365.

It is enough for us here to have shown the possibility of these solutions. For brevity, we pass over many devices that can ease these operations. But we cannot omit some very important consequences.

What precedes proves that all the auxiliary equations for solving the equation $x^\nu = 1$, if converted into congruences, have possible roots when the period

$$x + x^p + x^{pp} + \ldots + x^{p^{m-1}}$$

has not yet been split. Let us stay with the case where ν is prime; then m is a divisor of $\nu - 1$. Here, therefore, the auxiliary congruences have real roots if the number of periods found through them is a divisor of $\frac{\nu-1}{m}$. Thus, if $\frac{\nu-1}{m}$ is even, that is, if m is a divisor of $\frac{\nu-1}{2}$, or equivalently if $p^{\frac{\nu-1}{2}} \equiv 1 \pmod{\nu}$, or equivalently if p is a quadratic residue of the prime ν, the quadratic equation that divides the roots into two periods will have real roots modulo p. But in Chapter VI we showed that, on putting $\nu = 4n \pm 1$, this equation is always $xx + x \mp n = 0$. Hence we have the notable

THEOREM. *If the prime p is a quadratic residue of the prime $4n \pm 1$, the congruence*

$$xx + x \mp n \equiv 0 (\text{mod. } p)$$

will have real roots, and therefore so will the congruence

$$4xx + 4x \mp 4n \equiv 0 \quad \text{or} \quad (2x+1)^2 \mp \nu \equiv 0$$

that is, $\pm\nu$ will be a quadratic residue of p.

366.

This, then, is the third complete proof of the fundamental theorem of Chapter IV, all the more worthy of attention because the principles from which it is drawn are entirely different from those we used for the earlier proofs. .But from this same source, by the opposite route, let us derive a fourth. Namely, let ν be a prime of the form $4n \pm 1$, let p be any other prime, and let $\pm\nu$ be a quadratic residue of the prime p; we shall prove that p is a quadratic residue of ν.

Let p^m be the least power of p that is $\equiv 1 (\text{mod. } \nu)$. The elementary divisors of the function $\frac{x^\nu - 1}{x-1}$ modulo p will have degree m, so their total number will be $= \frac{\nu-1}{m}$. Now, since $\pm\nu Rp$, the congruence

$$xx + x \mp n \equiv 0 (\text{mod. } p)$$

will be solvable; let its roots be A, A'. Distribute the functions $x, xx, .. x^{\nu-1}$ into two classes, denoted by ξ, ξ'. We have

$$\xi + \xi' \equiv A + A' + (1 + x + xx + .. + x^{\nu-1})$$
$$\xi\xi' \equiv AA' + \lambda(1 + x + xx + .. + x^{\nu-1})$$

so that

$$(z - \xi)(z - \xi') - (z - A)(z - A')$$

will be divisible by every elementary divisor of the function $\frac{x^\nu - 1}{x-1}$. Each of these elementary divisors will therefore divide either $\xi - A$ and $\xi' - A'$, or $\xi - A'$ and $\xi' - A$. It follows (since A is not $\equiv A'$) that, if in place of x we substitute x^p, then ξ and ξ' do not change. For if ξ passed into ξ', and conversely, then $\xi - A$ and $\xi - A'$ would be divisible by the same prime function. Q. E. A. Finally, it follows that $\frac{\nu-1}{2}$ is divisible by m, or equivalently that $p^{\frac{\nu-1}{2}} - 1$ is divisible by ν. Thus p is a quadratic residue of ν. Q. E. D.

It is easy to derive all cases of the fundamental theorem from the two theorems.

367.

Although we have restricted ourselves here to the case where ν is prime, analogous theorems can also be determined without much difficulty when ν is composite; brevity does not now allow us to set this out more fully.

It is evident that similar observations can also be made about a greater number of periods. Thus, if $\frac{\nu-1}{m}$ is divisible by 3, that is, if p is a cubic residue of the prime ν, the equation that distributes the roots of the equation $x^\nu = 1$ into three periods, and whose determination a priori we taught in Chapter VI, will be solvable modulo p, and conversely. For example, the congruence $x^3 + xx - 2x - 1 \equiv 0$ can be solved modulo any prime of the form $7n \pm 1$, but cannot if it has another form.

It would not be difficult for us to enrich this chapter with many other observations, if the limits within which we must keep did not prevent it. For those who wish to go further, these principles can at least point the way.

368.

We say that a congruence $S \equiv 0$ has *equal* roots, or more generally equal divisors, if it is divisible by a power of some function.

Whether a given congruence has equal divisors is decided in the same way as in the theory of equations. Put

$$X \equiv \xi^m P$$

Clearly,

$$\frac{\mathrm{d}X}{\mathrm{d}x} \equiv \xi^{m-1}\left(mP\frac{\mathrm{d}\xi}{\mathrm{d}x} + \xi\frac{\mathrm{d}P}{\mathrm{d}x}\right)$$

so $\frac{\mathrm{d}X}{\mathrm{d}x}$ is divisible by ξ^{m-1}. More generally, let

$$X \equiv A^a B^b C^c \text{ etc.}$$

where A, B, C, etc., denote distinct prime functions. Then

$$\frac{\mathrm{d}X}{\mathrm{d}x} \equiv X\left(\frac{a\,\mathrm{d}A}{A\,\mathrm{d}x} + \frac{b\,\mathrm{d}B}{B\,\mathrm{d}x} + \frac{c\,\mathrm{d}C}{C\,\mathrm{d}x} + \text{etc.}\right)$$

It follows that, unless one of the numbers a, b, c, etc., is divisible by the modulus, $\frac{\mathrm{d}X}{\mathrm{d}x}$ is divisible by $A^{a-1}B^{b-1}C^{c-1}$, etc., but not by A^a, B^b, C^c, etc. Hence the following

THEOREM. *If the common divisor of greatest degree of the functions X and $\frac{\mathrm{d}X}{\mathrm{d}x}$ is ξ, all the prime factors of ξ will also occur in X, each with a multiplicity $+1$ greater than in ξ. Thus, if X and $\frac{\mathrm{d}X}{\mathrm{d}x}$ are coprime functions, X will have no equal factors.*

369.

Example I. We ask whether the function

$$x^5 + 3x^4 - 6x^3 + 3x - 4 \ldots (X)$$

has equal divisors modulo 17. We have

$$\frac{\mathrm{d}X}{\mathrm{d}x} \equiv 5x^4 - 5x^3 - xx + 3$$

From this we find that the functions X and $\frac{\mathrm{d}X}{\mathrm{d}x}$ are coprime, so X has no equal divisors.

Example II. Let

$$X \equiv x^5 + 6x^4 - 3x^3 - 4xx + 2x - 3 (\mathrm{mod.}\, 13)$$

Then

$$\frac{\mathrm{d}X}{\mathrm{d}x} \equiv 5x^4 - 2x^3 + 4xx + 5x + 2$$

The greatest common divisor of the functions X, $\frac{\mathrm{d}X}{\mathrm{d}x}$ is $\equiv 5xx + 7x + 7$, or, on multiplication by 8: $xx + 4x + 4$. Since this divisor is $\equiv (x+2)^2$, the function X is divisible by $(x+2)^3$, and the quotient (which is $xx + 11$) contains no further double divisor.

370. 371.

If, by the preceding sections, the function X has been expressed as $A^a B^b C^c$, etc., with A, B, C, etc., coprime and the numbers a, b, c, etc., unequal, the resolution can be carried further. Let X, then, be a function containing no further equal divisors. We saw above that $x^p - x$ is the product of all prime functions of degree one. Let ξ be the common divisor of greatest degree of the functions X and $x^p - x$; then ξ is the product of all divisors of X of degree one, and $\frac{X}{\xi}$ has no further divisors of this kind. If, however, the functions X and $x^p - x$ are found to be coprime, X will have no divisor of degree one, and hence the congruence $X \equiv 0$ will have no real roots. Further, since $x^{pp} - x$ is the product of all prime functions of degrees two and one, the common divisor of greatest degree of the functions $x^{pp} - x$ and $\frac{X}{\xi}$, namely ξ', contains all divisors of X of degree two. Continuing in this way, we see that X is thus resolved into factors ξ, ξ', ξ'', etc., which contain all divisors of degrees one, two, three, etc., respectively.

372.

But if a product of several prime functions of the same degree is given, the individual functions must be found by trial. This problem is closely analogous to that of finding the factors of composite numbers. Here, however, it is determined a priori whether the given function can still be split into factors.

Since here too the number of all possible factors is finite, we can use an aid similar to the one above. But we do not wish to dwell on this matter, for an experienced calculator who has thoroughly grasped the principles will readily find particular devices when needed.

We proceed to another topic, namely the consideration of congruences when the modulus is not a prime number, as we have always assumed up to now. Here the case where the modulus is a power of a prime number especially deserves attention, both in itself and because it is needed to remove certain doubts (§.§. . .).

373.

PROBLEM. *If the function X has been resolved modulo p into coprime factors ξ, ξ', ξ'', etc., resolve X modulo pp into similar factors Ξ, Ξ', Ξ'', etc., so that*

$$\xi \equiv \Xi, \quad \xi' \equiv \Xi', \quad \xi'' \equiv \Xi'', \text{ etc. } (\mathrm{mod.}\, p)$$

Solution. Let $X \equiv \xi\psi (\mathrm{mod.}\, p)$, or equivalently $X = \xi\psi + p\Sigma$. Put

$$\Xi = \xi + p\varphi, \quad \Psi = \psi + p\omega$$

Then

$$\Xi\Psi = X - p\Sigma + (\varphi\psi + \xi\omega)p + pp\varphi\omega$$

Thus, if $\Xi\Psi$ is to be $\equiv X (\mathrm{mod.}\, pp)$, $\varphi\psi + \xi\omega - \Sigma$ must necessarily be divisible by p. But since ψ and ξ are coprime functions modulo p, φ and ω can be determined so that this condition is fulfilled (§. 336), and moreover so that the degrees of φ and ω are respectively one less than the degrees of the functions ξ, ψ. Hence $X \equiv \Xi\Psi (\mathrm{mod.}\, pp)$. It is clear that Ψ can again be split in the same way into factors $\Xi'\Omega$, with the factor Ξ' being $\equiv \xi' (\mathrm{mod.}\, p)$, and so on, until finally

$$X \equiv \Xi\, \Xi'\, \Xi'' \text{ etc. } (\mathrm{mod.}\, pp). \quad \text{Q. E. Fac.}$$

374.

From this it is easy to prove that the function X can also be resolved into factors modulo p^3, p^4, etc. More generally, let

$$X \equiv PQ(\mathrm{mod.}\, p^m) \quad \text{or} \quad X = PQ + p^m R$$

and let the function P be coprime to Q modulo p. On putting

$$P' = P + Ap^m, \quad Q' = Q + Bp^m$$

we have

$$P'Q' = X - p^m R + (AQ + BP)p^m + ABp^{2m}$$

Hence, for any modulus p^ν (with ν being $> m$ and $< 2m + 1$), we have

$$P'Q' \equiv X, \quad \text{if} \quad R \equiv AQ + BP(\mathrm{mod.}\, p^{\nu - m})$$

From this we see that, if the function X has no equal divisors modulo p, it can be split into factors modulo p^k in the same way as modulo p. But if X has equal divisors, the matter becomes much more complicated and cannot even be fully dealt with by the preceding principles. Since we cannot communicate everything belonging to this subject, we shall consider only one case, which occurs very often and whose resolution is needed to settle certain doubts in what precedes. This is the case where only equal factors of degree one are considered. It can also be used directly to find the roots of congruences. We shall treat the matter in general on another occasion.

375.

Let, then, $X \equiv X'(x - a)^m(\mathrm{mod.}\, p)$, and let the function X' be coprime to $x - a$: we seek all divisors of degree one congruent to $x - a$ modulo p that divide X modulo pp, p^3, etc. (We assume that the function X is not divisible by $x - a$ without a modulus; otherwise $x - a$ would divide the function X modulo every modulus.) If we substitute $z + a$ for x, we obtain

$$Z \equiv Z'z^m(\mathrm{mod.}\, p) \quad \text{or} \quad Z = Z'z^m + pA$$

Now, if Z is divisible modulo pp by some divisor of the form $z + \alpha p$, then A must be of the form $zZ'' + pB$. Unless this is so, the investigation is already finished. Let us therefore put

$$Z \equiv Z'z^m + pZ''z(\mathrm{mod.}\, pp)$$
$$\text{or} \quad Z = Z'z^m + pZ''z + ppB$$

It is clear that Z is divisible by z and by every other divisor congruent to it modulo p.

To fix our attention, we put $m = 4$; it is easy to see that every other case can be treated in a similar way. Now, if Z is divisible modulo p^3 by some divisor of the form $z + \alpha p$, we have

$$0 \equiv -\alpha pp Z'' + ppB(\text{modd. } z + \alpha p, p^3)$$
$$\text{or} \quad \alpha Z'' \equiv B(\text{modd. } z, p)$$

There can now be three cases

1) If $Z'' \equiv 0(\text{modd. } z, p)$ and B is not $\equiv 0$, then clearly no value of α satisfies the congruence, and hence Z has, modulo p^3, no divisor of the form $z + \alpha p$. The investigation is therefore finished

2) If neither Z'' nor B is $\equiv 0(\text{modd. } z, p)$, then α has a unique value, namely

$$\alpha \equiv \frac{B}{Z''}(\text{modd. } z, p).$$

Thus there is a unique divisor $\equiv z + \alpha p(\text{mod. } pp)$ of Z modulo p^3, and we have

$$Z \equiv V(z + \alpha p) + p^3 W$$

Now let the divisor of $Z(\text{mod. } p^4)$ be $z + \alpha p + \beta pp$; then

$$0 \equiv$$

REMARKS ON THE ANALYSIS RESIDUORUM.

The two preceding treatises are taken from an extensive manuscript entitled Analysis Residuorum, which probably dates from 1797 or 1798; a complete revision of it later gave rise to the Disquisitiones Arithmeticae. The full title of Chapter Six reads:

Solution of the congruence $x^m - 1 \equiv 0$ and the equation $x^m - 1 = 0$; with elucidations on the theory of regular polygons.

The essential content of its second part (§§. 253–278) passed into the seventh section of the Disqq. Arithm.

Chapter Seven is also partly preserved: Various Applications of Certain Preceding Investigations (§§. 279–302). It is divided into the following subsections:

On Transformations of Common Fractions (§§. 279–281).

On the Conversion of Common Fractions into Decimals (§§. 282–292).

On the Solution of the Indeterminate Equation $xx = a + by$ (§§. 293–297).

On the Solution of the Indeterminate Equation $axx + byy = c$ (§§. 298–301).

On Finding the Divisors of Numbers (§. 302; the following sheets are missing).

All this was incorporated almost word for word into the sixth section of the Disqq. Arithm.

The two sections given here treat the subjects which, as appears from the preface and Articles 11, 44, 61, 62, 65, 84 of the Disqq. Arithm., were intended to form the eighth section of that work. It should, however, be noted that this plan was later changed again: among the manuscripts is a fragment headed Section Eight: A Fuller Consideration of Certain Investigations Concerning the Division of the Circle. This fragment begins with Art. 367 and was therefore intended to continue the Disqq. Arithm.; but the content of the few articles still preserved later passed into the treatise Summatio quarundam serierum singularium, and for this reason the fragment is excluded from the present edition.

In the preceding printing of the two parts of the Analysis Residuorum, the original text has essentially been retained faithfully, although its form cannot be called ready for printing; the following remarks indicate the most important alterations and add some elucidations.

§. 237. Compare Disqq. Arithm. artt. 61, 62.

§. 239. Compare Disqq. Arithm. artt. 53, 54, 65.

§. 241. If $n = 2^\nu$ and $\nu \geqq 3$, there is indeed no number ρ of the stated kind, but this does not substantially change the investigation as a whole.

§. 251. The difficulty noted here was presumably to be removed by introducing higher powers of p as moduli. Compare §§. 363, 372, 373.

§. 332. The assumption that the modulus is a prime number is retained through §. 372 inclusive.

§. 338. The incomplete citation may refer to Disqq. Arithm. art. 44.

§§. 344–346. Of the two proofs present in the manuscript, the first, introduced by the words "we now set about proving" and based on a closer examination of the expressions $(1^\alpha 2^\beta 3^\gamma \ldots)$, has here been omitted entirely in accordance with the author's own instruction ("The whole preceding proof, together with another proof of the preceding theorem which we intended to add, is to be suppressed, because we have discovered another that is infinitely simpler. It rests on this foundation"); moreover, in the printing above the second proof has been shortened by considering the expansion of $\frac{x\,\mathrm{d}P}{P\,\mathrm{d}x}$ instead of that of $\frac{x\,\mathrm{d}P}{\mathrm{d}x}$, thus also avoiding a reference in the original to the suppressed first proof.

§. 348. The expression radix prima is to be understood here in the same sense as radix propria in the treatise Summatio quarundam serierum singularum, art. 11. — The assertion that the coefficients $A'.B'\ldots$ of the expanded product are rational integers refers to Chapter Six, where, however, the theory of the equation $x^\tau - 1 = 0$ is treated only for the case where τ is prime; the form of the proof in §. 349 first leads to the following supplement. If the expanded product is put into the form (valid for all roots of the equation $\theta^\tau = 1$)

$$S = E + F\theta + \ldots + N\theta^{\tau-1}$$

then the coefficients $E, F \ldots N$ are polynomials in x with rational integer coefficients. Further, since the product remains unchanged when θ is replaced by θ^k, where k is any number coprime to τ, the same holds for the expression S. It follows without difficulty that all powers occurring in S of θ whose exponents s have one and the same greatest common divisor with τ must also have identical coefficients. Finally, since every sum of such powers θ^s is always an integer, it is evident that the expression S, and hence also the product in question, is a polynomial in x with integer coefficients, as was to be shown. This consideration also establishes the correctness of the remark at the end of the paragraph. Other reasons nevertheless suggest that the author already knew at that time the general theorem on the transformation of symmetric functions (Demonstratio nova altera theorematis omnem functionem etc. art. 4), from which the propositions above follow immediately.

§. 352. The notation $R \equiv S(\mathrm{mod.}\, P)$, or also $R \equiv S(\mathrm{mod.}\, P, p)$, means here and in what follows that the difference $R - S$ has, modulo p, the divisor P. — The incomplete citation may refer to Disqq. Arithm. art. 49.

§. 354. Multiplication by $x^\nu - 1$ shows that the sums of equal powers of the roots of the two equations $(P, \rho^{k\nu+t}) = 0$, $(P, \rho^t) = 0$ are congruent to one another (mod. p), and from this follows the congruence $(P, \rho^{k\nu+t}) \equiv (P, \rho^t)(\mathrm{mod.}\, p)$ as soon as $m < p$ (compare §. 244). But if $m \geqq p$, the coefficient of the power x^{m-p} in an equation can no longer be determined modulo p from the given power sums of its roots, because it carries the factor p in the formulas of NEWTON used for this purpose. Indeed, from the congruence of each pair of equal power sums of the roots of the equations $A = 0$, $B = 0$, one may in general infer only that $A \equiv \mathfrak{A}^p\mathfrak{C}$, $B = \mathfrak{B}^p\mathfrak{C}(\mathrm{mod.}\, p)$, where $\mathfrak{C}$ denotes the greatest common divisor of the two functions A, B modulo the prime number p, while $\mathfrak{A}$ and $\mathfrak{B}$ are entirely undetermined functions. Presumably the author derived the general validity of the proposition from the theory of the transformation of symmetric functions, and specifically from the following proposition: if, with respect to any modulus p, the difference $R(x) - S(x)$ is divisible by the function $P(x)$, and $a, b, c \ldots$ are the roots of the equation $P(x) = 0$, then the functions

$$(x - R(a))(x - R(b))(x - R(c)) \ldots \quad \text{and}$$
$$(x - S(a))(x - S(b))(x - S(c)) \ldots$$

are congruent to one another modulo p.

§. 355. It is shown in §. 368 that P and $\frac{\mathrm{d}P}{\mathrm{d}x}$ have no common divisor if P contains no factor more than once.

§§. 358, 359. The note placed below the text is taken from a separate sheet, which was probably intended to replace §. 359, already crossed out in the manuscript.

§. 360. An inaccuracy in the manuscript has been corrected in the statement of the theorem.

§. 361. Here the exponent $\frac{1}{k}$ in the notation $(\xi, \rho^{\frac{1}{k}})$ means any positive integer k' such that $kk' \equiv 1(\mathrm{mod.}\, \nu)$, where ν is the least positive integer for which $x^\nu - 1$ is divisible by ξ modulo p; it must be assumed here that ξ is not divisible by x modulo p, and also that k is coprime to ν. The correctness of the assertion that ξ' is divisible by $(\xi, \rho^{\frac{1}{k}})$ (mod. p) follows from §. 354.

§. 363. The concluding remark presumably concerns the introduction of moduli that are powers of the prime number p; compare §§. 251, 372, 373.

§. 367. The roots of the equation $x^3 + xx - 2x - 1 = 0$ are the periods of two terms into which the roots of the equation $\frac{x^7-1}{x-1} = 0$ divide. The same example also occurs on a separate sheet, where the principal result of §§. 362, 363 is stated under the title "the golden theorem".

§. 371. This paragraph was intended to contain an example, but it was not worked out.

R. Dedekind.

FURTHER DEVELOPMENT OF THE INVESTIGATIONS OF PURE EQUATIONS.

1.

The method by which, in Article 360 of the *Disquiss. Arithm.*, we showed how to solve the equation $x^n - 1 = 0$ forms a most fruitful and important theory, of which we could touch on only the first elements in that work. We therefore hope that mathematicians will welcome our taking up this subject here again, treating more fully what was only briefly sketched, in part with the proofs omitted, and pursuing in greater depth the developments made since that time.

The exponent n is assumed to be a prime number, and the number $n-1$ to be resolved into the factors $\alpha \times \beta \times \gamma$; further, we denote by g any primitive root modulo n. Let r denote an arbitrary root of the equation $x^n - 1 = 0$, and R an arbitrary root of the equation $x^\beta - 1 = 0$. Thus, denoting the circumference of the circle whose radius $= 1$ by P, and the imaginary quantity $\sqrt{-1}$ by i, all the roots of the equation $x^\beta - 1 = 0$, or all the values of R, will be represented by the formula

$$\cos \frac{kP}{\theta} + i \sin \frac{kP}{\theta}$$

where k ranges over the integers $0, 1, 2, 3 \ldots \beta - 1$. Further, all powers of any root R are evidently themselves roots, and if R is the root corresponding to a value of k coprime to β, all the powers $R^0, R, R^2, R^3 \ldots R^{\beta-1}$ are distinct from one another and thus exhaust the entire collection of roots. In this case we shall call R a *proper* root of the equation $x^\beta - 1 = 0$; conversely, a root R corresponding to a value of k that is not coprime to β will be called *improper*. It is readily seen that, if δ is the greatest common divisor of the numbers k and β, then $R^{\frac{\beta}{\delta}\cdot} = 1$, while all the powers $R^0, R, R^2, R^3 \ldots R^{\frac{\beta}{\delta}-1}$ are distinct from one another, so that R is a proper root of the equation $x^{\frac{\beta}{\delta}} - 1 = 0$. The same holds for the equation $x^n - 1 = 0$, but all its roots are necessarily proper except for the root 1.

2.

With these preliminaries, our investigation will chiefly concern functions of the following form, consisting of $\beta\gamma$ terms,

$$r + Rr^{g^\alpha} + R^2 r^{g^{2\alpha}} + R^3 r^{g^{3\alpha}} \ldots + R^{\beta\gamma-1} r^{g^{\alpha\beta\gamma-\alpha}}$$

which for brevity we shall denote by the symbol $[r, R]$. The individual terms of such an expression are products of powers of r and powers of R; the exponents of the former form a geometric progression, and those of the latter an arithmetic progression. The exponents

$$1, \quad g^\alpha, \quad g^{2\alpha}, \quad g^{3\alpha} \ldots\ldots g^{\alpha\beta\gamma-\alpha}$$

are all incongruent to one another modulo n, and hence those powers of r are distinct from one another; continued further, however, they would begin the same series again, since $g^{\alpha\beta\gamma} \equiv 1 (\mathrm{mod.}\, n)$, and hence $r^{g^{\alpha\beta\gamma}} = r$. The other factors,

$$1, \quad R, \quad R^2, \quad R^3 \ldots . R^{\beta\gamma-1}$$

form γ equal periods, since $R^\beta = 1$, $R^{\beta-1} = R$, etc. It follows that the function $[r, R]$ can also be represented as

$$\begin{aligned}
&r + r^{g^{\alpha\beta}} + r^{g^{2\alpha\beta}} \ldots\ldots + r^{g^{\alpha\beta\gamma-\alpha\beta}} \\
&+ R\big(r^{g^\alpha} + r^{g^{\alpha\beta+\alpha}} + r^{g^{2\alpha\beta+\alpha}} \ldots\ldots + r^{g^{\alpha\beta\gamma-\alpha\beta+\alpha}}\big) \\
&+ R^2\big(r^{g^{2\alpha}} + r^{g^{\alpha\beta+2\alpha}} + r^{g^{2\alpha\beta+2\alpha}} \ldots\ldots + r^{g^{\alpha\beta\gamma-\alpha\beta+2\alpha}}\big) \\
&+ \text{etc.} \\
&+ R^{\beta-1}\big(r^{g^{\alpha\beta-\alpha}} + r^{g^{2\alpha\beta-\alpha}} + r^{g^{3\alpha\beta-\alpha}} \ldots\ldots + r^{g^{\alpha\beta\gamma-\alpha}}\big).
\end{aligned}$$

or, introducing the notation of Article 343 of the Disq. Ar.,

$$[r, R] = (\gamma, 1) + R(\gamma, g^\alpha) + R^2(\gamma, g^{2\alpha}) \ldots + R^{\beta-1}(\gamma, g^{\alpha\beta-\alpha})$$

3.

If we take unity for the root r, we have

$$[1, R] = 1 + R + R^2 + R^3 \ldots + R^{\beta\gamma-1} = \gamma(1 + R + R^2 + R^3 \ldots + R^{\beta-1})$$

whose value will be $= \beta\gamma$ if R is also taken to be the root 1, but $= 0$ for every other value of R. Conversely, leaving r indeterminate and putting $R = 1$, we obtain $[r, 1] = r + r^{g^\alpha} + r^{g^{2\alpha}} + r^{g^{3\alpha}} \ldots + r^{g^{\alpha\beta\gamma-\alpha}}$, or, using the notation introduced in the Disq. Ar., $[r, 1] = (\beta\gamma, 1)$; that is, it will consist of a period of $\beta\gamma$ roots, one of which is r itself. Whenever $\alpha = 1$, this period will contain all the roots $r, r^2, r^3 \ldots r^{n-1}$, with only their order changed.

Note also the following relations, whose reason is evident:

$$[r, R] = R[r^{g^\alpha}, R] = R^2[r^{g^{2\alpha}}, R] \quad \text{or generally} \quad = R^k[r^{g^{\alpha k}}, R]$$

where k denotes any positive integer. It follows that the function $[r^m, R]$ either is $= [1, R]$, namely if m is divisible by n, or can be reduced in the remaining cases to the form $R^\mu[r^{g^\nu}, R]$, with $\nu < \alpha$. For if m is not divisible by n, it will be congruent modulo n to some power of g, whose exponent is conveniently expressed by $\operatorname{ind}. m$, as in the Disq. Ar.; thus, putting $\operatorname{ind}. m = \lambda\alpha + \nu$, which can evidently be done with $\nu < \alpha$, we obtain $[r^m, R] = [r^{g^{\lambda\alpha+\nu}}, R] = R^{-\lambda}[r^{g^\nu}, R]$. We must therefore set $\mu = -\lambda$ or, if you want a positive exponent, $\mu \equiv -\lambda (\operatorname{mod}. \beta)$.

4.

THEOREM. *If r', like r, denotes an arbitrary root of the equation $x^n - 1 = 0$, and R', like R, an arbitrary root of the equation $x^\beta - 1 = 0$, the product will be*

$$\begin{aligned}[r, R] \times [r', R'] = \\ [rr', RR'] + R[r^{g^\alpha} r', RR'] + R^2[r^{g^{2\alpha}} r', RR'] \\ + R^3[r^{g^{3\alpha}} r', RR'] \ldots + R^{\beta\gamma-1}[r^{g^{\alpha\beta\gamma-\alpha}} r', RR']\end{aligned}$$

Proof. By multiplying $[r, R]$ by the individual terms of $[r', R']$, the product can be represented in the form

$$\begin{aligned}&[r, R]r' + RR'[r^{g^\alpha}, R]r'^{g^\alpha} + R^2R'^2[r^{g^{2\alpha}}, R]r'^{g^{2\alpha}} \\ &+ R^3R'^3[r^{g^{3\alpha}}, R]r'^{g^{3\alpha}} \ldots \\ &+ R^{\beta\gamma-1}R'^{\beta\gamma-1}[r^{g^{\alpha\beta\gamma-\alpha}}, R]r'^{g^{\alpha\beta\gamma-\alpha}}\end{aligned}$$

Then collecting the first terms of the individual parts, after duly expanding them, gives $[rr', RR']$; similarly, collecting the second terms gives $R[r^{g^\alpha} r', RR']$, and so on, finally producing the stated form of the product. Q. E. D.

Moreover, simply interchanging r, R with r', R' shows that the same product can also be put in the form

$$[rr', RR'] + R'[rr'^{g^\alpha}, RR'] + R'^2[rr'^{g^{2\alpha}}, RR'] + R'^3[rr'^{g^{3\alpha}}, RR'] \ldots + R'^{\beta\gamma-1}[rr'^{g^{\alpha\beta\gamma-\alpha}}, RR']$$

It follows further that, if r'', r''', etc., also denote arbitrary roots of the equation $x^n - 1 = 0$, and R'', R''', etc., arbitrary roots of the equation $x^\beta - 1 = 0$, the product of the functions $[r, R], [r', R'], [r'', R''], [r''', R'''],$ etc., however many there are, will equal the sum

$$\Sigma R'^{k'} R''^{k''} R'''^{k'''} \text{ etc.}$$
$$[rr'^{g^{\alpha k'}} r''^{g^{\alpha k''}} r'''^{g^{\alpha k'''}} \text{ etc.},\ RR'R''R''' \text{ etc.}]$$

For k', k'', k''', etc., we substitute all the numbers $0, 1, 2, 3 \ldots \beta\gamma - 1$, combined with one another in every possible different way. This gives altogether $\beta^{\mu-1}\gamma^{\mu-1}$ terms, if μ denotes the number of those functions multiplied together.

5.

The formula by which we expressed the product of any number of functions in the preceding article is general and assumes no relation between the roots r, r', r'', r''', etc., or between R, R', R'', R''', etc. We can readily deduce that, if the roots r', r'', r''', etc., may be regarded as powers of r, and the roots R', R'', R''', etc., as powers of R, each term of the product will have the form $R^M[r^m, R^\lambda]$, where the exponent λ is the same for every term, namely $R^\lambda = RR'R''R'''$, etc. Thus, by what we noted in Article 3, such a product reduces to the following form:

$$A[1, R^\lambda] + B[r, R^\lambda] + B'[r^g, R^\lambda] + B''[r^{g^2}, R^\lambda] + B'''[r^{g^3}, R^\lambda] + \text{etc.} + B^{(\alpha-1)}[r^{g^{\alpha-1}}, R^\lambda]$$

where each of the coefficients A, B, B', B'', B''', etc., will have the form

$$h + h'R + h''R^2 + h'''R^3 + \text{etc.} + h^{(\beta-1)}R^{\beta-1}$$

and h, h', h'', h''', etc., denote fixed integers.

The simplest case is that in which we put $r = r' = r'' = r'''$, etc., and also $R = R' = R'' = R'''$, etc.; our product then becomes the power $[r, R]^\lambda$, which therefore always reduces to the form given above.

6.

Thus, putting $\lambda = \beta$, the power $[r, R]^\beta$ takes the form

$$\begin{aligned} &A[1,1] + B[r,1] + B'[r^g,1] + \text{etc.} + B^{\alpha-1)}[r^{g^{\alpha-1}},1] \\ &= \beta\gamma A + B(\beta\gamma,1) + B'(\beta\gamma,g) + B''(\beta\gamma,g^2) + \text{etc.} \\ &\quad + B^{(\alpha-1)}(\beta\gamma,g^{\alpha-1}) = \theta' \end{aligned}$$

If, then, not only the value of the root R (and hence the values of the coefficients A, B, B', etc.) but also the values of the individual sums of $\beta\gamma$ terms $(\beta\gamma,1),(\beta\gamma,g)$, etc., are assumed known, the value of θ' will be known at once, and from it $[r, R]$ can be found by the formula $\sqrt[\beta]{\theta'}$. This expression admits β distinct values, so it might seem doubtful which one should be taken; but it is easily shown that this choice is entirely arbitrary whenever R is a *proper* root of the equation $x^\beta - 1 = 0$. For in this case those β values of the radical expression $\sqrt[\beta]{\theta'}$ are evidently

$$[r,R], \quad [r^{g^\alpha},R], \quad [r^{g^{2\alpha}},R] \ldots . [r^{g^{\alpha\beta-\alpha}},R]$$

since, by Article 3, the powers of these functions with exponent β are equal to one another, while the functions themselves are proportional to the β distinct roots of the equation $x^\beta - 1 = 0$. But as long as only the sums of $\beta\gamma$ terms $(\beta\gamma,1),(\beta\gamma,g)$, etc., are known, the root r itself is determined only to the extent that it must lie in the collection $(\beta\gamma,1)$, and it remains arbitrary which member of this collection we take for r. These roots are $r, r^{g^\alpha}, r^{g^{2\alpha}}$, etc., and therefore we can also take any of the functions $[r,R],[r^{g^\alpha},R],[r^{g^{2\alpha}},R]$, etc., for $[r, R]$.

These conclusions would not hold if R were not a proper root of the equation $x^\beta - 1 = 0$. For if R is assumed to be a proper root of the equation $x^{\beta'} - 1 = 0$, with β' a divisor of β, it is readily seen that

$$[r,R] = [r^{g^{\alpha\beta'}},R], \quad [r^{g^\alpha},R] = [r^{g^{\alpha\beta'+\alpha}},R] \text{ etc.}$$

so that in the collection of β functions $[r,R],[r^{g^\alpha},R] \ldots [r^{g^{\alpha\beta-\alpha}},R]$ only β' distinct functions occur, and therefore among the values of the expression $\sqrt[\beta]{\theta'}$ no more than β' are admissible, while the remaining $\beta - \beta'$ are extraneous. It is readily seen, however, that in this case we need not go as far as the power with exponent β of the function $[r, R]$ since already the power $[r, R]^{\beta'}$ reduces to our form

$$\beta\gamma A + B(\beta\gamma,1) + B'(\beta\gamma,g) + B''(\beta\gamma,g^2) \text{ etc.}$$

Thus we shall obtain $[r, R]$ by an expression such as $\sqrt[\beta']{\theta}$, and it will make no difference which value of this expression we take.

7.

In the same way as $[r, R]$, we can also determine the functions $[r, R^2]$, $[r, R^3]$, etc., or in general $[r, R^k]$. For if substituting in θ', in place of R, the powers R^2, R^3, etc., R^k, is assumed to give the functions θ'', θ''', etc., $\theta^{(k)}$, then evidently $[r, R^2]^\beta = \theta''$, $[r, R^3]^\beta = \theta'''$, etc., and in general $[r, R^k]^\beta = \theta^{(k)}$; hence these functions too can be expressed by radicals, $[r, R^2] = \sqrt[\beta]{\theta''}$, etc. But it is not suitable to use these radical expressions when a quantity is to be expressed by a function of $[r, R]$, $[r, R^2]$, etc. For since their individual values are not fully determined, it would remain doubtful which values could be combined with one another; yet this is evidently by no means arbitrary. It is readily seen that, as soon as a definite value is taken for $[r, R]$, all of $[r, R^2]$, $[r, R^3]$, etc., must also acquire fully determined values, which the radical expressions do not indicate. We must therefore set these expressions aside and seek others by which $[r, R^2]$, $[r, R^3]$, etc., can be expressed *rationally* in terms of $[r, R]$ and known quantities. We readily accomplish this as follows.

By the theorem of Article 4 and what we showed in Article 5, the product $[r, R^k] \times [r, R]^{\beta-k}$ also reduces to a form such as

$$\beta\gamma A + B(\beta\gamma, 1) + B'(\beta\gamma, g) + B''(\beta\gamma, g^2) + \text{etc.} \\ + B^{(\alpha-1)}(\beta\gamma, g^{\alpha-1})$$

where A, B, B', B'', etc., are rational functions of R. Thus, putting the products

$$[r, R^2] \times [r, R]^{\beta-2} = \vartheta'' \\ [r, R^3] \times [r, R]^{\beta-3} = \vartheta''' \\ [r, R^4] \times [r, R]^{\beta-4} = \beta'''' \\ \text{etc.}$$

the quantities $\vartheta'', \vartheta''', \vartheta''''$, etc., can also be determined rationally, and

$$[r, R^2] = \frac{\vartheta''}{\theta'}[r, R]^2 \\ [r, R^3] = \frac{\vartheta'''}{\theta'}[r, R]^3 \\ [r, R^4] = \frac{\vartheta''''}{\theta'}[r, R]^4 \\ \text{etc.}$$

These expressions therefore give the values of the functions $[r, R^2]$, $[r, R^3]$, etc., rationally, provided that we do not have $[r.R] = 0$; if this equality held, they would become indeterminate. But we can rigorously prove that $[r, R] = 0$ can never occur whenever r denotes a root different from 1, although we must reserve the presentation of this proof for another occasion, so as not to become too lengthy here.

8.

What we set out in the preceding articles is useful when the aim is to descend from periods of $\beta\gamma$ terms to periods of γ terms. For it is readily seen that, if R denotes a proper root, we have

$$\begin{aligned}
\beta(\gamma,1) &= (\beta\gamma,1) + [r,R] + [r,R^2] + [r,R^3] + \text{etc.} + [r,R^{\beta-1}] \\
\beta(\gamma,g^\alpha) &= (\beta\gamma,1) + R^{\beta-1}[r,R] + R^{\beta-2}[r,R^2] \\
&\quad + R^{\beta-3}[r,R^3] + \text{etc.} + R[r,R^{\beta-1}] \\
\beta(\gamma,g^{2\alpha}) &= (\beta\gamma,1) + R^{2\beta-2}[r,R] + R^{2\beta-4}[r,R^2] \\
&\quad + R^{2\beta-6}[r,R^3] + \text{etc.} + R^2[r,R^{\beta-1}] \\
&\text{etc.}
\end{aligned}$$

If here, for the individual $[r,R]$, $[r,R^2]$, etc., we took the radical expressions $\sqrt[\beta]{\theta'}$, $\sqrt[\beta]{\theta''}$, etc., the value of each series would be uncertain among $\beta^{\beta-1}$ values; by contrast, when the rational expressions for $[r,R^2]$, etc., are taken, it will be subject to no ambiguity other than that which is unavoidable by the nature of the matter. This observation seems to have escaped the attention of the illustrious LAGRANGE, who thought he had *simplified* our method given in Article 360 of the Disquis. arithm., where we had deliberately passed over radical expressions and proposed only rational ones, by substituting the former for the latter (Traité de la résolution numérique des équations; edition 2, p. 311).

Moreover, it hardly needs mentioning here that, as soon as the values of the periods $(\gamma,1)$, (γ,g^α), etc., or of just one of them, have been found, the values of all the remaining periods of γ terms can be deduced rationally from them. Thus the descent from periods of $\beta\gamma$ terms to periods of γ terms requires solving the equations $x^\beta = 1$, $x^\beta = \theta'$, and the remaining operations are performed rationally.

9.

All this had already been treated in almost the same way in the Disquis. Ar.; some things, however, had been added there with their proof omitted, and we think it appropriate to supply that proof here. We stated there that finding the value of the radical quantity $\sqrt[\beta]{\theta'}$, which, since θ' is an imaginary quantity, seems to require dividing both a ratio and an angle into β parts, depends on the latter alone, and that the former can always be reduced to extracting a single square root. We prove this as follows.

Denoting the imaginary quantity $\sqrt{-1}$ by i, as above, and writing $\theta' = P + iQ$ and $\sqrt[\beta]{\theta'} = p + iq$, using any one value of the radical and taking P, Q, p, q to be real, it is known that, if the positive quantities E, e and the angles F, f are determined so that $P = E\cos F$, $Q = E\sin F$, $p = e\cos f$, $q = e\sin f$, then $e = \sqrt[\beta]{E}$, and f equals one of the angles

$$\frac{1}{\beta}F, \quad \frac{1}{\beta}(F + 360^0), \quad \frac{1}{\beta}(F + 720^0) \ldots . \frac{1}{\beta}(F + (\beta - 1)360^0)$$

Thus f· ger is determined by division of the angle F into β parts, but we can avoid extracting the root $\sqrt[\beta]{E}$ as follows. Every product $r^k R^K$ has the same real part as $r^{-k}R^{-K}$, while the imaginary parts containing the factor i in these products are equal and opposite. It follows at once that $[r^{-1}, R^{-1}] = p - iq = e(\cos f - i\sin f)$, and hence

$$[r, R] \times [r^{-1}, R^{-1}] = e^2$$

But by the theorem of Article 4 this product becomes

$$\begin{aligned} &= [1,1] + R[r^{g^\alpha - 1}, 1] + R^2[r^{g^{2\alpha}-1}, 1] + \text{etc.} \\ &\quad + R^{\beta\gamma-1}[r^{g^{\alpha\beta\gamma-\alpha}-1}, 1] \\ &= \beta\gamma + R(\beta\gamma, g^\alpha - 1) + R^2(\beta\gamma, g^{2\alpha} - 1) + \text{etc.} \\ &\quad + R^{\beta\gamma-1}(\beta\gamma, g^{\alpha\beta\gamma-\alpha} - 1) \end{aligned}$$

This quantity can be determined if R and all the periods of $\beta\gamma$ terms are assumed known. Determining e therefore requires only the extraction of a square root.

In the special case $\alpha = 1$, the individual periods $(\beta\gamma, g^\alpha - 1)$, $(\beta\gamma, g^{2\alpha} - 1)$, etc., are evidently $= r + r^2 + r^3 + r^4 + \text{etc.} + r^{n-1}$, and hence

$$\begin{aligned} ee &= \beta\gamma + (R + R^2 + R^3 + \text{etc.} + R^{\beta\gamma-1})(r + r^2 + r^3 + \text{etc.} + r^{n-1}) \\ &= \beta\gamma + 1 = n \end{aligned}$$

provided that r and R are assumed to denote roots different from 1, and therefore we always have $e = \sqrt{n}$ (Disq. arithm., end of Article 360).

10.

So far we have conducted our investigation in the greatest generality, to cover all values of the numbers α, β, γ; from now on, however, we shall pass to the more restricted case $\alpha = 1$, which will open the way to most fruitful and elegant investigations. Thus the symbol $[r, R]$ will denote the function

$$r + Rr^g + R^2 r^{g^2} + R^3 r^{g^3} + \text{etc.} + R^{n-2} r^{g^{n-2}}$$

where n is a prime number, r an arbitrary root of the equation $x^n - 1 = 0$ (the root 1 not excluded), R an arbitrary root of the equation $x^\beta - 1 = 0$, with β denoting a given divisor of $n - 1$, and finally g an integer that is a fixed primitive root modulo n. For brevity we shall also write

$$1 + r + r^2 + r^3 + \text{etc.} + r^{n-1} = s$$
$$1 + R + R^2 + R^3 + \text{etc.} + R^{n-2} = S$$

from which it is evident that s becomes $= n$ for $r = 1$, but $s = 0$ for every other value of r, and similarly $S = n - 1$ for $R = 1$, but $S = 0$ for every other value of R.

By Article 3 we therefore have $[1, R] = S$, $[r, 1] = s - 1$; further, for every integer value of m not divisible by n, $[r^m, R] = R^{-\operatorname{ind} m}[r, R]$, or more generally $[r^m, R^M] = R^{-M \operatorname{ind} m}[r, R^M]$, where $\operatorname{ind} m$ is the exponent of the power of g congruent modulo n to m. Applying this transformation to what we showed in Article 5, we find that a product of two or more functions such as $[r^h, R^H]$ reduces to the form

$$A[1, R^\lambda] + B[r, R^\lambda]$$

where A and B are rational functions of R with integer coefficients, and λ is the sum of all the values of H. It will be of great importance to put such transformations into an efficient algorithm; for this purpose we must first consider more closely the nature of the product of two functions.

11.

The product $[r, R^{\mu}] \times [r, R^{\nu}]$ becomes, by the theorem of Article 4, =

$$[r^2, R^{\mu+\nu}] + R^{\mu}[r^{g+1}, R^{\mu+\nu}] + R^{2\mu}[r^{g^2+1}, R^{\mu+\nu}] + R^{3\mu}[r^{g^3+1}, R^{\mu+\nu}] + \text{etc.} + R^{(n-2)\mu}[r^{g^{n-2}+1}, R^{\mu+\nu}]$$

Among the $n-1$ exponents $2, g+1, g^2+1, g^3+1$, etc., $g^{n-2}+1$, exactly one is divisible by n, namely $g^{\frac{1}{2}(n-1)}+1$, so that the corresponding term of our sum will be $R^{\frac{1}{2}(n-1)\mu}[1, R^{\mu+\nu}]$. This term will be $= 0$ whenever we do not have $R^{\mu+\nu} = 1$, and $= (n-1)R^{\frac{1}{2}(n-1)\mu} = \pm(n-1)$ for $R^{\mu+\nu} = 1$. The remaining parts of our sum, whose sum we shall put $= \Omega$, are transformed as follows:

$$\begin{aligned}
[r^2, R^{\mu+\nu}] &= R^{-(\mu+\nu)\operatorname{ind} 2}[r, R^{\mu+\nu}] \\
R^{\mu}[r^{g+1}, R^{\mu+\nu}] &= R^{\mu-(\mu+\nu)\operatorname{ind}(g+1)}[r, R^{\mu+\nu}] \\
R^{2\mu}[r^{g^2+1}, R^{\mu+\nu}] &= R^{2\mu-(\mu+\nu)\operatorname{ind}(g^2+1)}[r, R^{\mu+\nu}] \\
R^{3\mu}[r^{g^3+1}, R^{\mu+\nu}] &= R^{3\mu-(\mu+\nu)\operatorname{ind}(g^3+1)}[r.R^{\mu+\nu}] \\
&\text{etc.}
\end{aligned}$$

From this we obtain

$$\text{I.} \qquad \Omega = [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind} x - (\mu+\nu)\operatorname{ind}(x+1)}$$

if for x we successively substitute the values $1, g, g^2, g^3 \ldots . g^{n-2}$, except for $g^{\frac{1}{2}(n-1)}$, or, which evidently amounts to the same thing, if for x we substitute the values $1, 2, 3, 4 \ldots . n-2$, since these values are congruent to those modulo n, though their order is changed.

Taking the integer y inverse to x modulo n, that is, determined so that $xy \equiv 1 \pmod{n}$, we have $\operatorname{ind} x \equiv -\operatorname{ind} y \pmod{n-1}$ and $\operatorname{ind}(x+1) + \operatorname{ind} y \equiv \operatorname{ind}(xy+y) \equiv \operatorname{ind}(1+y) \pmod{n-1}$; hence

$$\begin{aligned}
\mu \operatorname{ind} x - (\mu+\nu)\operatorname{ind}(x+1) &\equiv -\mu \operatorname{ind} y - (\mu+\nu)\{\operatorname{ind}(y+1) - \operatorname{ind} y\} \\
&\equiv \nu \operatorname{ind} y - (\mu+\nu)\operatorname{ind}(y+1)
\end{aligned}$$

Therefore, since the inverses of $1, 2, 3 \ldots n-2$ coincide with these same numbers, with only their order changed, we also have

$$\text{II.} \qquad \Omega = [r, R^{\mu+\nu}] \times \Sigma R^{\nu \operatorname{ind} y - (\mu+\nu)\operatorname{ind}(y+1)}$$

substituting for y successively the numbers $1, 2, 3 \ldots n-2$. The same formula follows directly from I, since the numbers μ, ν can evidently be interchanged.

Finally, taking the integer z inverse to $x+1$ modulo n, or $xz+z \equiv 1 \pmod{n}$, we have $\operatorname{ind}(1-z) \equiv \operatorname{ind} x + \operatorname{ind} z \pmod{n-1}$, $\operatorname{ind}(x+1) \equiv -\operatorname{ind} z \pmod{n-1}$, and hence

$$\begin{aligned}\mu \operatorname{ind} x - (\mu+\nu)\operatorname{ind}(x+1) &\equiv \mu(\operatorname{ind}(1-z) - \operatorname{ind} z) + (\mu+\nu)\operatorname{ind} z \\ &\equiv \mu \operatorname{ind}(1-z) + \nu \operatorname{ind} z\end{aligned}$$

Thus, as x runs through the values $1, 2, 3 \ldots n-2$, the number z must run through the values $2, 3, 4 \ldots n-1$, though in another order, and we obtain the third expression

$$\text{III.} \qquad \Omega = [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind}(1-z)+\nu \operatorname{ind} z}$$

substituting for z successively the values $2, 3, 4 \ldots n-1$, or, if you prefer,

$$\begin{aligned}\text{IV.} \qquad \Omega &= [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind}(n+1-z)+\nu \operatorname{ind} z} \\ &= [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind} z+\nu \operatorname{ind}(n+1-z)}\end{aligned}$$

Since $\operatorname{ind}(1-z) = \frac{1}{2}(n-1) + \operatorname{ind}(z-1)$, our product can also be represented as follows:

$$\begin{aligned}[r, R^{\mu}] \times [r, R^{\nu}] &= R^{\frac{1}{2}(n-1)\mu}\{[1, R^{\mu+\nu}] + [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind}(z-1)+\nu \operatorname{ind} z}\} \\ &= R^{\frac{1}{2}(n-1)\nu}\{[1, R^{\mu+\nu}] + [r, R^{\mu+\nu}] \times \Sigma R^{\mu \operatorname{ind} z+\nu \operatorname{ind}(z-1)}\}\end{aligned}$$

where z is always understood to take the values $2, 3, 4 \ldots . n-1$.

Moreover, in all these formulas the numbers

$$\mu \operatorname{ind} x - (\mu+\nu)\operatorname{ind}(x+1), \quad \nu \operatorname{ind} y - (\mu+\nu)\operatorname{ind}(y+1), \quad \mu \operatorname{ind}(1-z) + \nu \operatorname{ind} z$$

etc., can evidently be replaced by their least residues modulo β.

If $\mu + \nu \equiv 0 \pmod{\beta}$, we have

$$\begin{aligned}[r, R^{\mu}][r, R^{\nu}] = (n-1)&R^{\frac{1}{2}(n-1)\mu} \\ &+ (r + r^2 + r^3 + \cdots + r^{n-1}) \\ &\times (1 + R^{\mu} + R^{2\mu} + R^{3\mu} + \cdots + R^{(n-2)\mu} - R^{\frac{1}{2}(n-1)\mu})\end{aligned}$$

12.

The product $[1, R^\mu] \times [r, R^\nu]$ becomes, by the theorem of Article 4,

$$= [r, R^{\mu+\nu}] + R^\mu [r, R^{\mu+\nu}] + R^{2\mu} [r, R^{\mu+\nu}] + \text{etc.} + R^{(n-2)\mu} [r, R^{\mu+\nu}]$$
$$= [r, R^{\mu+\nu}] \times (1 + R^\mu + R^{2\mu} + R^{3\mu} + \text{etc.} + R^{(n-2)\mu})$$
$$= [r, R^{\mu+\nu}] \times \frac{n-1}{\beta} (1 + R^\mu + R^{2\mu} + R^{3\mu} + \text{etc.} + R^{(\beta-1)\mu})$$

From this the product $[1, R^\mu] \times [1, R^\nu]$ expands into

$$\frac{n-1}{\beta} [1, R^{\mu+\nu}] \times (1 + R^\mu + R^{2\mu} + R^{3\mu} + \text{etc.} + R^{(\beta-1)\mu})$$

The general product $[r^m, R^\mu] \,.\, [r^{m'}, R^{\mu'}]$ can now be found without difficulty. For $[r^m, R^\mu] = R^{-\mu \operatorname{ind} m} [r, R^\mu]$ when the value of m is not divisible by n, and $= [1, R^\mu]$ when it is divisible; and since a similar transformation holds for the other factor $[n^{m'}, R^{\mu'}]$, the multiplication reduces either to the problem of the preceding article or to the cases considered in this article.

13.

Now that we have shown how to expand a product of two factors, expanding a product of more factors will present no difficulty. Once the product $[r, R^\mu] \times [r, R^\nu]$ has been reduced to the form $A[1, R^{\mu+\nu}] + B[r, R^{\mu+\nu}]$, it is evident that, on introducing a third factor $[r, R^\pi]$, the product becomes $= C[1, R^{\mu+\nu+\pi}] + D[r, R^{\mu+\nu+\pi}]$ if we put

$$[r, R^{\mu+\nu}][r, R^\pi] = c[1, R^{\mu+\nu+\pi}] + d[r, R^{\mu+\nu+\pi}]$$

and

$$C = Bc$$
$$D = Bd + A\{1 + R^{\mu+\nu} + R^{2\mu+2\nu} + \text{etc.} + R^{(n-2)(\mu+\nu)}\}$$

Thus $[r, R]^\lambda$ can easily be reduced to the form $A[1, R^\lambda] + B[r, R^\lambda]$.

As an example, we shall expand the powers of the function $[r, R]$ for $n = 11$, $\beta = 5$, setting $g = 2$. The following indices then correspond to the numbers:

numbers	1.2.3.4.5.6.7.8.9.10
indices	0.1.8.2.4.9.7.3.6. 5

Thus, to expand the square $[r, R]^2$ by formula I of Article 11, we have

$$\mu = 1, \quad \nu = 1$$

values of x ...	1. 2.3.4. 5. 6.7. 8. 9
$\operatorname{ind} x$ ...	0. 1.8.2. 4. 9.7. 3. 6
$2 \operatorname{ind}(x+1)$ ...	2.16.4.8.18.14.6.12.10
Least residues of	$\operatorname{ind} x - 2 \operatorname{ind}(x+1)$
modulo 5 ...	3. 0.4.4. 1. 0.1. 1. 1

from which we deduce

$$\Omega = [r, R^2] \times \{2 + 4R + R^3 + 2R^4\}$$

and

$$1^0. \qquad [r, R]^2 = [1, R^2] + [r, R^2] \times \{2 + 4R + R^3 + 2R^4\}$$

The same expression follows from formula III of Article 11, namely

values of z ...	2.3.4.5.6.7.8.9. 10
$\operatorname{ind} z$ ...	1.8.2.4.9.7.3.6. 5
$\operatorname{ind}(n+1-z)$ ...	5.6.3.7.9.4.2.8. 1
least residues of	$\operatorname{ind} z + \operatorname{ind}(n+1-z)$
modulo 5 ...	1.4.0.1.3.1.0.4. 1

In exactly the same way we find

$$2^0. \qquad [r, R^2] \,.\, [r, R] = [1, R^3] + [r, R^3] \times \{2 + R + 4R^2 + 2R^3\}$$
$$3^0. \qquad [r, R^3] \,.\, [r, R] = [1, R^4] + [r, R^4] \times \{2 + 4R + R^3 + 2R^4\}$$

Finally, we obtain

$$4^0. \qquad [r, R^4] \,.\, [r, R] = [1, 1] + [r, 1] \times \{1 + 2R + 2R^2 + 2R^3 + 2R^4\}$$

Multiplying equation 1^0 by $[r, R]$ and substituting for $[r, R^2] \,.\, [r, R]$ its value from 2^0, and using

$$[1, R^2] \,.\, [r, R] = [r, R^3] \,.\, \{2 + 2R + 2R^2 + 2R^3 + 2R^4\}$$

we deduce

$$\begin{aligned}[r, R]^3 = {} & [1, R^3] \times \{2 + 4R + R^3 + 2R^4\} \\ & + [r, R^3] \times \{12 + 22R + 18R^2 + 24R^3 + 15R^4\}\end{aligned}$$

and in the same way

$$\begin{aligned}[r, R]^4 = {} & [1, R^4] \times \{12 + 22R + 18R^2 + 24R^3 + 15R^4\} \\ & + [r, R^4] \times \{164 + 170R + 205R^2 + 180R^3 + 190R^4\} \\ [r, R]^5 = {} & [1, 1] \times \{164 + 170R + 205R^2 + 180R^3 + 190R^4\} \\ & + [r, 1] \times \{1836 + 1830R + 1795R^2 + 1820R^3 + 1810R^4\} \\ = {} & 1640 + 1700R + 2050R^2 + 1800R^3 + 1900R^4 \\ & + (1836 + 1830R + 1795R^2 + 1820R^3 + 1810R^4)(s - 1) \\ = {} & 918Ss - 98S - (6R + 41R^2 + 16R^3 + 26R^4)s \\ & + 66R + 451R^2 + 176R^3 + 286R^4\end{aligned}$$

14.

The calculation in the preceding articles' was carried out so that it could be extended to all values of r and R. It becomes considerably shorter if we regard R from the outset as a proper root of the equation $x^\beta - 1 = 0$. Under this assumption the product $[r, R^\mu] \times [r, R^\nu]$ reduces to the form $B[r, R^{\mu+\nu}]$ whenever $\mu + \nu$ is not divisible by β; but when $\mu + \nu$ is divisible by β, that product becomes $= (n-1)R^{\frac{1}{2}(n-1)\mu} + [r, 1]\Sigma R^{\mu \operatorname{ind} x}$, substituting for $\operatorname{ind} x$ all the numbers $0, 1, 2, 3 \ldots . n-2$ except $\frac{1}{2}(n-1)$. It follows easily that in this case, if μ and therefore also ν are not divisible by β, we have

$$[r, R^\mu] \,.\, [r, R^\nu] = R^{\frac{1}{2}(n-1)\mu}\{n - 1 - [r, 1]\}$$

and hence $= 0$ for $r = 1$, and $= nR^{\frac{1}{2}(n-1)\mu}$ for every other value of r. Moreover, since $R^{\frac{1}{2}(n-1)\mu}$ becomes $= +1$ or $= -1$ according as $\frac{n-1}{\beta} \,.\, \mu$ is even or odd, our product becomes $= n$ in the former case and $= -n$ in the latter.

It further follows that we can put

$$\begin{aligned} [r, R]^2 &= A'[r, R^2] \\ [r, R^2] \,.\, [r, R] &= A''[r, R^3] \\ [r, R^3] \,.\, [r, R] &= A'''[r, R^4] \end{aligned}$$

and so on, up to

$$[r, R^{\beta-2}] \,.\, [r, R] = A^{(\beta-2)}[r, R^{\beta-1}]$$

from which we have

$$\begin{aligned} [r, R]^2 &= A'[r, R^2] \\ [r, R]^3 &= A'A''[r, R^3] \\ [r, R]^4 &= A'A''A'''[r, R^4] \end{aligned}$$

and so on. Finally,

$$[r, R]^\beta = \pm nA'A''A''' \ldots A^{(\beta-2)}$$

where the upper or lower sign is to be taken according as $\frac{n-1}{\beta}$ is even or odd.

It is therefore evident that, once the value of $[r, R]$ has been found, the remaining functions

$$[r, R^2] = \frac{[r, R]^2}{A'}, \quad [r, R^3] = \frac{[r, R]^3}{A'A''} \quad \text{etc.}$$

can be determined here much more readily than in the cases where the condition $\alpha = 1$ does not hold, as we already noted in the *Disq. Ar.* (Article 360, III). A fuller consideration of the nature of the functions A', A'', etc., makes these operations still easier.

15.

In Article 9 we showed that the value of the function $[r, R]$ can be reduced to the form $\sqrt{n}(\cos f + i \sin f)$, and in the same way the functions $[r, R^2]$, $[r, R^3]$, etc., up to $[r, R^{\beta-1}]$, can be reduced to a similar form. Let us put

$$\begin{aligned} [r, R] &= \sqrt{n}(\cos f' + i \sin f') \\ [r, R^2] &= \sqrt{n}(\cos f'' + i \sin f'') \\ [r, R^3] &= \sqrt{n}(\cos f''' + i \sin f''') \\ &\text{etc.} \end{aligned}$$

Then

$$\begin{aligned} A' &= \sqrt{n}(\cos(2f' - f'') + i \sin(2f' - f'')) \\ A'' &= \sqrt{n}(\cos(f' + f'' - f''') + i \sin(f' + f'' - f''')) \\ A''' &= \sqrt{n}(\cos(f' + f''' - f'''') + i \sin(f' + f''' - f'''')) \\ &\text{etc.} \end{aligned}$$

It is evident from this that, if the functions A', A'', A''', etc., are reduced to the forms

$$\begin{aligned} A' &= a'(\cos b' + i \sin b') \\ A'' &= a''(\cos b'' + i \sin b'') \\ A''' &= a'''(\cos b''' + i \sin b''') \\ &\text{etc.} \end{aligned}$$

with all of a', a'', a''', etc., positive, we shall have

$$\begin{aligned} a' &= a'' = a''' \text{ etc.} = \sqrt{n} \\ f' &= \frac{1}{\beta}(b' + b'' + b''' + \text{etc.} + b^{(\beta-2)}) \end{aligned}$$

if $\frac{n-1}{\beta}$ is even, or

$$f' = \frac{1}{\beta}(180^0 + b' + b'' + \text{etc.} + b^{(\beta-2)})$$

if $\frac{n-1}{\beta}$ is odd, and then

$$\begin{aligned} [r, R] &= \sqrt{n}(\cos f' + i \sin f') \\ [r, R^2] &= \sqrt{n}(\cos(2f' - b') + i \sin(2f' - b')) \\ [r, R^3] &= \sqrt{n}(\cos(3f' - b' - b'') + i \sin(3f' - b' - b'')) \\ &\text{etc.} \end{aligned}$$

Finally, by the formulas of Article 8,

$$\left(\frac{n-1}{\beta},1\right) = -\frac{1}{\beta} + \frac{\sqrt{n}}{\beta}\{\cos f'' + \cos(2f' - b') + \cos(3f' - b' - b'') + \text{etc.}$$
$$+ \cos((\beta-1)f' - b' - b'' - b''' - \text{etc.} - b^{(\beta-2)})\}$$
$$+ \frac{i\sqrt{n}}{\beta}\{\sin f' + \sin(2f' - b') + \sin(3f' - b' - b'') + \text{etc.}\}$$

and similarly we obtain the values of the functions $\left(\frac{n-1}{\beta},g\right)$, $\left(\frac{n-1}{\beta},g^2\right)$, $\left(\frac{n-1}{\beta},g^3\right)$, etc., if in this formula f' is replaced respectively by $f' - \frac{360^0 k}{\beta}$, $f' - 2\frac{360^0 k}{\beta}$, $f' - 3\frac{360^0 k}{\beta}$, etc., assuming $R = \cos\frac{360^0 k}{\beta} + i\sin\frac{360^0 k}{\beta}$.

16.

A further simplification follows from the observation below. Since by Article 14 we have

$$\pm[r,R][r,R^{\beta-1}] = [r,R^2][r,R^{\beta-2}] = \pm[r,R^3][r,R^{\beta-3}] \text{ etc.} = n$$

taking [in the first, third, etc., product] the upper or lower sign according as $\frac{n-1}{\beta}$ is even or odd, in the former case we must have

$$\cos(f' + f^{(\beta-1)}) = \cos(f'' + f^{(\beta-2)}) = \cos(f''' + f^{(\beta-3)}) \text{ etc.} = 1$$

in the latter

$$-\cos(f' + f^{(\beta-1)}) = \cos(f'' + f^{(\beta-2)}) = -\cos(f''' + f^{(\beta-3)}) \text{ etc.} = 1$$

and in both cases

$$\sin(f' + f^{(\beta-1)}) = \sin(f'' + f^{(\beta-2)}) = \sin(f''' + f^{(\beta-3)}) \text{ etc.} = 0$$

We may therefore put, in the former case,

$$f^{(\beta-1)} = -f', \quad f^{(\beta-2)} = -f'', \quad f^{(\beta-3)} = -f''' \quad \text{etc.}$$

in the latter

$$f^{(\beta-1)} = 180^0 - f', \quad f^{(\beta-2)} = -f'', \quad f^{(\beta-3)} = 180^0 - f''' \quad \text{etc.}$$

From this it follows that, in the former case,

$$b^{(\beta-2)} = b', \quad b^{(\beta-3)} = b'', \quad b^{(\beta-4)} = b''' \quad \text{etc.}$$
$$A^{(\beta-2)} = A', \quad A^{(\beta-3)} = A'', \quad A^{(\beta-4)} = A''' \quad \text{etc.}$$

and in the latter

$$b^{(\beta-2)} = b' - 180^0, \quad b^{(\beta-3)} = b'' + 180^0, \quad b^{(\beta-4)} = b''' - 180^0 \quad \text{etc.}$$
$$A^{(\beta-2)} = -A', \qquad A^{(\beta-3)} = -A'', \qquad A^{(\beta-4)} = -A''' \quad \text{etc.}$$

so that the number of functions A', A'', A''', etc., is reduced by half. It further follows that, in the former case,

$$f' = \frac{1}{\beta}(2b' + 2b'' + \text{etc.} + 2b^{(\frac{1}{2}\beta-1)})$$

$$\begin{aligned}\left(\frac{n-1}{\beta}, 1\right) = -\frac{1}{\beta} + \frac{\sqrt{n}}{\beta}\{&2\cos f' + 2\cos(2f' - b') \\ &+ 2\cos(3f' - b' - b'') + \text{etc.} \\ &+ 2\cos((\tfrac{1}{2}\beta - 1)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{8}\beta-2)}) \\ &+ \cos(\tfrac{1}{2}\beta f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-1)})\}\end{aligned}$$

(where the last term is evidently $= \cos 0 = 1$) or

$$f' = \frac{1}{\beta}(2b' + 2b'' + \text{etc.} + 2b^{(\frac{1}{2}(\beta-3))} + b^{(\frac{1}{2}(\beta-1))})$$

$$\begin{aligned}\left(\frac{n-1}{\beta}, 1\right) = -\frac{1}{\beta} + \frac{\sqrt{n}}{\beta}\{&2\cos f' + 2\cos(2f' - b') \\ &+ 2\cos(3f' - b' - b'') + \text{etc.} \\ &+ 2\cos(\tfrac{1}{2}(\beta - 1)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}(\beta-3))})\}\end{aligned}$$

according as β is even or odd; and in the latter case,

$$f' = \frac{1}{\beta}(2b' + 2b'' + \text{etc.} + 2b^{(\frac{1}{2}\beta-1)})$$

$$\begin{aligned}\left(\frac{n-1}{\beta}, 1\right) = -\frac{1}{\beta} + \frac{\sqrt{n}}{\beta}\{&2\cos(2f' - b') + 2\cos(4f' - b' - b'' - b''') + \text{etc.} \\ &+ 2\cos((\tfrac{1}{2}\beta - 2)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-3)}) \\ &+ \cos(\tfrac{1}{2}\beta f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-1)})\} \\ + i\frac{\sqrt{n}}{\beta}\{&2\sin f' + 2\sin(3f' - b' - b'') + \text{etc.} \\ &+ 2\sin((\tfrac{1}{2}\beta - 1)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-2)})\}\end{aligned}$$

or

$$f' = \frac{1}{\beta}(2b' + 2b'' + \text{etc.} + 2b^{(\frac{1}{2}\beta-1)} + 180^0)$$

$$\begin{aligned}\left(\frac{n-1}{\beta}, 1\right) = -\frac{1}{\beta} + \frac{\sqrt{n}}{\beta}\{&2\cos(2f' - b') + 2\cos(4f' - b' - b'' - b''') + \text{etc.}\\ &+ 2\cos((\tfrac{1}{2}\beta - 1)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-2)})\}\\ + i\frac{\sqrt{n}}{\beta}\{&2\sin f' + 2\sin(3f' - b' - b'') + \text{etc.}\\ &+ 2\sin((\tfrac{1}{2}\beta - 2)f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-3)})\\ &+ \sin(\tfrac{1}{2}\beta f' - b' - b'' - \text{etc.} - b^{(\frac{1}{2}\beta-1)})\}\end{aligned}$$

according as $\frac{1}{2}\beta$ is even or odd. What we noted above also holds for the remaining periods of $\frac{n-1}{\beta}$ terms. In general, therefore, we conclude that determining these periods requires the division of the whole circle into β parts, on which the construction of the angles b', b'', b''', etc., depends rationally; then the division of the angle $b' + b'' + b''' + \text{etc.}$ into β parts; and finally the square root $\sqrt{n}$. If we set $\beta = \frac{1}{2}(n-1)$ at once, those periods evidently coincide with twice the cosines of the angles $\frac{360^0}{n}$, $2\frac{360^0}{n}$, $3\frac{360^0}{n}$, etc., up to $\frac{1}{2}(n-1)\frac{360^0}{n}$, so that dividing the circle into n parts depends on dividing the whole circle into $\frac{1}{2}(n-1)$ parts, dividing an angle that can be constructed after that division into $\frac{1}{2}(n-1)$ parts, and the radical quantity $\sqrt{n}$. If we intend to proceed as far as the sines of the angles $\frac{360^0}{n}$, etc., one further operation will be needed.